

Anatomy of Lower Limb

- Introduction
- Bones of the lower limb
- Superficial structures of lower limb
- Thigh and gluteal region
- Popliteal fossa
- Leg
- Foot
- *Joints and ligaments of lower limb*

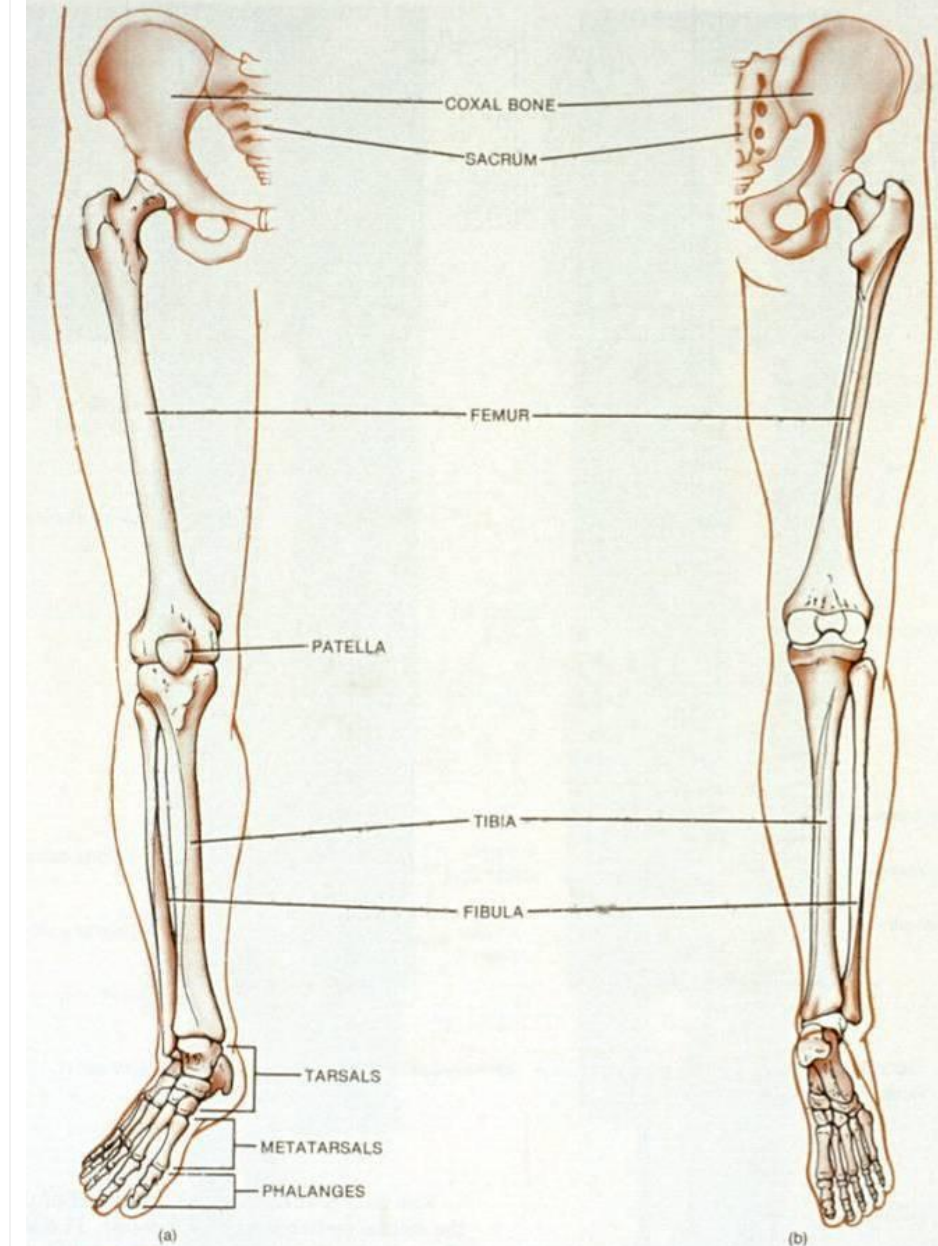


Regions

- **Gluteal region:** buttocks and hip region
- Thigh (femoral region)
- Knee region
- Leg
- Ankle
- Foot

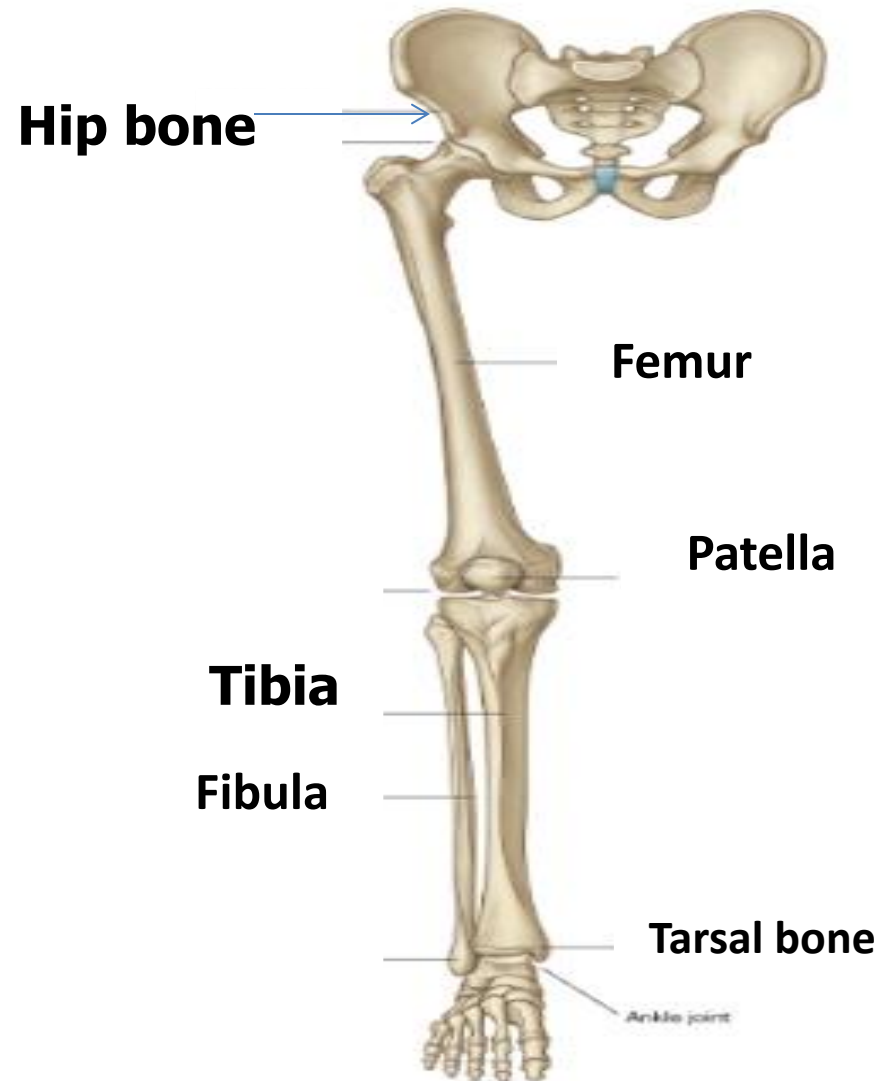
Function

- Locomotion
- Support body weight
- Maintain balance

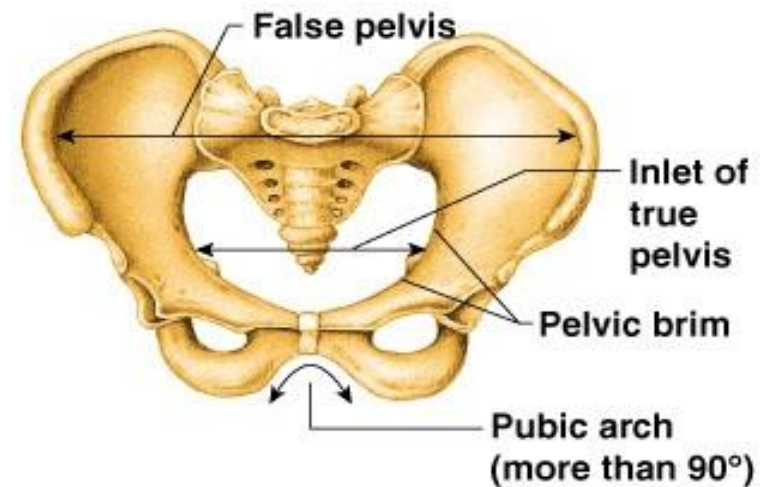
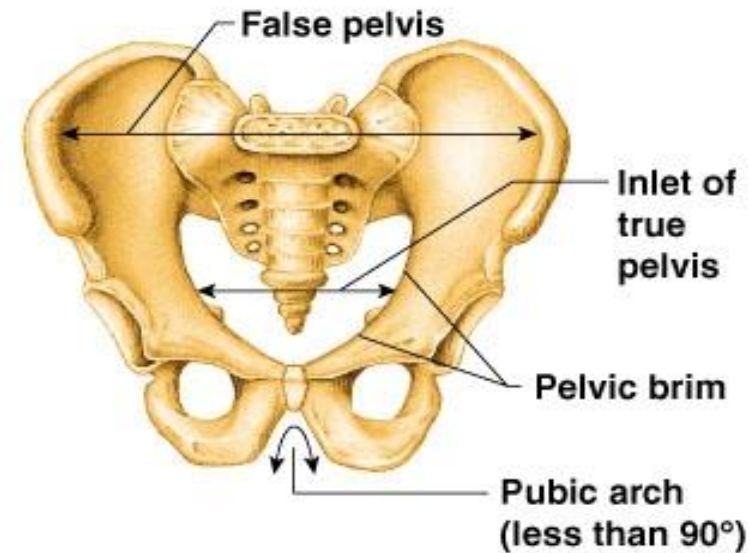
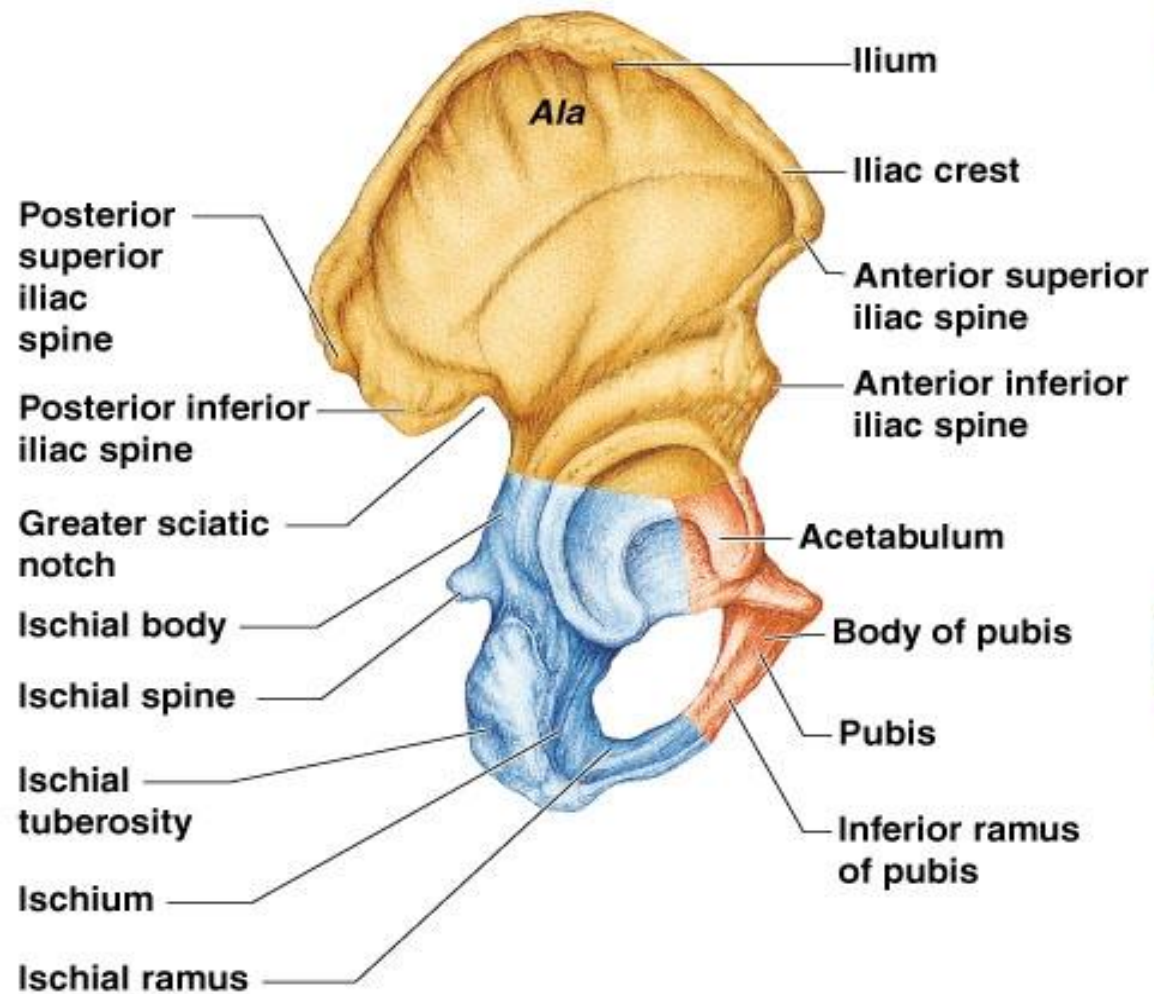


Bone of Lower limb

- **Hip region**
 - Hip bone
- **Thigh**
 - Femur
- **Knee**
 - Patella
- **Leg**
 - Tibia
 - Fibula
- **Foot**
 - 7 Tarsal bones
 - 5 Metatarsal bones
 - 14 Phalanges



Bone of the Lower Limb (Pelvic Girdle)



Femur

- The longest, heaviest & strongest bone in the body
- To support the erect bipedal posture, femurs are obliquely placed
- **Proximal end**

□ Head

- rounded
- covered with articular cartilage except a medial pit, **fovea** for ligament of head (ligamentum teres)
- **articulates with acetabulum**

□ Neck

- trapezoidal, narrow end supports head and broad base continuous with shaft

❑ Trochanters

- where neck joins shaft are 2 blunt elevations - Trochanters (greater & lesser)
- **intertrochanteric line (anteriorly) &**
- **intertrochanteric crest (posteriorly) in between**

❑ Shaft

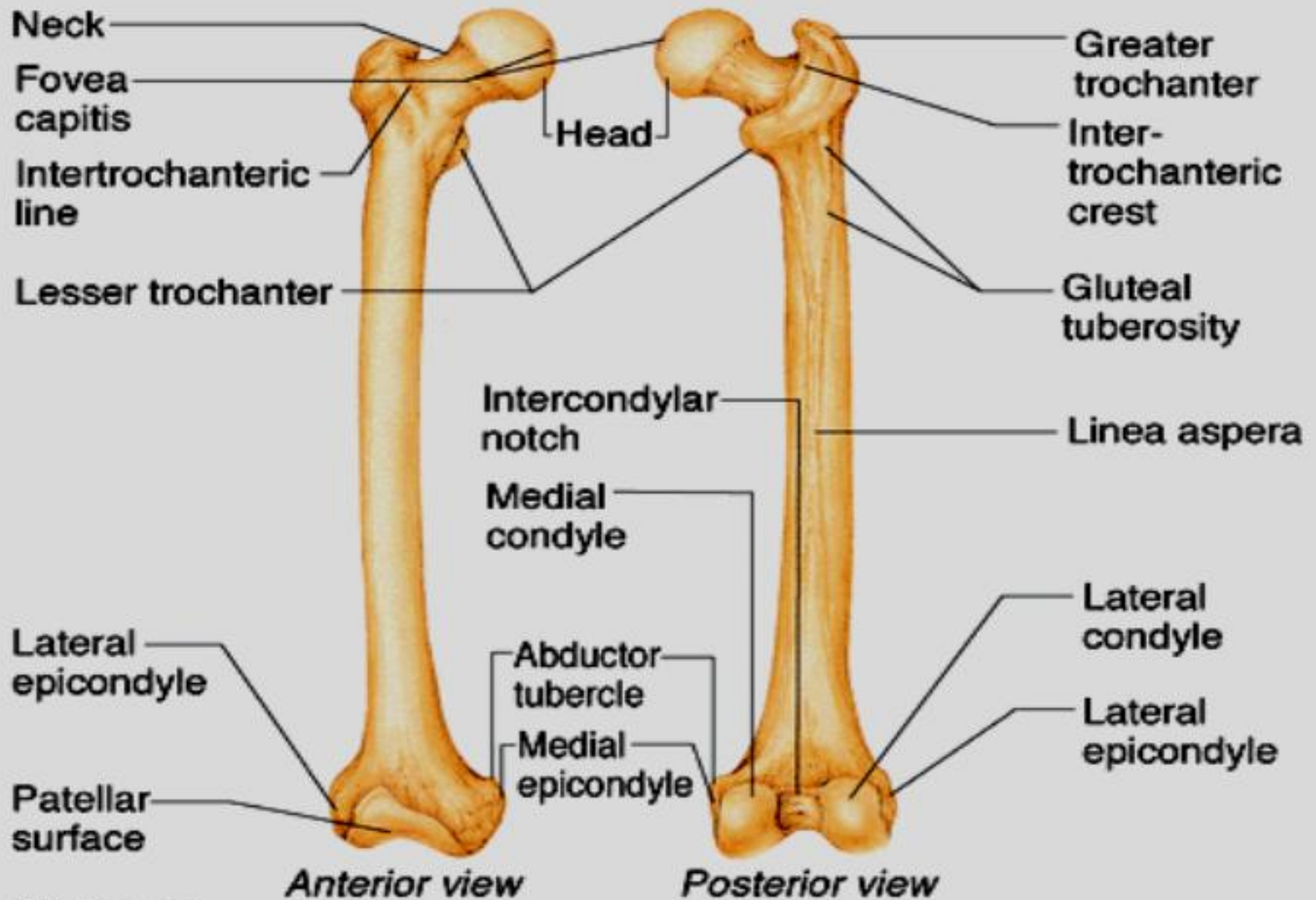
- possesses linea aspera; double edge ridge on posterior aspect

❑ Distal end

- **has medial & lateral condyles**
 - articulates with tibial condyle to form knee joint
- patellar surface for articulation with the patella

• Femoral fracture

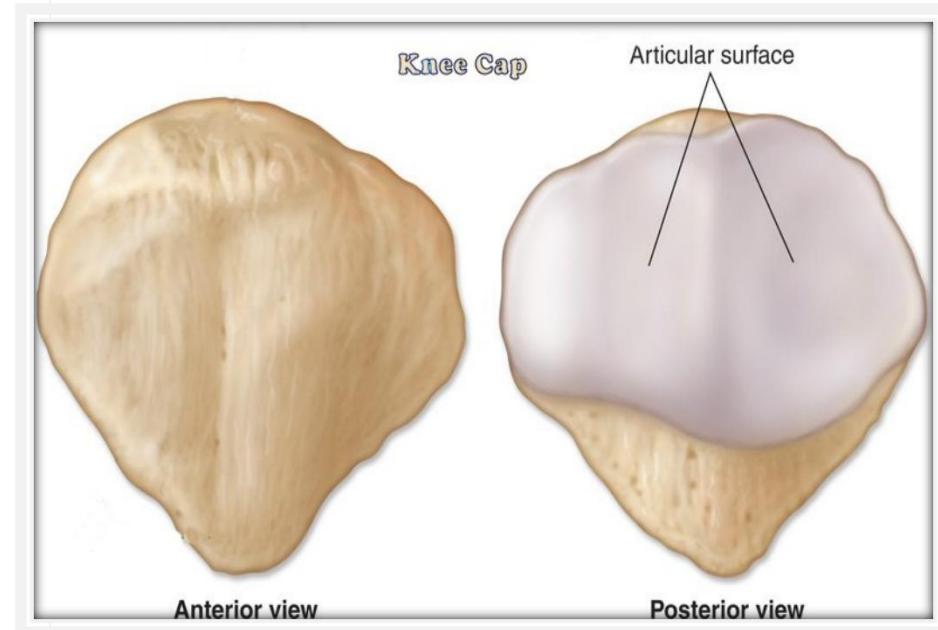
- neck is the most commonly fractured site



(b) Femur

Patella (knee cap)

- Large sesamoid bone, formed intratendinously
- Located anterior to knee joint
- Triangular in shape
- Articulate with patellar surface of femur
- **Anterior surface** is convex
- **Posterior surface** is concave = 2 (medial & lateral)
- **Base** – superior border
- **Lateral and medial border** converge to form **apex**

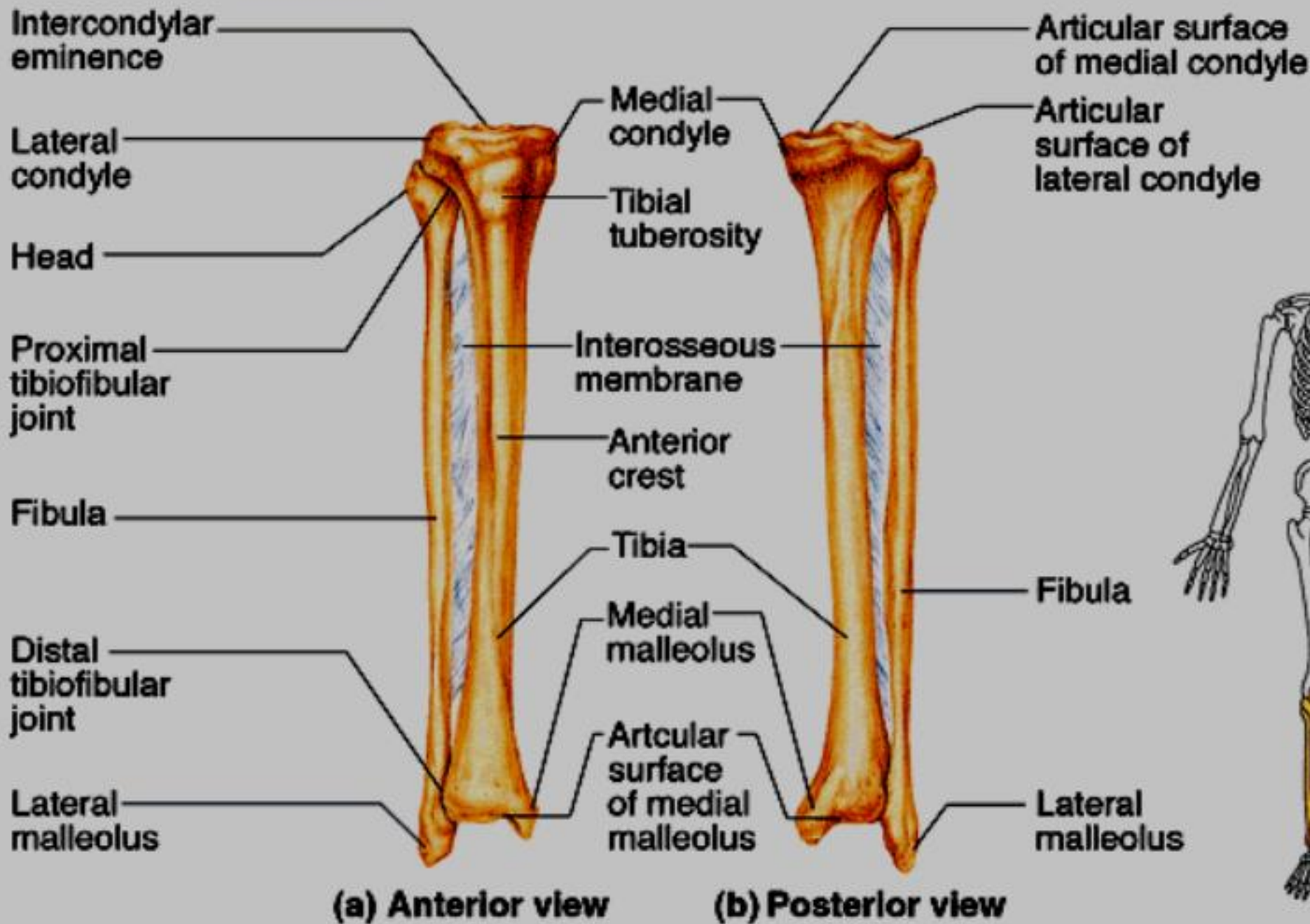


Tibia

- **Large, weight bearing**
- Articulate
 - proximally with condyles of femur at knee joint
 - distally with the talus of ankle
 - laterally at its proximal and distal ends with fibula
- has **medial & lateral condyles**
- **medial & lateral tibial tuberosity**
- **Medial malleolus** - inferiorly directed projection from medial side of distal end
- Shaft is narrowest at the junction of its **inferior and middle thirds**, which is the **most frequent site of fracture**

Fibula

- Long, slender bone on the lateral side of leg
- Serves mainly for **muscle attachment**
- **It doesn't bear body weight)**
- has head and neck
- Distal end enlarges to form **lateral malleolus**
- Fractures commonly occur just proximal to lateral malleolus and often are associated with dislocation of ankle joint
- Also at **neck of the fibula** is common site of fracture
- Fibula is a common source of **for bone grafting**



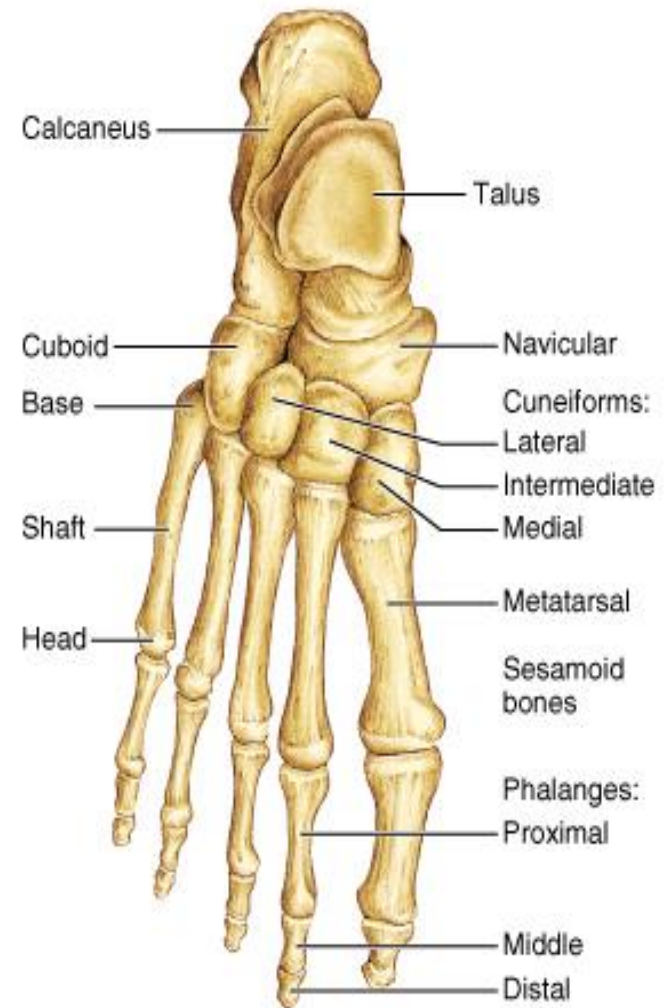
Bones of Foot

❑ Tarsal Bones

- **7 bones** which form the ankle joint; **talus, calcaneus, navicular, 3 cuneiforms & cuboid**
- The Only **talus** articulate with leg bones

❑ Calcaneus

- Heel bone
- Largest and strongest bone in the foot
- Articulate with talus and cuboid



(a) Superior view

Copyright © Pearson Education, Inc., publishing as Benjamin Cummings.

❑ Talus

- Rests on anterior 2/3 of calcaneus
- Superior surface, trochlea of talus, bears weight of the **body and articulates with two malleoli**

❑ Navicular

- Flattened, boat shaped
- Located between talar head and cuneiforms

❑ Cuboid

- Most lateral

❑ Cuneiforms: medial, intermediate and lateral

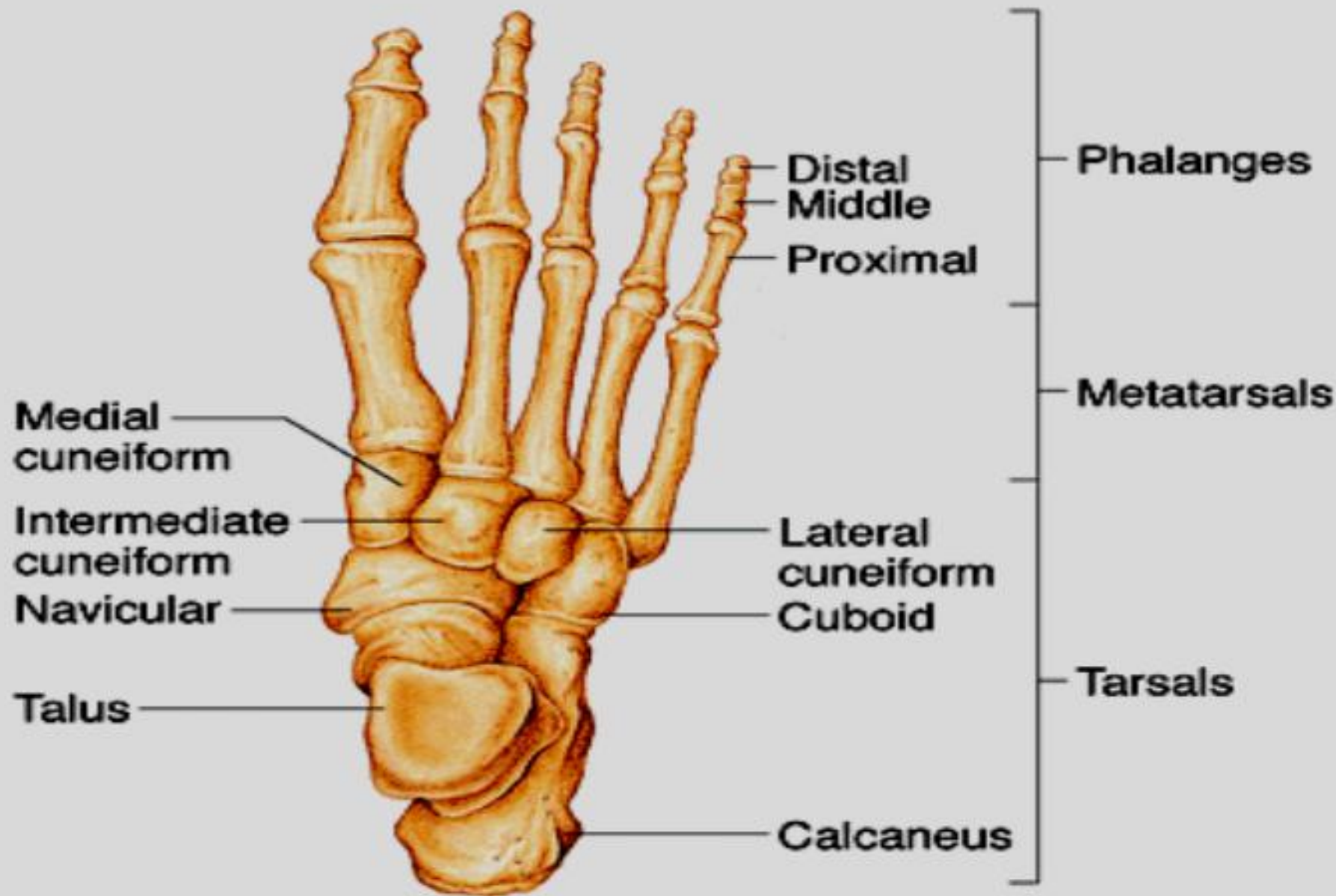
- Wedge shaped
- Each articulate with navicular posteriorly and metatarsal anteriorly

☐ Metatarsal Bones

- 5 bones which form middle part of the foot
- Each bone has base, body & head
- Bases articulate with cuneiform and cuboid bones
- Heads articulate with proximal phalanges

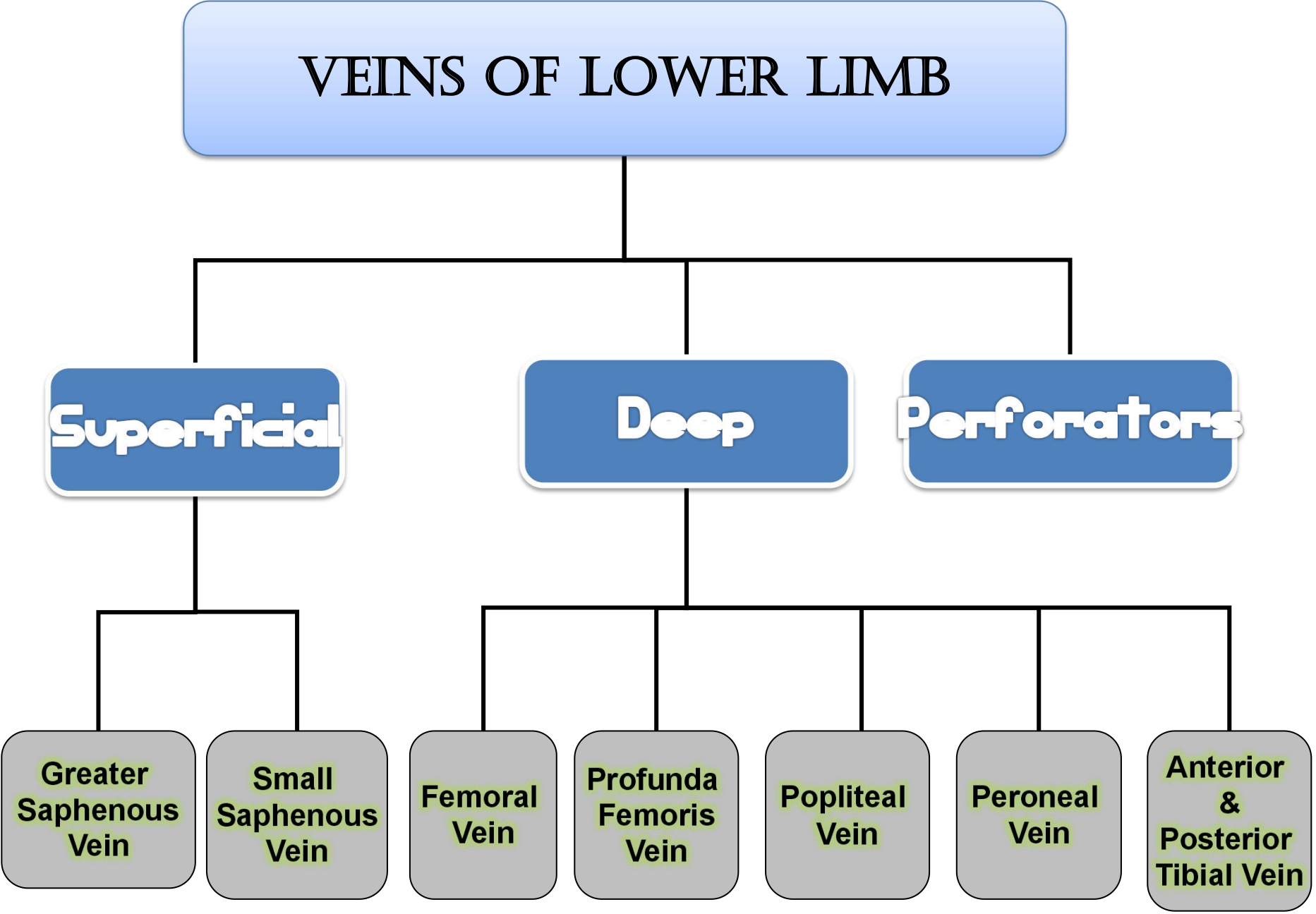
☐ Phalanges

- 14 bones
- Lateral four digits have proximal, middle & distal phalanx
- Great toe (hallux) has only proximal & distal phalanx
- Each phalanx has base, body & head



(a) Superior view

VEINS OF LOWER LIMB



```
graph TD; A[VEINS OF LOWER LIMB] --> B[Superficial]; A --> C[Deep]; A --> D[Perforators]; B --> E[Greater Saphenous Vein]; B --> F[Small Saphenous Vein]; C --> G[Femoral Vein]; C --> H[Profunda Femoris Vein]; C --> I[Popliteal Vein]; C --> J[Peroneal Vein]; C --> K[Anterior & Posterior Tibial Vein];
```

The diagram is a hierarchical flowchart titled 'VEINS OF LOWER LIMB'. The title is in a light blue box at the top. A vertical line descends from the title and branches into three horizontal lines leading to three blue boxes: 'Superficial', 'Deep', and 'Perforators'. From the 'Superficial' box, a vertical line descends and branches into two vertical lines leading to two grey boxes: 'Greater Saphenous Vein' and 'Small Saphenous Vein'. From the 'Deep' box, a vertical line descends and branches into five vertical lines leading to five grey boxes: 'Femoral Vein', 'Profunda Femoris Vein', 'Popliteal Vein', 'Peroneal Vein', and 'Anterior & Posterior Tibial Vein'. The 'Perforators' box has no further branches.

Superficial

Deep

Perforators

**Greater
Saphenous
Vein**

**Small
Saphenous
Vein**

**Femoral
Vein**

**Profunda
Femoris
Vein**

**Popliteal
Vein**

**Peroneal
Vein**

**Anterior
&
Posterior
Tibial Vein**

Superficial veins of the lower limb

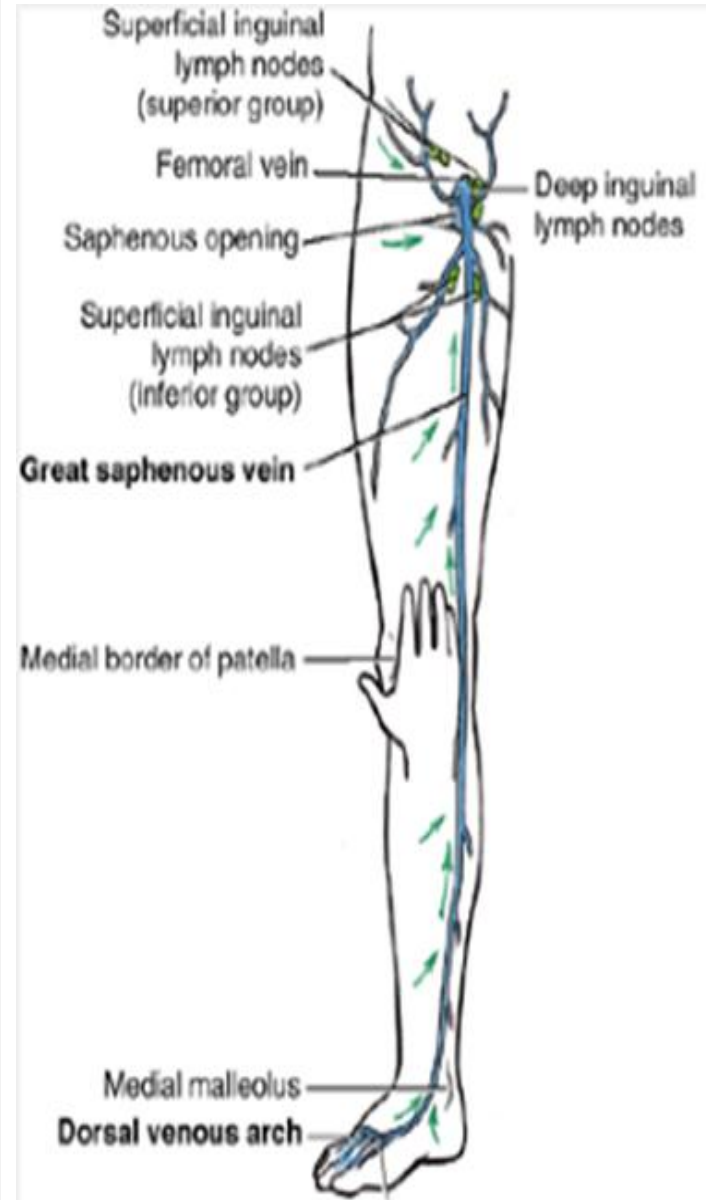
□ Great saphenous vein

• Origin

- It arises from the union of dorsal digital vein of great toe and dorsal venous arch on the dorsum of the foot

• Course

- runs anterior to the medial malleolus
- crosses the lower 1/3 of the medial surface of tibia
- runs 1 cm posterior to the medial border of the tibia and **inter in saphenous opening and join with femoral vein**



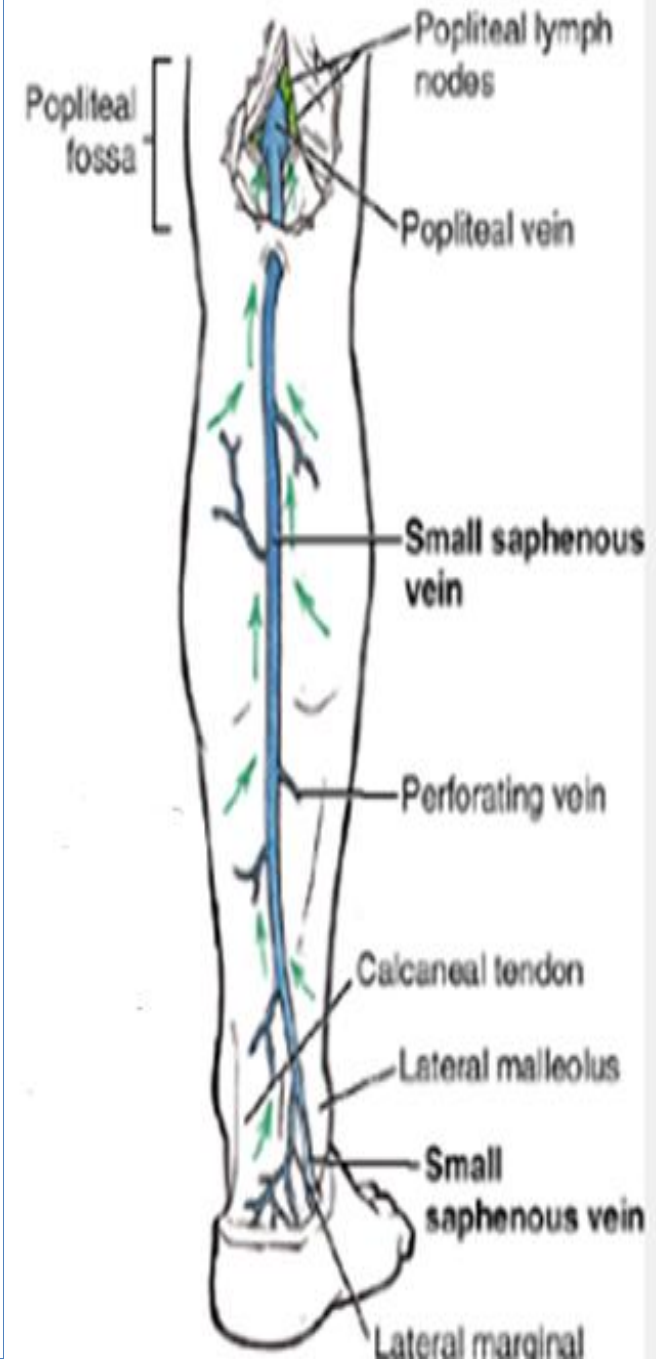
□ Small saphenous vein

• Origin

- Arises on lateral side of the foot from the union of dorsal digital vein of little toe with dorsal venous arch

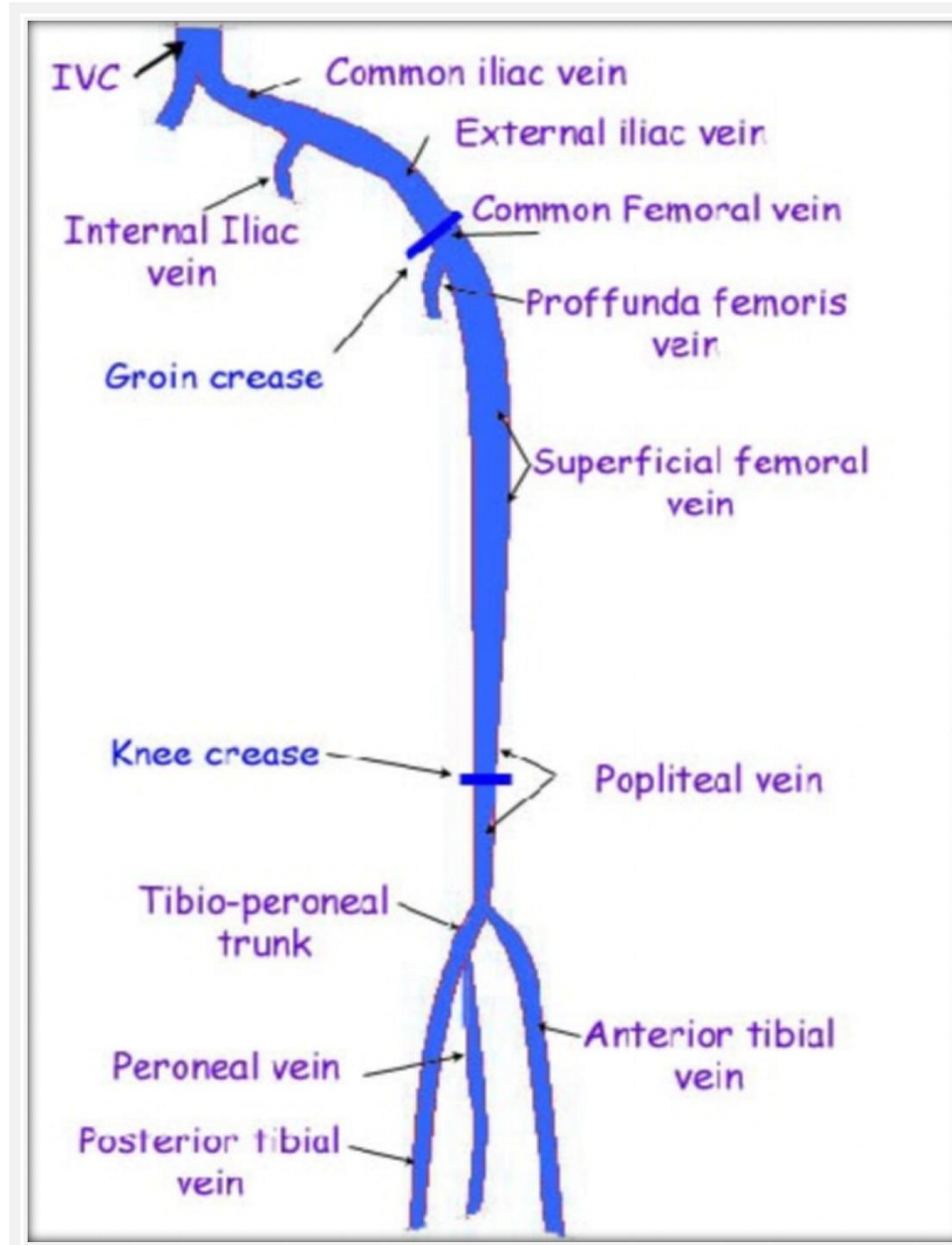
• Course

- Ascend posterior to lateral malleolus
- Pass along lateral border of **calcaneal tendon**
- Incline to midline of fibula and penetrates deep fascia
- Empties into **popliteal vein in popliteal fossa**



Deep veins of the lower limb

- **Accompany arteries**
- Usually paired, frequently interconnecting veins
- Contained within vascular sheath with the artery
 - **arterial pulsations help move the blood**
- In the leg they flow to **popliteal vein**, which become **femoral vein** in thigh which become **external iliac vein** in pelvis



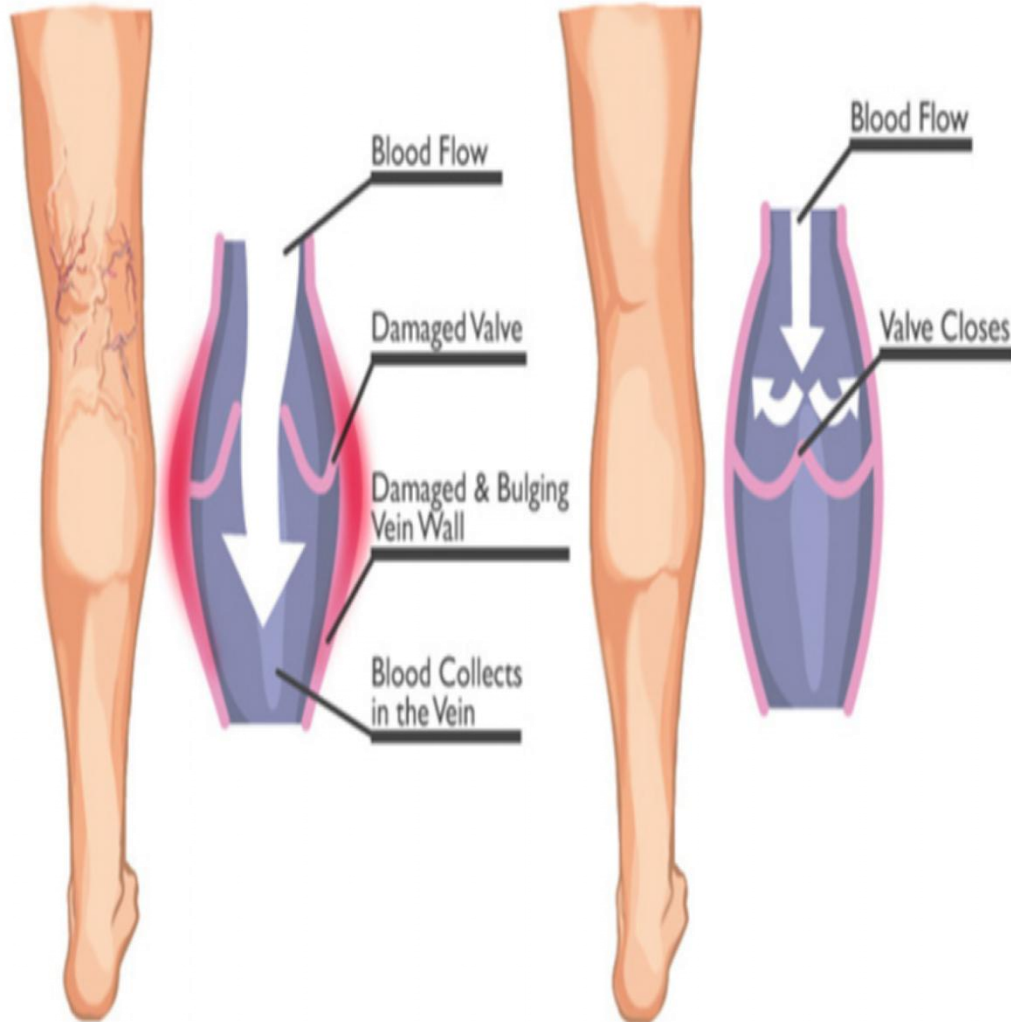
□ Varicose veins

- Normal flow of blood in the venous system depends upon the presence of competent valves, which prevent reflux.
- Venous return is supplemented with contraction of the muscles in the lower limb, which pump the blood toward the heart.
- When venous valves become incompetent they tend to place extra pressure on more distal valves
- This condition produces **dilated tortuous superficial veins (varicose veins)**.
- Some individuals have a genetic predisposition to developing varicose veins.

Varicose veins

VARICOSE VEINS

HEALTHY VEINS

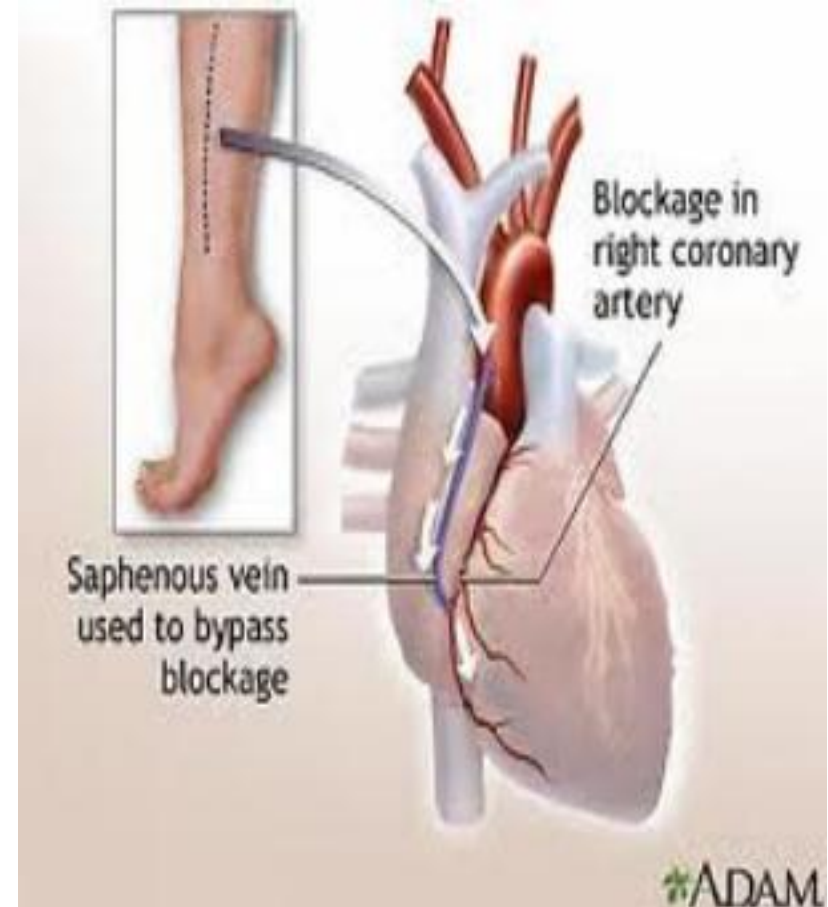


Great Saphenous Vein Cut Down

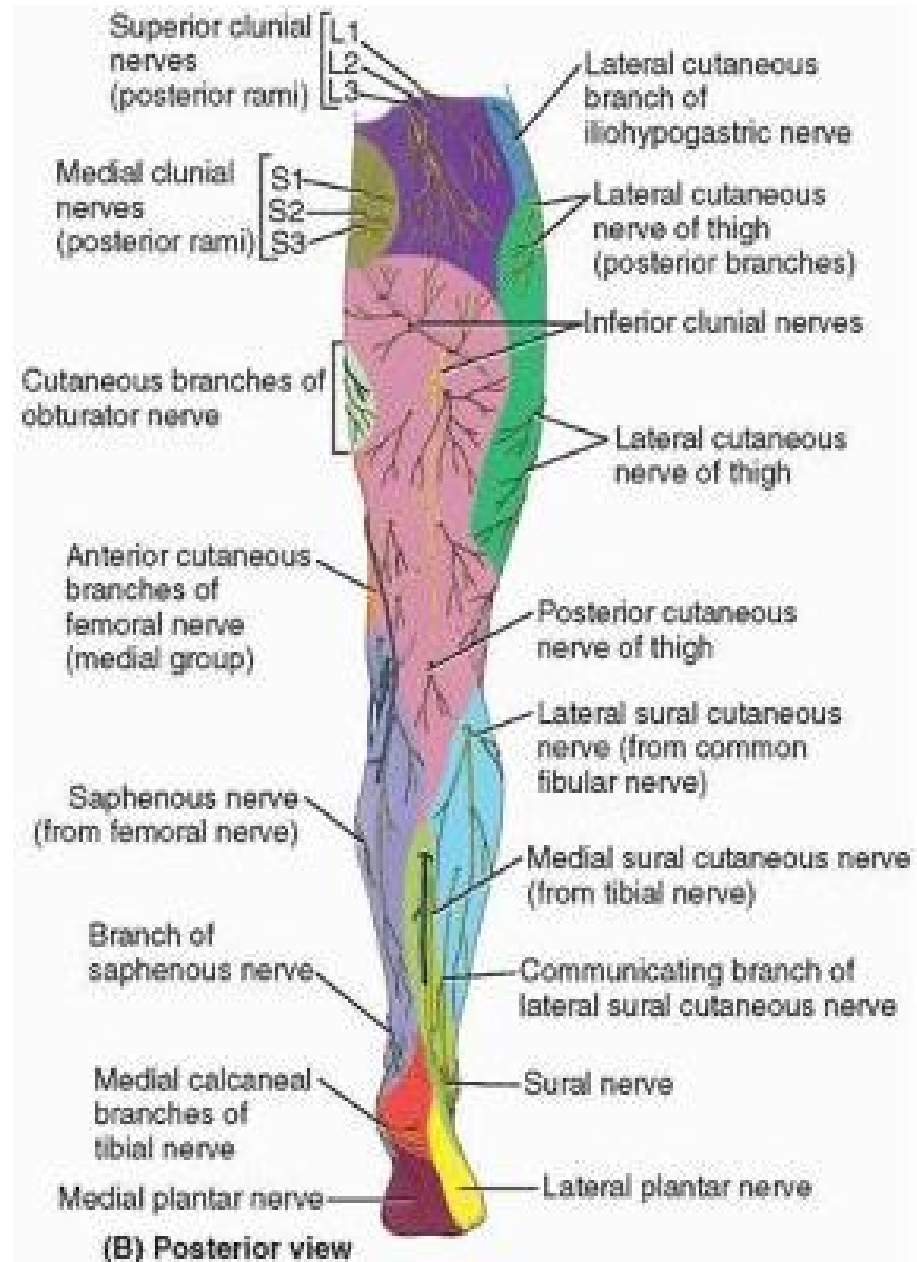
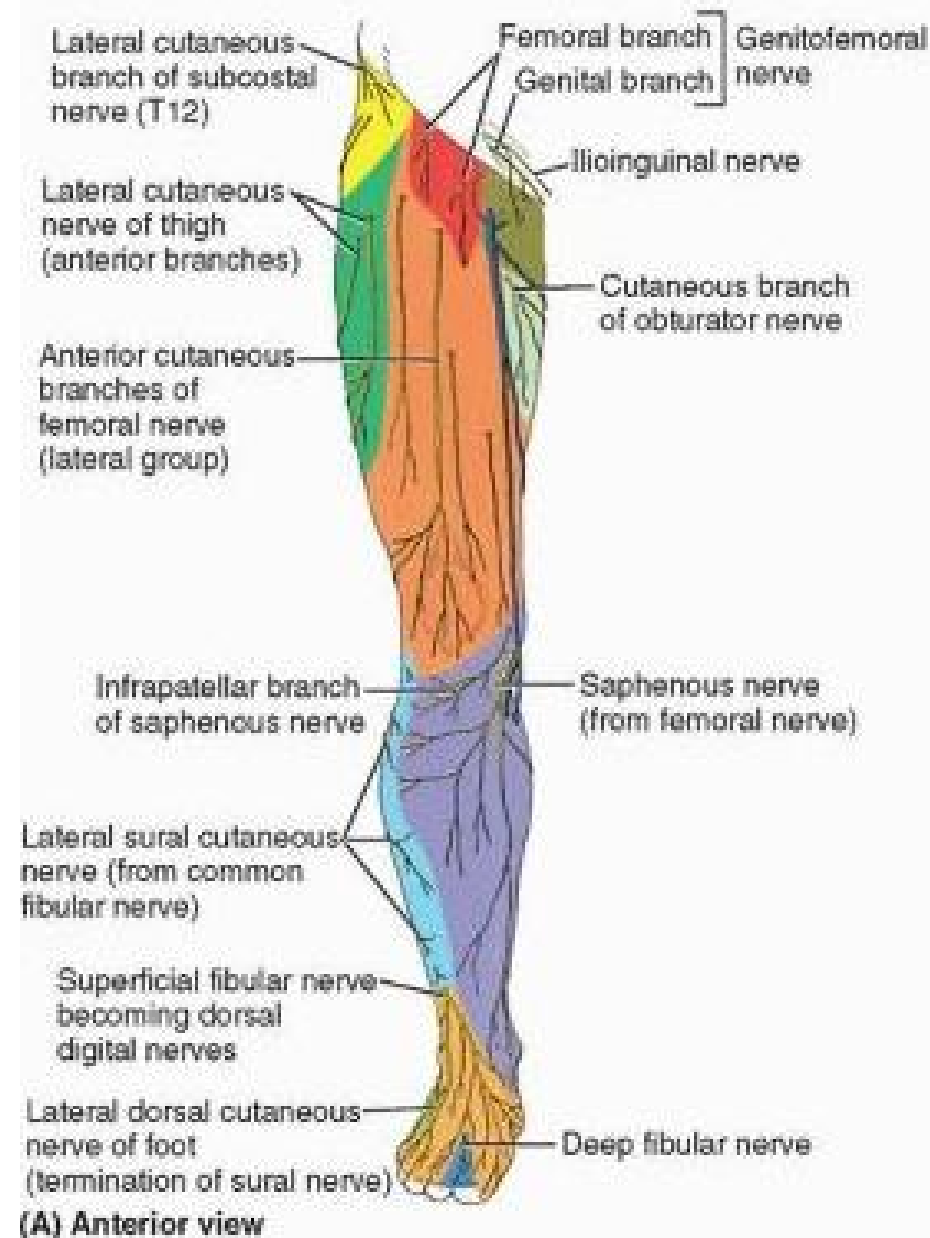
- Usually performed at the ankle. Thus, it would be called Ankle Vein Cut Down.
- The saphenous nerve branches are blocked with anaesthesia.
- The great saphenous vein can also be entered through the groin – Groin Vein Cut Down
- The branches of Ilioinguinal nerve and intermediate cutaneous nerve of the thigh

Great Saphenous Vein in Coronary Bypass Surgery

- In patients with occlusive coronary disease caused by atherosclerosis, the diseased arterial segment can be bypassed by inserting a graft consisting of a portion of Great saphenous vein.
- Vein segment is reversed to prevent obstruction by valves

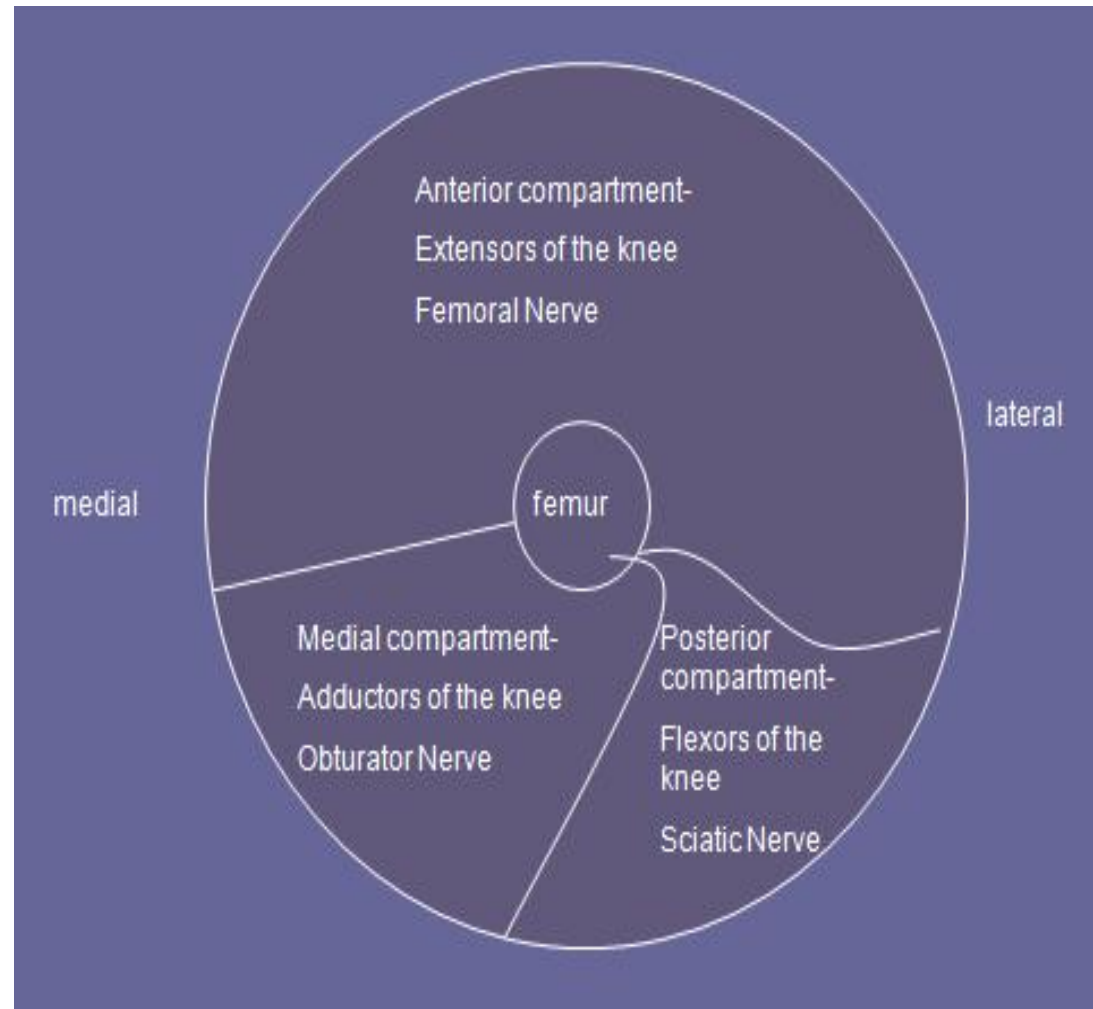


Cutaneous innervation of lower limb



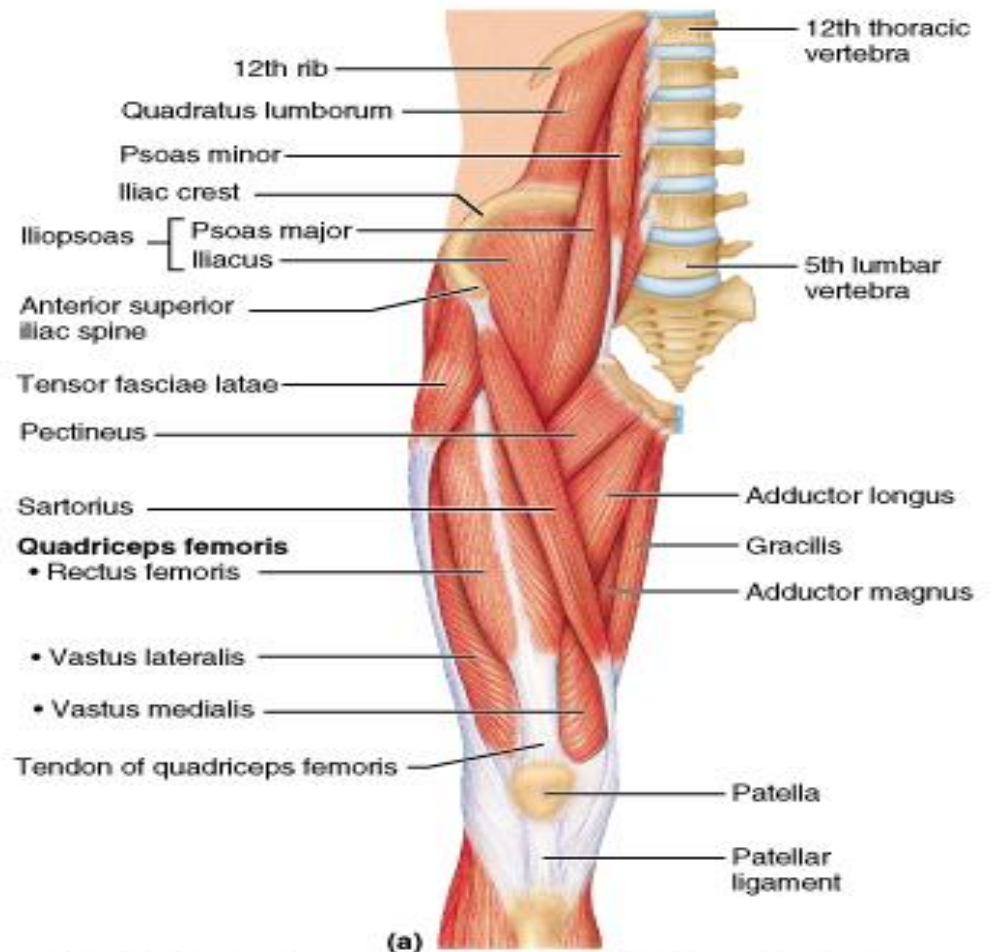
Thigh and gluteal region

- Thigh muscles are organized into 3 compartments by intermuscular septa
 - **Anterior or extensor**
 - **Medial or adductor**
 - **Posterior or flexor**



Anterior thigh muscles

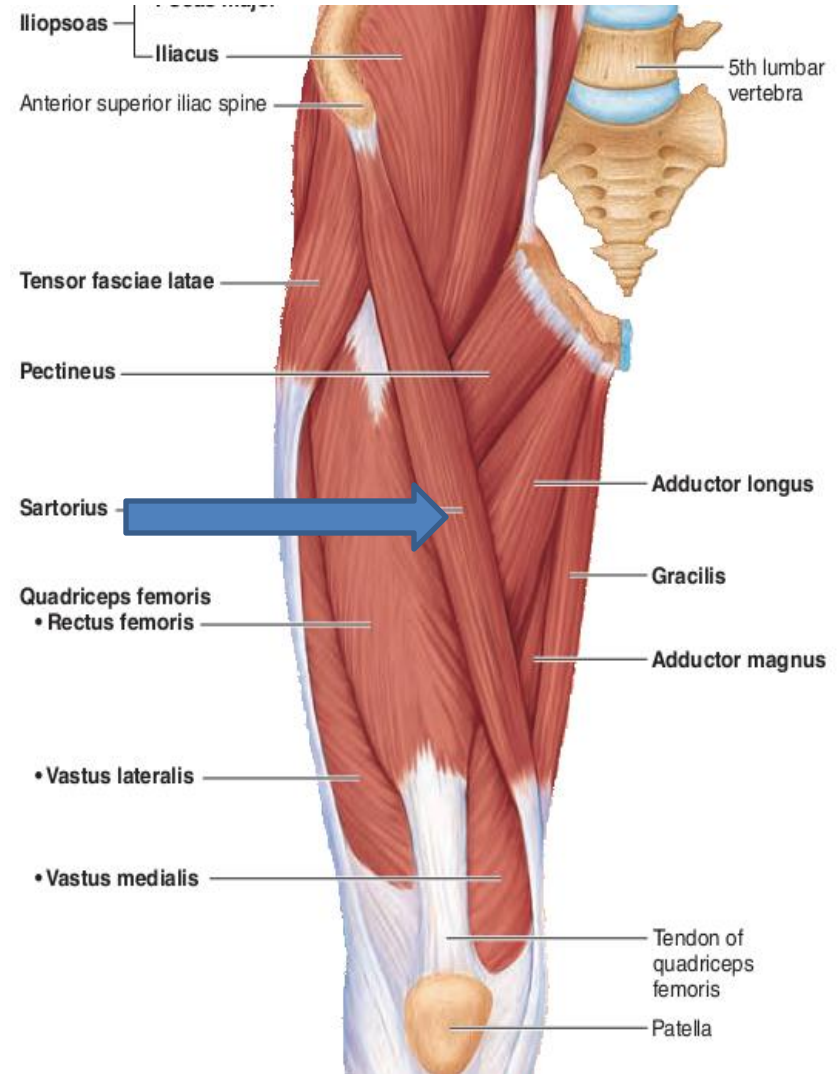
- Found in anterior compartment of thigh
- ❖ Flexors of hip and extensors of knee
- Sartorius
- Quadriceps femoris
 - ✓ Rectus femoris
 - ✓ Vastus medialis
 - ✓ Vastus lateralis
 - ✓ Vastus intermedius



Copyright © 2001 Benjamin Cummings, an imprint of Addison Wesley Longman, Inc.

Sartorius

- Long, ribbon-like muscle
- The most superficial muscle in the anterior thigh
- Passes obliquely (lateral to medial) across the superoanterior part of the thigh
- Innervation
 - **femoral nerve**
- **Action**
 - flexes, abducts and laterally rotates thigh at hip joint
 - flexes leg at knee joint
 - when acting bilaterally, the muscles bring the lower limbs into the cross-legged sitting position; "**tailor's muscle**"

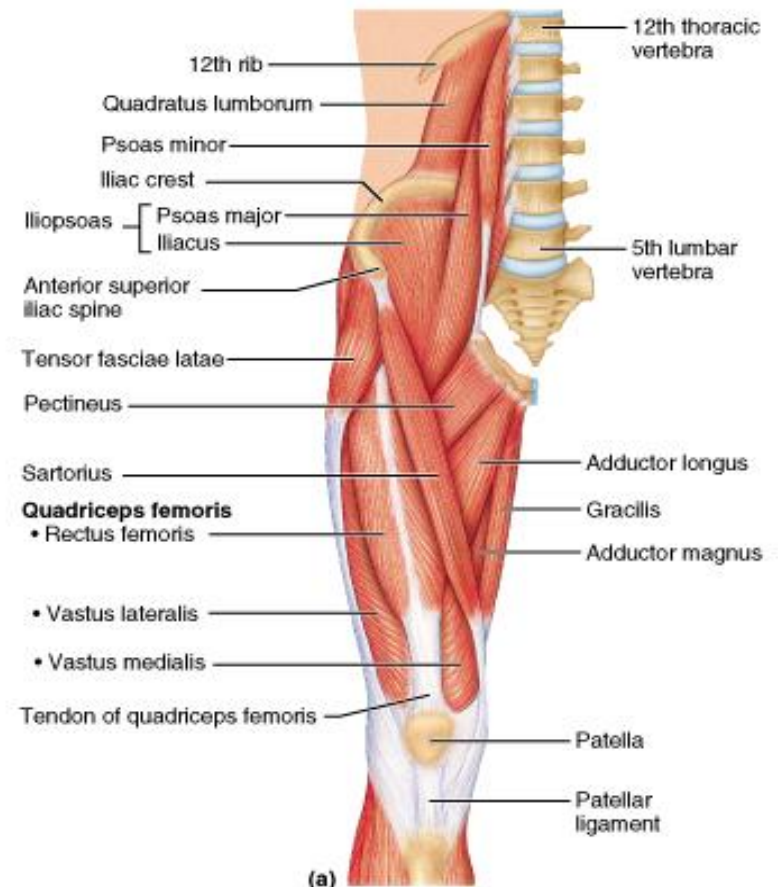


Quadriceps femoris

- Four headed muscles
- Forms the main bulk of the anterior thigh

The four parts

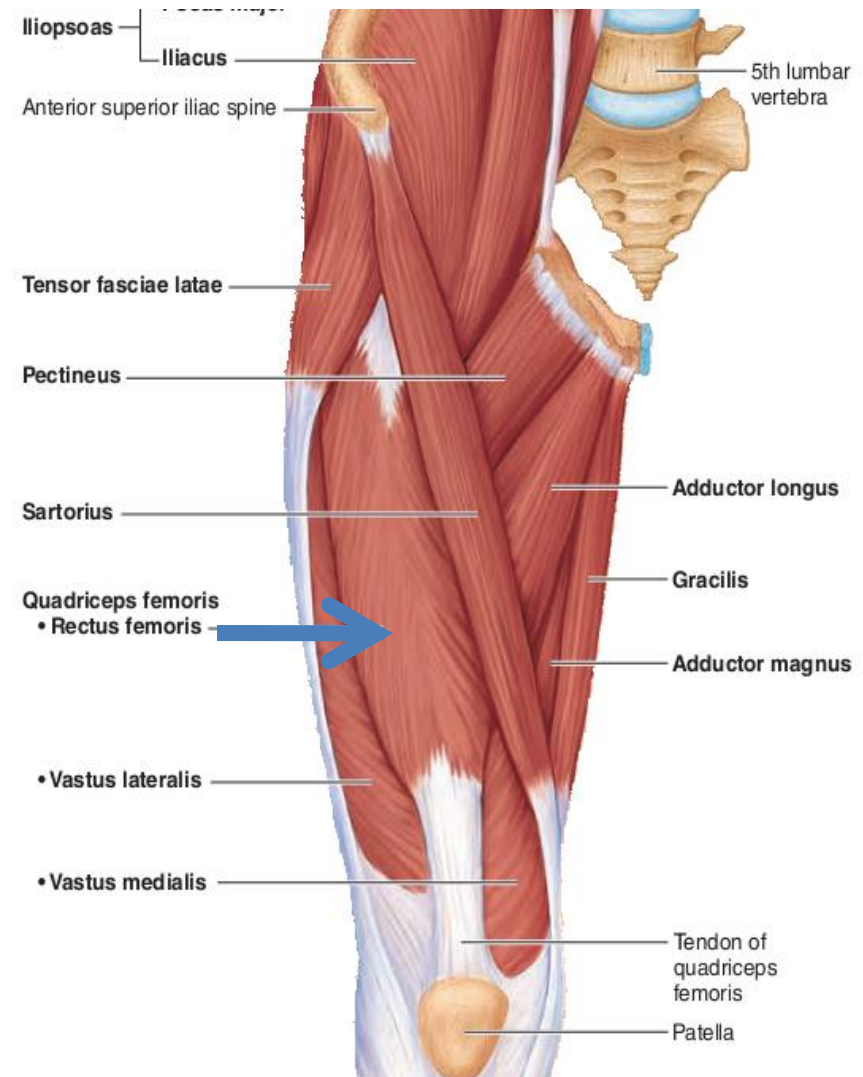
- **Rectus femoris: anterior**
- **Vastus lateralis: lateral**
- **Vastus medialis: medial**
- **Vastus intermedius: deep to rectus femoris between vastus lateralis and medialis**
- **All insert into tibial tuberosity by a common distal tendon - quadriceps tendon**



Copyright © 2001 Benjamin Cummings, an imprint of Addison Wesley Longman, Inc.

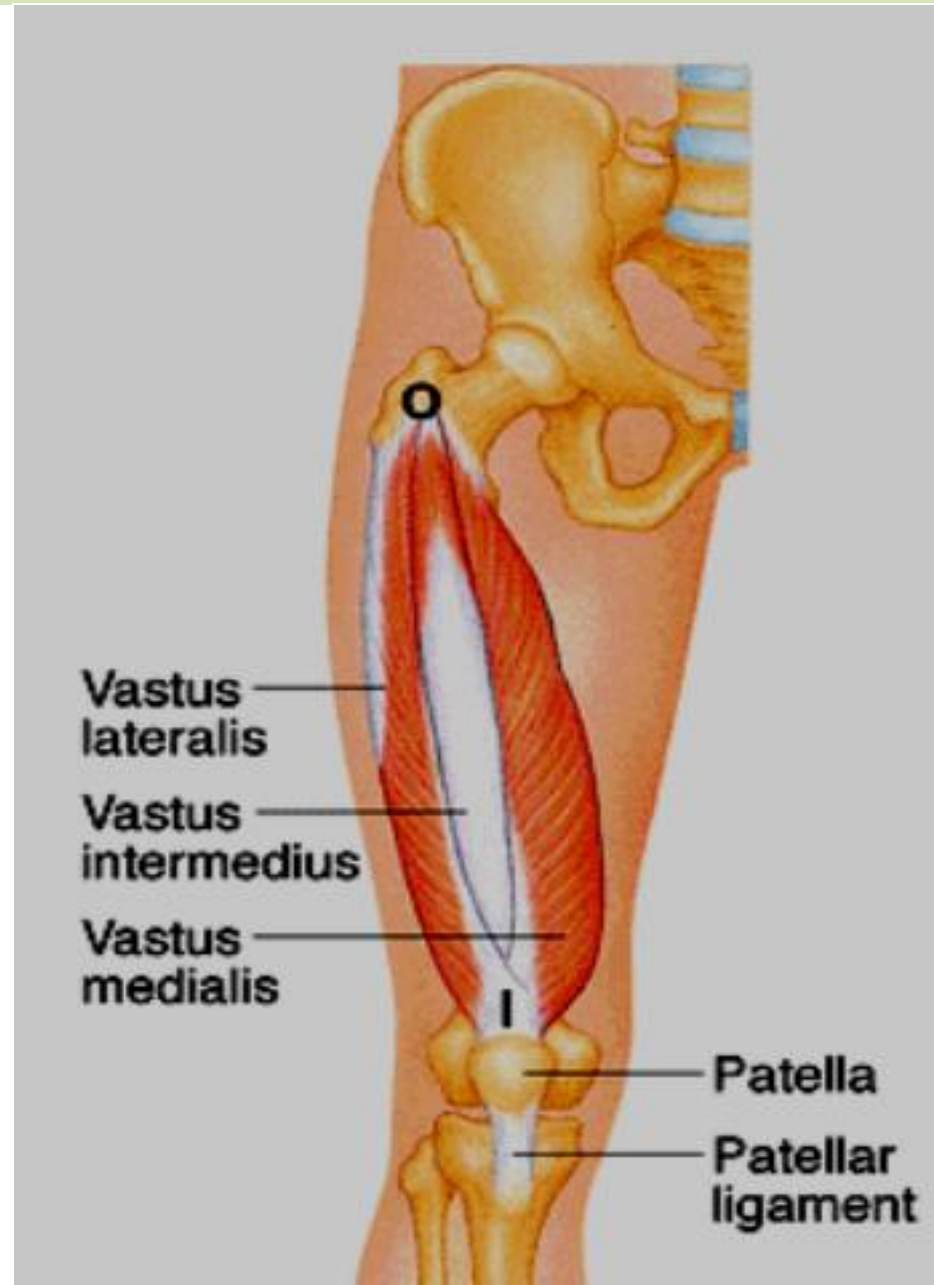
Rectus femoris

- Origin - anterior inferior iliac spine (by its straight head) and acetabular margin (by its reflected head)
 - Thus, it can also flex the hip joint
- Insertion - patella and tibial tuberosity
- **Action** - extends knee and flexes thigh at hip
- **Innervation** - femoral nerve



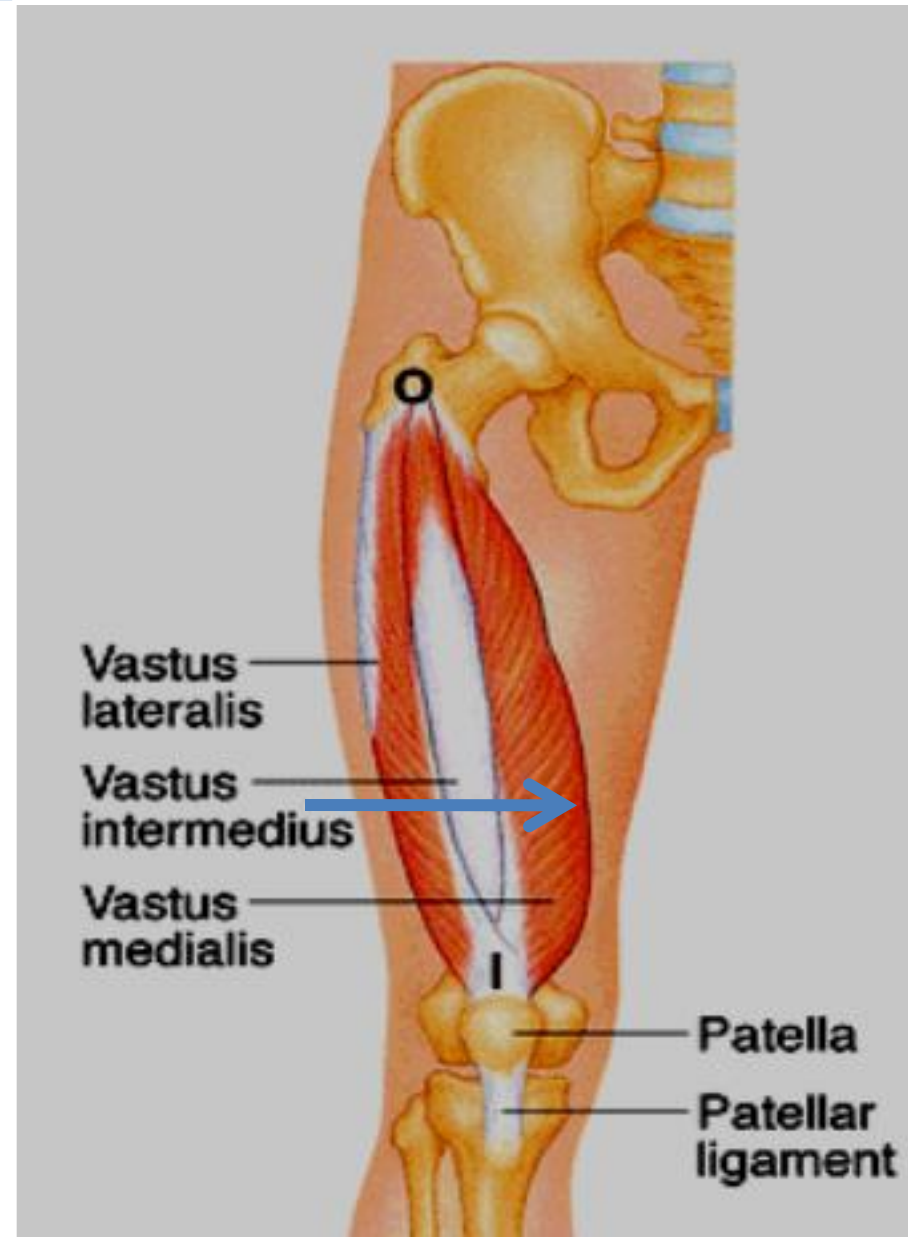
Vastus lateralis

- Origin - lateral lip of the linea aspera; greater trochanter, intertrochanteric line
- Insertion - patella and tibial tuberosity
- Action - extends knee
- **Used for intramuscular injection**
- **Innervation - femoral nerve**



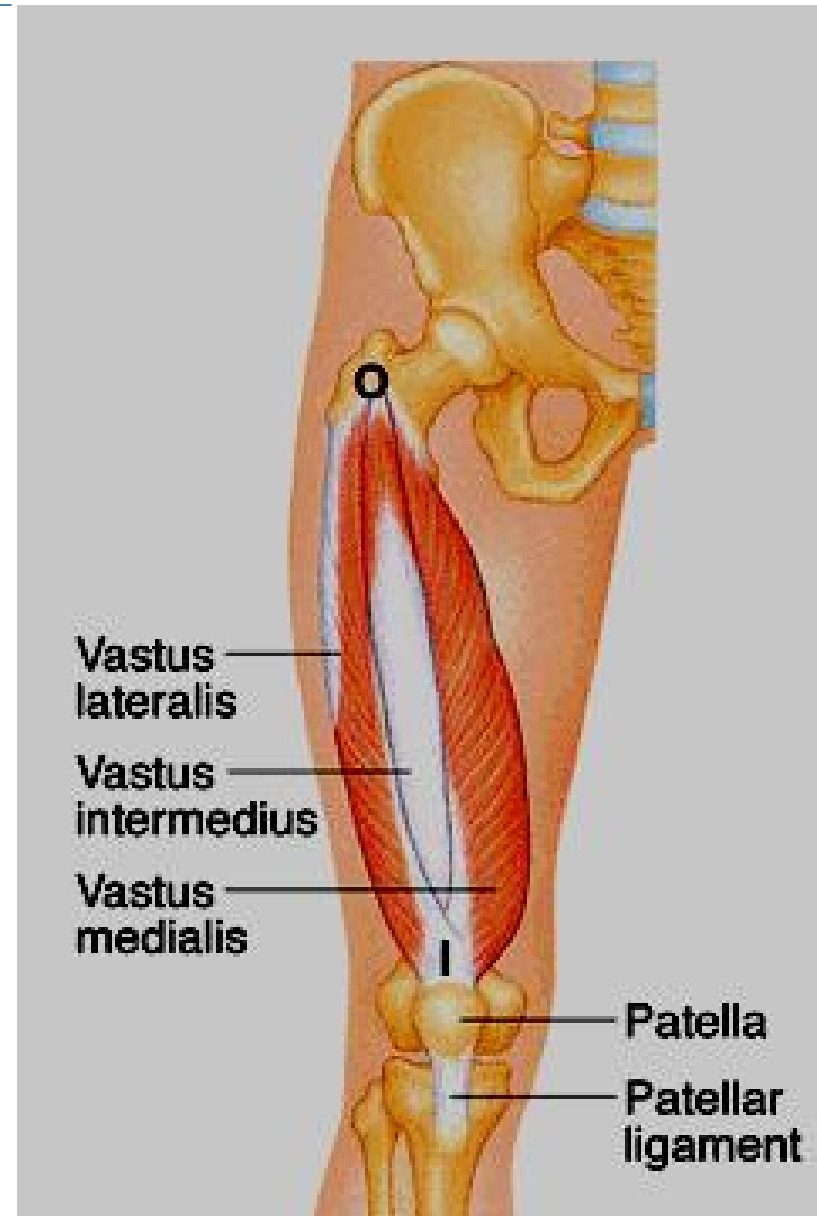
Vastus medialis

- Origin - medial lip of the linea aspera; intertrochanteric line
- Insertion - patella and tibial tuberosity
- Action - extends knee
- **Innervation - femoral nerve**



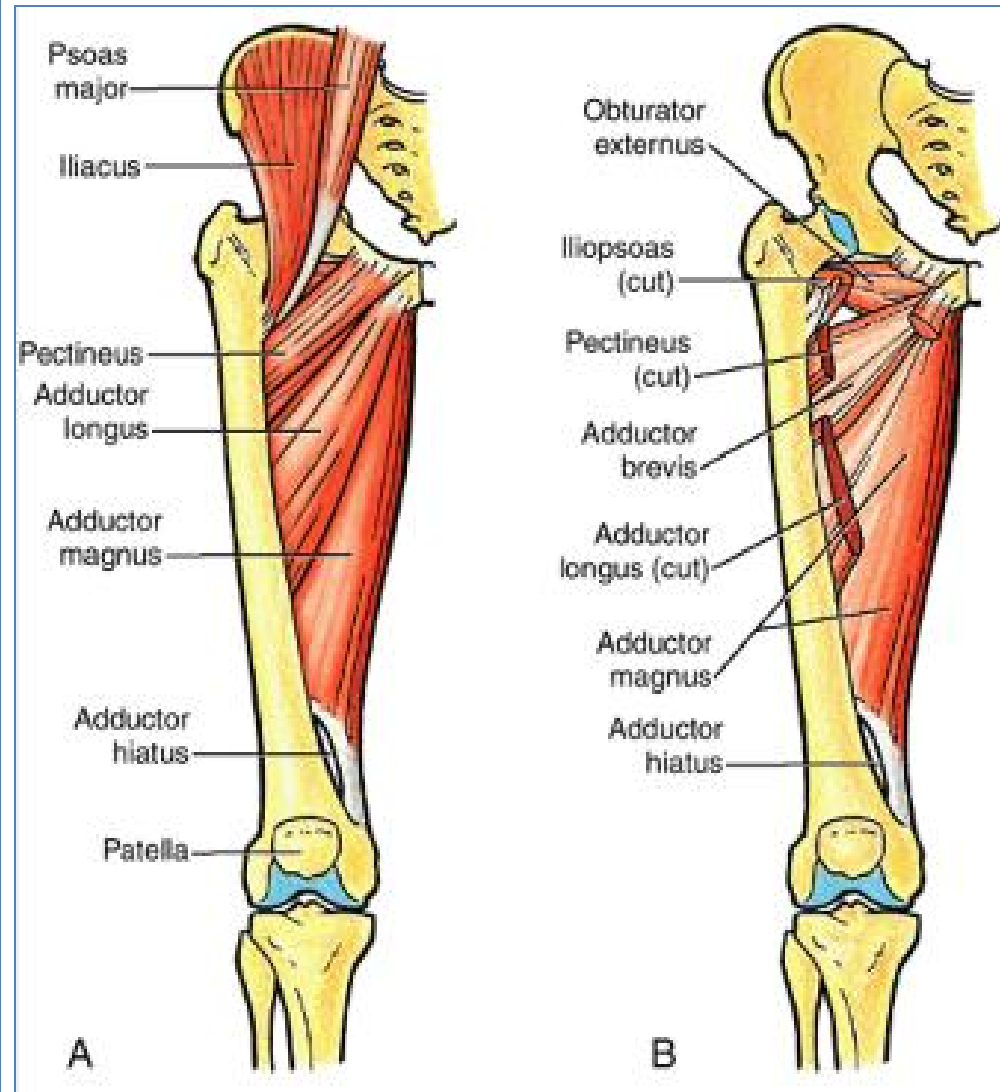
Vastus intermedius

- **Origin** - anterior and lateral surfaces of proximal shaft of femur
 - It is intermedius b/n Vastus medialis and vastus lateralis
- **Insertion** - patella and tibial tuberosity
- **Action** - extend knee
- **Innervation** - **femoral nerve**



Muscles of the Hip

- Medial muscles of the hip
 - Psoas major and minor
 - Iliacus
- Iliacus
- Origin - iliac fossa
- Insertion - lesser trochanter of femur
- Action - hip flexion
- **Innervation - femoral branch in the abdomen**



Psoas major

- Origin - transverse processes and bodies of T12 and all lumbar vertebrae
- Insertion - lesser trochanter of femur
- Action – **chief hip flexion**
- Innervation - **branches from lumbar plexus**



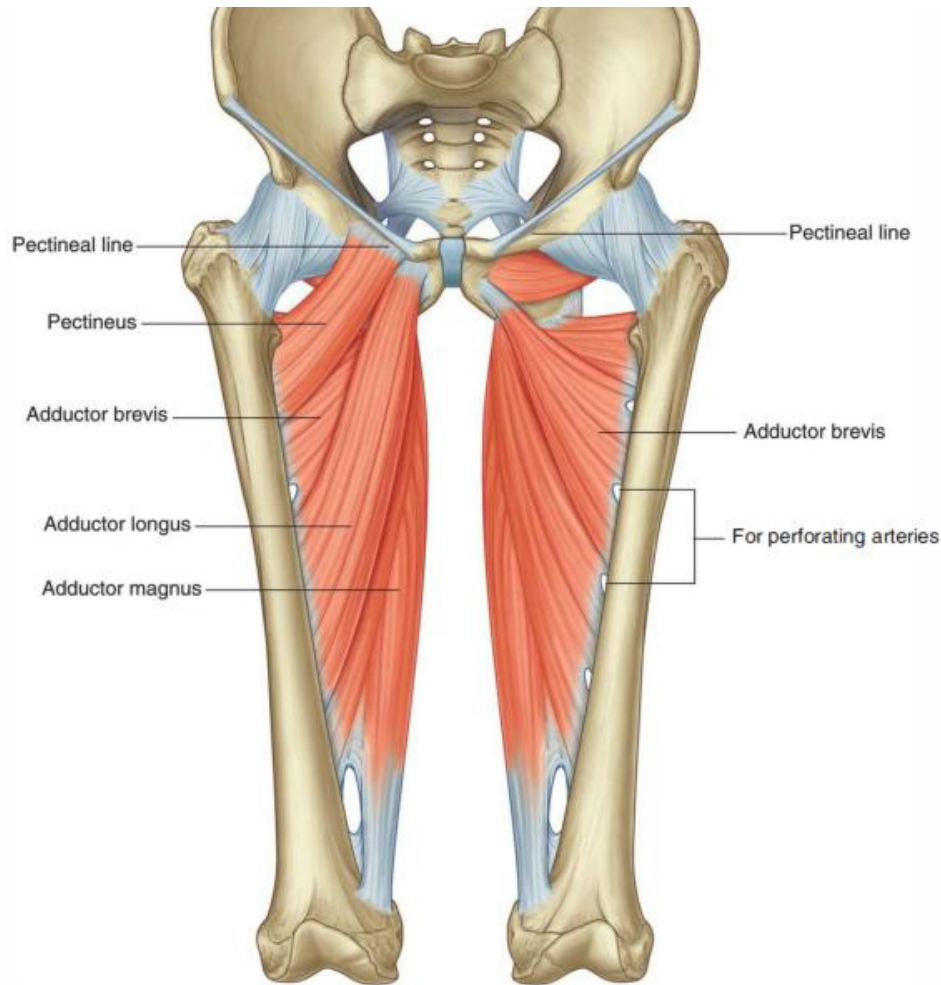
Psoas minor

- Origin - transverse processes and bodies of T12 and L1
- **It may absent**
- Insertion - rim of acetabulum
- Action - flexes lumbar vertebrae



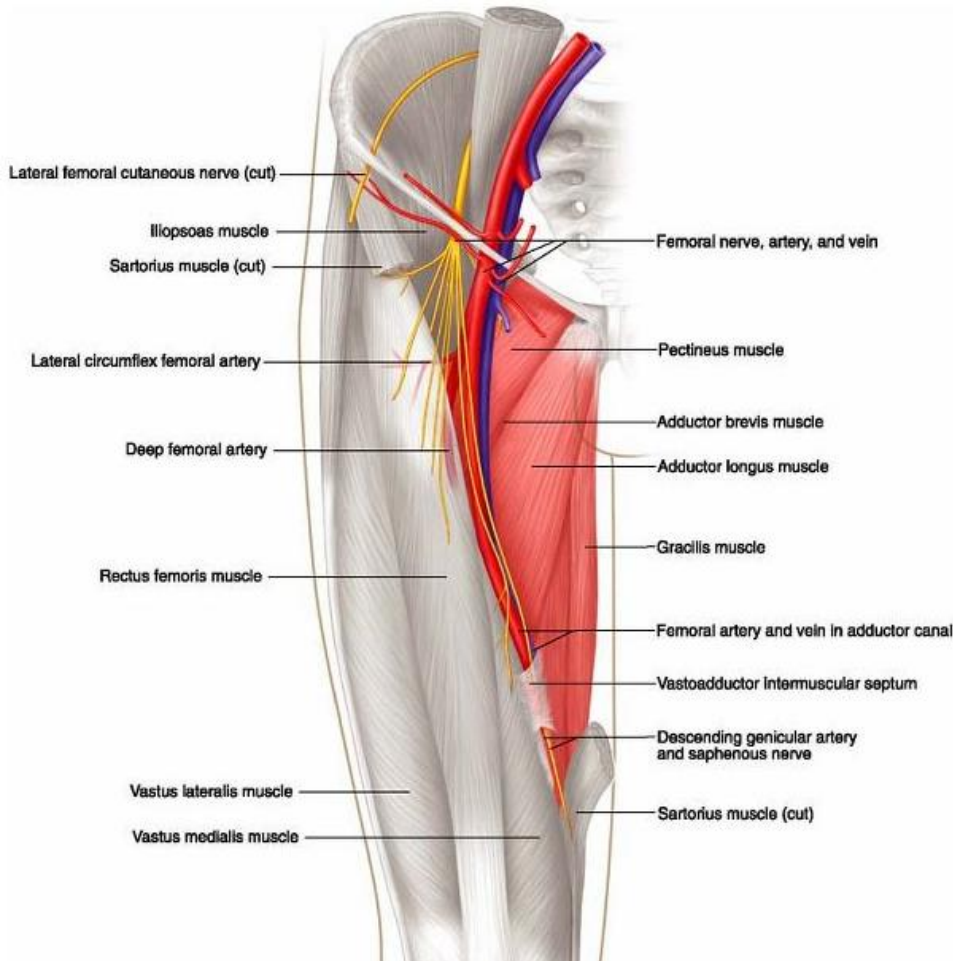
Medial thigh muscles

- Collectively called **adductor group**
- Found in medial compartment
- **Adductor muscles are**
 - ❖ Adductor magnus
 - ❖ Adductor longus
 - ❖ **Adductor brevis**
 - ❖ **Pectineus(by two)**
 - ❖ **Gracilis**
 - ❖ **Obturator externus**
 - **All innervated by obturator nerve**



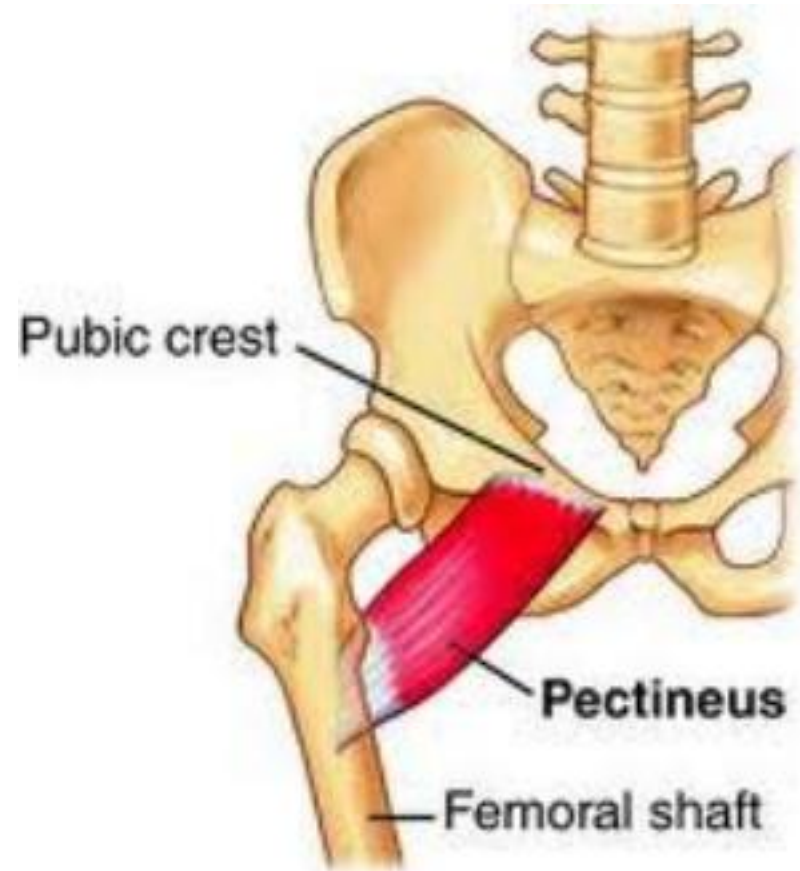
Medial thigh muscles

- All are supplied by **obturator nerve** except **pectineus** and **hamstring part of adductor magnus**
- **Three layers**
 - Superficial - pectineus, gracilis and adductor longus
 - Middle - adductor brevis
 - Deep - adductor magnus & obturator externus



Pectineus

- Origin - pectineal line of pubis
- Insertion - lesser trochanter of femur
- Action - adducts, flexes, and laterally rotates thigh
- **It has dual innervation**
- Innervation - **femoral nerve & obturator nerve**



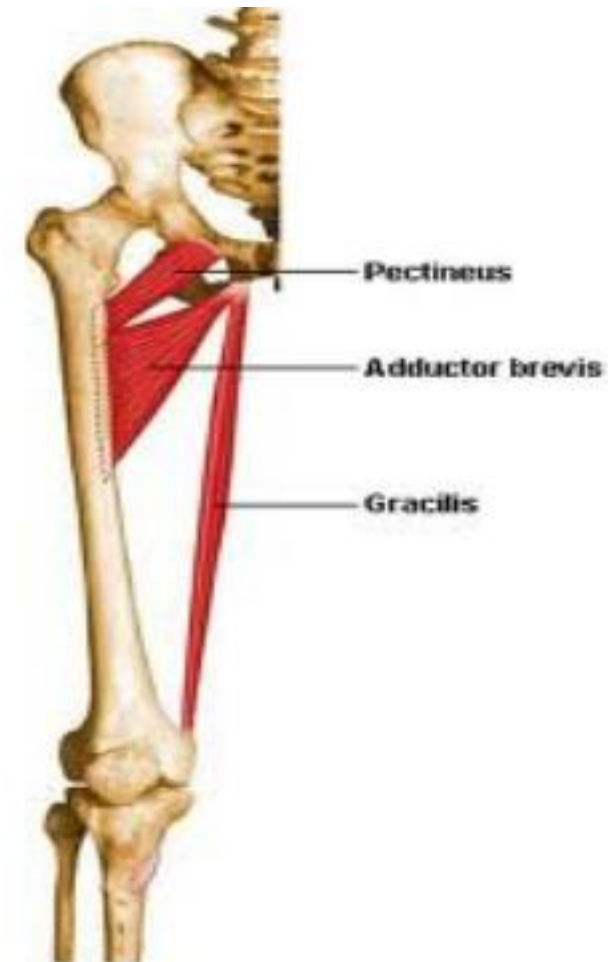
Adductor longus

- Origin - pubis near pubic symphysis
- Insertion - linea aspera medial lip of middle 1/3
- Action - adducts thigh
- Innervation - **Obturator nerve (anterior branch)**



Gracilis

- Origin - inferior ramus and body of pubis
- Insertion - medial surface of tibia just inferior to medial condyle
- Action - adducts the thigh, flexes and medially rotates the leg
- It is important for **grafting of muscles**
- Innervation - **anterior branch of obturator nerve**



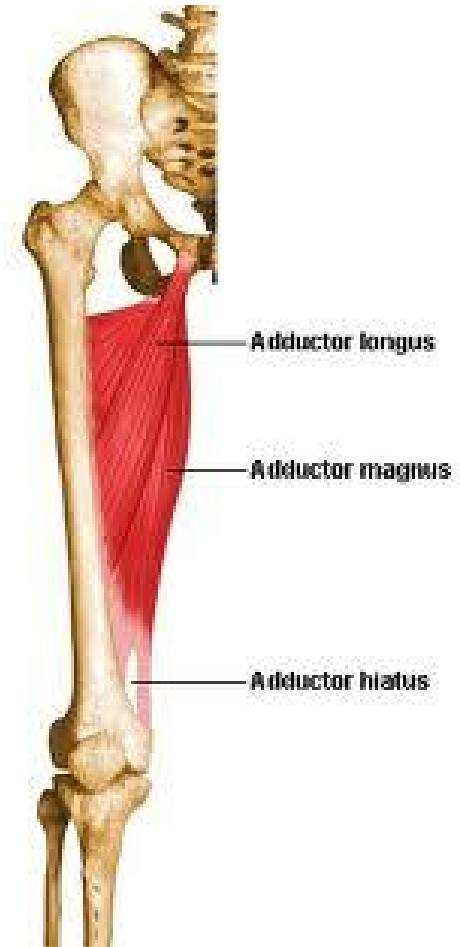
Adductor brevis

- Origin - body and inferior ramus of pubis
- Insertion - **linea aspera** (medial lip)
- Action – adducts, laterally rotates, and flex (weak) the thigh
- **It is important land mark for divide the obturator nerve in to anterior & posterior division**
- Innervation - **anterior branch of obturator nerve**



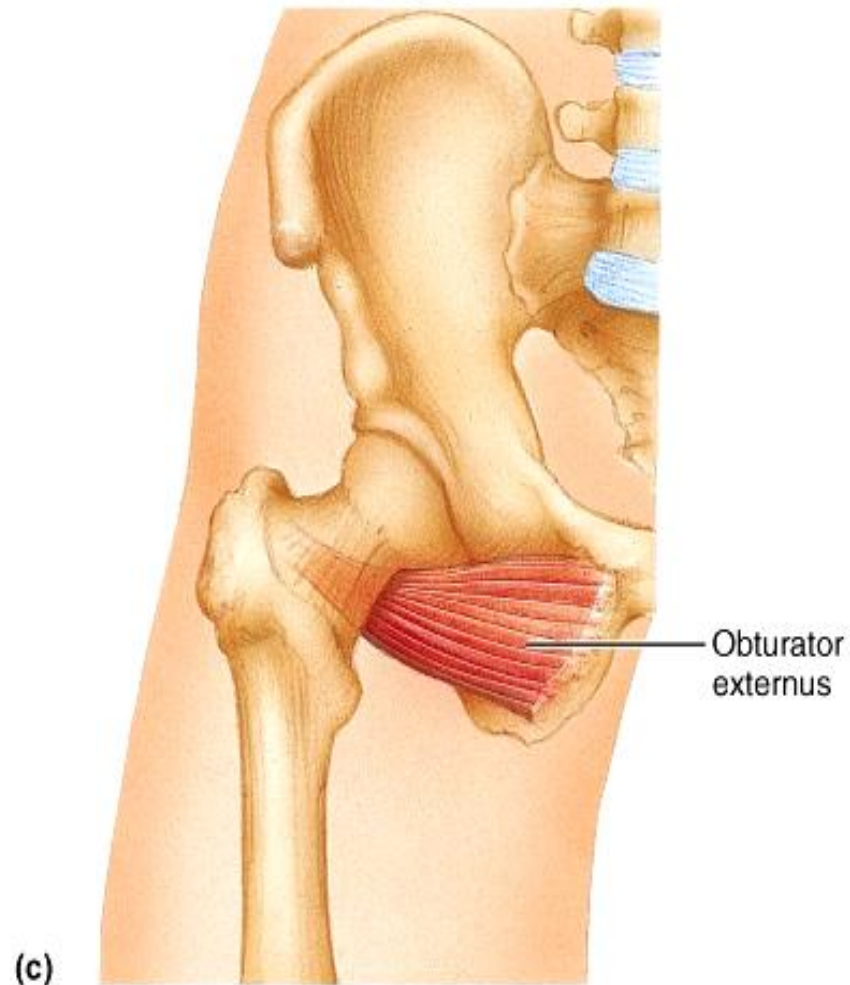
Adductor magnus

- Origin
 - Adductor part: ischial and pubic rami
 - Hamstring part: ischial tuberosity
- Insertion
 - **Adductor part**: linea aspera (medial lip) and supracondylar line
 - **Hamstring part**: adductor tubercle
- Action - pubic part adducts and ischial part flexor of the hip
- Innervation
 - Adductor part: **obturator nerve**
 - Hamstring part: **sciatic nerve(tibia branch)**



Obturator externus

- Origin:- margins of obturator foramen and obturator membrane
- Insertion:- trochanteric fossa of femur
- **Innervation:- obturator nerve**
- Action:- lateral rotators of the thigh; steadies head of femur

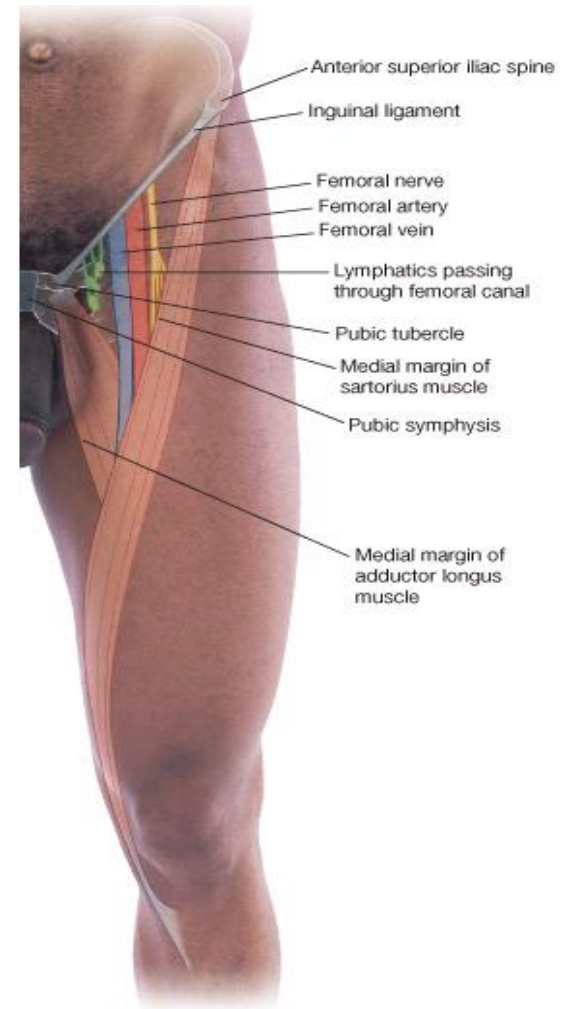


Copyright © 2001 Benjamin Cummings, an imprint of Addison Wesley Longman, Inc.

Femoral triangle

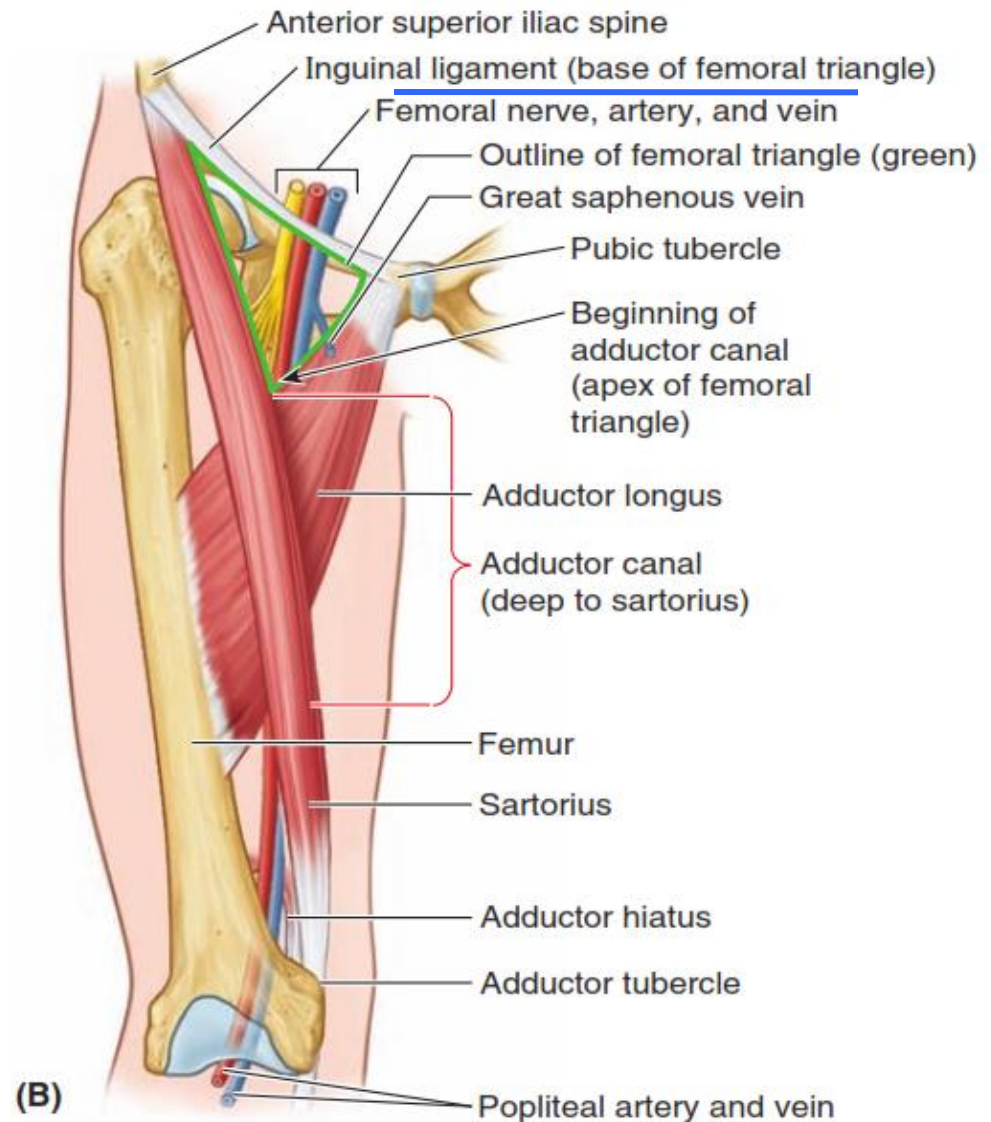
□ Boundaries

- Medial – **lateral boarder of adductor longus**
- Lateral - **medial boarder of sartorius**
- Superior - **inguinal ligament**
- Floor - pectineus, psoas and iliacus
- Roof - formed by
 - superficial fascia, containing superficial inguinal lymph nodes and great saphenous vein with its tributaries



□ Contents

- femoral sheath
- femoral artery and its branch
- femoral vein and its tributaries
- femoral nerve and its branches
- deep inguinal lymph nodes
- profunda femoris artery and vein and their circumflex branches

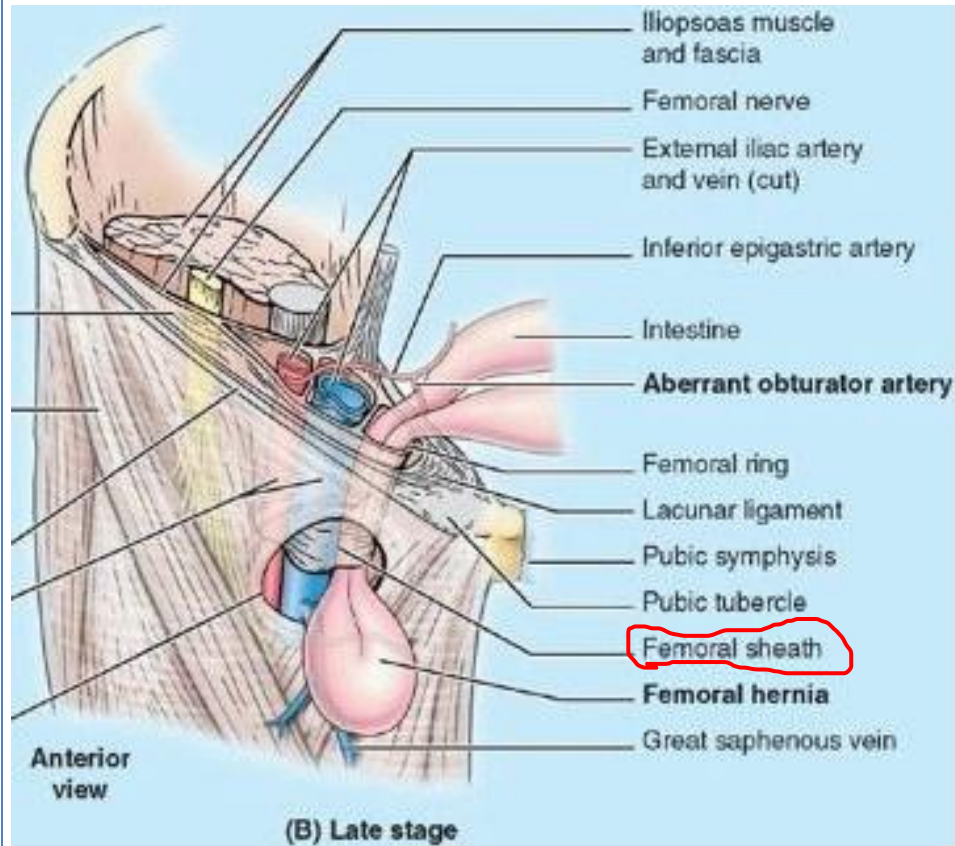


Femoral sheath

- Prolongation of extraperitoneal areolar tissue
- **Has 3 compartments and two septa**
 - lateral for the **femoral artery** and genital branch of genitofemoral nerve
 - middle for the **femoral vein**
 - medial for the **femoral canal**
 - contain lymph vessels and deep lymph nodes
 - **It is the site of femoral hernia**
 - It is most common in female

Femoral canal

- medial part of the femoral sheath
- small, vertically placed gap
- larger in female due to the greater width of the female pelvis
 - femoral hernias are more common in female



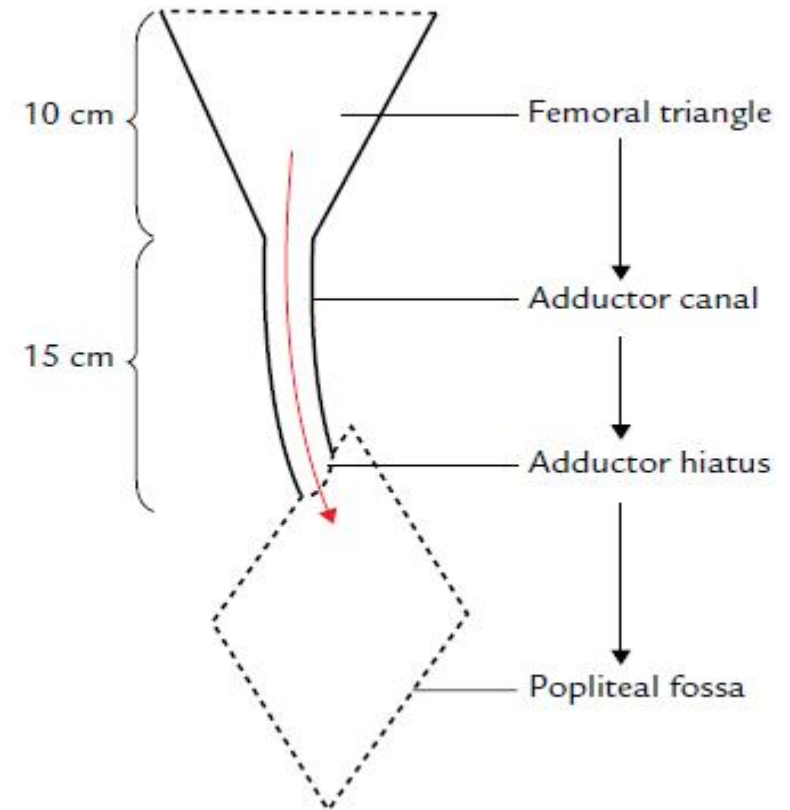
Femoral canal

□ Boundaries

- **Anteriorly** - inguinal ligament
- **Medially** - lacunar ligament
- **Laterally** - femoral vein
- **Posteriorly** - pectineal ligament

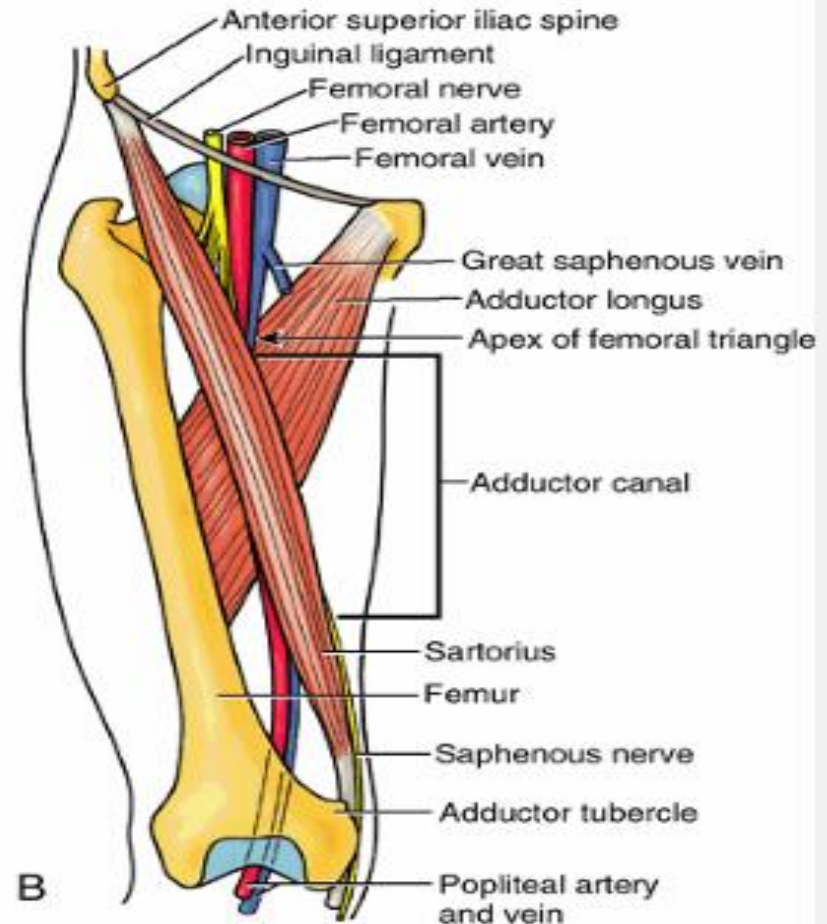
□ Contents

- **fat**
- **lymph nodes**
- **lymph vessels**



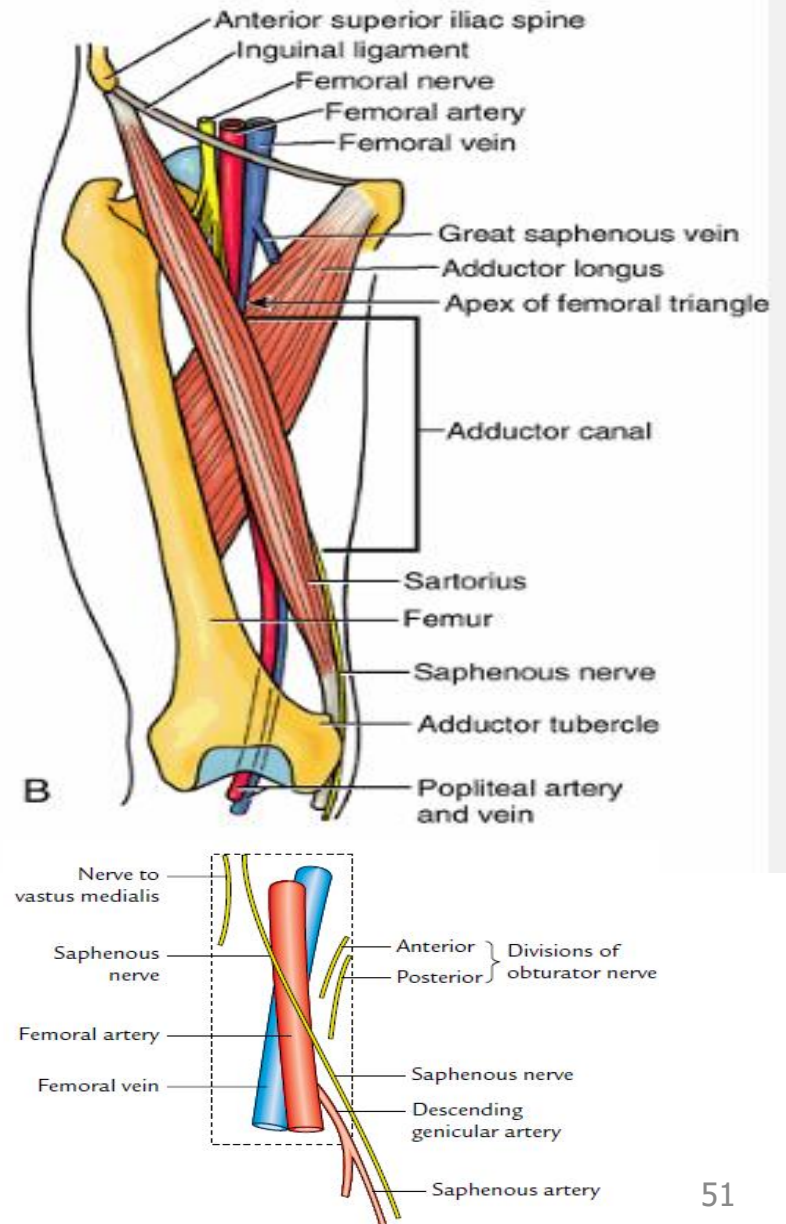
Adductor canal

- Leads on from the apex of the femoral triangle
- Ends 10cm above the **adductor tubercle**
- Triangular
- **Its walls (boundaries) are:**
 - **Posteriorly** - adductor longus and magnus
 - **Anterolaterally** - vastus medialis
 - **Anteromedial** - sartorius



Adductor canal...

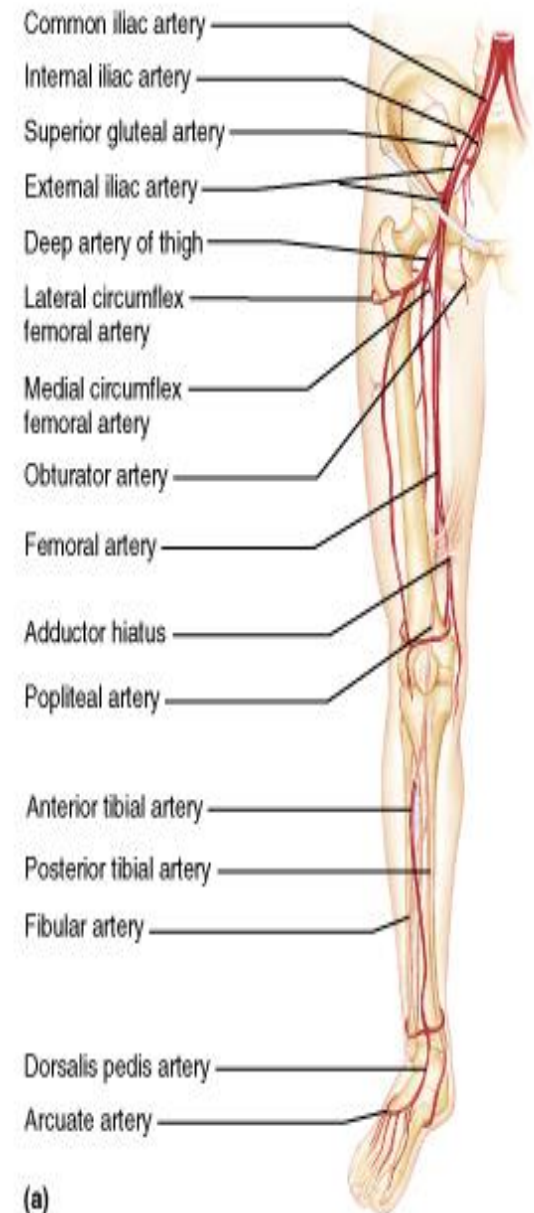
- **Contents of the adductor canal**
 - femoral artery
 - femoral vein
 - saphenous nerve
 - nerve to vastus medialis



Blood vessels

Femoral artery

- Origin
 - Main continuation of **external iliac** distal to inguinal ligament
- Distribution: anterior and anteromedial thigh
- Branches
 - the femoral artery gives off:
 - ❑ **Superficial branches of femoral artery**
 - **Superficial external pudendal artery**
 - **Superficial epigastric artery**
 - **Superficial circumflex iliac artery**



(a)

Deep femoral (profunda femoris) artery

- Origin: arises posterolaterally from the femoral
- And it passes deep to adductor longus
- Distribution: perforating branches supply muscles in medial, posterior and lateral part of anterior compartment

• Branches of deep femoral artery

– Medial circumflex femoral

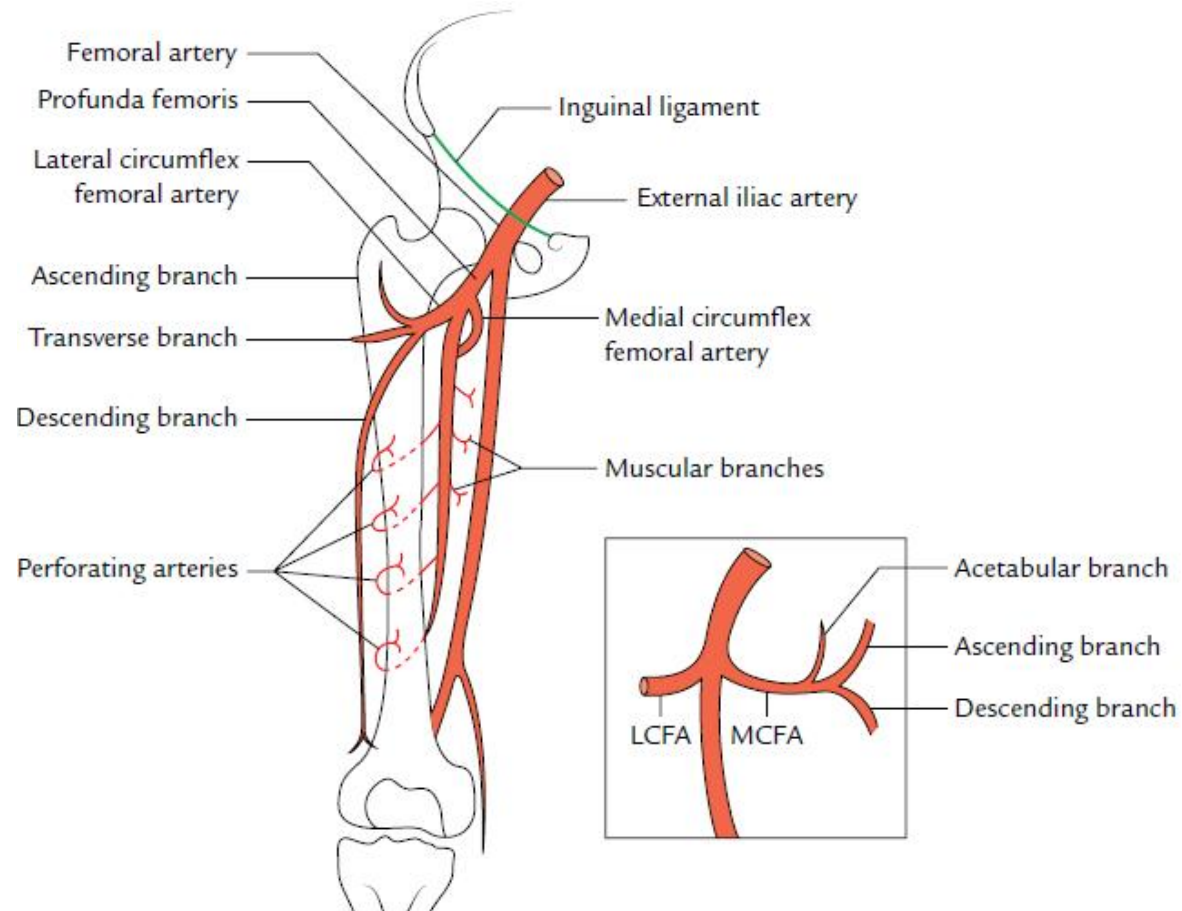
- Passes medially and posteriorly between the pectineus and iliopsoas
- It divides into 2 branches; **transverse and ascending branches**

– Lateral circumflex femoral

- Passes laterally deep to sartorius and rectus femoris muscle
- Divide into ascending, transverse and descending branches

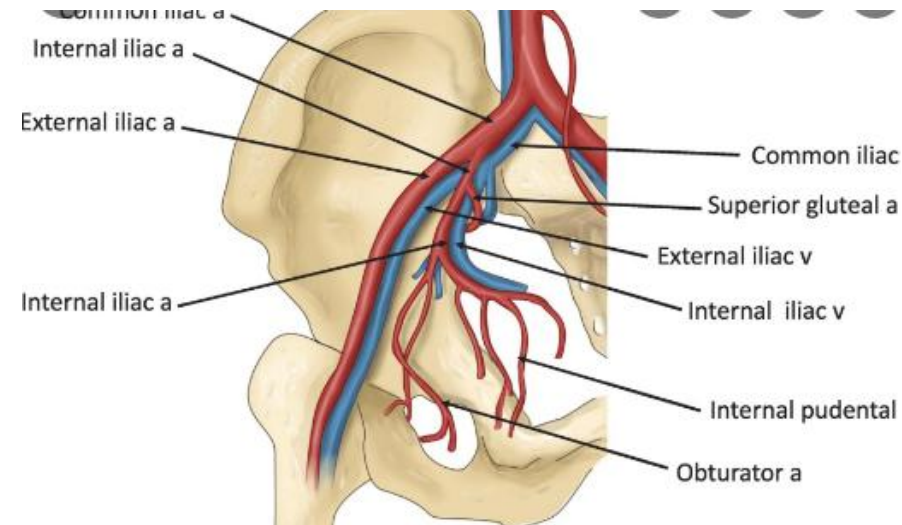
Deep femoral (profunda femoris) artery

- **Four perforating branches** and terminates at the fourth perforating



Obturator artery

- Arises from **internal iliac artery**
- **Passes through obturator foramen**
- Enters medial compartment of the thigh
- Divides into anterior and posterior branches
- Supplies obturator externus and adductors of thigh
- Posterior branch gives off acetabular branch to head of femur



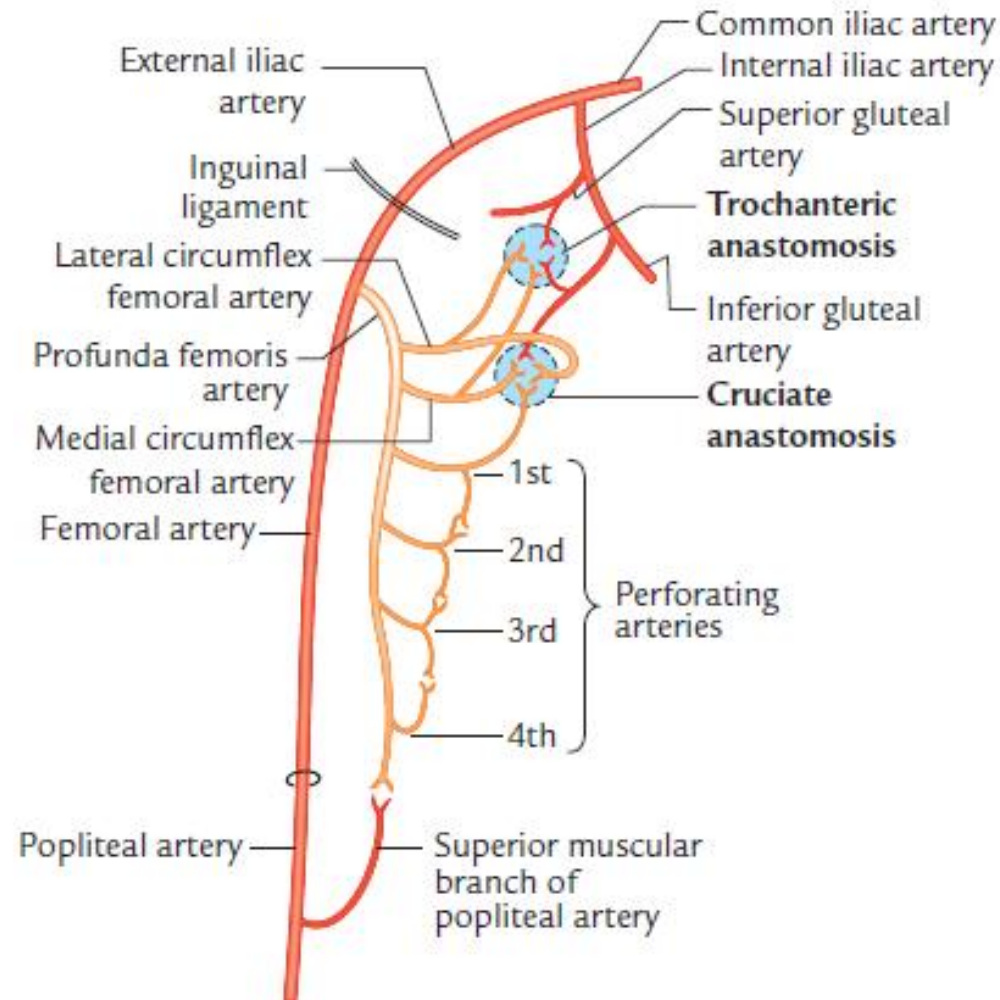
Arterial anastomosis on hip joint

❑ Trochanteric anastomosis

- Superior gluteal artery
- Inferior gluteal artery
- Medial circumflex femoral artery
- Lateral circumflex femoral artery

❑ Cruciate anastomosis

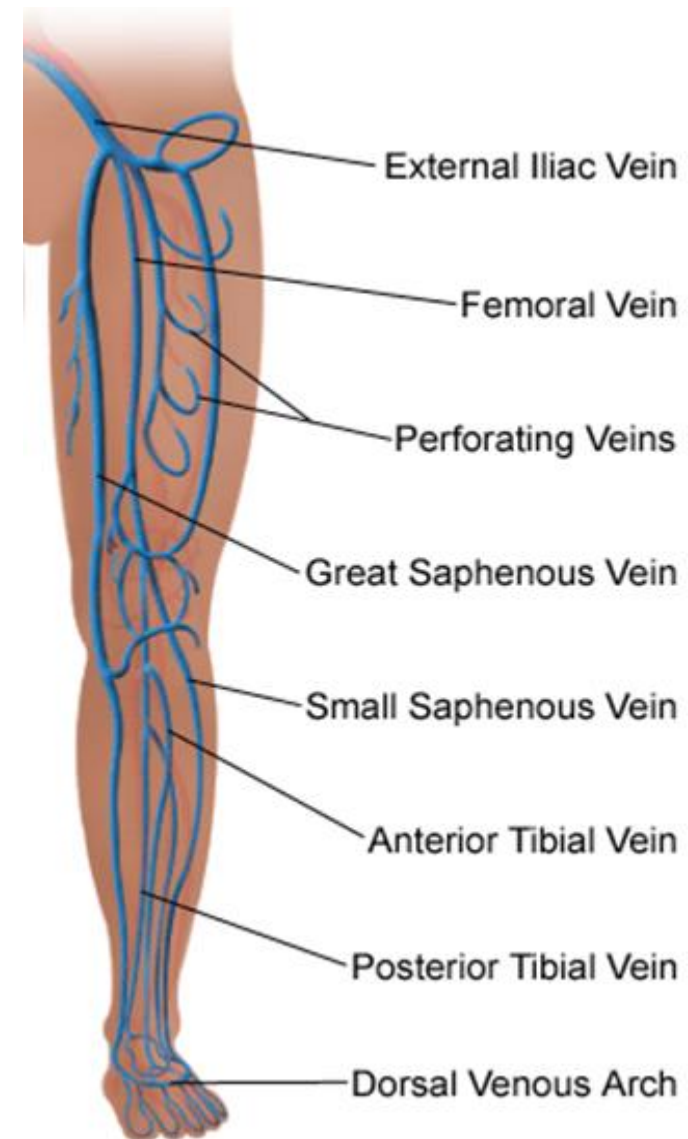
- Inferior gluteal artery
- Medial circumflex femoral artery
- Lateral circumflex femoral artery
- 1st perforating branch



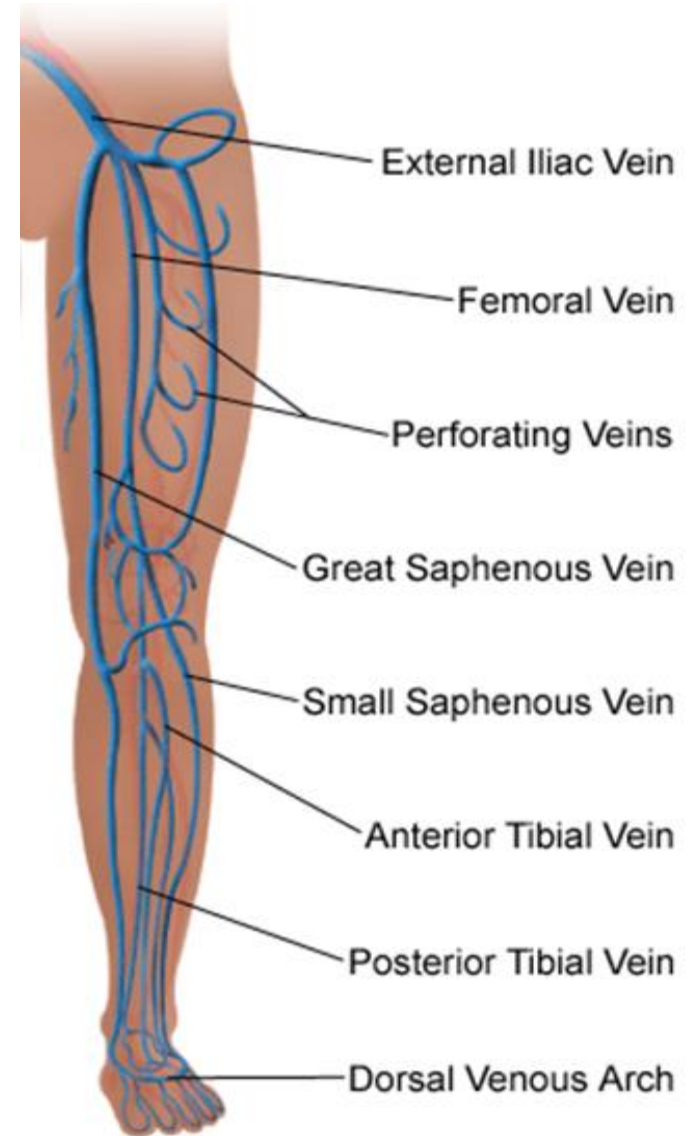
Deep veins

Femoral vein

- A continuation of **popliteal vein** at adductor hiatus
- Course:
 - ascends through adductor canal
 - lies medial and then posterior to femoral artery
 - enters femoral sheath lateral to femoral canal
 - ends posterior to inguinal ligament, becomes **external iliac vein**



- **Tributaries**
 - Descending genicular
 - Profunda femoris
 - Lateral circumflex femoral
 - Medial circumflex femoral
 - Deep external pudendal
 - Great saphenous

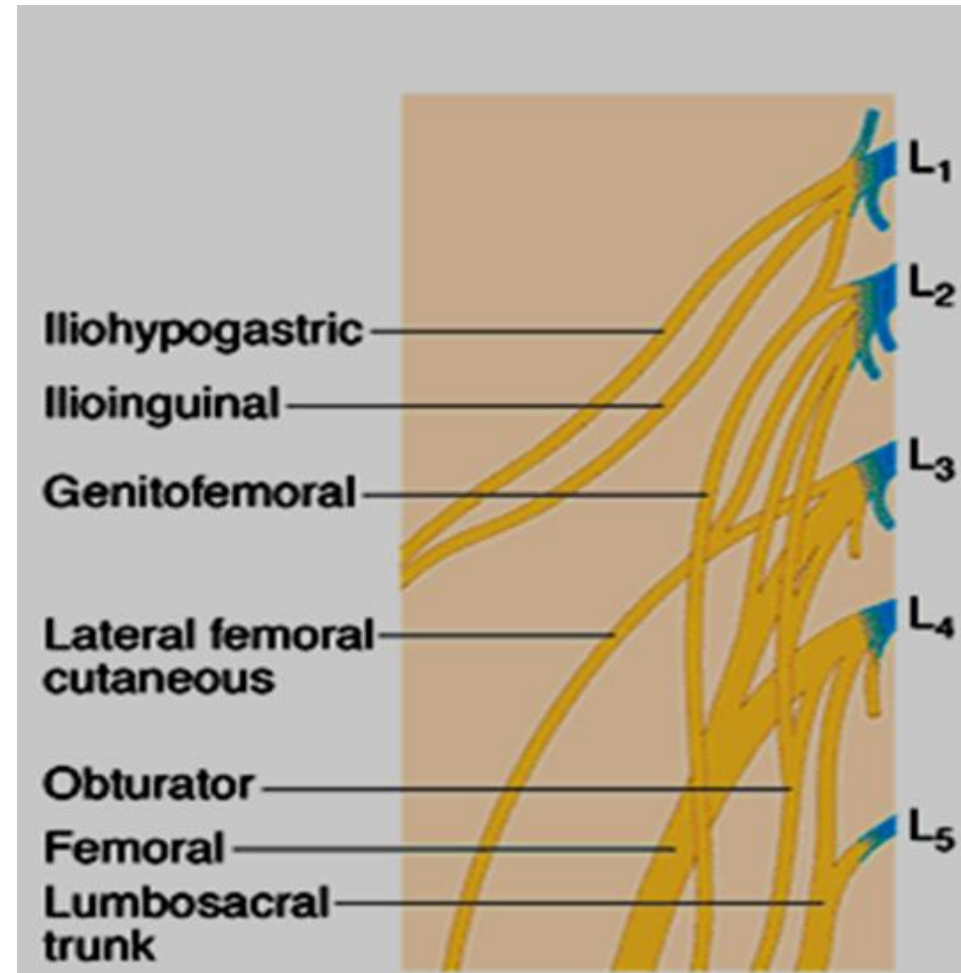


Lumbar plexus

- The plexus originates from the anterior rami of (L1 - L4)
- The trunks of the plexus traverse psoas major and emerge from its **lateral border**
 - There are two exceptions:
 - **Obturator nerve appears at the medial border of psoas tendon**
 - **Genitofemoral nerve emerges on the anterior aspect of the muscle by piercing the psoas major.**



- **Branches of the lumbar plexus**
 - **Obturator nerve(L2,4)**
 - **Femoral nerve (L2-4)**
 - **Lumbosacral trunk(L4,5)**
 - **Ilioinguinal nerve (L1)**
 - **Iliohypogastric nerve(L1)**
 - **Genitofemoral nerve(L1,2)**
 - **Lateral cutaneous nerve of the thigh(L2,3)**



Obturator nerve (L2- L4)

- Emerges from the **medial border of the psoas major**
- Passes into the lesser pelvis
- Passing inferior to the superior pubic ramus (through the ***obturator foramen***) to the medial thigh
- Divides into **anterior and posterior divisions** past the obturator foramen and supplies adductor muscles
- **The nerve to gracilis becomes the only cutaneous branch**
- Has articular branches for **both the hip and knee joint**

- **Two divisions – anterior and posterior**

- **Anterior division**

- Adductor longus
 - Adductor brevis
 - Gracilis
 - Articular branch to hip joint
 - Cutaneous branch to skin overlying adductor canal & knee joint

- **Posterior division**

- Obturator externus
 - Adductor magnus (adductor part)

Femoral nerve (L2 - L4)

- Emerges from the **lateral border of the psoas major**
- Lies in front of the iliacus; innervates it
- Passes deep to the inguinal ligament to the anterior thigh lies **lateral to the artery**
- Breaks up into motor and sensory branches
- Supply the flexors of the hip and extensors of the knee
- **Two of the branches follow the artery on its lateral side into the adductor canal:**
 - nerve to vastus medialis (motor)
 - saphenous nerve (sensory)

❑ Branches

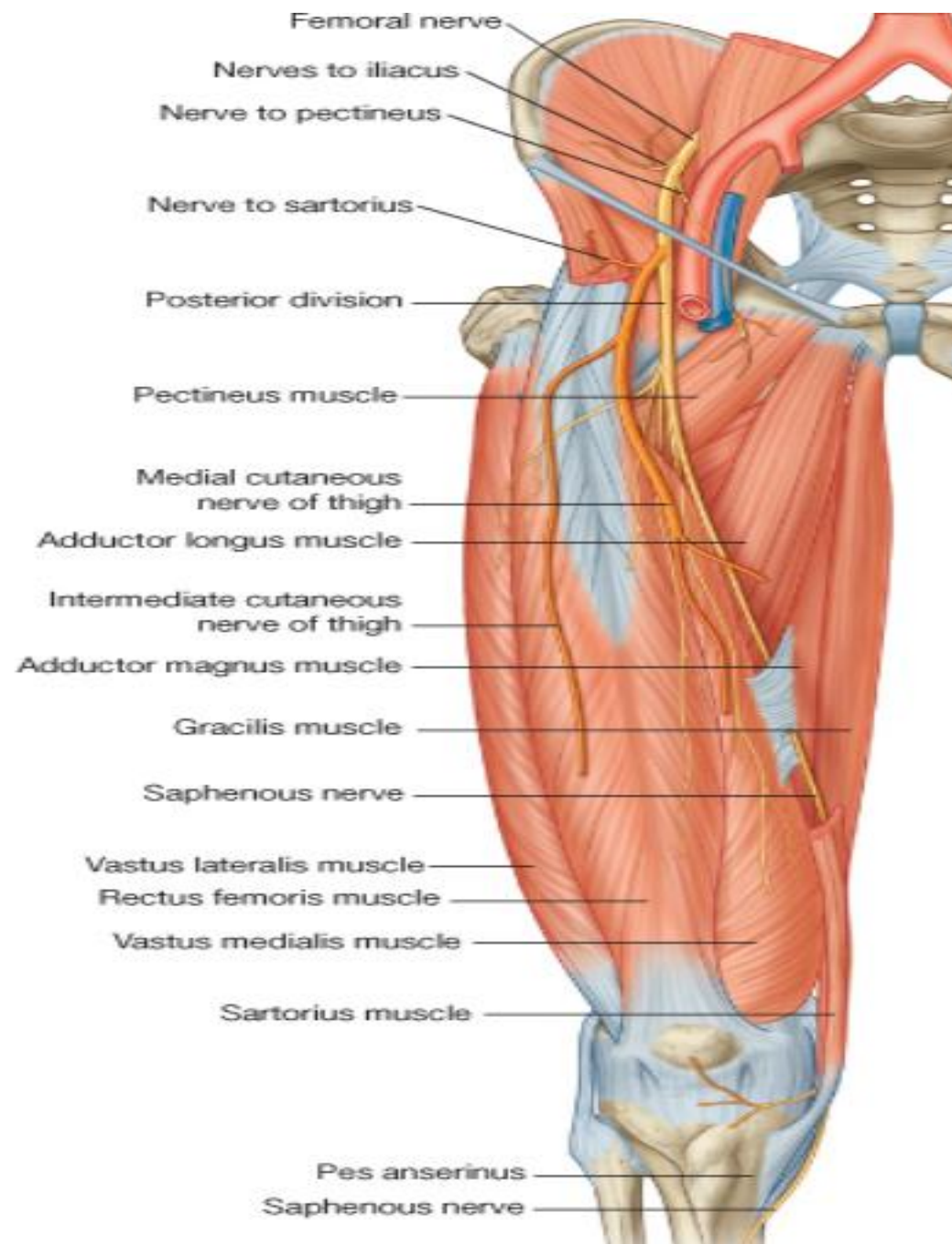
❑ Muscular

- to the anterior compartment of the thigh (quadriceps, sartorius and pectineus)

❑ Cutaneous

- **medial and intermediate cutaneous nerves** of the thigh
 - the lateral aspect is supplied by the lateral cutaneous nerve of the thigh (L2–3)
- **saphenous nerve**
 - traverses the adductor canal
 - supply the skin of the medial side of the leg, ankle and foot to the great toe

❑ Articular - to the **hip and knee joints**



Lumbosacral trunk (L4 & L5)

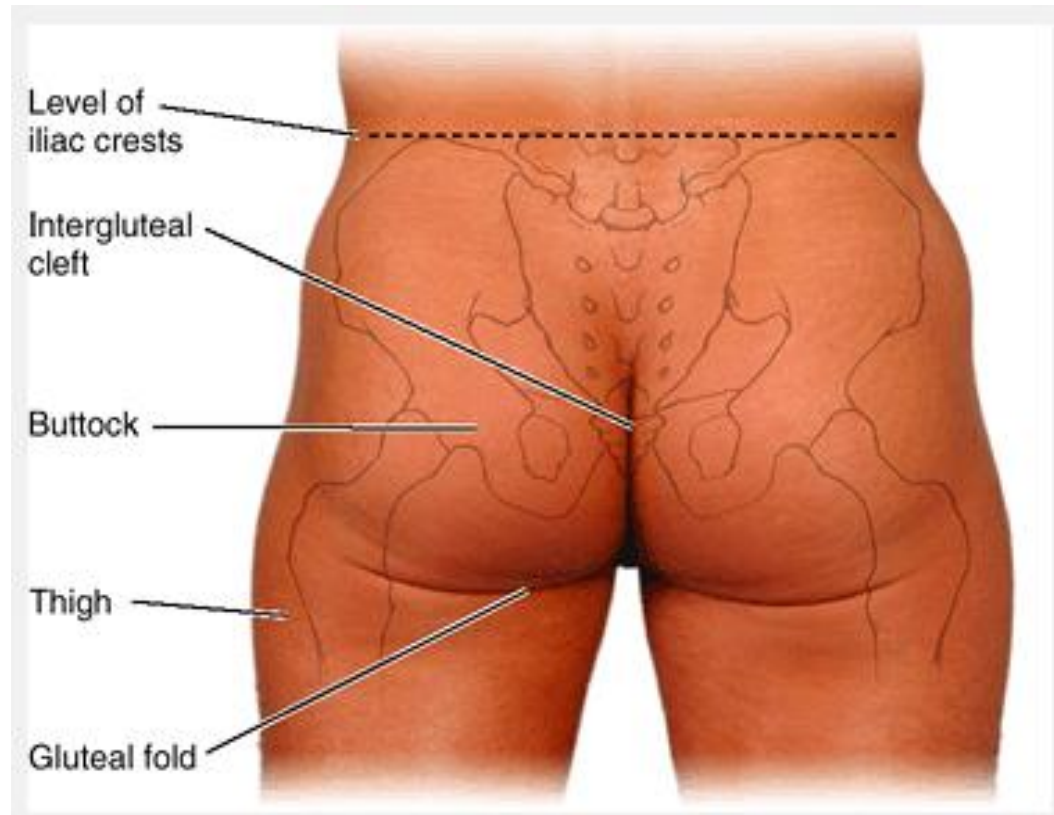
- Passes over the ala (wing) of the sacrum
- Descends into the pelvis
- Participate in the formation of the sacral plexus with the anterior rami of S1 - S4 nerves

Genitofemoral nerve (L1 & L2)

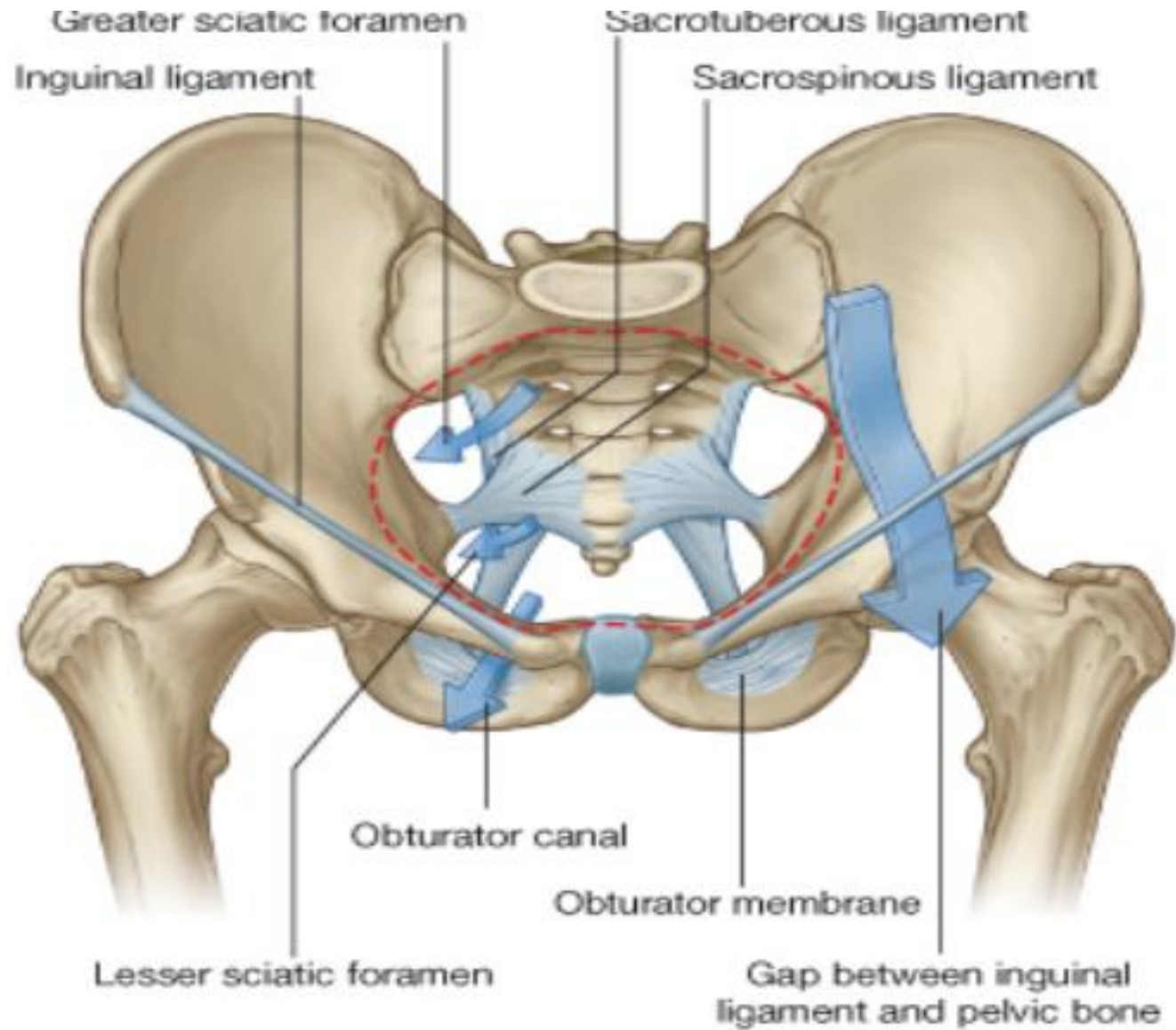
- **Pierces the psoas major** and runs inferiorly on its anterior surface, deep to the psoas fascia
- Divides lateral to the common and external iliac arteries into femoral and genital branches
- The lateral cutaneous nerve of the thigh, or lateral femoral cutaneous nerve (L2, L3)
- Runs inferolaterally on the iliacus and enters the thigh deep to the inguinal ligament/iliopubic tract, just medial to the ASIS
- Supplies skin on the anterolateral surface of the thigh

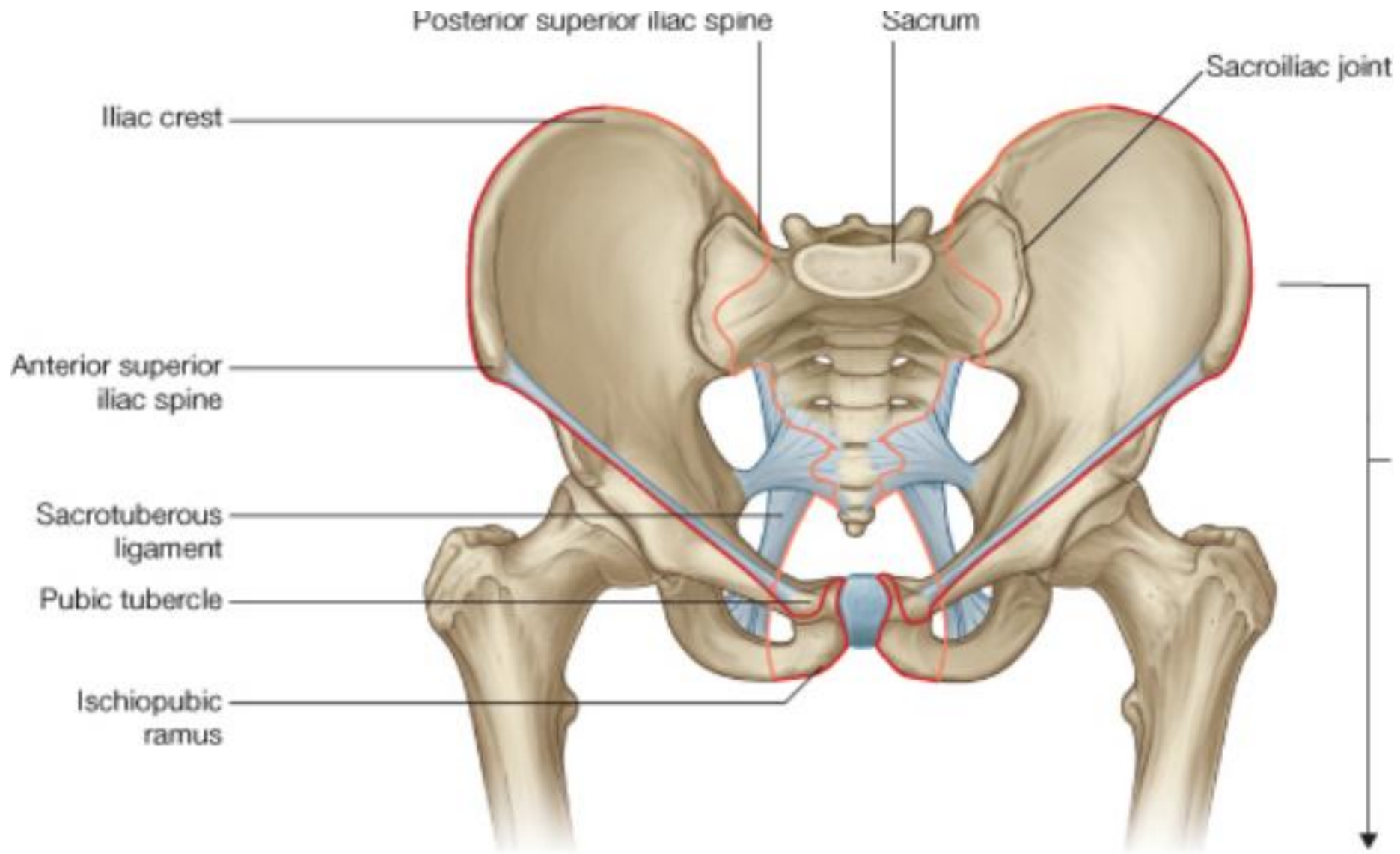
The Gluteal Region

- The gluteal muscles (gluteus maximus, medius, and minimus and tensor of the fascia lata are superficial muscles)
- Other are deep muscles
- Piriformis
- Two gemilli
- Quadratus femoris
- Obturator internus

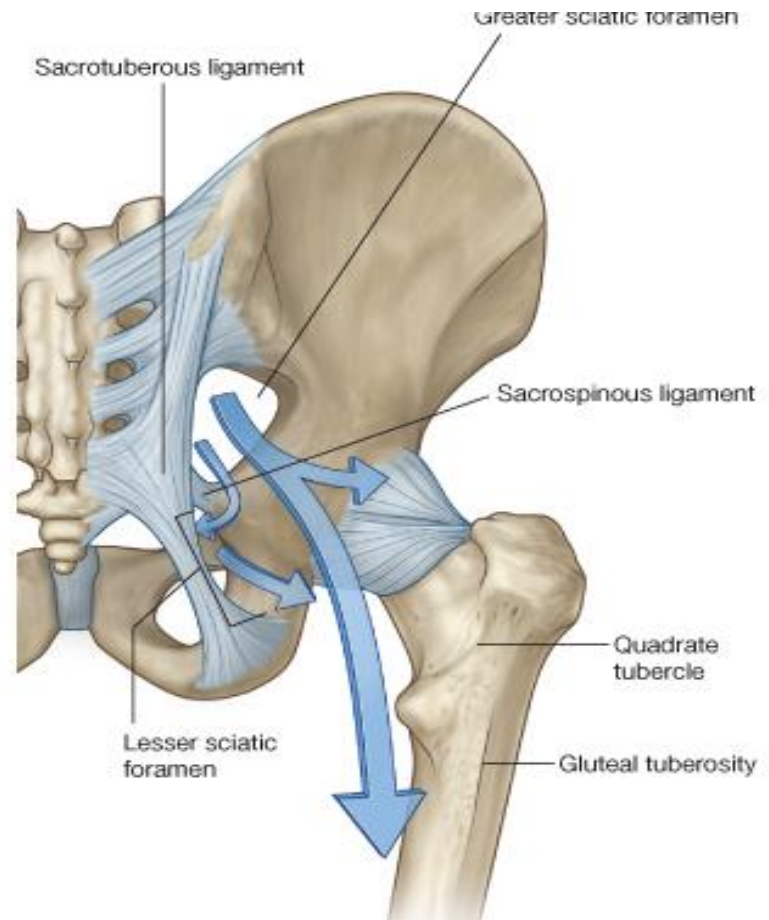


Gluteal Ligaments





- **Greater sciatic foramen**
 - Passageway for structures entering or leaving the pelvis
 - all lower limb arteries and nerves leave the pelvis and enter the gluteal region
- **Lesser sciatic foramen**
 - Passageway for structures entering or leaving the **perineum**

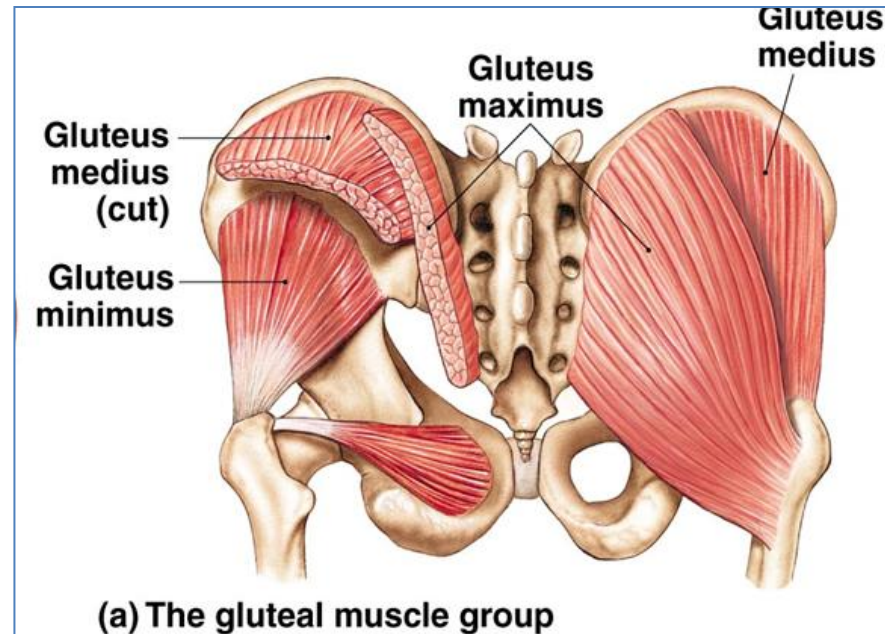


Gluteal Muscles

- **Organized into 2 layers:** superficial and deep
- ❑ **Superficial layer: 4**
 - Three gluteal muscles (max. ,medius ,minimus)
 - Tensor of fascia lata
- ❑ **Deep layer: 5**
 - piriformis
 - obturator internus
 - superior & inferior gemilli
 - quadratus femoris

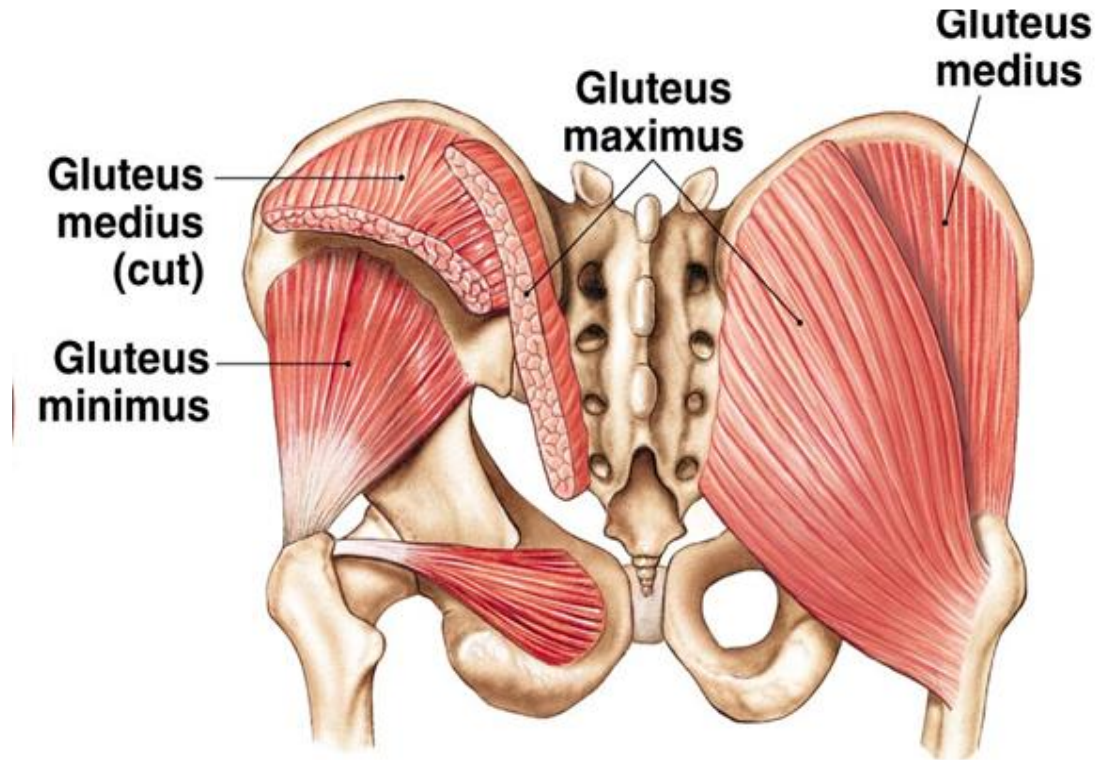
Gluteus maximus

- Origin - dorsal surface of ilium, sacrum and coccyx
- Insertion - iliotibial tract and gluteal tuberosity of femur
- Action - **thigh extension, lateral rotation**
- Innervation: **inferior gluteal nerves**



Gluteus medius

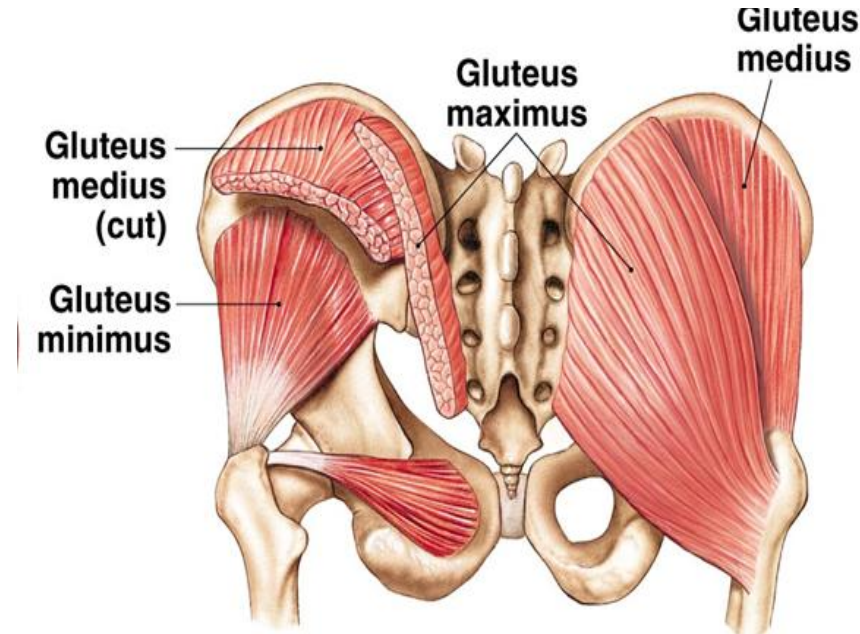
- Origin - between anterior and posterior gluteal lines on lateral surface of ilium
- Insertion - greater trochanter of femur
- **Action - abducts and medially rotates thigh**
- **Innervation - superior gluteal nerve**



(a) The gluteal muscle group

Gluteus minimus

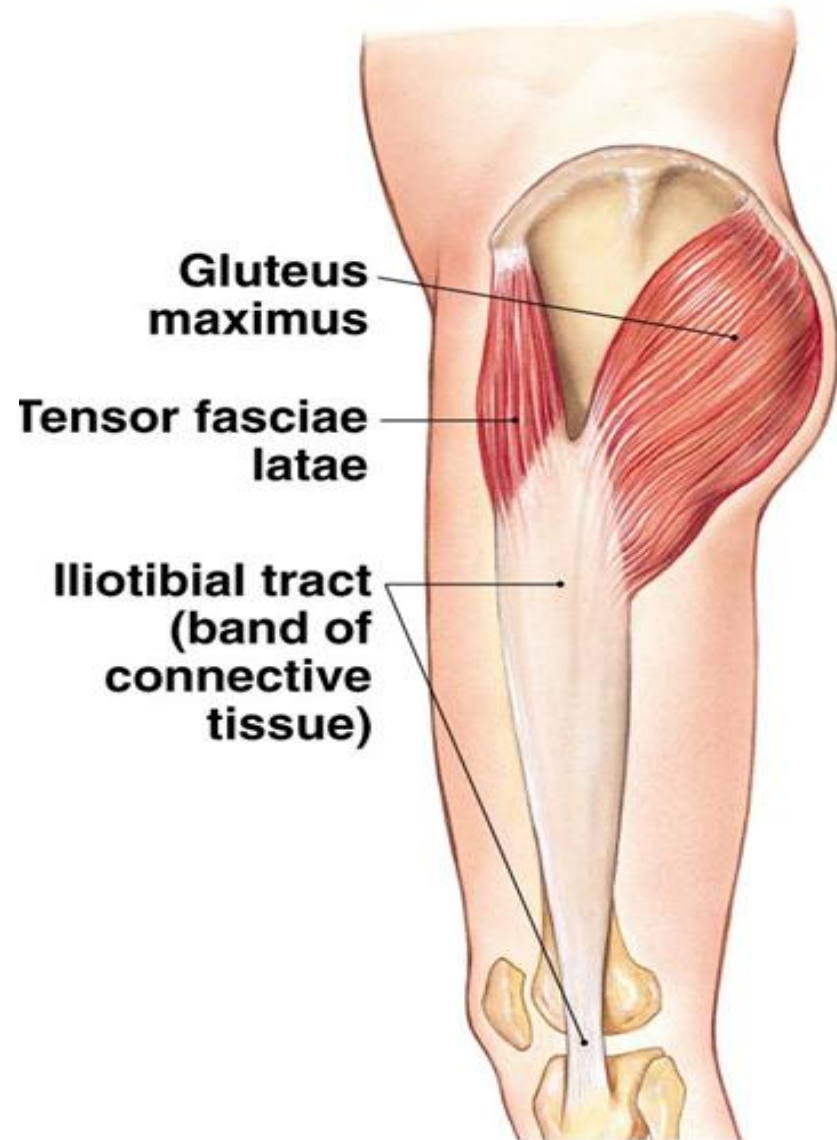
- Origin - between anterior and inferior gluteal lines on external surface of ilium
- Insertion - greater trochanter of femur
- Action - **abducts and medially rotates thigh and with gluteus medius prevent tilting of the pelvis**
- Innervation - **superior gluteal nerve**



(a) The gluteal muscle group

Tensor fascia latae

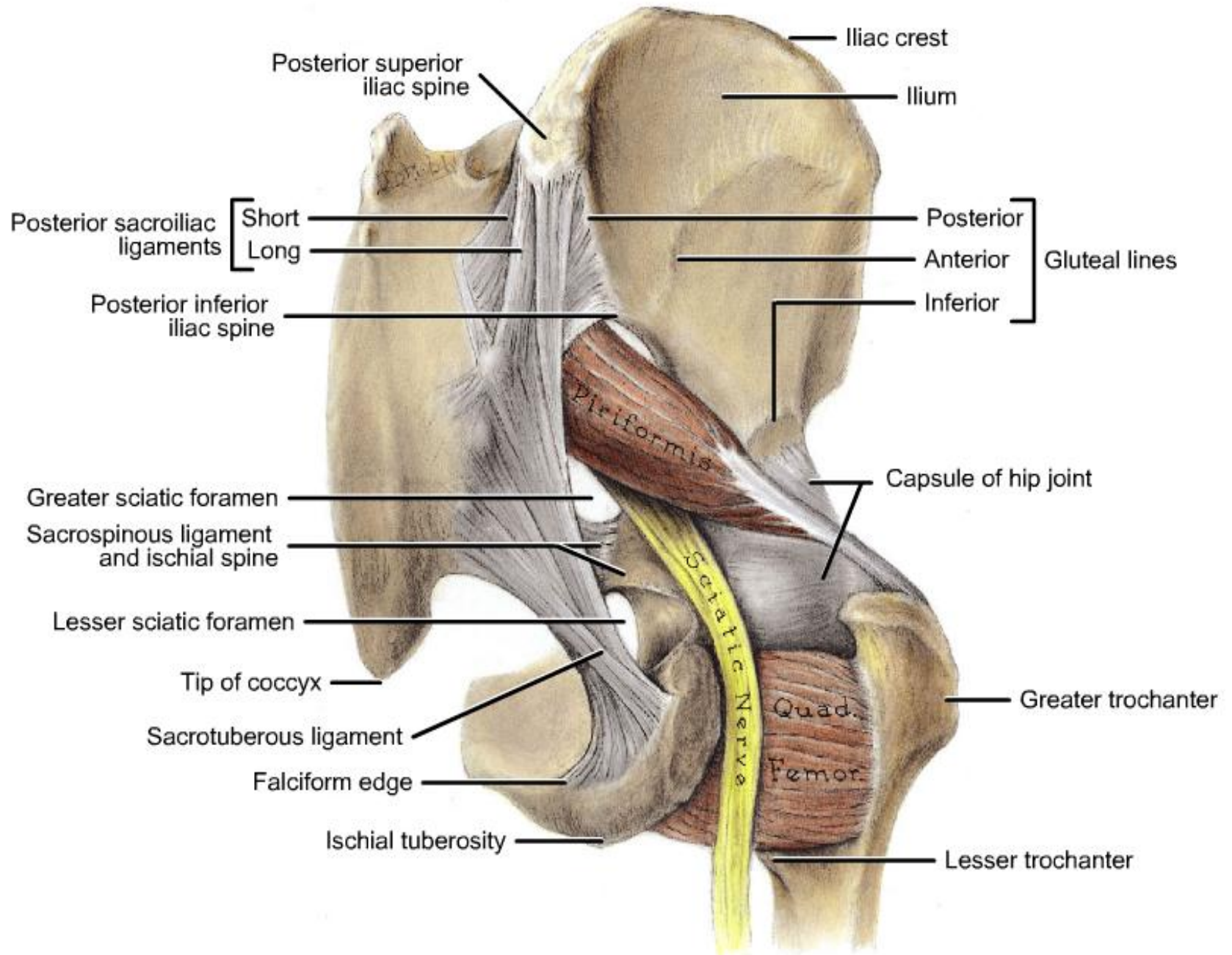
- Origin - anterior superior iliac spine and the outer lip of the iliac crest.
- Insertion - iliotibial tract
- Action - **flexes and abducts thigh**
- Innervation - **superior gluteal nerve**



Lateral rotators of the hip joint(deep muscles)

Piriformis

- **Key muscle** – divide the greater sciatic foramen in to 2
 - **structures passing above the muscle**
 - superior gluteal nerve and vessels
 - **structures passing below the muscle**
 - nerves - inferior gluteal, sciatic, posterior femoral cutaneous, pudendal, nerve to quadratus femoris and obturator internus tendone
 - vessels - inferior gluteal and internal pudendal vessles
- Origin: pelvic surface of sacrum
- Insertion: **greater trochanter**
- Innervation - **nerve to piriformis**



Quadratus femoris

- Origin: ischial tuberosity
- Insertion: intertrochanteric crest of femur
- Innervation: **nerve to quadratus femoris** (L4, 5, S1)

Obturator internus

- Origin: pelvic surface of the obturator membrane
- Insertion: **greater trochanter**
- Innervation: **nerve to obturator internus** (L5, S1, 2)

Superior gemellus

- Origin: ischial spine
- Insertion: **greater trochanter**
- Innervation: **nerve to obturator internus**

Inferior gemellus

- Origin: ischial tuberosity
- Insertion: **greater trochanter**
- Innervation: **nerve to quadratus femoris**

Superior
gemellus

Obturator
internus

Inferior
gemellus

Gluteus medius (cut)

Gluteus minimus

Piriformis

Obturator externus

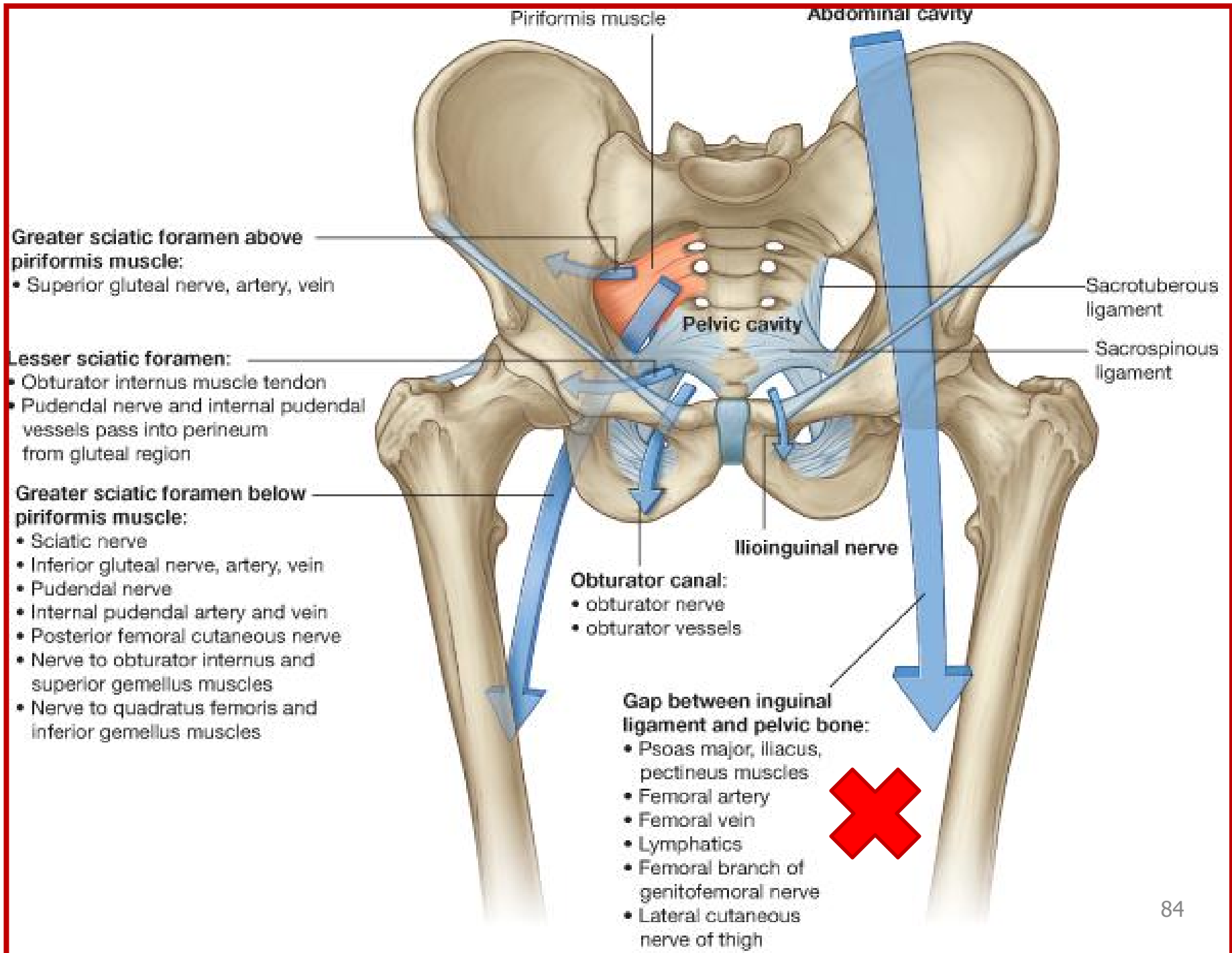
Quadratus femoris

Gluteus maximus
(cut)

(b)

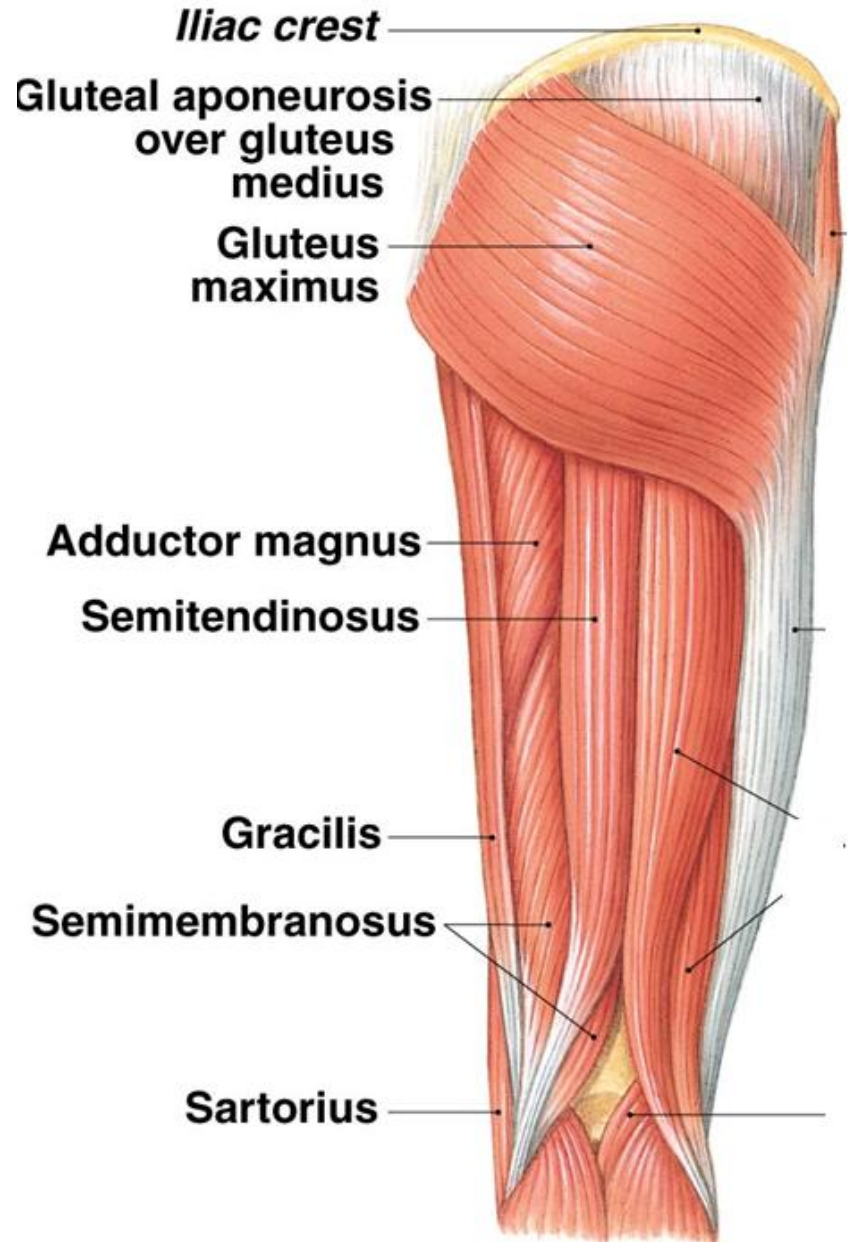
- **Two groups of nerve and vessels emerge inferior to piriformis**
 - ❑ **A group of 5 structures**
 - **Sciatic nerve**
 - lies between ischial tuberosity and greater trochanter
 - Its "safe" side for gluteal injection is the suprolateral side where it has no branches
 - It may divide in this region into common peroneal and tibial nerves
 - **Inferior gluteal nerve**
 - **Inferior gluteal artery**
 - **Inferior gluteal vein**
 - **Posterior femoral cutaneous nerve**

- A group of 3 structures
 - Nerve to obturator internus
 - Internal pudendal vessels
 - Pudendal nerve
- They runs through the **lesser sciatic foramen** into the ischioanal fossa
- The obturator internus is the only structure entering the gluteal region via the **lesser sciatic foramen**



Posterior thigh muscles

- **Three of the four muscles** in posterior thigh are **hamstrings**
 - Semitendinosus
 - Semimembranosus
 - Biceps femoris (long head)
 - Hamstring part of adductor magnus also
- **Short head of biceps femoris**
 - It not hamstring as it crosses only knee joint
 - innervated by **fibular division of sciatic nerve**

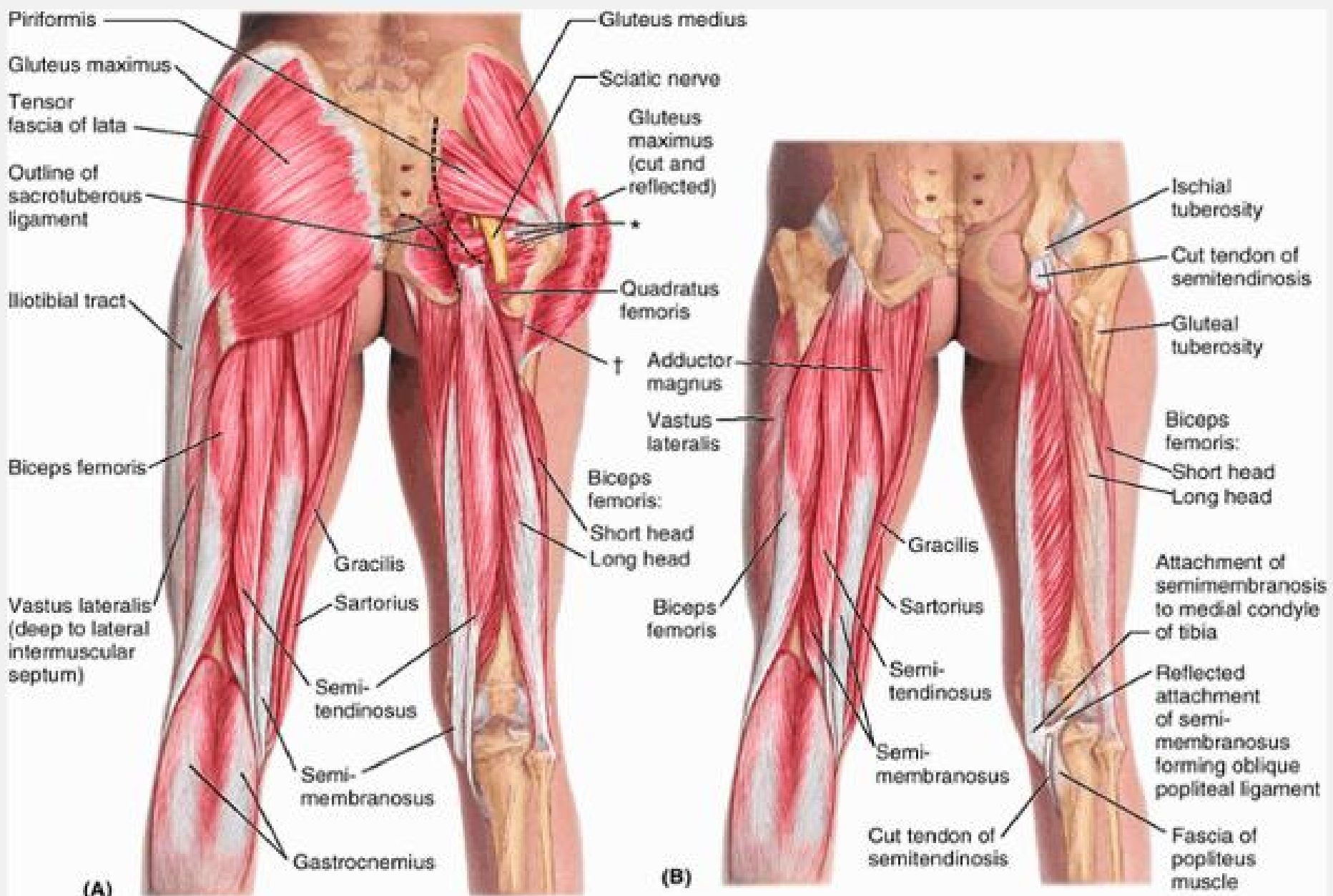


Hamstring muscles

- Arise from **ischial tuberosity**
- Insert into tibia or fibula
- Innervated by the **tibial division of sciatic nerve**
 - The nerve to the short head of the biceps femoris comes from the lateral (peroneal) division whereas branches to other hamstrings arise from the medial (tibial) division
- **Extend hip and flex knee joint**
- The semimembranosus and semitendinosus also rotate the leg medially, being attached to the medial head of the tibia

Biceps femoris

- Origin
 - Long head - ischial tuberosity
 - Short head - linea aspera
- Insertion: head of fibula and lateral condyle of tibia
- Action
 - **Long head** - extends and laterally rotates the thigh and flexes knee
 - **Short head** - flex the knee and laterally rotates leg
- Innervation
 - **Long head** - tibial division of sciatic
 - **Short head** - fibular division of sciatic nerve



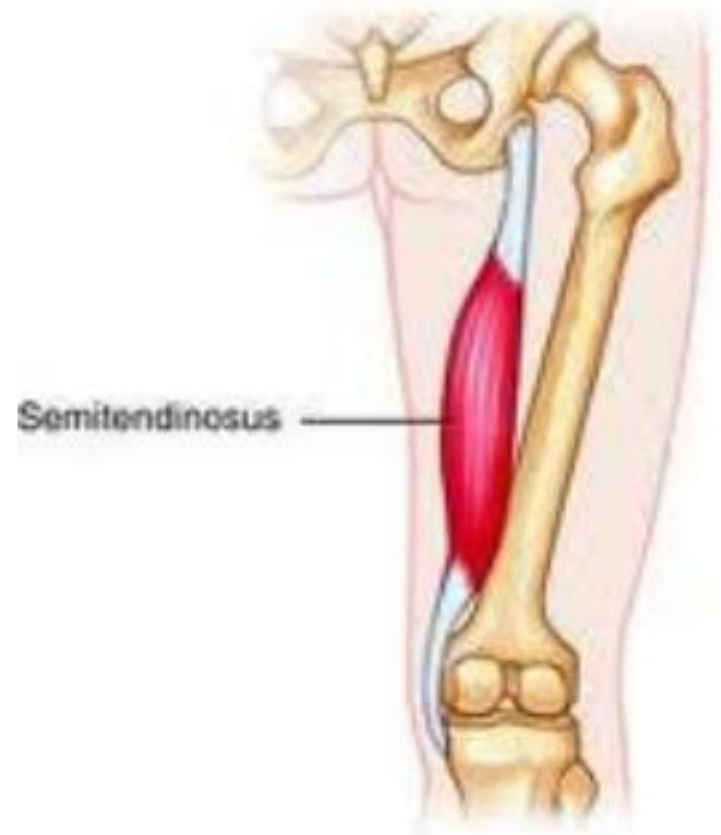
Posterior views

*Attachment of inferior half of deep part of gluteus maximus to gluteal tuberosity

*Triceps coxae:
Tendon of obturator internus
Superior gemellus
Inferior gemellus

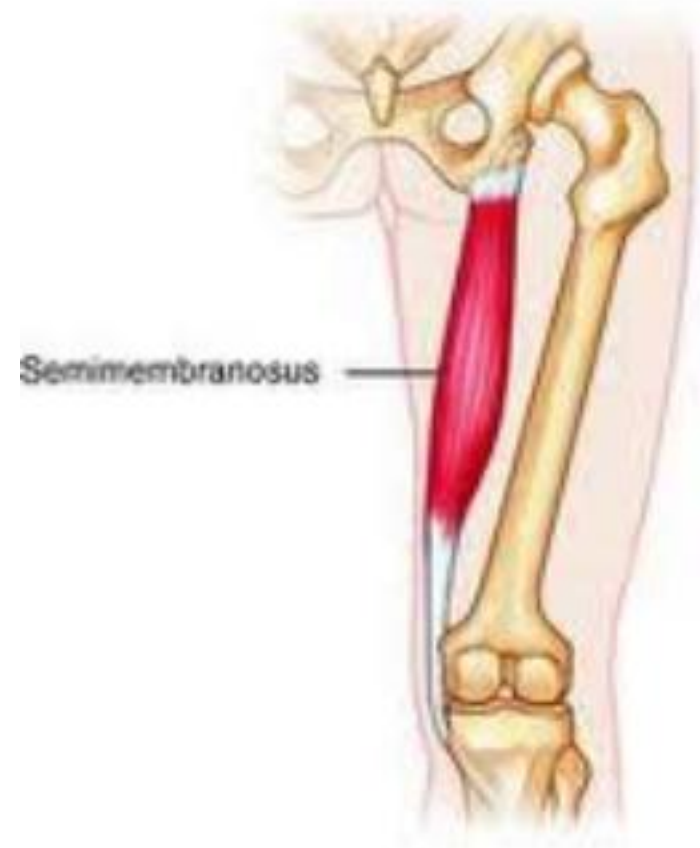
Semitendinosus

- Origin - ischial tuberosity
- Insertion - medial aspect of upper tibial shaft
- **It is lateral to Semimembranosus**
- **Action - extends thigh at hip; flexes knee, medially rotates leg**
- **Innervation - tibial division of sciatic nerve**



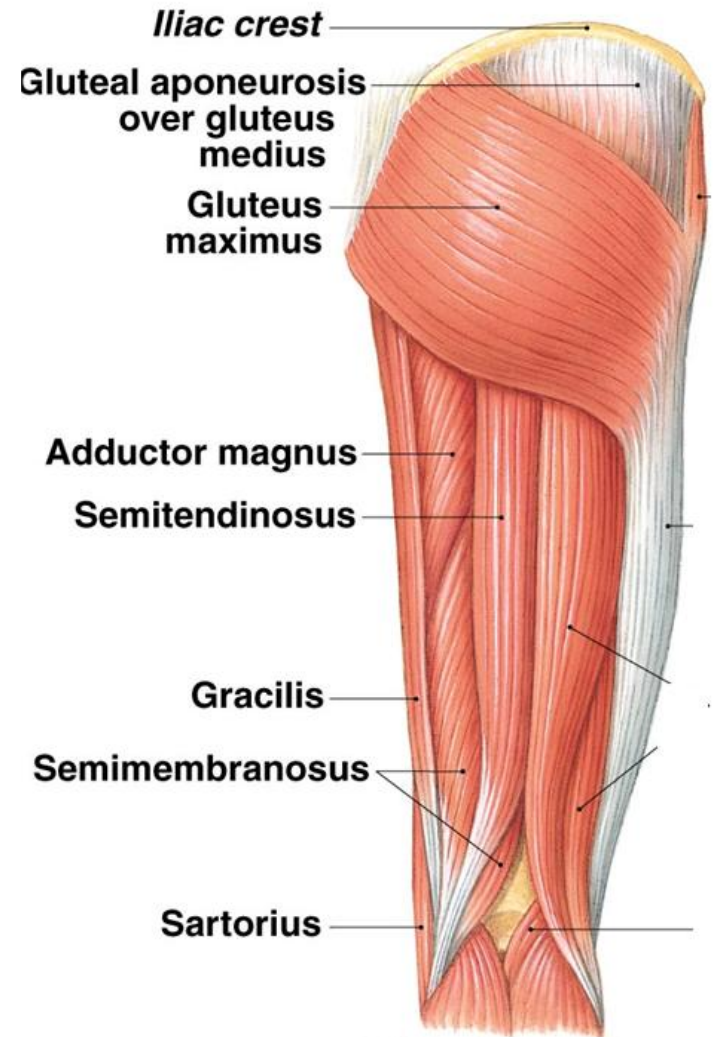
Semimembranosus

- Origin - ischial tuberosity
- Insertion - medial condyle of tibia
- **It is medial to semi tendinous**
- **Action - extends thigh and flexes knee, medially rotates leg**
- **Innervation - tibial division of sciatic nerve**



Hamstring injuries

- **Hamstrings strains (pulled and/or torn hamstrings) are common in people who run and/or kick hard (e.g., in running, jumping and quick-start sports such as baseball and soccer)**
- The violent muscular exertion required to excel in these sports may tear part of the proximal attachments of the hamstrings to the ischial tuberosity
- **Hamstring injuries may result from inadequate warming up before practice or competition**



Clunial (superficial gluteal) nerves

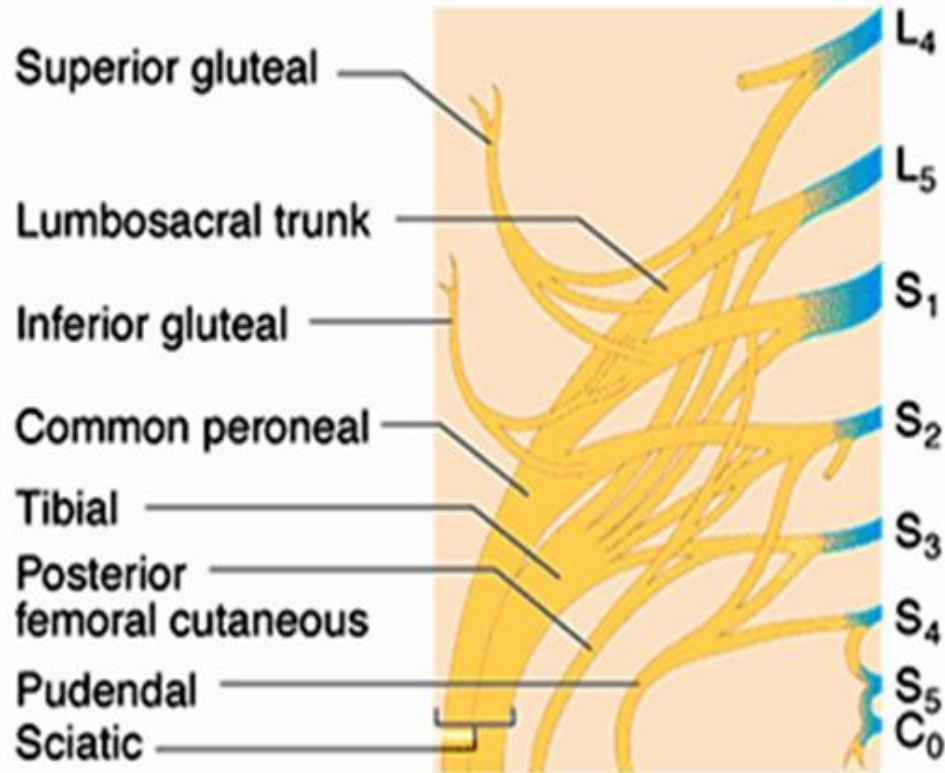
- The skin of the gluteal region is richly innervated by superior, middle, and inferior clunial nerves
- These superficial nerves supply the skin over the iliac crest, between the posterior superior iliac spines and over the iliac tubercles
- Consequently, these nerves are vulnerable to injury when bone is taken from the ilium for grafting

Deep gluteal nerves

- ❑ The deep gluteal nerves are:
 - superior &
 - inferior gluteal nerves
 - sciatic nerve
 - nerve to quadratus femoris
 - posterior cutaneous nerve of the thigh
 - nerve to obturator internus
 - pudendal nerve
- All of these nerves are branches of the **sacral plexus** and leave the pelvis through the **greater sciatic foramen**
- Except for the **superior gluteal nerve**, they all emerge **inferior to the piriformis**

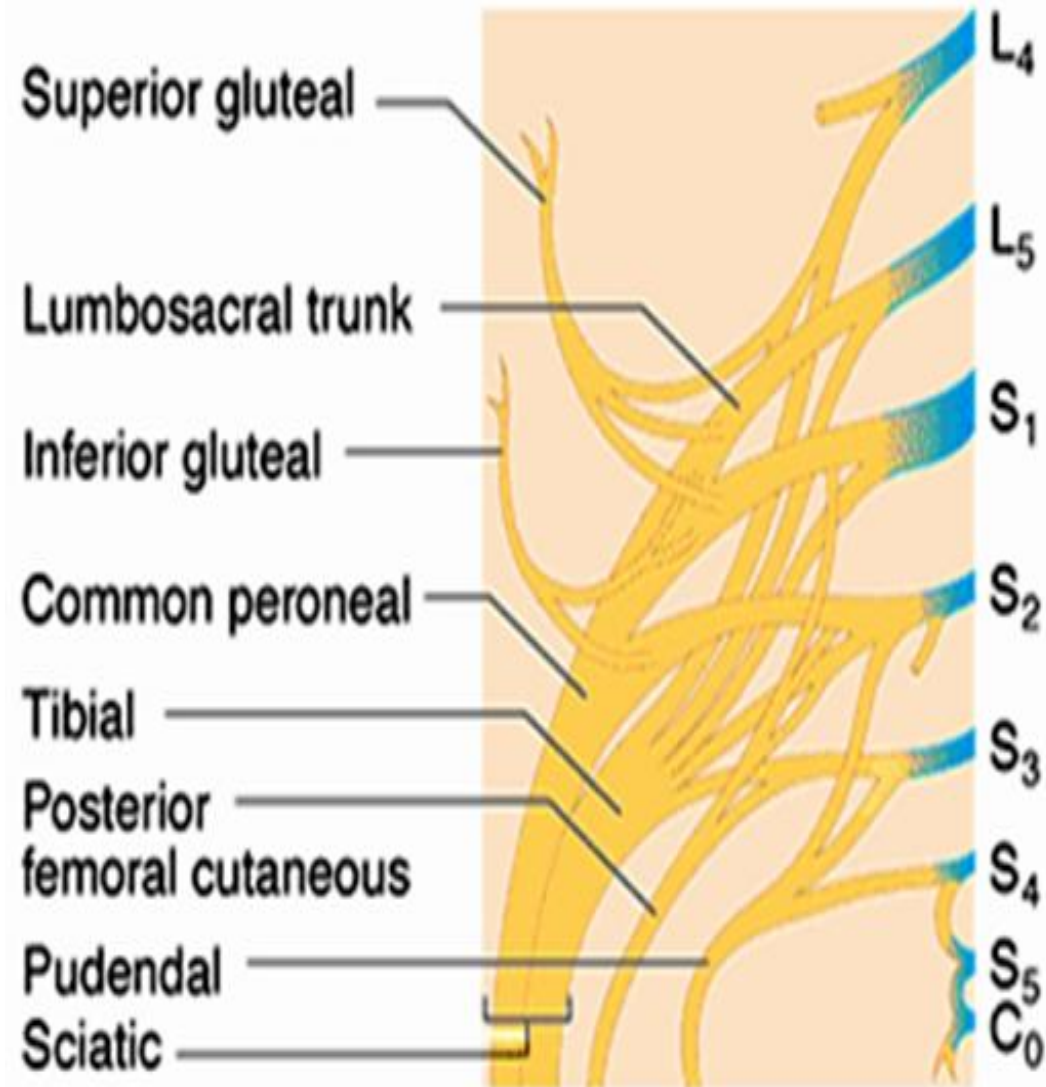
Sacral plexus

- **Originates from the anterior rami of (L4-S4)**
- **L4 is shared by both plexuses, a branch from it joining L5 to form the lumbosacral trunk**
- The sacral nerves emerge from the anterior sacral foramina and unite in front of piriformis where they are joined by the lumbosacral trunk



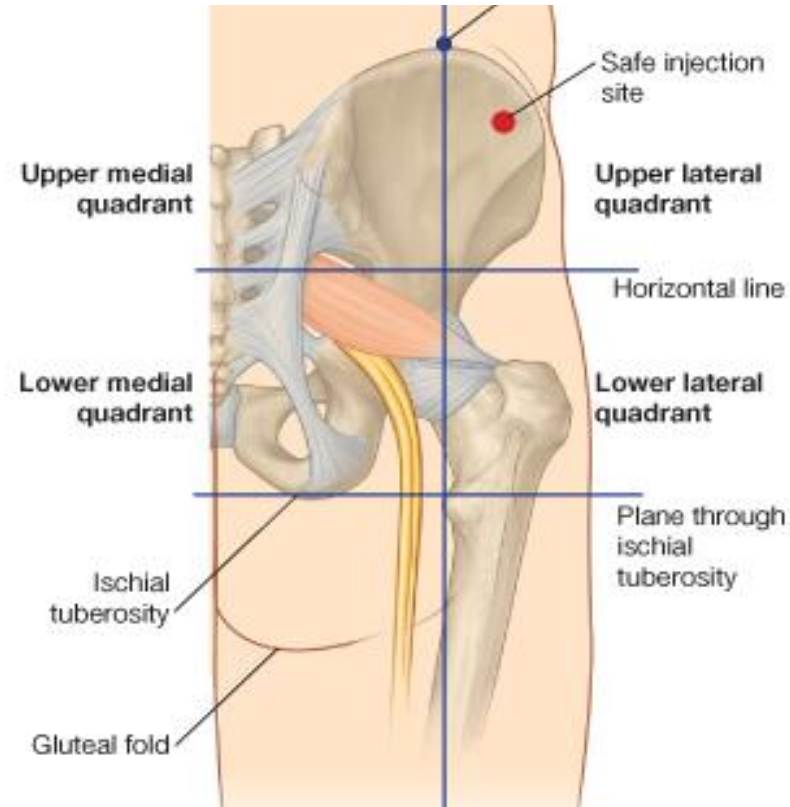
❑ Branches of sacral plexus

- Superior gluteal nerve -
- Inferior gluteal nerve
- Nerve to quadratus femoris
- Nerve to piriformis
- Nerve to obturator internus
- Posterior cutaneous nerve of the thigh
- Pudendal nerve
- Sciatic nerve



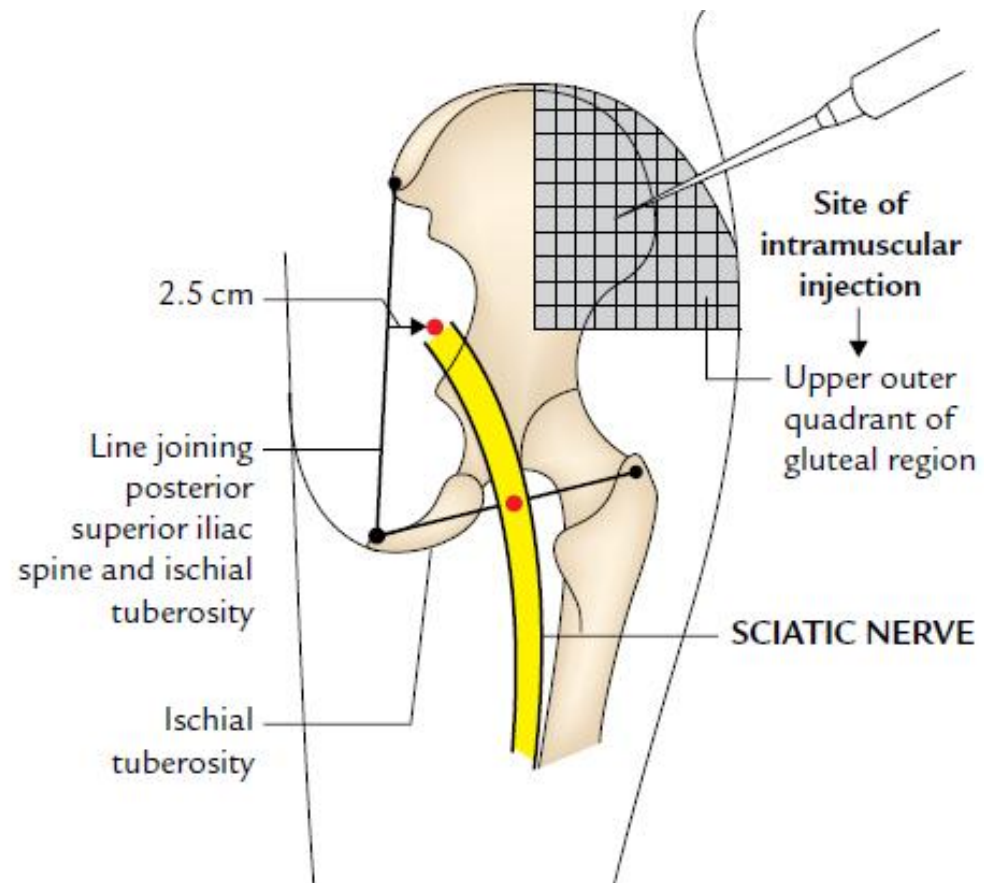
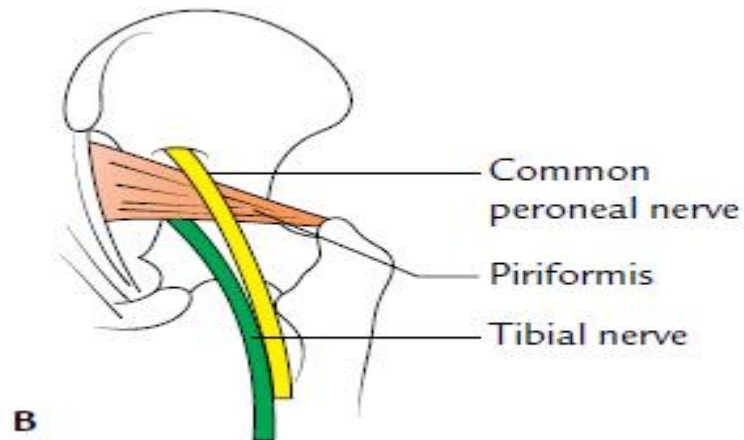
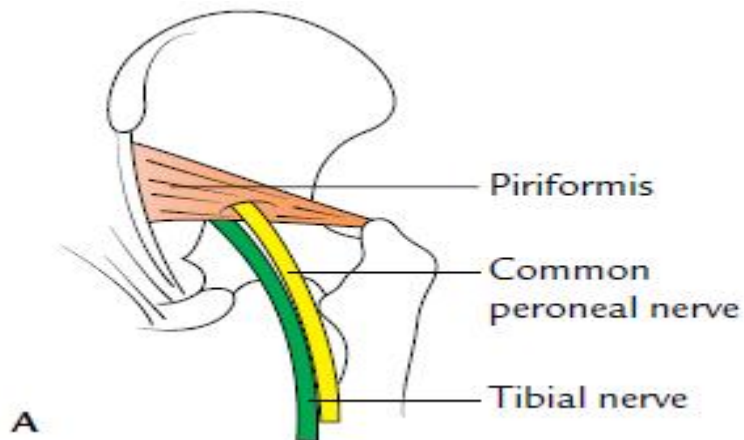
Sciatic nerve (L4-S3)

- **The largest nerve in the body**
- The continuation of the main part of the sacral plexus
 - The rami converge at the inferior border of the piriformis to form the sciatic nerve
- Terminate in the middle of the thigh by dividing into **tibial and common fibular nerves at apex of popliteal fossa**



- **The sciatic nerve is really two nerves which are loosely bound together in the same connective tissue sheath**
 - **tibial nerve**
 - **common fibular nerve -**
- The tibial and common fibular nerves usually separate approximately halfway or more down the thigh; however, in approximately **12% of people**, the nerves separate as they leave the pelvis
 - In these cases, the **tibial nerve passes inferior to the piriformis**, and the **common fibular nerve pierces this muscle or passes superior to it**

Pattern of Sciatic nerve (L4-S3)



□ Distribution

- The sciatic nerve supplies no structures in the gluteal region
- It supplies the posterior thigh muscles, all leg and foot muscles, and the skin of posterior thigh, most of the leg and the foot
- Motor for hamstring muscles
 - The nerve to the short head of the biceps femoris comes from the lateral (peroneal) division whereas branches to other hamstrings arise from the medial (tibial) division
- It also supplies the articular branches to all joints of the lower limb

Injury to sciatic nerve

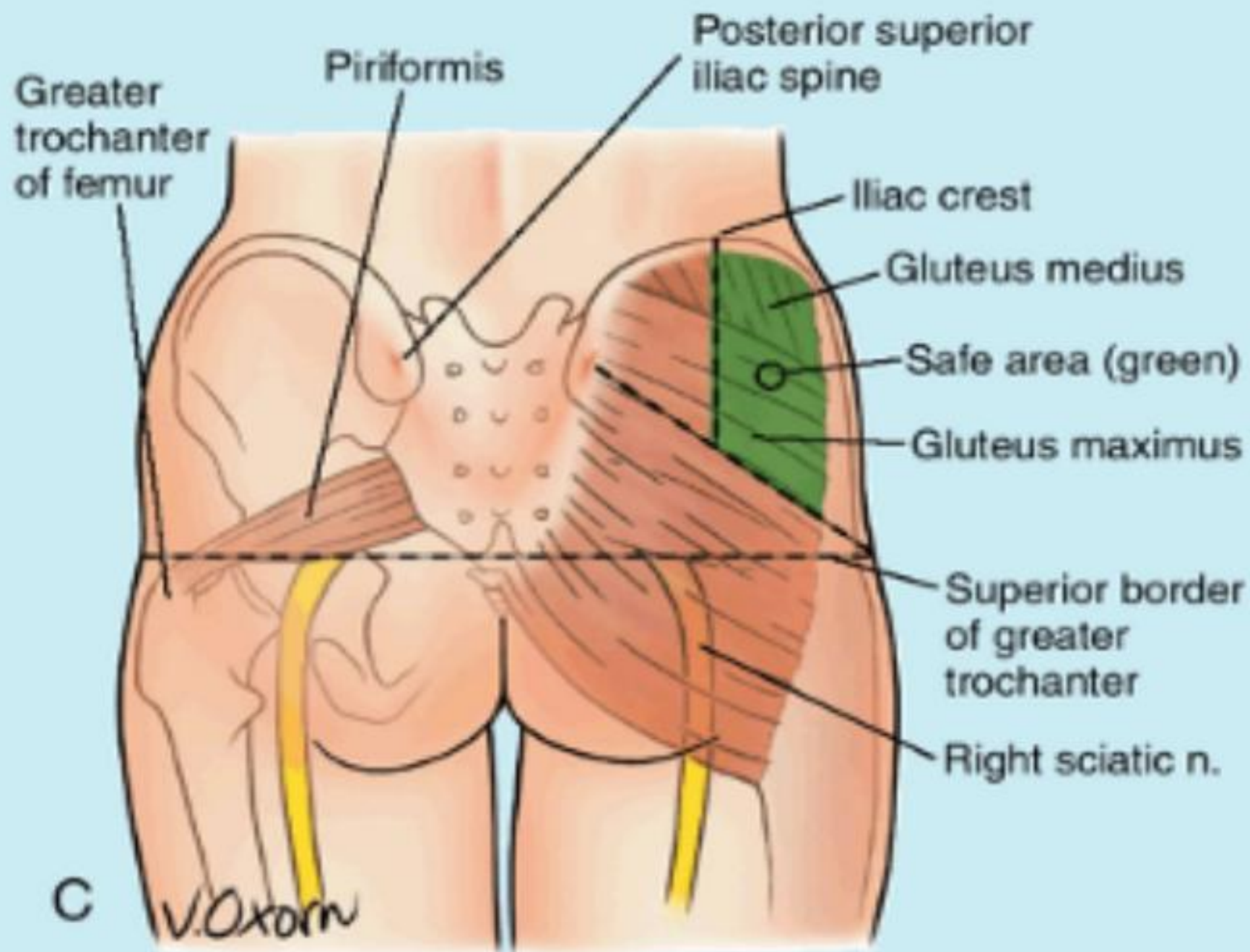
- **Piriformis syndrome (sciatica)**

- A pain in the buttock may result from compression of the sciatic nerve by the piriformis muscle
- Individuals involved in sports that require excessive use of the gluteal muscles (e.g., ice skaters, cyclists, and rock climbers)
- Complete section of the sciatic nerve is occur
- The leg is useless because extension of the hip is impaired, as is flexion of the leg
- All ankle and foot movements are also lost

Intragluteal injections

- The gluteal region is a common site for intramuscular injection of drugs
- The gluteal region is a favorable injection site because the muscles are thick and large; consequently, they provide a substantial volume for absorption of injected substances by intramuscular veins
- Complications of improper technique include nerve injury, hematoma, and abscess formation

- Injections into the buttock are safe only in the **superolateral quadrant** of the buttock or superior to a line extending from the PSIS to the superior border of the greater trochanter (approximating the superior border of the gluteus maximus)
- Intramuscular injections can also be given safely into the **anterolateral part** of the thigh, where the needle enters the tensor of the fascia lata as it extends distally from the iliac crest and ASIS
- **Deltoid , umblicus, vastus lateralis also important for intramuscular injection**



Superior gluteal nerve (L4, 5, S1)

- Runs along with the superior gluteal vessels, through the greater sciatic foramen (superior to the piriformis)
- Runs laterally across the lateral surface of the ilium between the gluteus medius and minimus, along with the deep branch of the superior gluteal artery
- It divides into a **superior** branch that supplies the gluteus medius and an **inferior** branch that continues to pass between the gluteus medius and the gluteus minimus to supply both muscles and the tensor of the fascia lata

Injury to superior gluteal nerve

- Injury to the superior gluteal nerve results in a characteristic motor loss, resulting in a disabling **gluteus medius limp**, to compensate for weakened abduction of the thigh by the gluteus medius and minimus, and/or a **gluteal gait**, a compensatory list of the body to the weakened gluteal side
- As a result of this compensation, the center of gravity is placed over the supporting lower limb
- Medial rotation of the thigh is also severely impaired.
- Two types of gluteal gait
 - **Waddling gait** = bilateral hip dislocation
 - **Dipping gait** = unilateral

- When a person with paralysis of the superior gluteal nerve is asked to stand on one leg, the pelvis descends on the intact side, indicating that the gluteus medius on the contralateral (supported) side is weak or non-functional
- This observation is referred to clinically as a **positive Trendelenburg test**



A. The pelvis is shown during normal gait

B. A positive Trendelenburg test indicates paralysis of superior gluteal nerve

Inferior gluteal nerve

- Leaves the pelvis through the greater sciatic foramen, inferior to the piriformis and superficial to the sciatic nerve, accompanied by multiple branches of the inferior gluteal artery and vein
- It also divides into several branches, which provide motor innervation to the overlying **gluteus maximus**

Nerve to quadratus femoris

- Leaves the pelvis anterior to the sciatic nerve and obturator internus and passes over the posterior surface of the hip joint
- It supplies an articular branch to this joint and innervates the **inferior gemellus and quadratus femoris**

Posterior cutaneous nerve of the thigh

- Supplies more skin than any other cutaneous nerve
- Its fibers from the anterior divisions of S2 and S3 supply the skin of the perineum
- Unlike most nerves bearing the name cutaneous, the main part of this nerve lies deep to the deep fascia (fascia lata), with only its terminal branches penetrating the subcutaneous tissue for distribution to the skin

Pudendal nerve

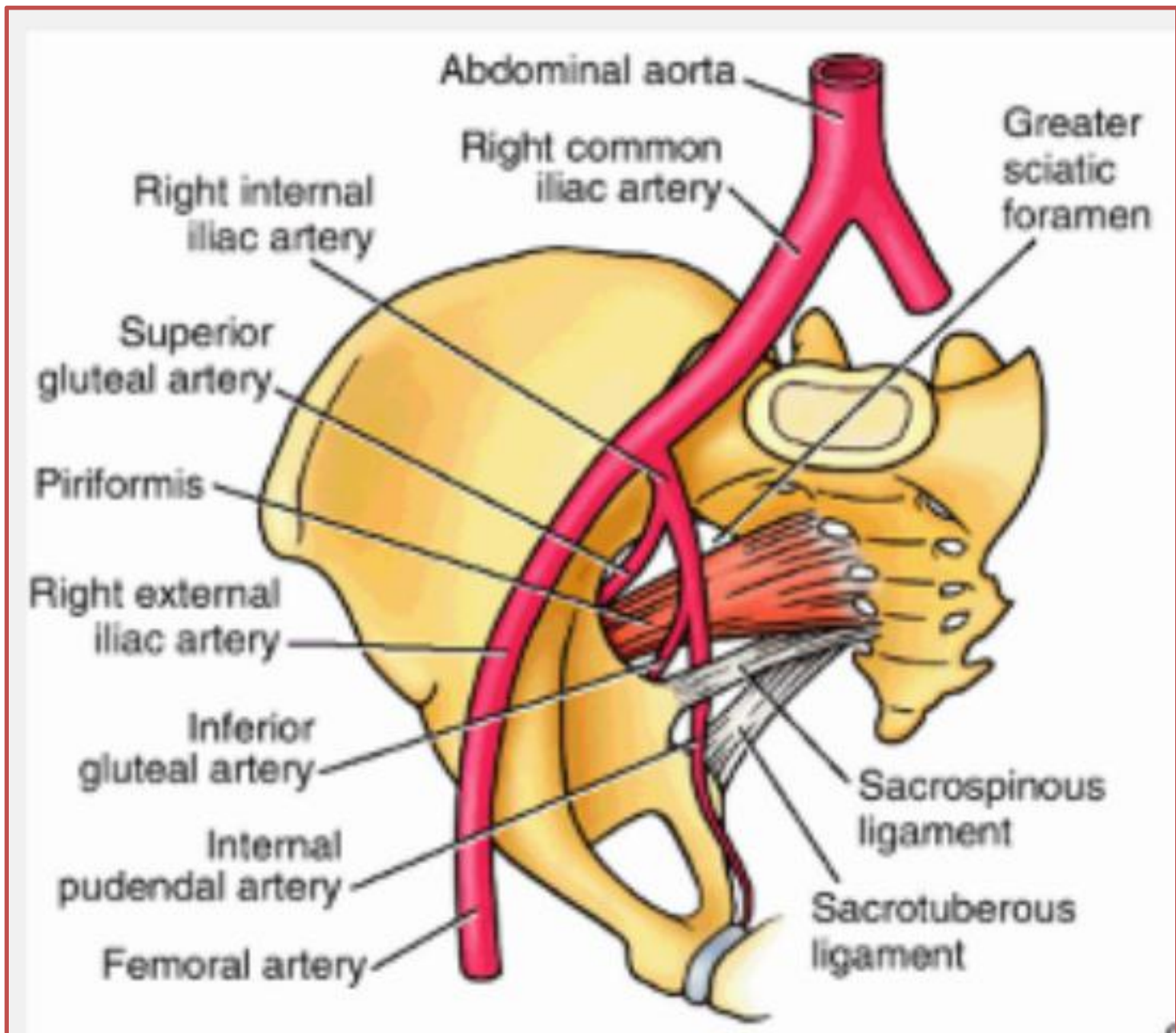
- The pudendal nerve is the most medial structure to exit the pelvis through the greater sciatic foramen inferior to the piriformis muscle
- It descends posterolateral to the sacrospinous ligament and enters the perineum through the lesser sciatic foramen
- Supply structures in the **perineum**
- it supplies **no structures in the gluteal region or posterior thigh**

Nerve to obturator internus

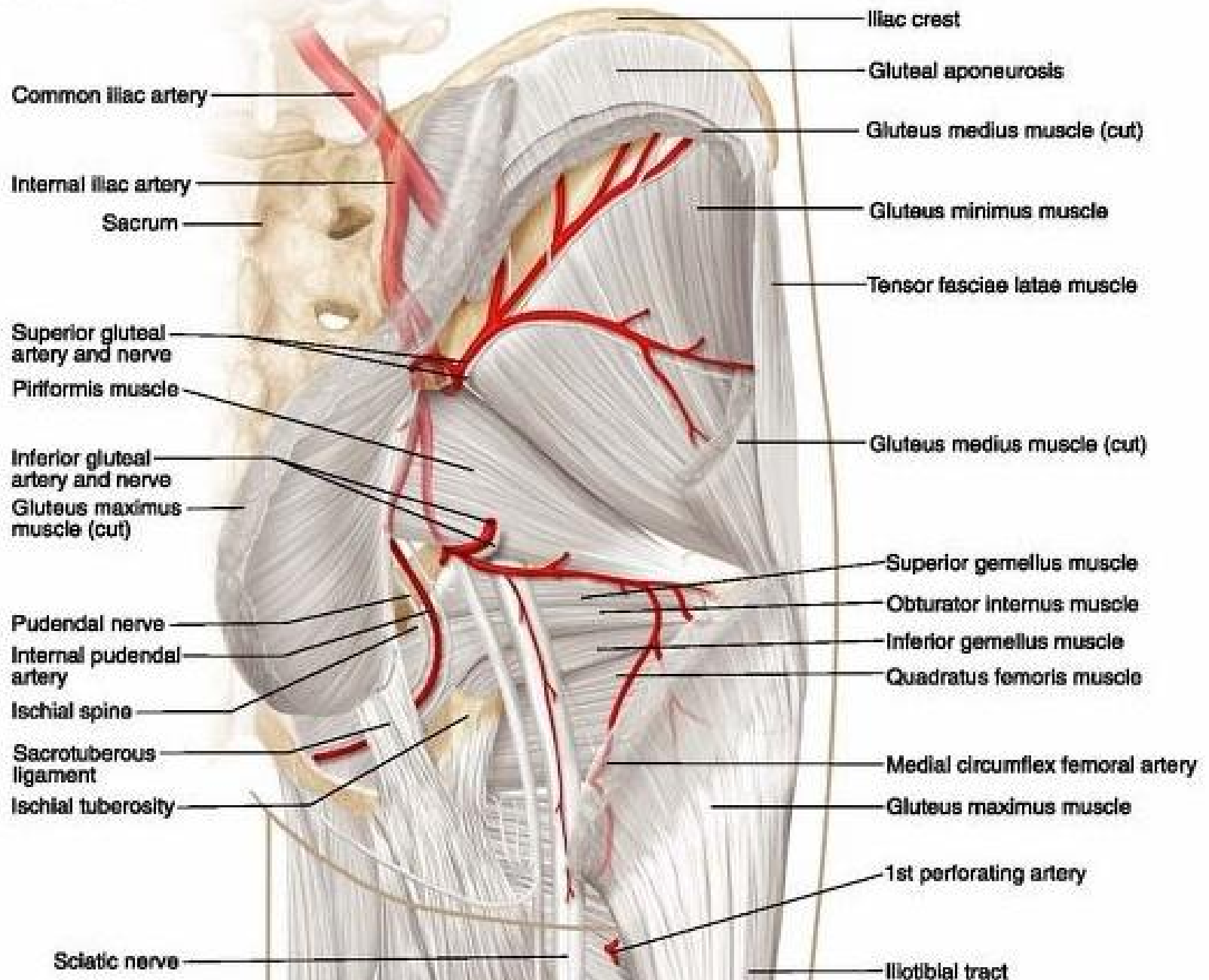
- Arises from the anterior divisions of the anterior rami of the L5 - S2 nerves and parallels the course of the pudendal nerve
- As it passes around the base of the ischial spine, it supplies the superior gemellus
- **After entering the perineum via the lesser sciatic foramen, it supplies the obturator internus muscle & superior gemellus.**

Arteries of the gluteal region and posterior thigh

- The arteries of the gluteal region arise, directly or indirectly, from the **internal iliac arteries**
- The major branches of the internal iliac artery that supply or traverse the gluteal region are
 - superior gluteal artery
 - inferior gluteal artery
 - internal pudendal artery
- the posterior compartment of the thigh **has no major artery exclusive to the compartment;**
- it receives blood from multiple sources:
- **inferior gluteal, medial circumflex femoral, perforating, and popliteal arteries**



A. Arteries



Superior gluteal artery

- **The largest branch of the internal iliac artery**
- Leaves the pelvis through the greater sciatic foramen, superior to the piriformis
- it divides immediately into **superficial and deep branches**
- The **superficial branch** supplies the gluteus maximus and skin over the proximal attachment of this muscle
- The **deep branch** supplies the gluteus medius, gluteus minimus, and tensor of the fascia lata
- *The superior gluteal artery anastomoses with the inferior gluteal and medial circumflex femoral arteries*

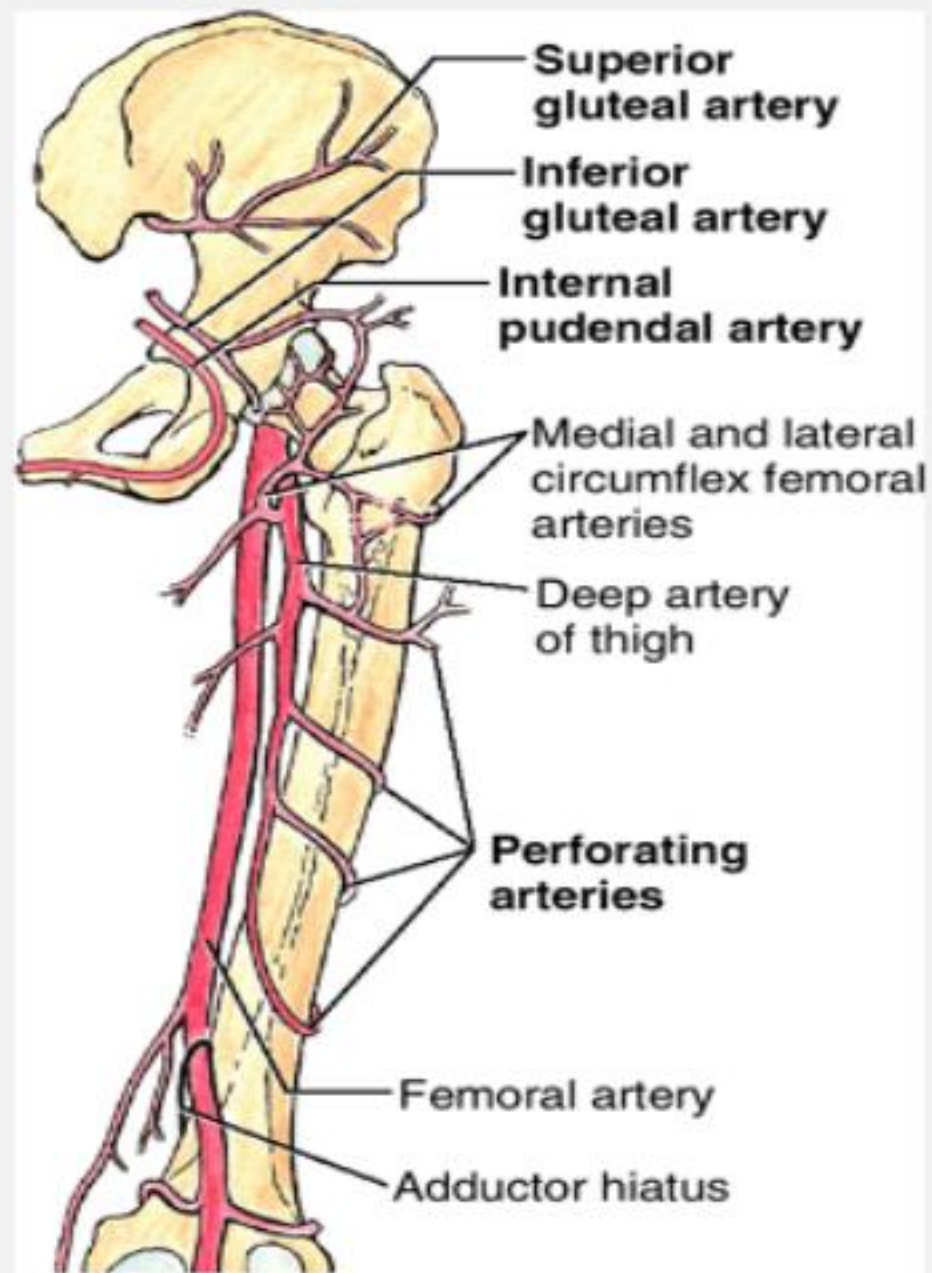
Inferior gluteal artery

- Arises from the internal iliac artery
- Leaves the pelvis through the greater sciatic foramen, inferior to the piriformis
- Enters the gluteal region deep to the gluteus maximus
- Descends medial to the sciatic nerve
- Supplies *gluteus maximus, obturator internus, quadratus femoris, and superior parts of the hamstrings.*

- It anastomoses with the superior gluteal artery and frequently participates in the cruciate anastomosis of the thigh
- involving the first perforating arteries of the deep artery of the thigh and the medial and lateral circumflex femoral arteries
- Developmentally, the **inferior gluteal artery** is the major artery of the posterior compartment, traversing its length and becoming continuous with the popliteal artery

Internal pudendal artery

- Arises from the **internal iliac artery**
- It entering the gluteal region through the greater sciatic foramen inferior to the piriformis
- It enters the perineum through the lesser sciatic foramen
- Like the pudendal nerve, it supplies the skin, external genitalia, and muscles in the perineal region
- It does not supply any structures in the gluteal region or posterior thigh



Posterior view

Veins of the gluteal region and posterior thigh

- The gluteal veins are tributaries of the internal iliac veins that drain blood from the gluteal region
- **Superior and inferior gluteal veins**
 - accompany the corresponding arteries through the greater sciatic foramen, superior and inferior to the piriformis, respectively
 - they communicate with tributaries of the femoral vein, thereby providing alternate routes for the return of blood from the lower limb if the femoral vein is occluded or has to be ligated

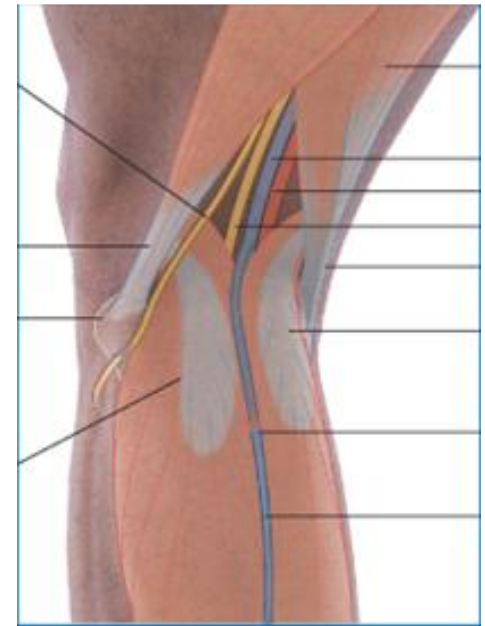
- **Internal pudendal veins**
 - accompany the internal pudendal arteries
 - join to form a single vein that enters the internal iliac vein
 - these veins drain blood from the external genitalia or pudendum
- **Perforating veins**
 - accompany the arteries of the same name
 - drain blood from the posterior compartment of the thigh into the deep vein of the thigh
 - the perforating veins, like the arteries, usually also communicate inferiorly with the popliteal vein and superiorly with the inferior gluteal vein

Popliteal fossa

- The popliteal fossa is a fat-filled diamond-shaped space posterior to the knee
- All important vessels and nerves from the thigh to the leg pass through this fossa
- Distal continuation of the adductor canal

□ Boundaries

- Laterally & inferiorly – lateral head of gastrocnemius
- Medially & inferiorly- medial head of gastrocnemius
- Laterally & superiorly - biceps femoris
- Medially & superiorly – semimembranosus



❑ Floor is formed by

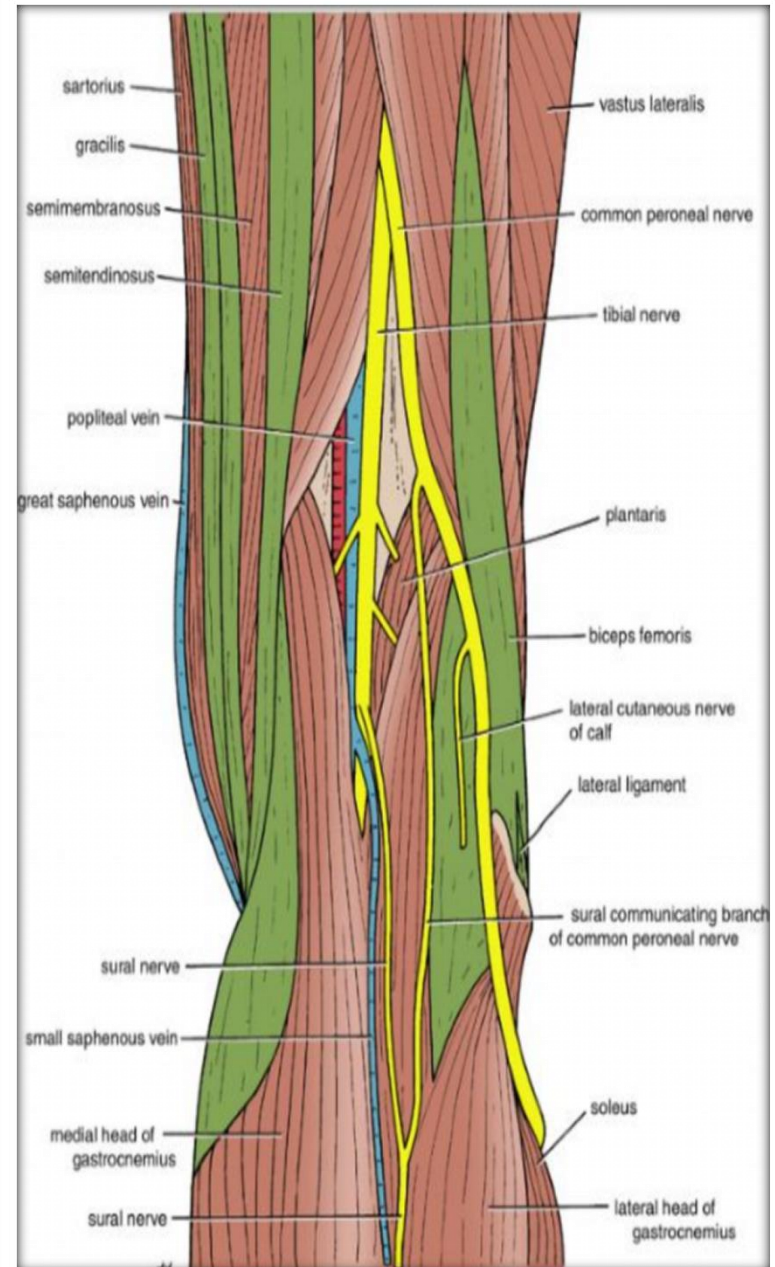
- popliteal surface of the femur
- capsule of the knee joint
- fascia over the popliteus muscle

❑ Roof

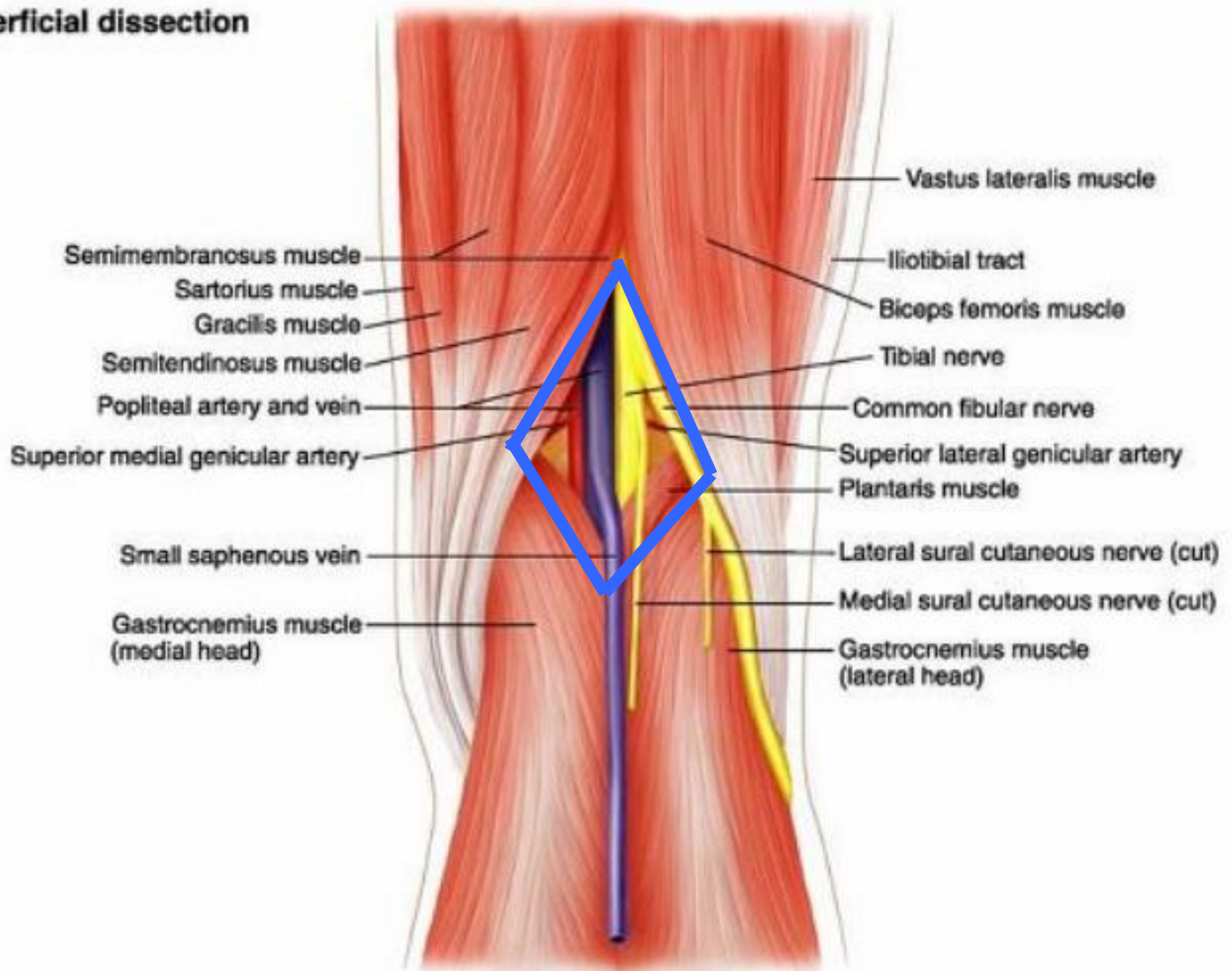
- deep fascia which is pierced by the small saphenous vein as this enters the popliteal vein

❑ Contents

- **popliteal vessels (artery, vein and lymphatics)**
- **tibial and common peroneal (fibular) nerves**
- **small saphenous vein**
- **popliteal lymph nodes**



A. Superficial dissection



Popliteal artery

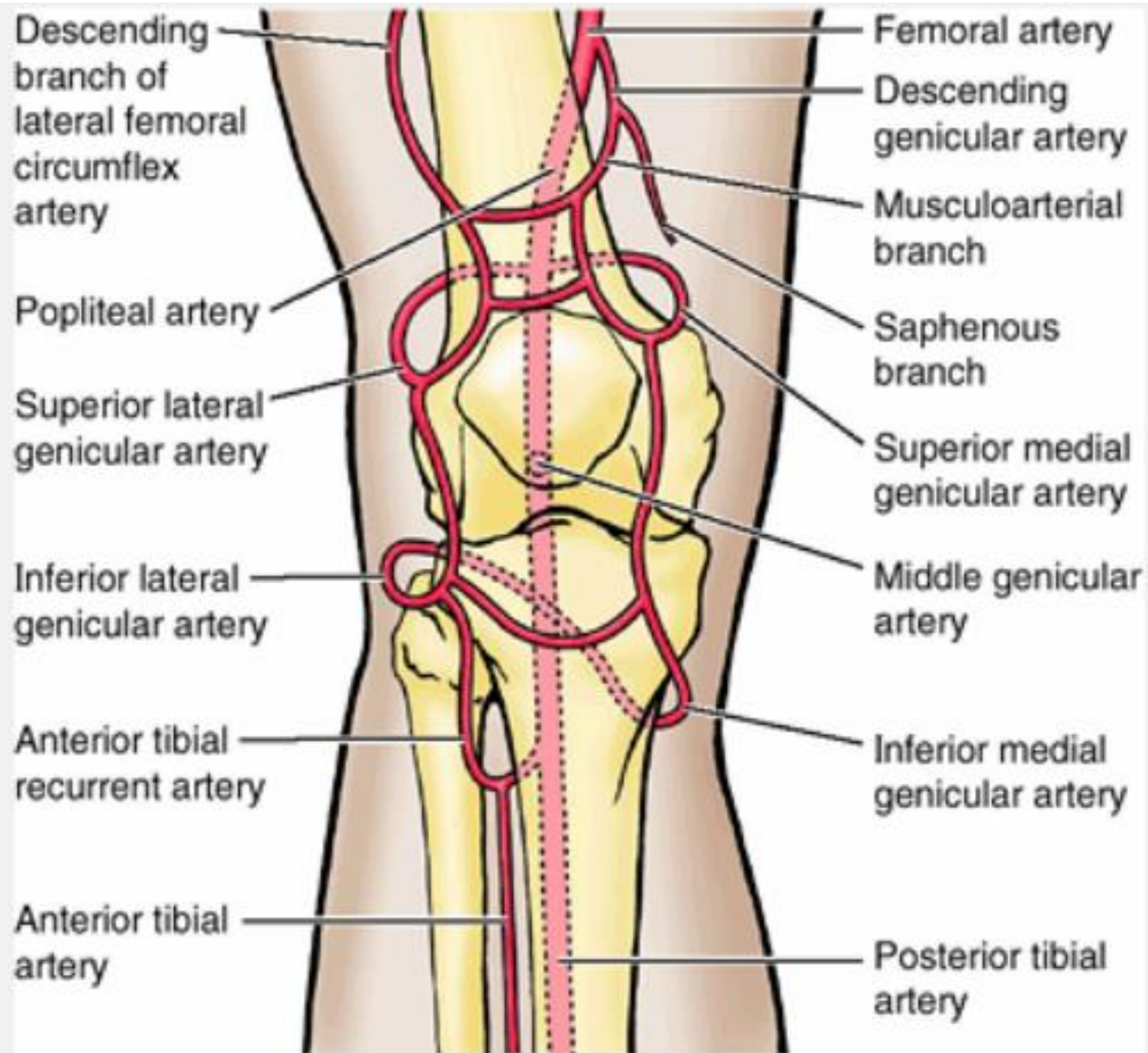
- Continuation of the femoral artery
- Begins where the femoral artery passes through the adductor hiatus
- Passes through the popliteal fossa
- Ends at the inferior border of the popliteus by dividing into anterior and posterior tibial arteries
- The deepest structure in the popliteal fossa runs close to the joint capsule of the knee joint

❑ Genicular branches

❑ Five genicular branches of the popliteal artery supply the joint capsule and ligaments of the knee joint

- superior lateral
- superior medial
- Middle
- inferior lateral
- inferior medial

– They participate in the formation of the **genicular anastomosis** (L. genu, knee), a network of vessels around the knee that provide collateral circulation capable of maintaining blood supply to the leg during full knee flexion, which may kink the popliteal artery



Anterior view

Popliteal vein

- Begins as an assembly of venae comitantes below the knee
- Formed at distal end of popliteus
- Terminate in the adductor hiatus
- Course in relation to the artery
 - **lower part - medial**
 - **middle – posterior**
 - **upper – posterolateral**
- **Tributaries**
 - main – vena comitantes of anterior and posterior tibial artery
 - muscular and cutaneous
 - genicular veins
 - small saphenous vein

Popliteal pulse

- Because the popliteal artery is deep in the popliteal fossa, it may be difficult to feel the popliteal pulse
- Palpation of this pulse is commonly performed by placing the person in the prone position with the knee flexed to relax the popliteal fascia and hamstrings
- The pulsations are best felt in the inferior part of the fossa
- **Weakening or loss of the popliteal pulse is a sign of femoral artery obstruction**

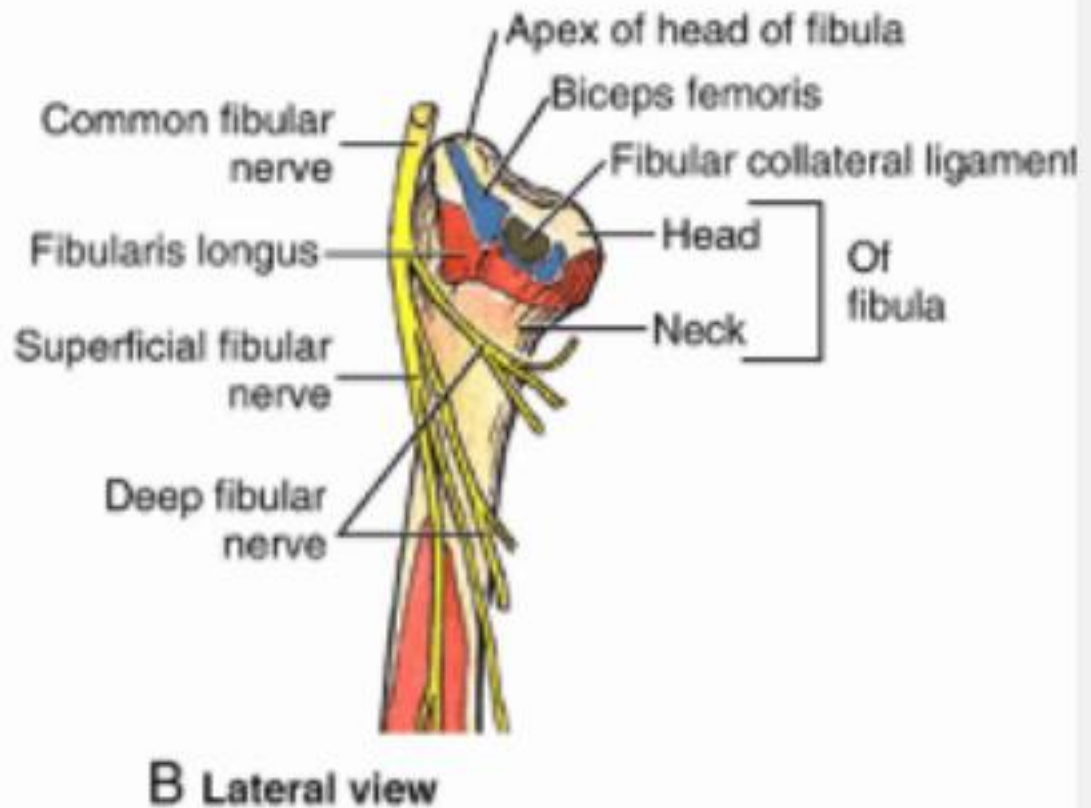
Nerves in popliteal fossa

- The sciatic nerve usually ends at the **superior angle** of the **popliteal fossa** by dividing into tibial and common fibular nerves
- **Tibial nerve**
 - The medial, larger terminal branch of the sciatic nerve
 - The **most superficial of the three main central components** of the popliteal fossa (**nerve, vein, and artery**)

- **Common fibular nerve**

- The lateral, smaller terminal branch of the sciatic nerve
- Begins at the superior angle of the popliteal fossa and follows closely the medial border of the biceps femoris and its tendon along the superolateral boundary of the popliteal fossa
- Leaves the fossa by passing superficial to the lateral head of the gastrocnemius and then passes over the posterior aspect of **the head of the fibula**

- Winds around the fibular neck, where it is vulnerable to injury
- Here it divides into its terminal branches, the **superficial and deep fibular nerves**

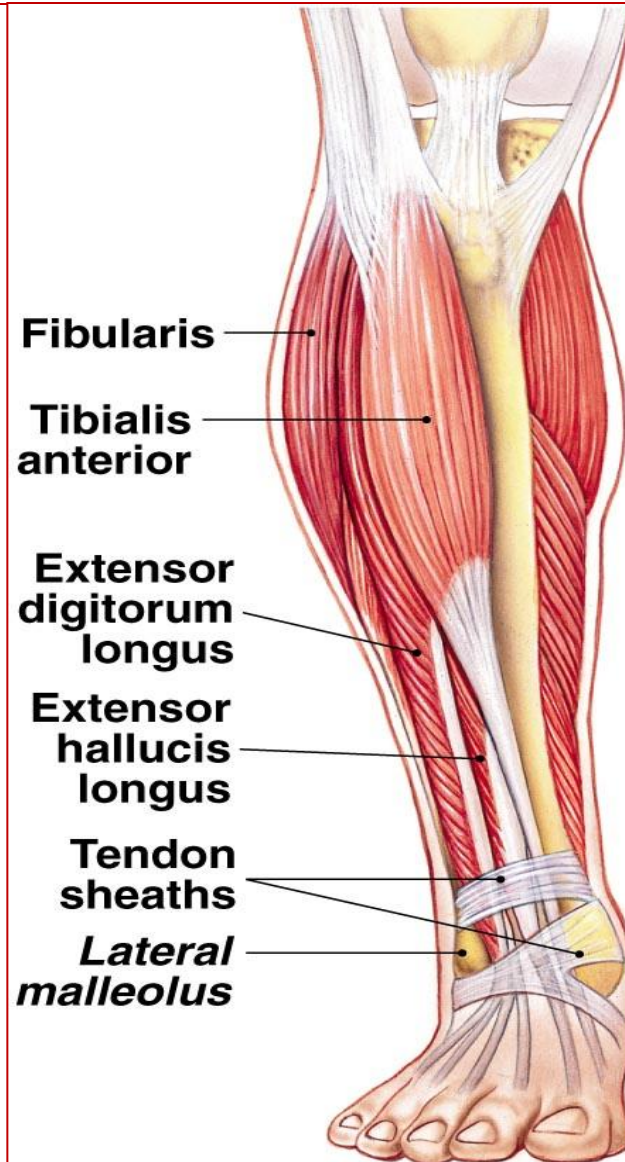


Leg

- The leg or crus is the portion of the lower limb between the knee and the ankle
- **It is divided into 3 compartments by anterior and posterior intermuscular septa and interosseous membrane**
 - **anterior crural compartment**
 - **posterior crural compartment**
 - **lateral (fibular or peroneal) compartment**

Anterior compartment of leg

- Dorsiflexor (extensor) compartment
- Located anterior to the interosseous membrane
- **It contain four muscles**
- **Anterior compartment is innervated by deep fibular nerve**
- **All supply by anterior tibial artery**



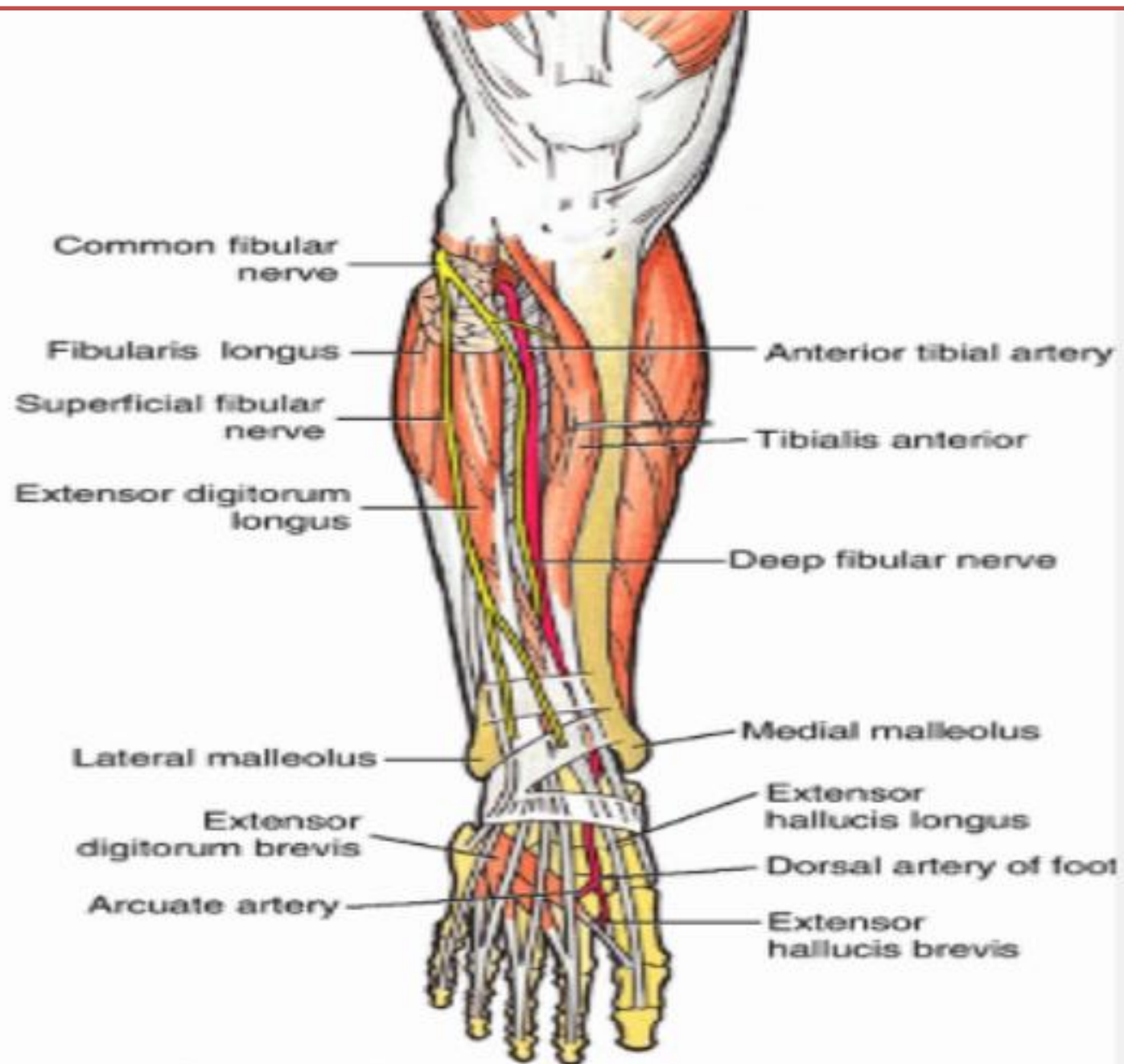
(c) Anterior view, right leg

- **Deep fibular nerve**

- one of the two terminal branches of the common fibular nerve
- the nerve of the anterior compartment
- arises between the fibularis longus muscle and the neck of the fibula
- **after entering the compartment, the nerve accompanies the anterior tibial artery**

- **Anterior tibial artery**

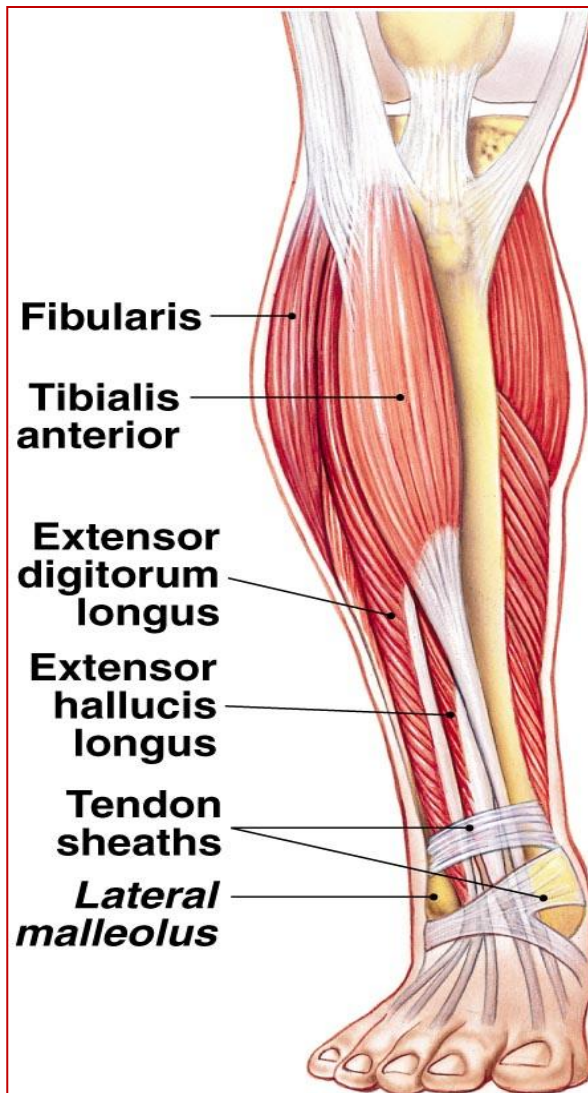
- Supplies structures in the anterior compartment
- The **smaller terminal branch of the popliteal artery**
- Begins at the inferior border of the popliteus muscle
- Passes between the tibialis anterior and the extensor digitorum longus
- Ends at the ankle joint, midway between the malleoli, where **it becomes the dorsal artery of the foot (L. arteria dorsalis pedis)**



D Anterior view

Muscles of anterior compartment

- Primarily **toes extensors** and **dorsiflexors** of ankle joint
 - Tibialis anterior (**inversion**)
 - Extensor digitorum longus
 - Extensor hallucis longus
 - Peroneus tertius (**eversion**)
- **Nerve supply: deep fibular**
- **Arterial supply by anterior tibial artery**



(c) Anterior view, right leg

Copyright © 2007 Pearson Education, Inc., publishing as Benjamin Cummings

Lateral compartment of leg

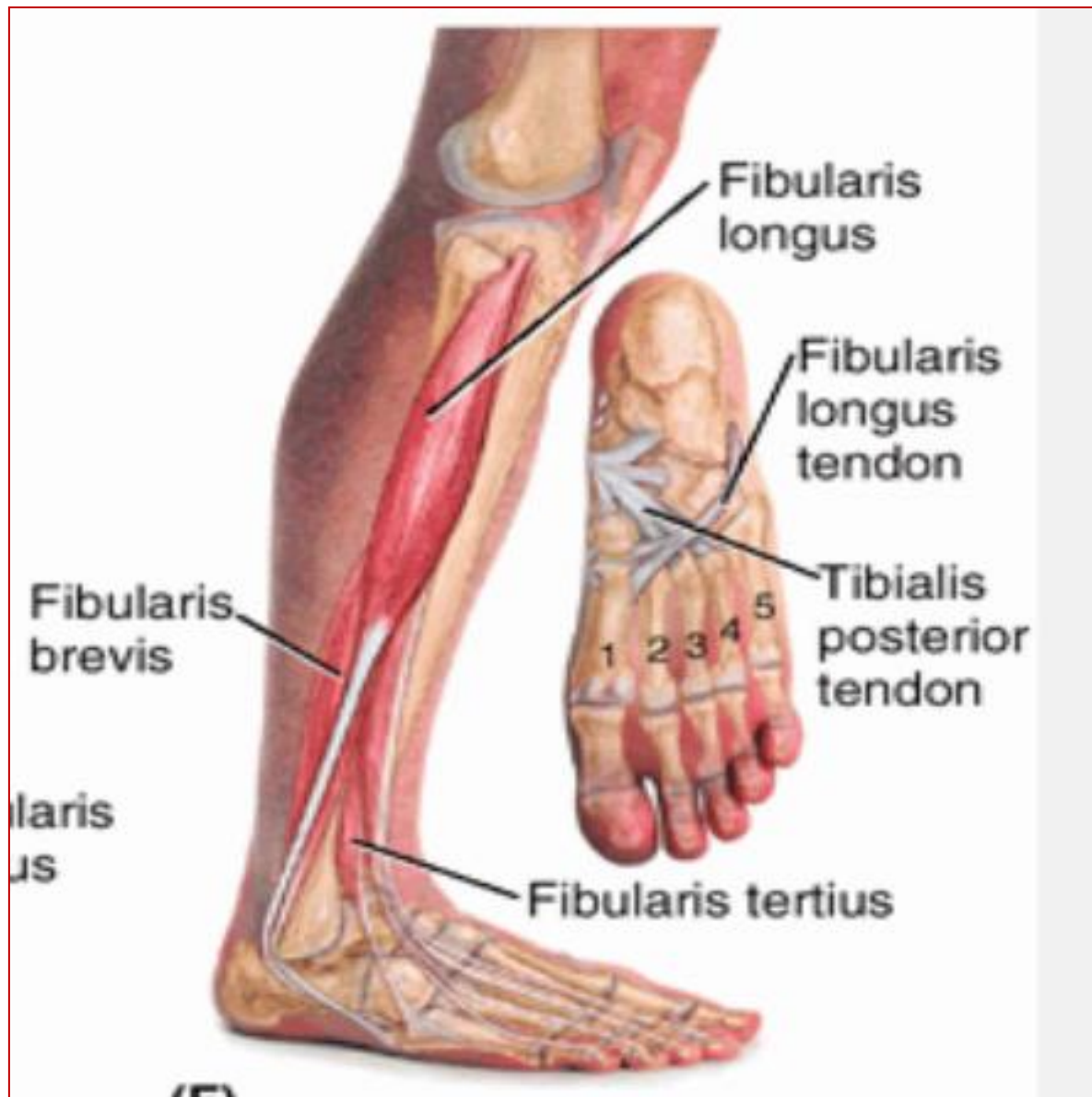
Lateral compartment

- There are only two muscles in the lateral compartment
 - **Fibularis longus**
 - Its name is due to long tendon and superficial to the brevis
 - **fibularis brevis**
 - It is because it has short tendon ,deep to fibularis longus

NB : the lateral compartment of leg muscle are innervated by **superficial fibular nerve** terminal branch of the common fibular nerve

The action of lateral compartment

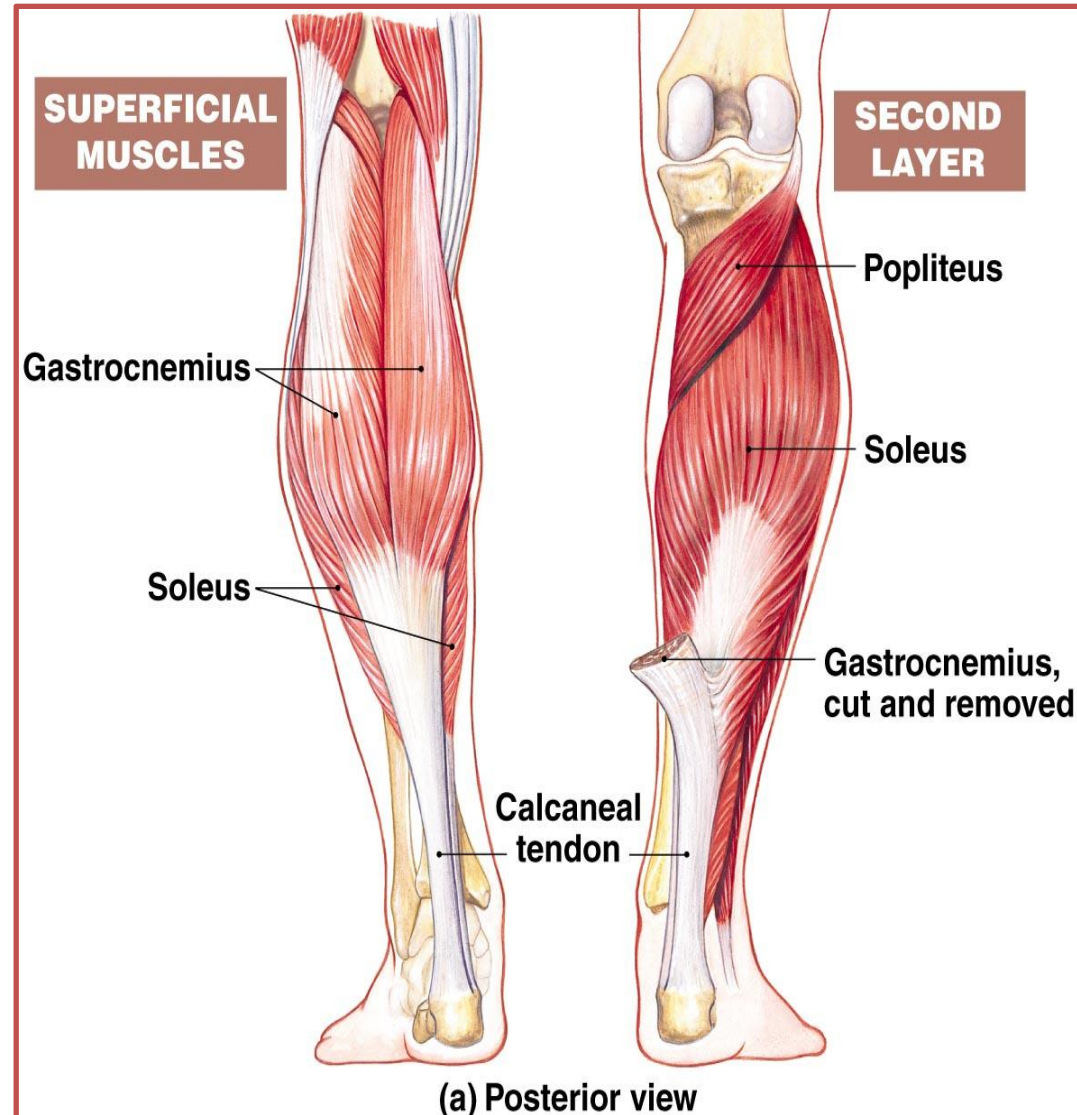
- **Plantar flexor and evertor of the foot, fibular abduction and maintain transverse arch**
- The muscles are arterial supplied **proximally** by perforating branches of the **anterior tibial artery** and **distally** by perforating branches of the **fibular artery branches from posterior tibial artery**
- These perforating arteries have accompanying veins (venae comitantes)

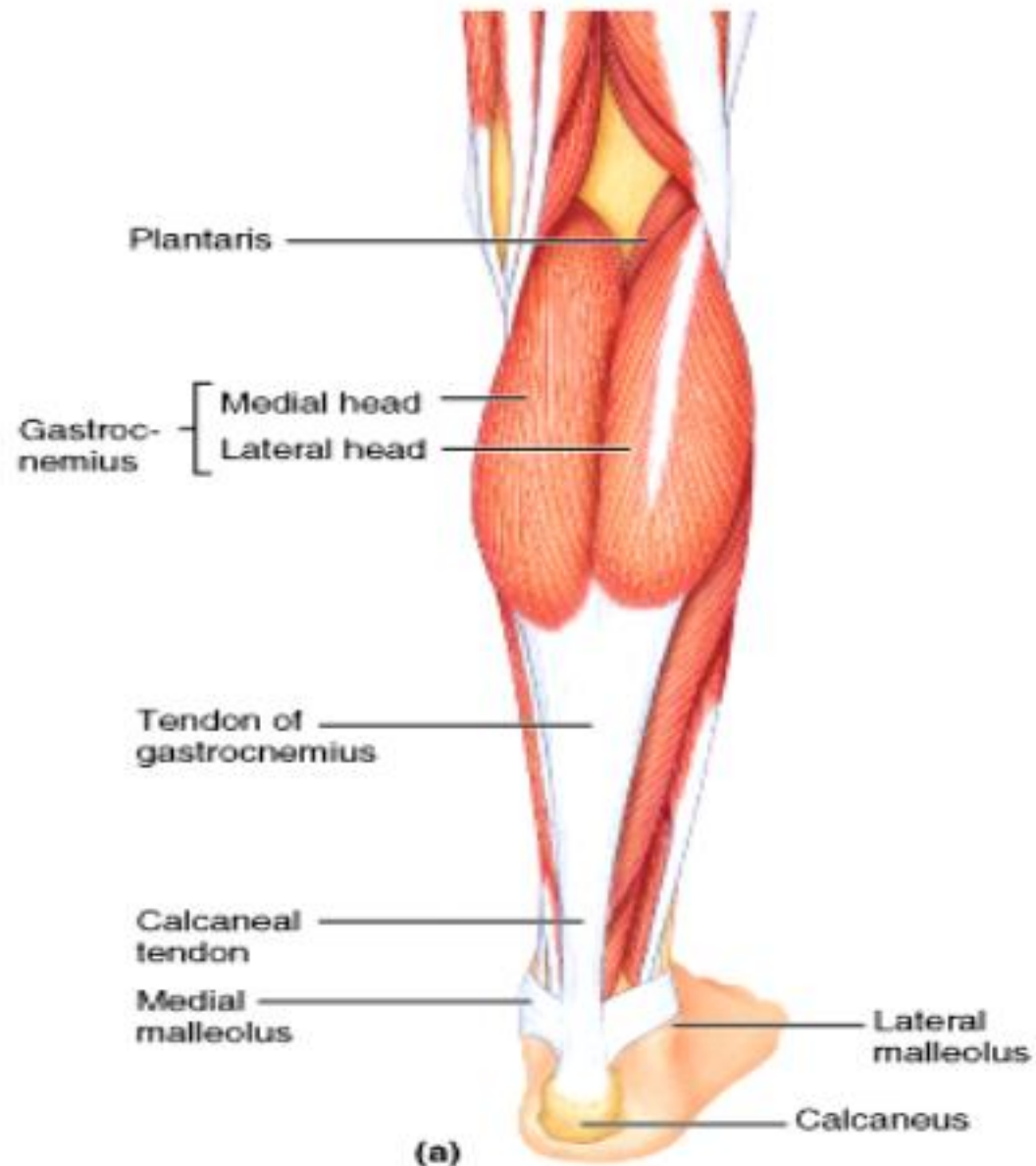


Muscles of posterior compartment of leg

- **Action :** Plantar flexors and flex the toes
- **Superficial group**
- Arranged in two layers
- has 3 muscles:
 - **Gastrocnemius**
 - **Soleus**
 - **Plantaris**

❑ **These three muscles form triceps surae**

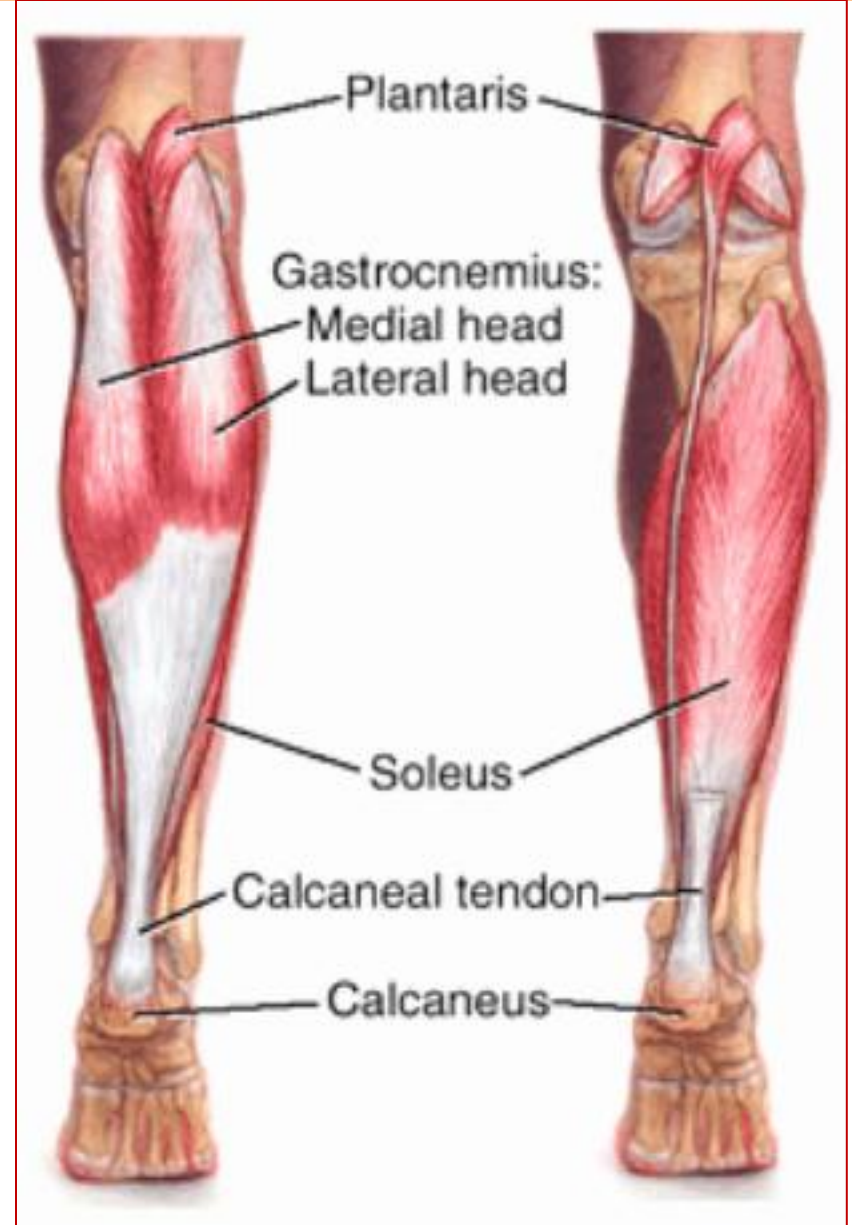




Copyright © 2001 Benjamin Cummings, an imprint of Addison Wesley Longman, Inc.

Gastrocnemius strain

- Gastrocnemius strain (**tennis leg**) is a painful calf injury resulting from partial tearing of the **medial belly of the gastrocnemius** at or near its musculotendinous junction
- It is caused by overstretching the muscle by concomitant full extension of the knee and dorsiflexion of the ankle joint



- **Deep group of muscles contains**

- **Popliteus**

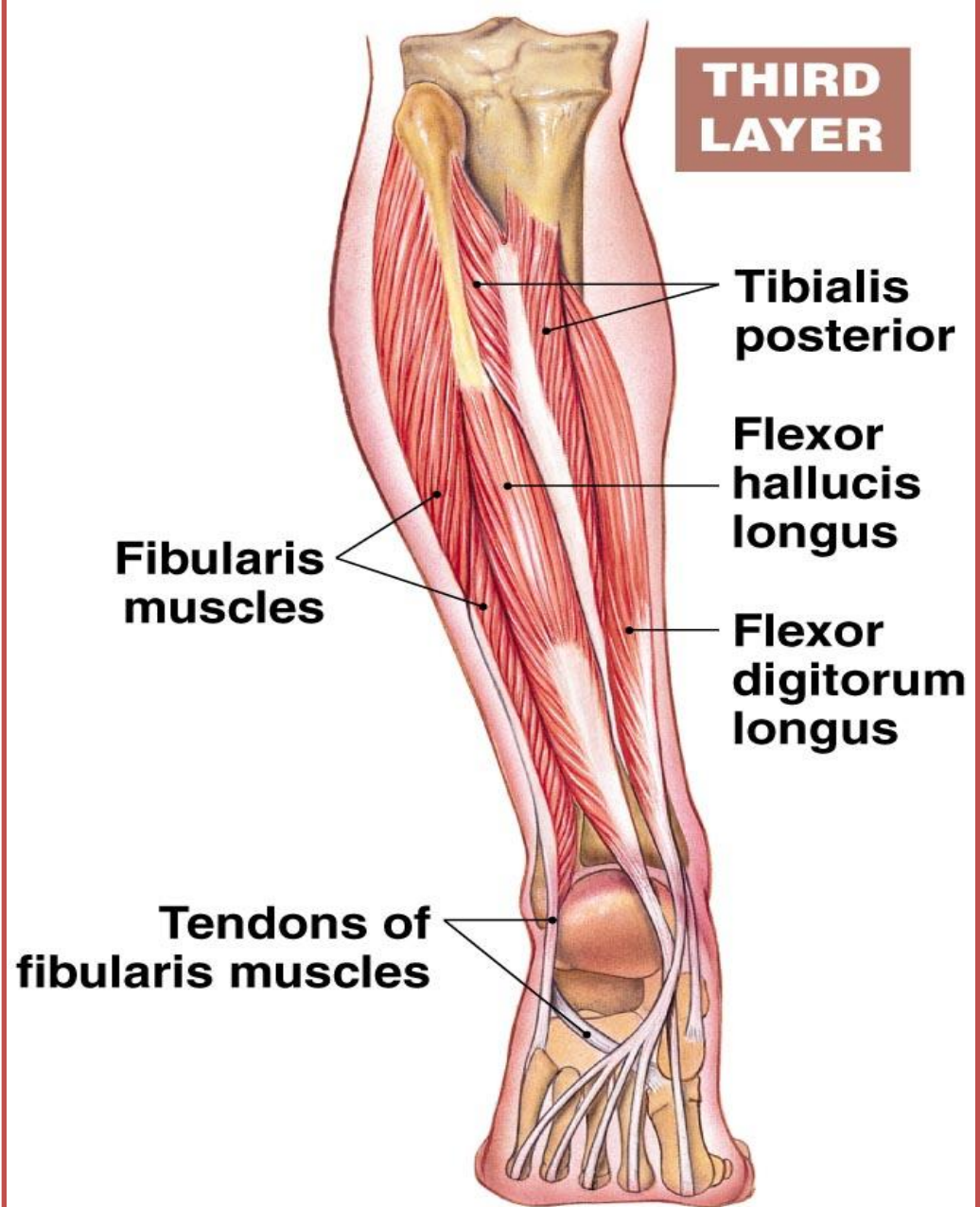
(unlock the knee joint)

- **Flexor hallucis longus**

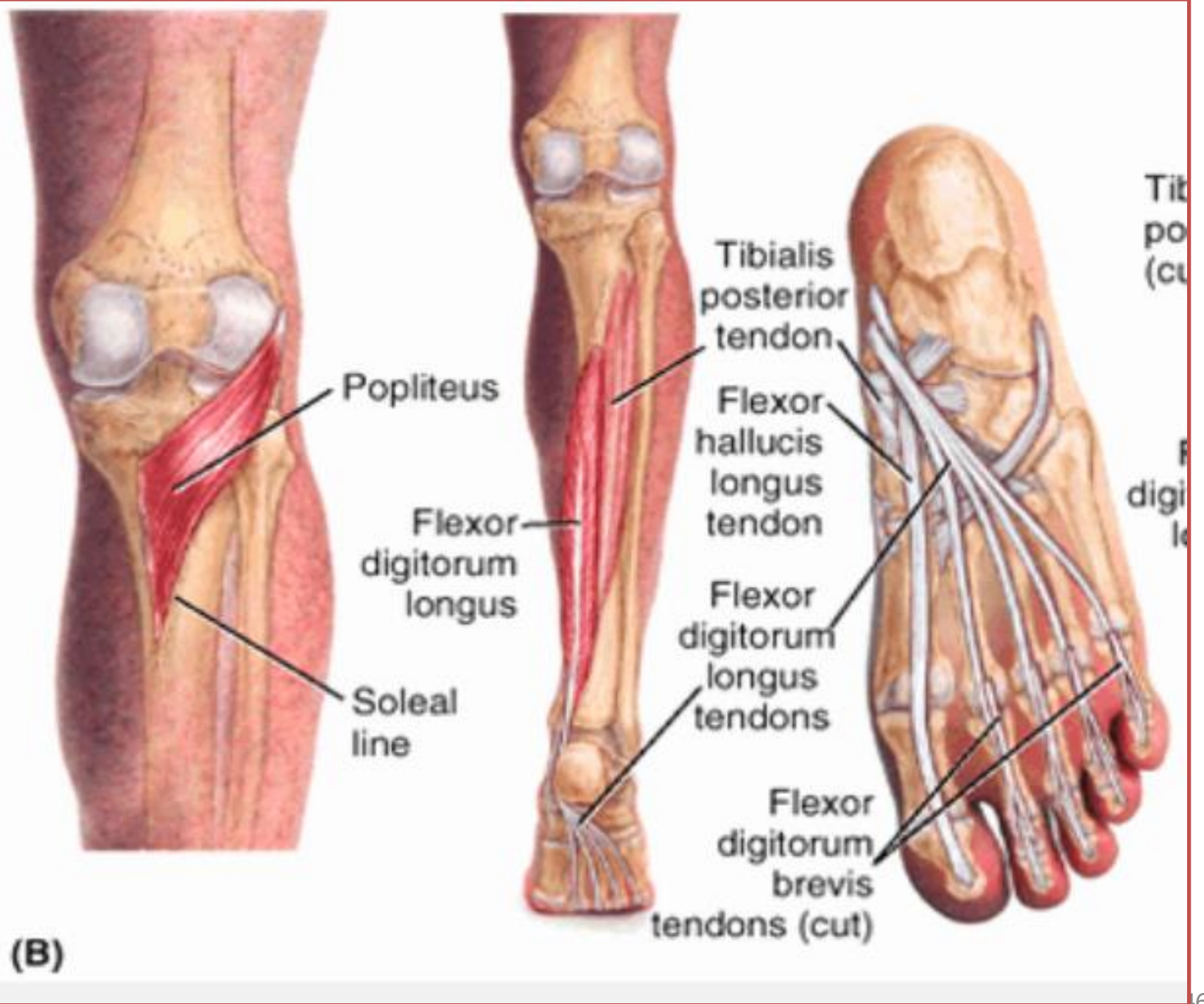
Invert great toe

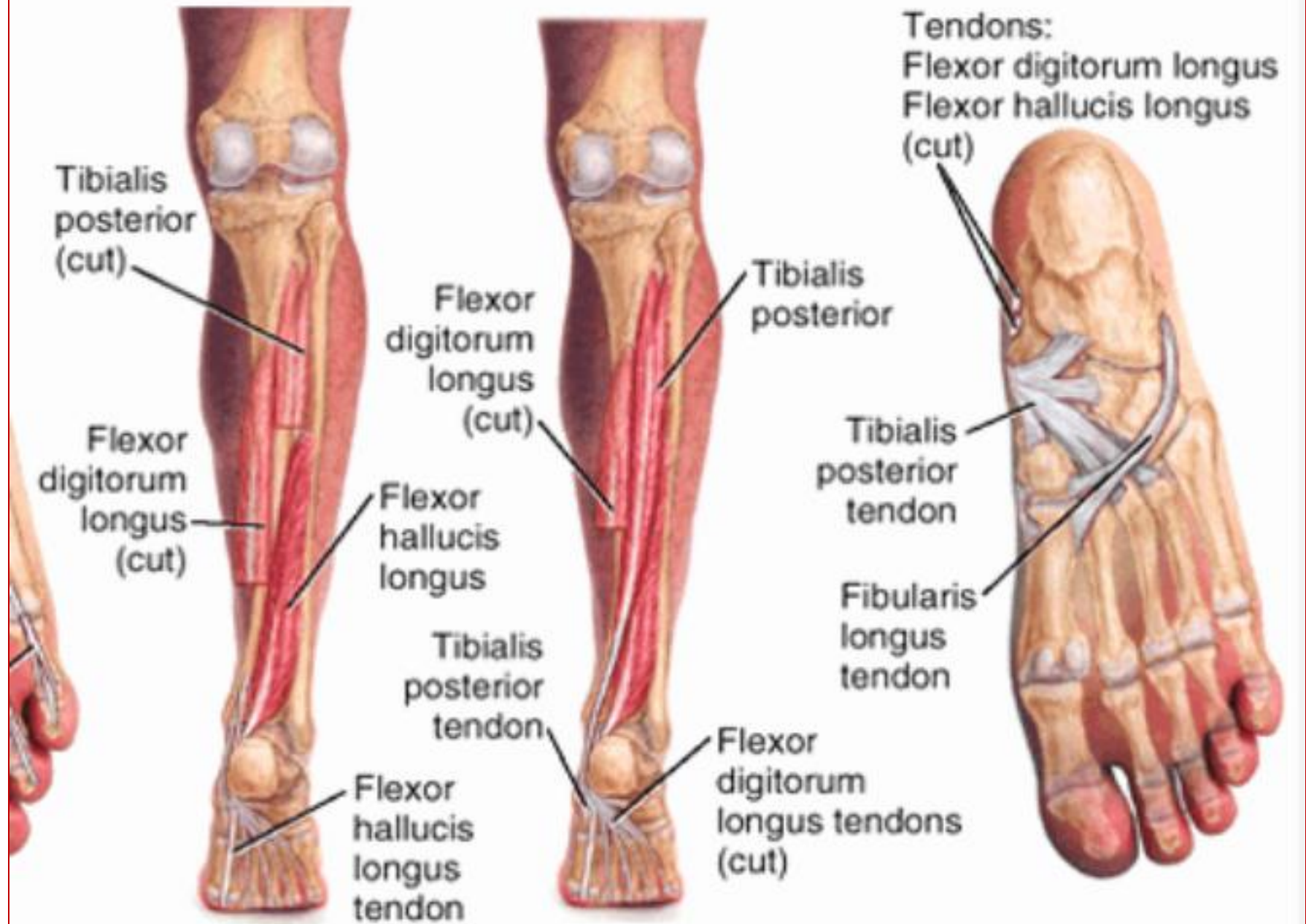
- **Flexor digitorum longus**

- **Tibialis posterior**



(b) Posterior view, left leg

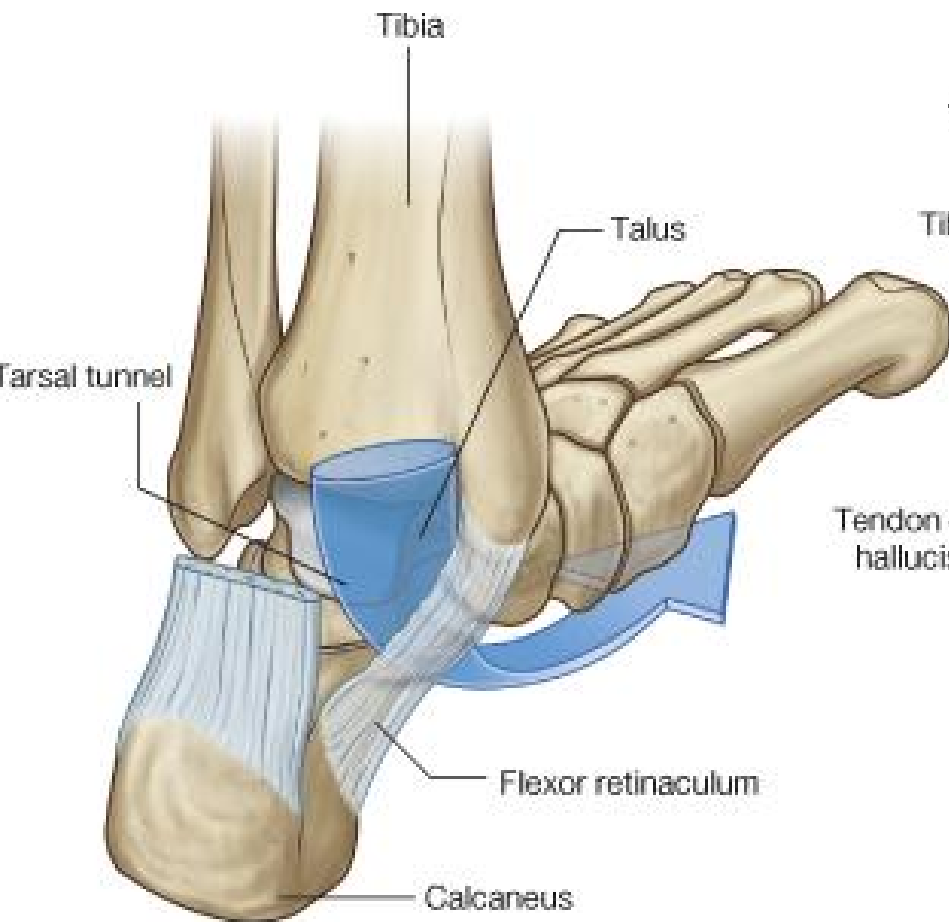




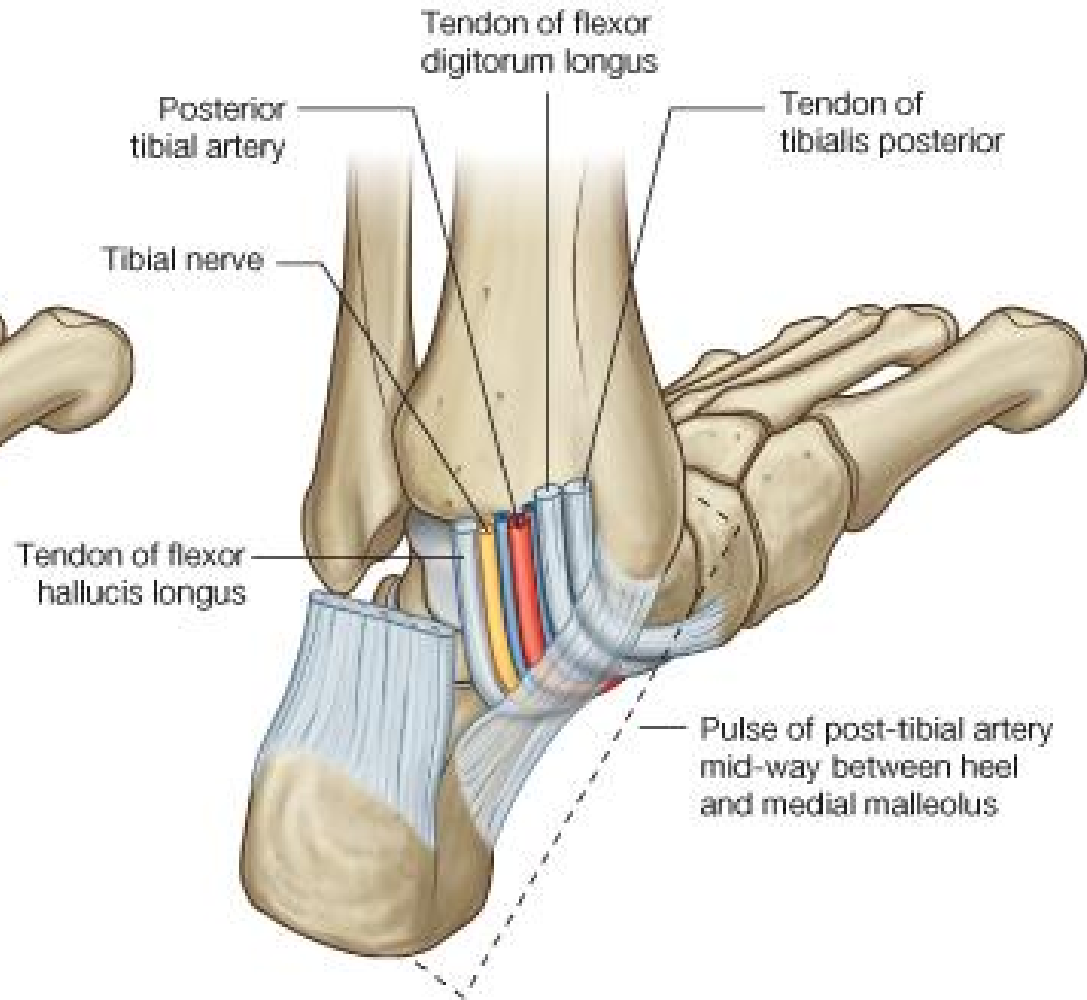
Tarsal tunnel

- **Medial malleolus to calcaneus**
- **Content – from medial to lateral**
 - Tendon of tibialis posterior
 - Tendons of flexor digitorum longus
 - Termination of posterior tibial artery and beginning of lateral and medial plantar arteries
 - Termination of posterior tibial nerve and beginning of medial and lateral plantar nerves
 - Tendon of flexor hallucis longus

A

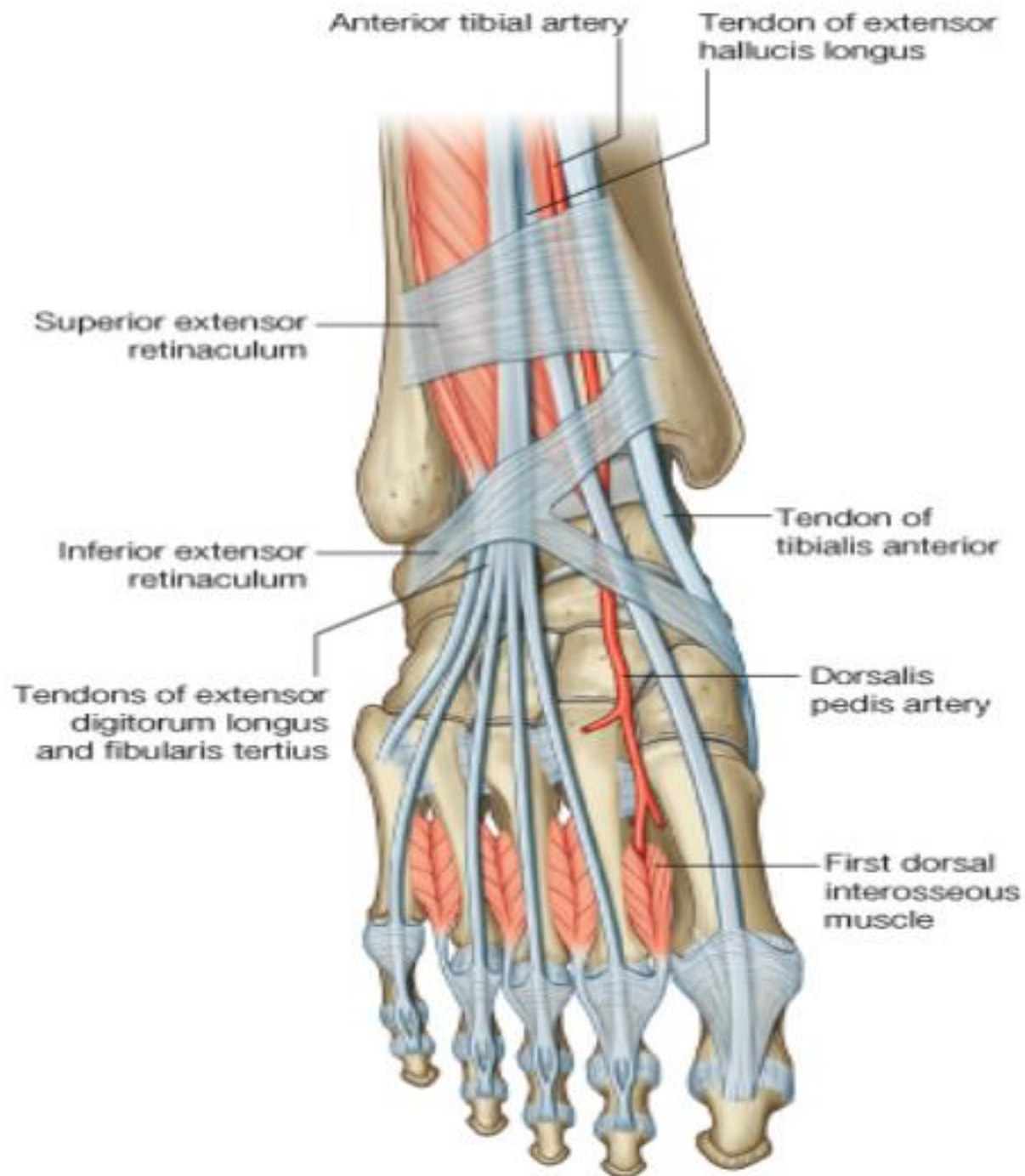


B



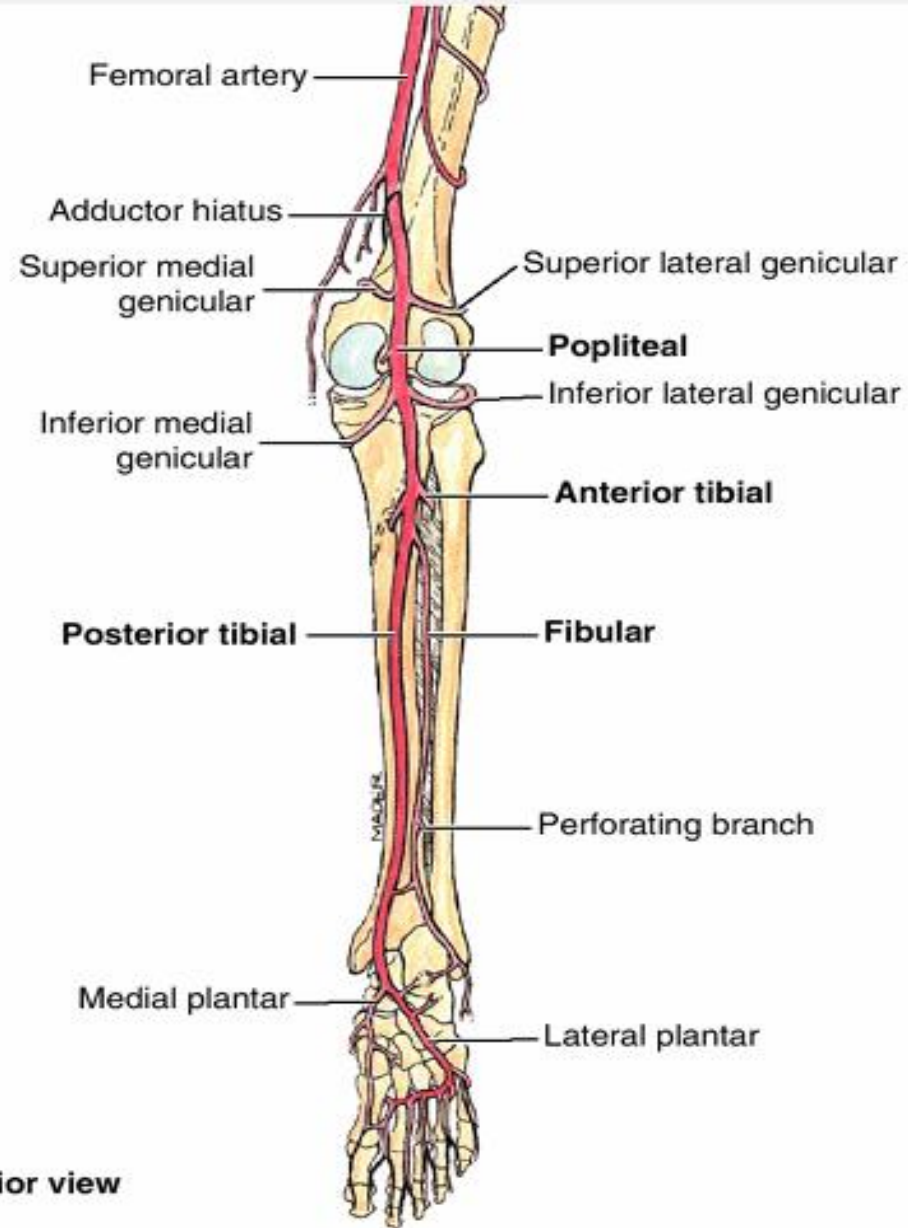
Extensor retinaculum

- Has superior and inferior portions
- Prevents the tendons of peroneus tertius, extensor hallucis longus and extensor digitorum longus from bowstringing during contraction
- There is also a superior and an inferior peroneal (fibular) retinaculum for the tendons of the peroneus brevis and longus



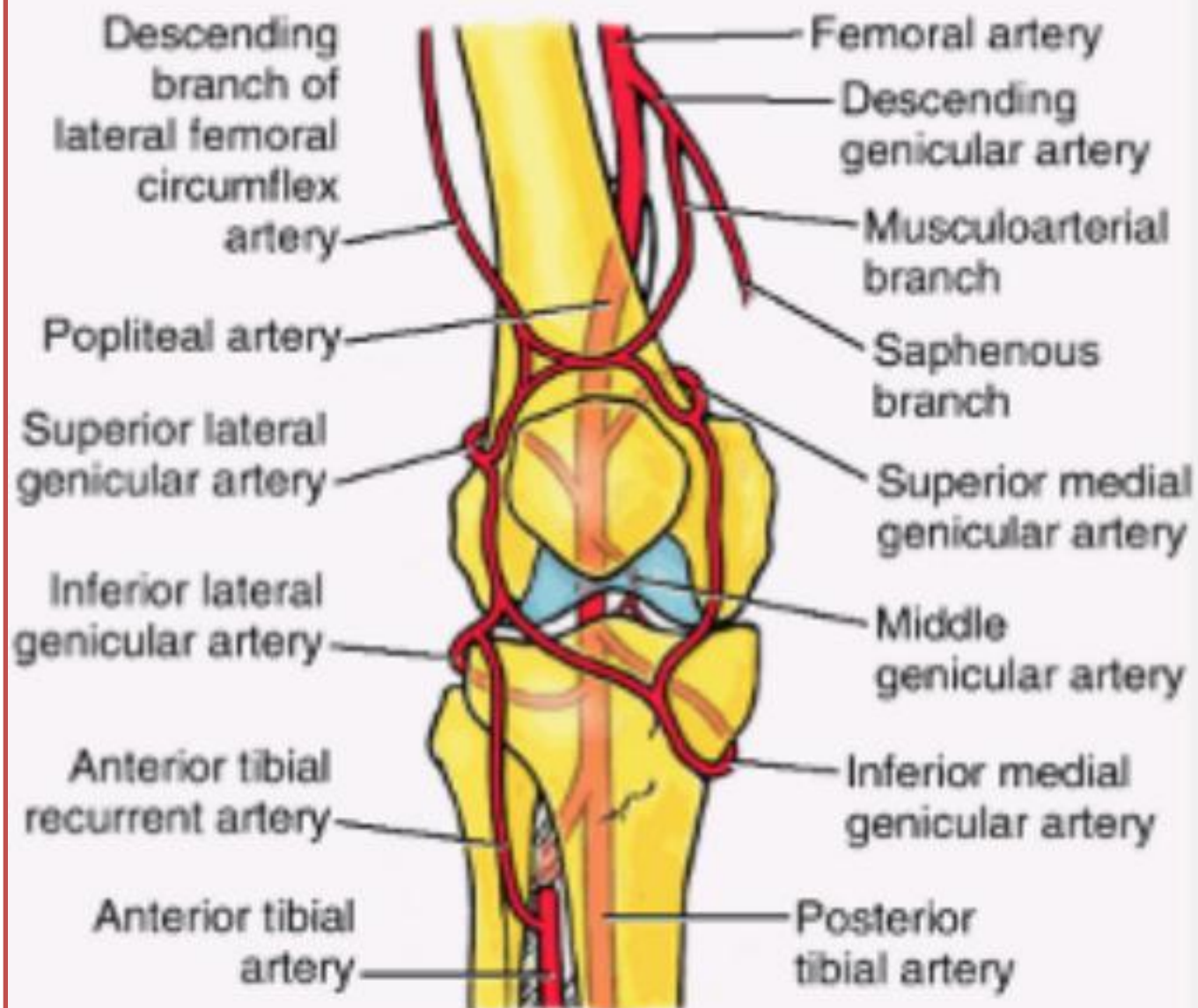
Arteries of the leg

- Popliteal
- Anterior tibial
- Posterior tibial
- Peroneal artery
- Perforating branch



Popliteal artery

- Origin: continuation of the femoral artery at adductor hiatus
- Course: passes through popliteal fossa to leg
- Ends at inferior border of popliteus by dividing into anterior and posterior tibial arteries
- Anterior tibial artery become **dorsalis pedis artery**
- Posterior tibial artery gives **fibular artery**
- Posterior tibial artery become **medial & lateral plantar artery**



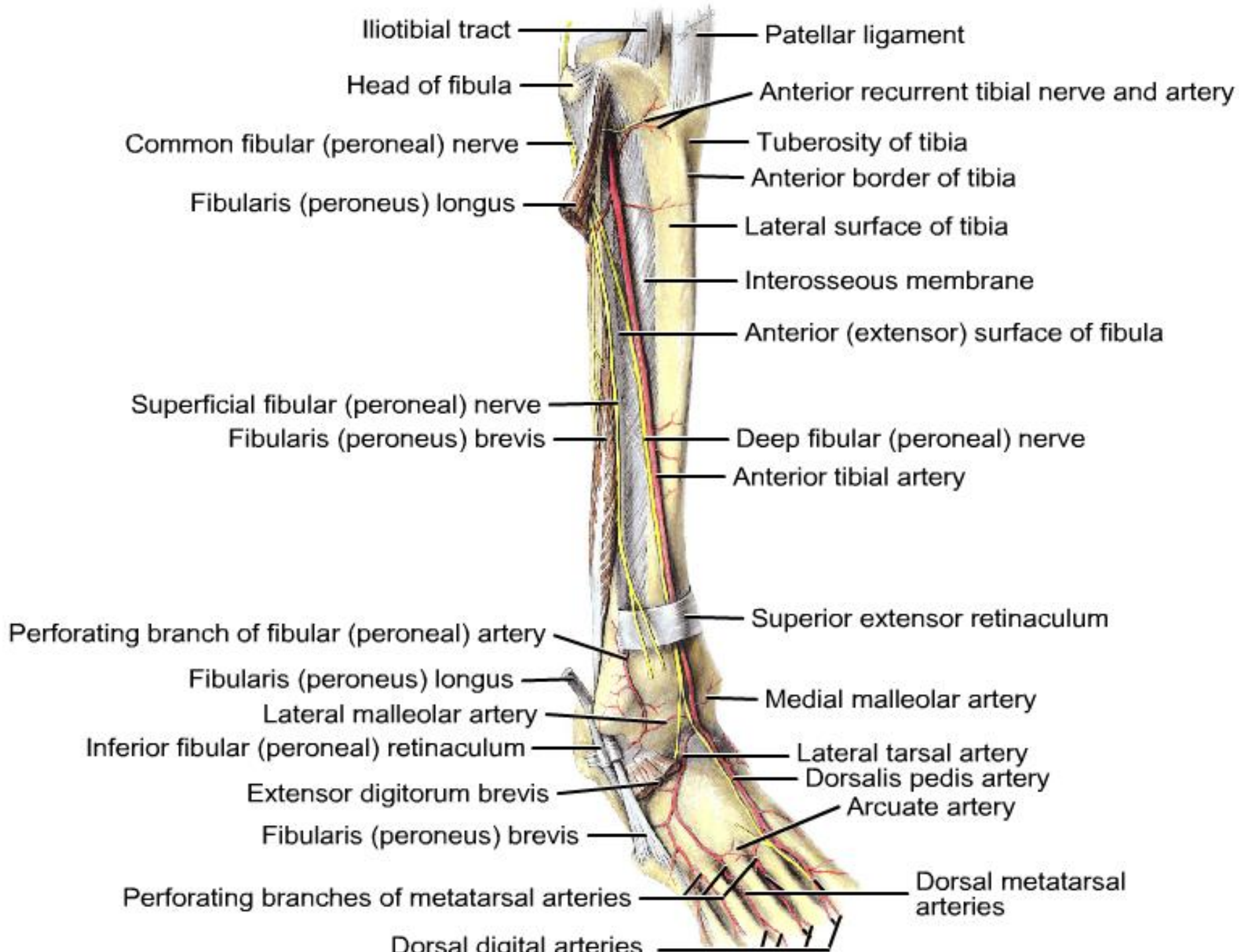
D Anterior view

Anterior tibial artery

- Artery of the anterior compartment, lying medial to the head of the fibula
- Origin: popliteal artery
- Course
 - Reach the anterior compartment of the leg through gap in superior part of interosseous membrane medial to the neck of the fibula
 - Descends on the membrane
 - Continues as **dorsalis pedis artery** after the extensor retinaculum

■ Branches

- anterior tibial recurrent artery
- medial and lateral malleolar arteries
- medial and lateral tarsal arteries
- arcuate artery
- dorsal metatarsal arteries

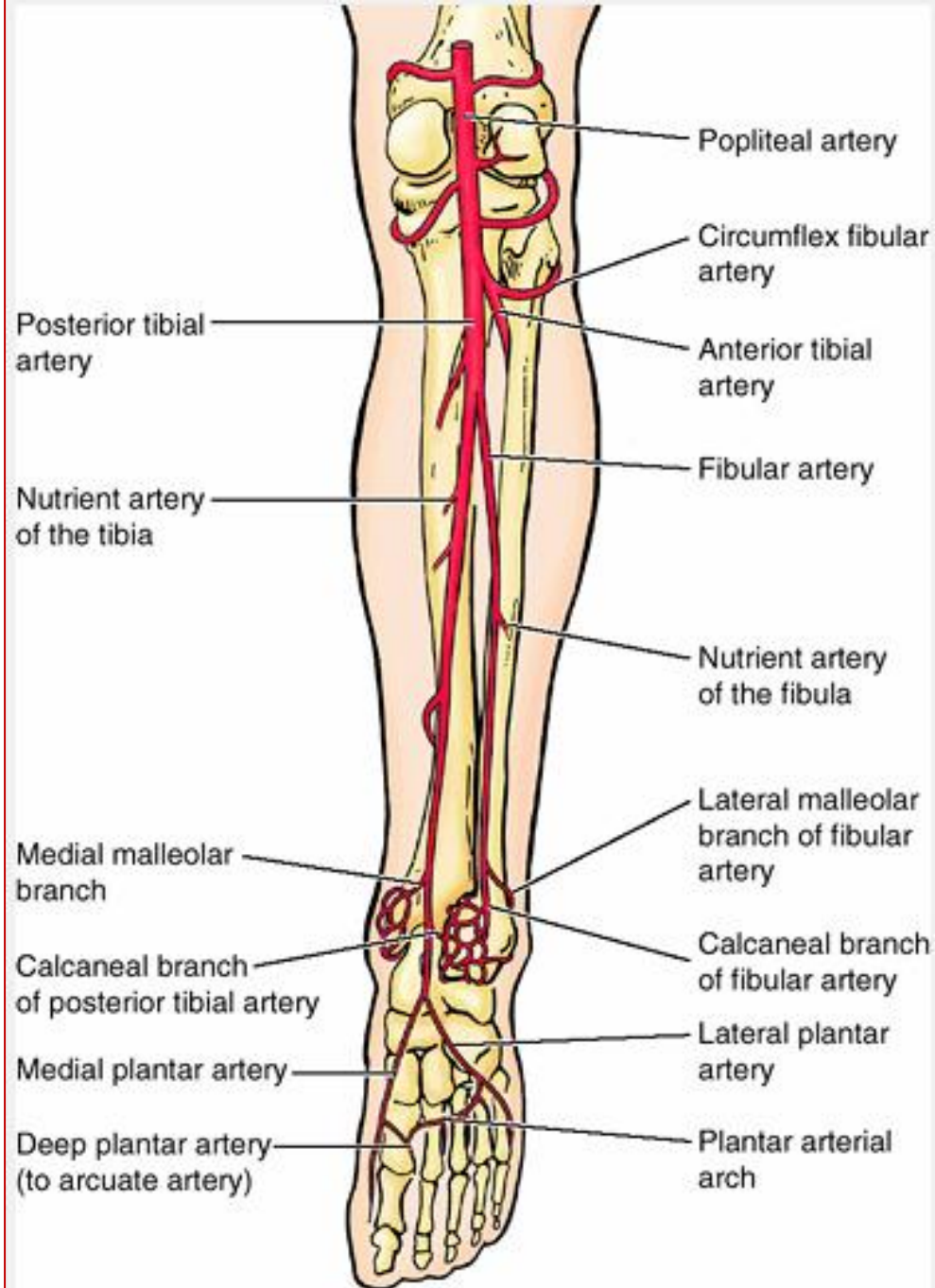


Posterior tibial artery

- Origin: popliteal; begins at the superior border of the soleus; distal border of popliteus
- Course: Passes through posterior compartment
- ends deep to the flexor retinaculum by dividing into **medial and lateral plantar arteries**

- **Branches**

- Circumflex fibular branch joins anastomoses around knee
- Peroneal
- Nutrient artery for tibia
- Muscular to posterior and lateral compartments of leg
- Communicating – with peroneal
- **Medial malleolar**
- Calcaneal
- **Termination – divide into medial and lateral plantar**



Posterior view with foot plantar flexed

Posterior tibial pulse

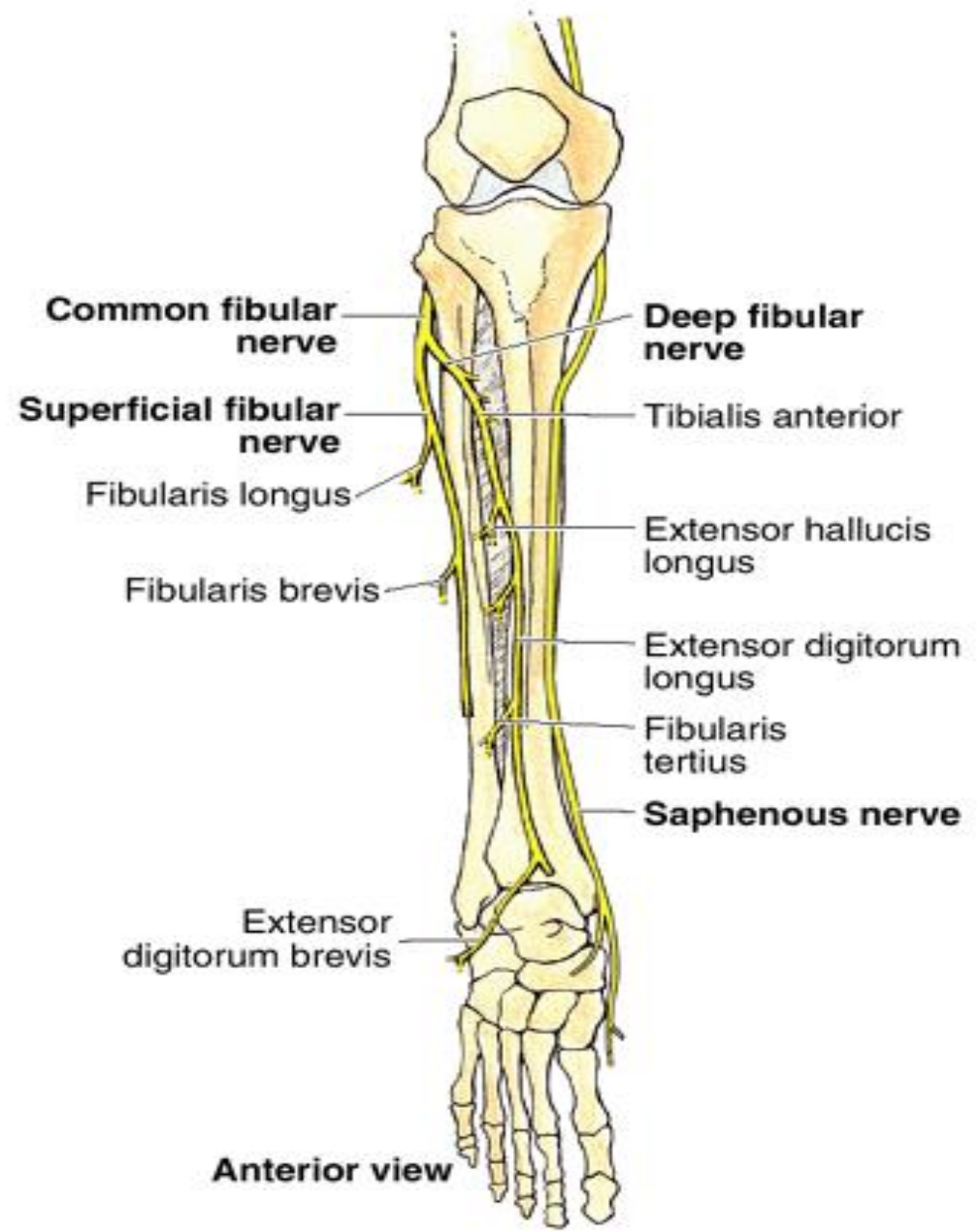
- The posterior tibial pulse can usually be palpated between the posterior surface of the medial malleolus and the medial border of the calcaneal tendon
- Because the posterior tibial artery passes deep to the flexor retinaculum, it is important when palpating this pulse to have the person relax the retinaculum by inverting the foot
- Failure to do this may lead to the wrong conclusion that the pulse is absent

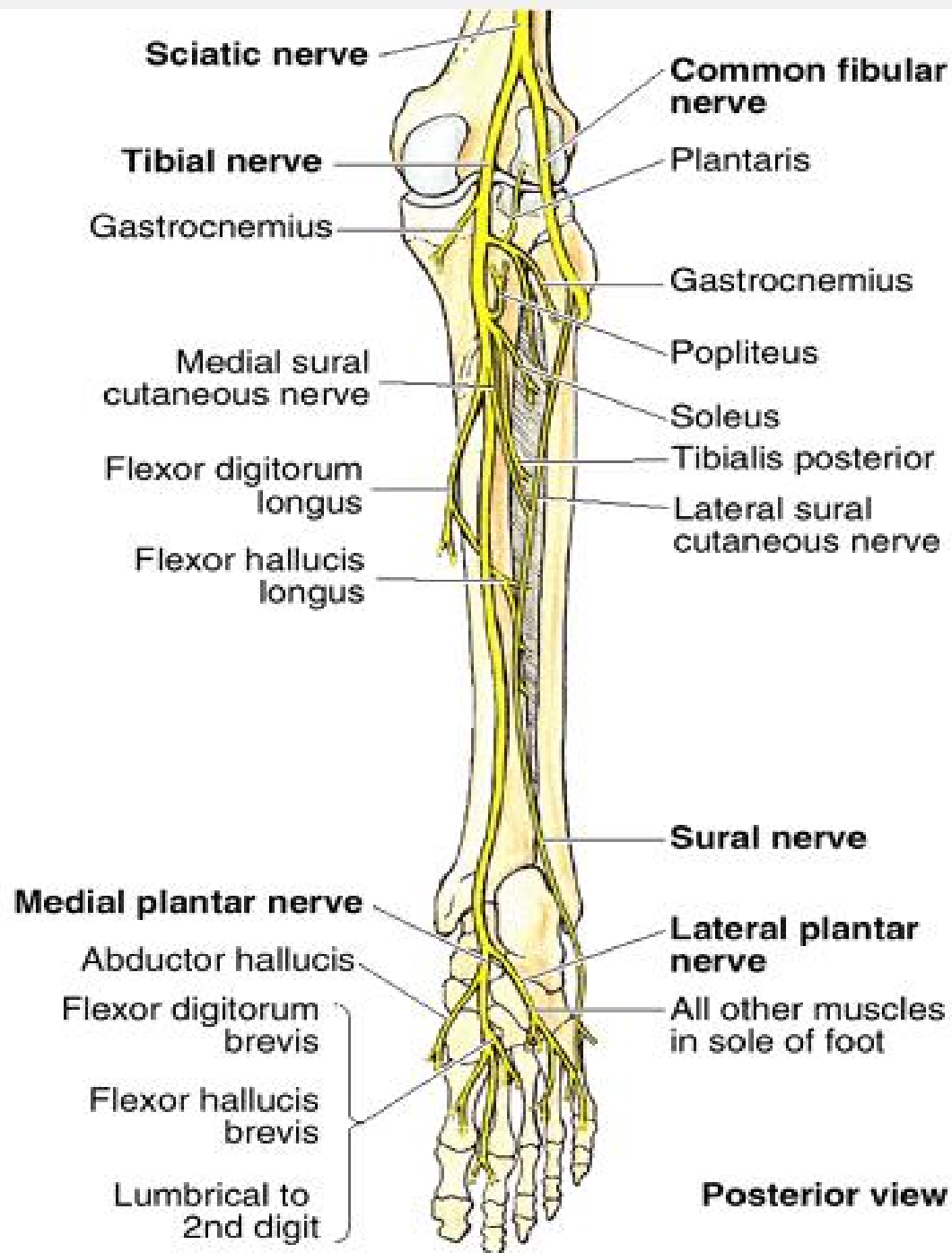
Peroneal (Fibular) artery

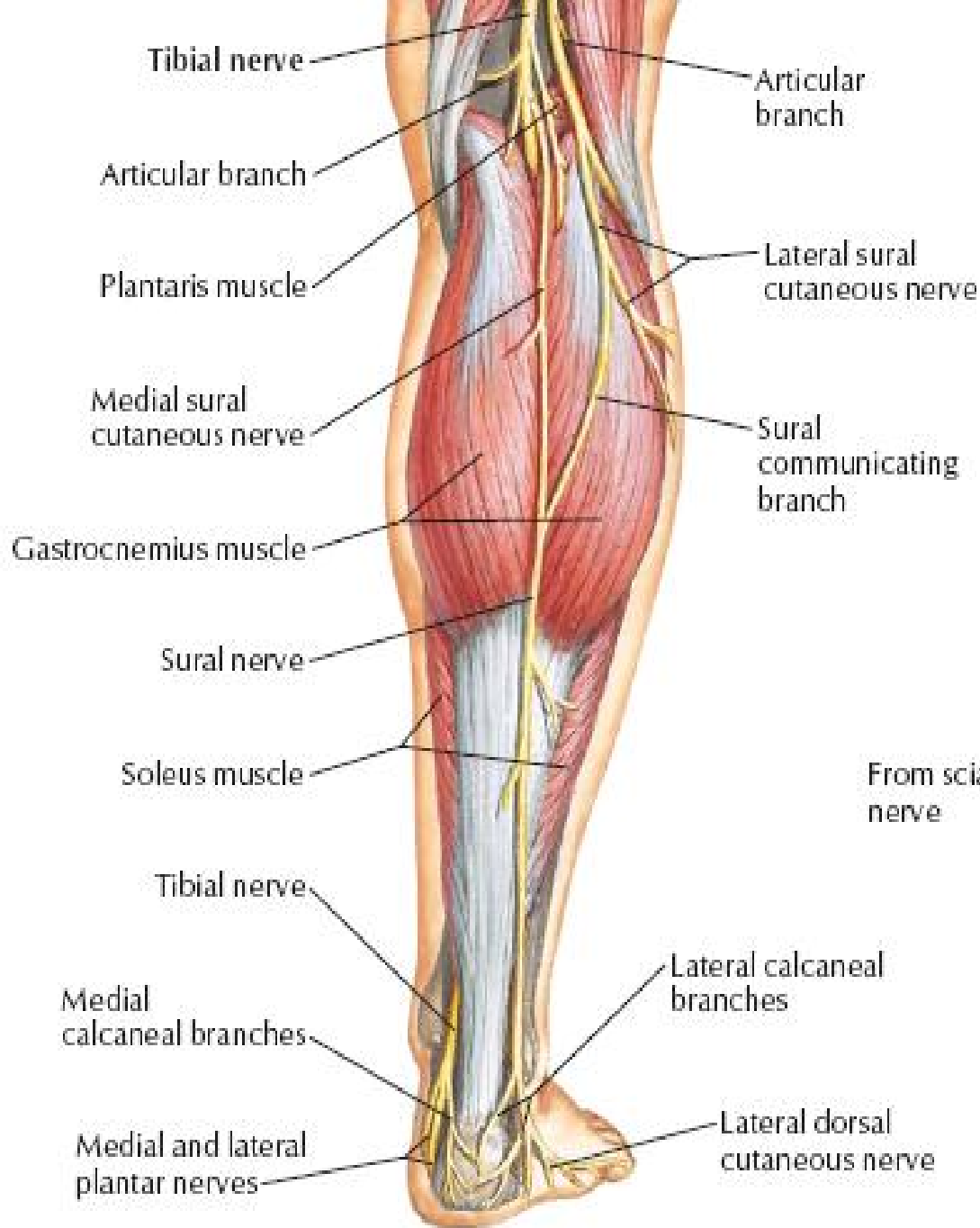
- Origin: posterior tibial artery
- Course: runs in posterior compartment adjacent to posterior intermuscular septum
- ends as the lateral calcaneal artery on the lateral surface of the calcaneus
- Branches
 - Muscular to posterior compartment of leg
 - Nutrient to fibula
 - Perforating branches to lateral compartment of leg
 - Terminal calcaneal

Nerves of the leg

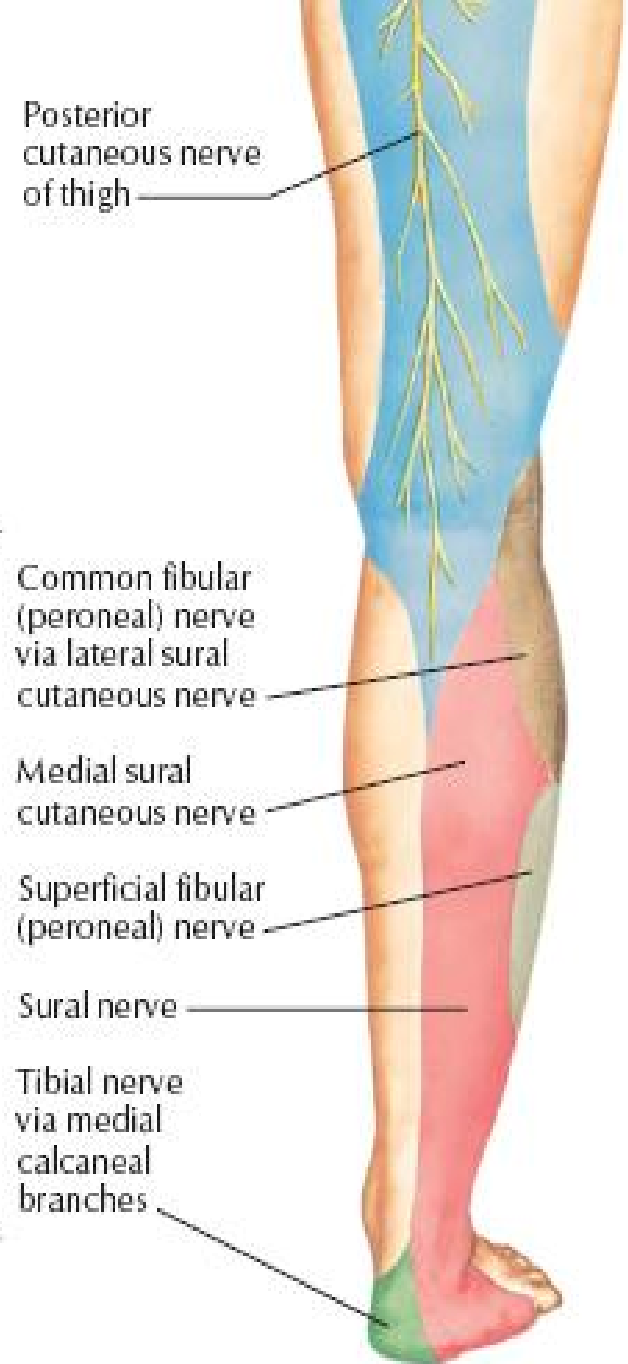
- Saphenous (medial skin)
- Sural nerve(lateral skin)
- Tibial nerve
- Common fibular
- Superficial fibular
- Deep fibular







From sciatic nerve



Tibial nerve

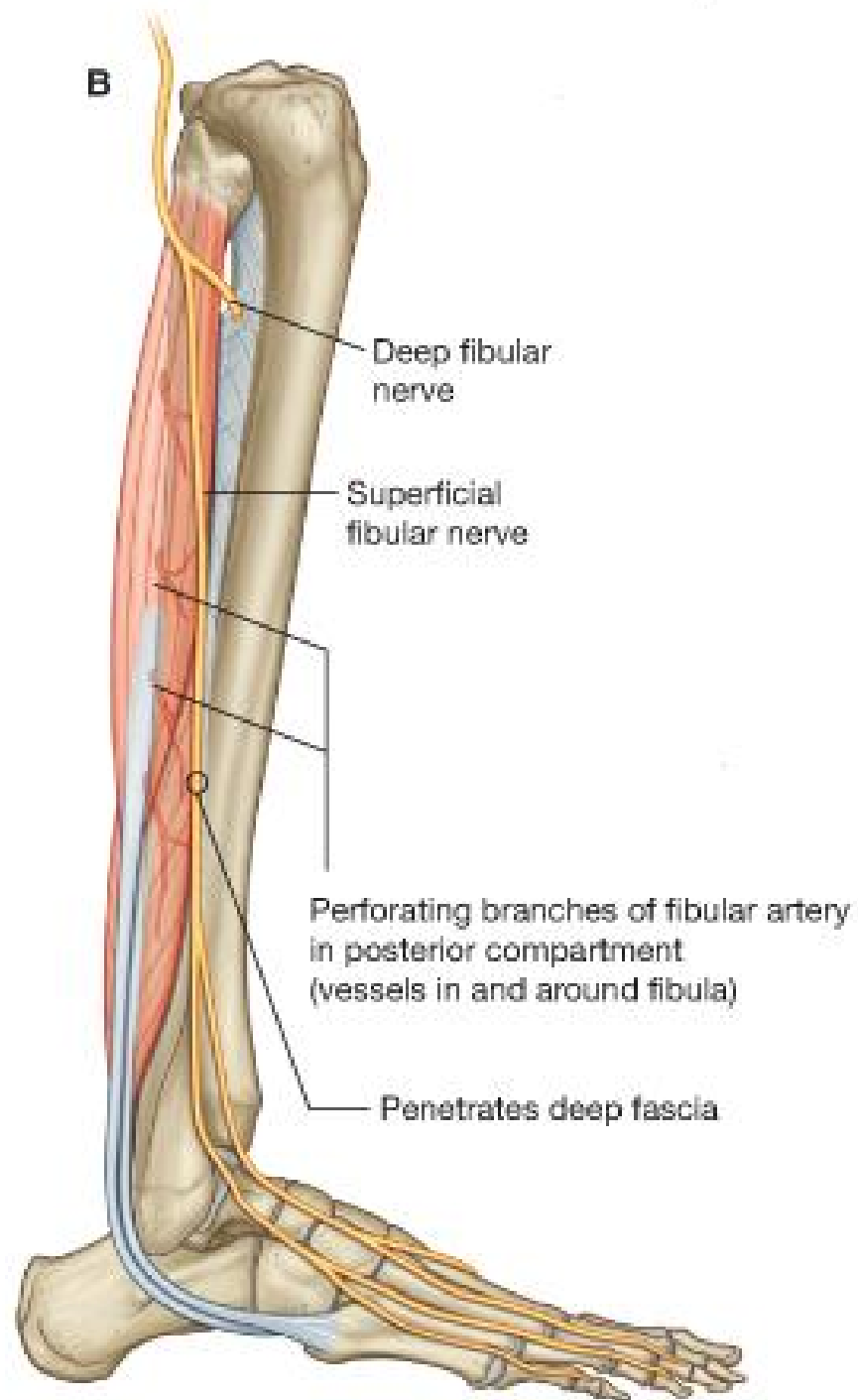
- Origin: sciatic nerve
- Course:
 - Descends through popliteal fossa and lies on popliteus
 - Runs inferiorly on the posterior surface of the tibialis posterior fascia and then on the tibia accompanies the posterior tibial artery
 - runs deep to the flexor retinaculum
 - divides into the medial and lateral plantar nerves
- Distribution: Posterior compartment muscles of the leg and knee joint

- **Branches**
 - **cutaneous branches** to the heel (medial calcaneal nerves)
 - **muscular branches** to muscles of the posterior compartment of the leg
 - medial and lateral heads of gastrocnemius
 - plantaris
 - soleus
 - popliteus and others
 - **articular branches to the ankle joint**

Common peroneal (fibular) nerve

- Origin: sciatic nerve
- Course
 - separates from the tibial nerve halfway down the thigh
 - enter popliteal fossa at superior angle
 - descend in the lateral boundary of the fossa medial to biceps femoris
 - leave the fossa through the lateral angle crossing plantaris and lateral head of gastrocnemius
 - wraps around the neck of the fibula
 - divides into deep peroneal and superficial peroneal (fibular) nerves lateral to the head of the fibula
- Distribution: skin on lateral leg and knee joint

- **Branches**
 - **Cutaneous**
 - Lateral sural cutaneous nerve - posteriolateral leg
 - **Muscular**
 - no in the popliteal fossa
 - give motor branch to short head of biceps before entering popliteal fossa
 - **Articular**
 - superior lateral, inferior lateral and recurrent genicular
 - **Terminal**
 - deep peroneal and superficial peroneal

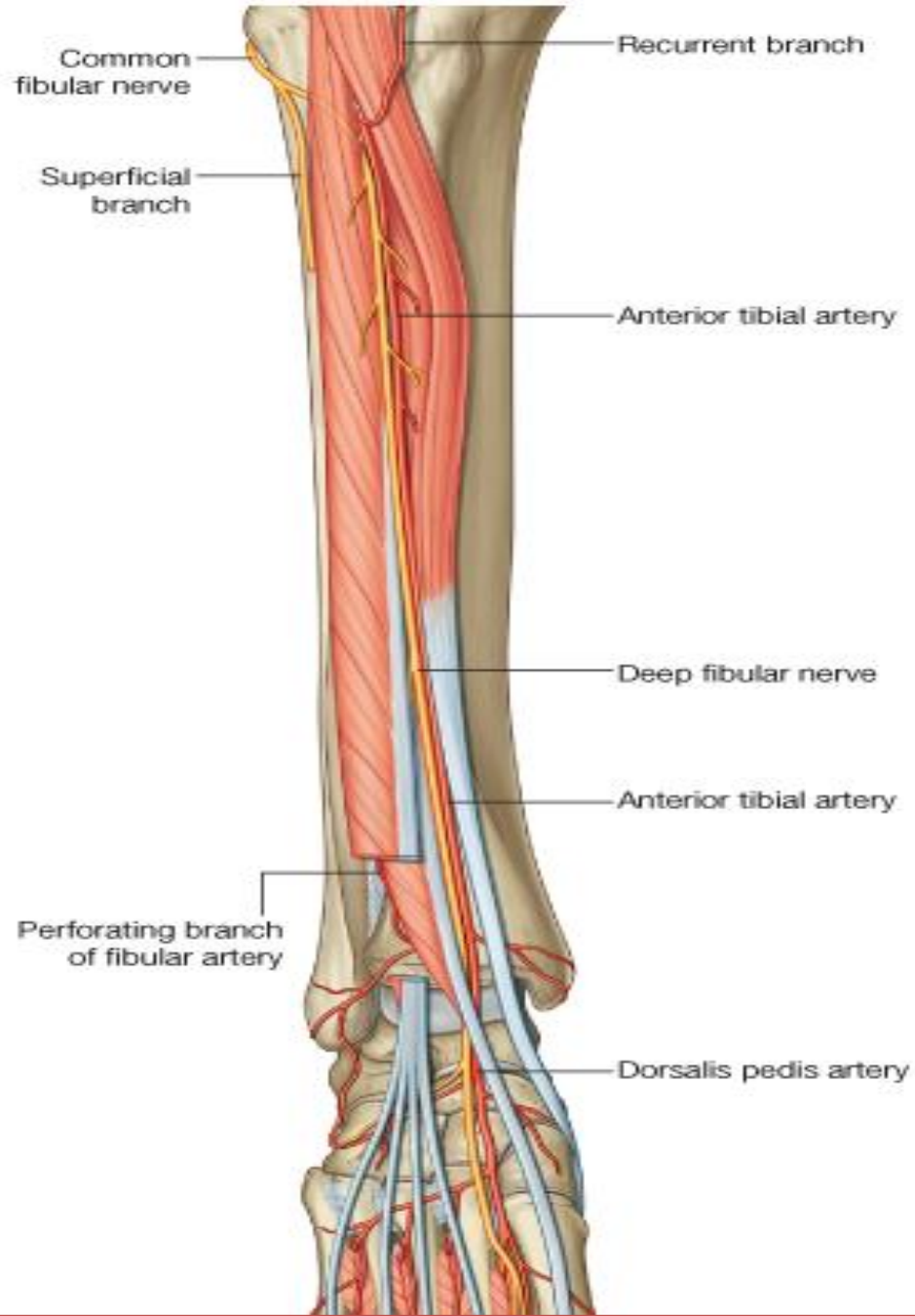


Superficial fibular nerve

- Origin: common fibular nerve
- Course
 - Arises between fibularis longus and neck of fibula
 - Descends in lateral compartment
 - Pierces deep fascia at lateral third of leg to become cutaneous
- Distribution: fibularis longus, brevis and skin on distal lateral leg and dorsum of foot

Deep fibular nerve

- Origin: common fibular nerve
- Course
 - Arises between fibularis longus and neck of fibula
 - Passes through extensor digitorum longus and descend on interosseous membrane
 - Crosses distal tibia and enters dorsum of foot
 - accompanies anterior tibial artery and then the dorsalis pedis artery
 - Distribution: anterior muscles of leg, dorsum of foot and skin of first interdigital cleft and joints



Injury to common fibular nerve

- It is the most commonly injured nerve in the lower limb, mainly because it **winds superficially around the neck of the fibula**, leaving it vulnerable to direct trauma
- It also be severed during **fracture of the fibular neck** or severely stretched when the knee joint is injured or dislocated
- Severance of the common fibular nerve results in paralysis of **all muscles in the anterior and lateral compartments** of the leg
- The loss of dorsiflexion of the ankle causes *foot drop*

Foot

Introduction

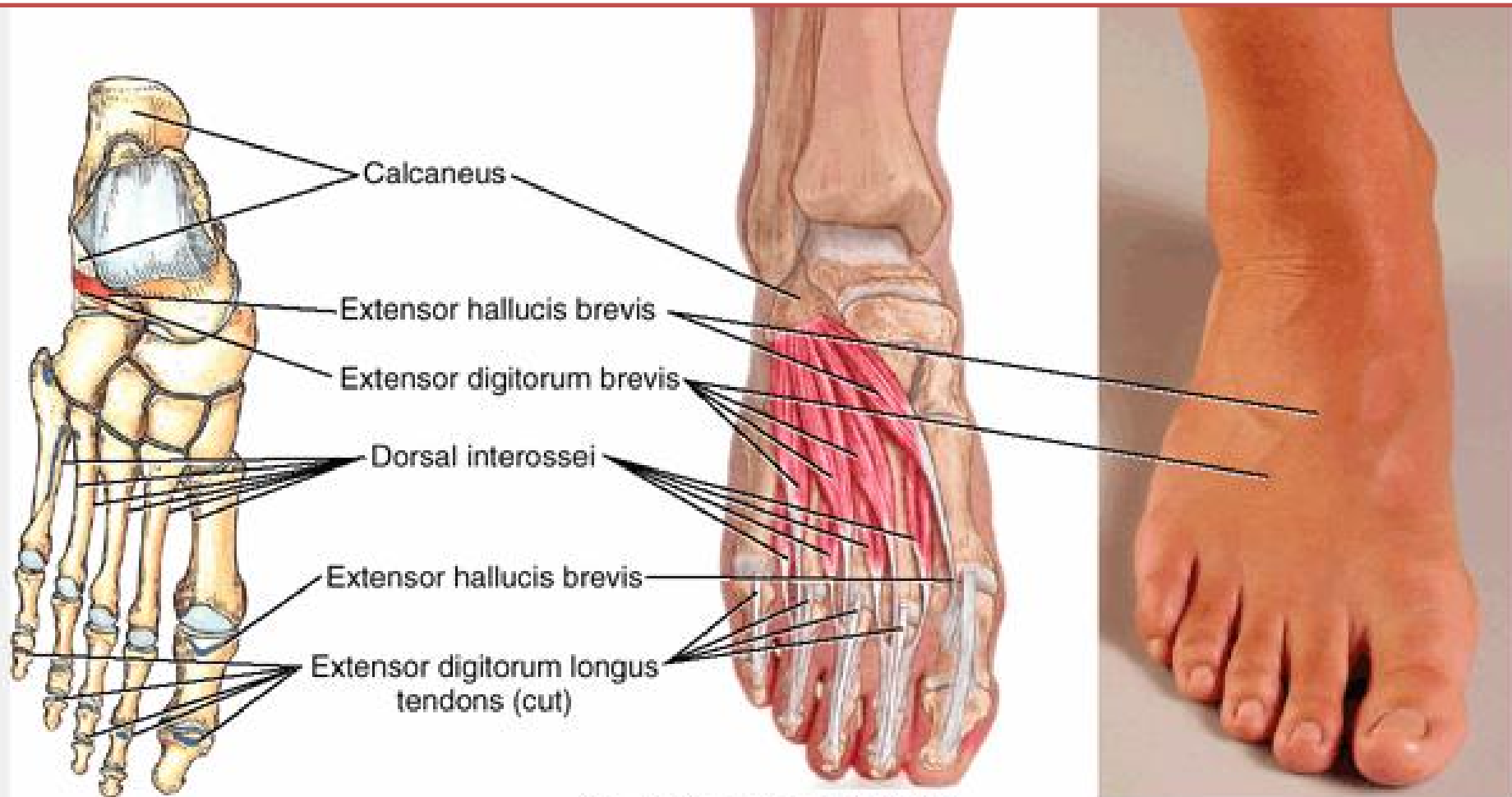
- **Foot is a region of lower limb distal to ankle**
- Provides platform for supporting the weight of the body
- The foot and its bones may be considered in terms of three anatomical and functional parts:
 - **The hindfoot:** talus and calcaneus
 - **The midfoot:** navicular, cuboid, and cuneiforms
 - **The forefoot:** metatarsals and phalanges

Regions of the foot

- **The plantar region** (sole): the part contacting the ground
- **The dorsal region** (dorsum of the foot): the part directed superiorly
- The heel region (heel): the sole underlying the calcaneus
- The ball of the foot: the sole underlying the heads of the medial two metatarsals

Dorsum of the foot

- **Two Muscles**
 - **Extensor digitorum brevis**
 - **Extensor hallucis brevis** (medial slip of the extensor digitorum brevis)
- **Nerve supply – deep fibular nerve**



(C) Anterosuperior views

Plantar muscles

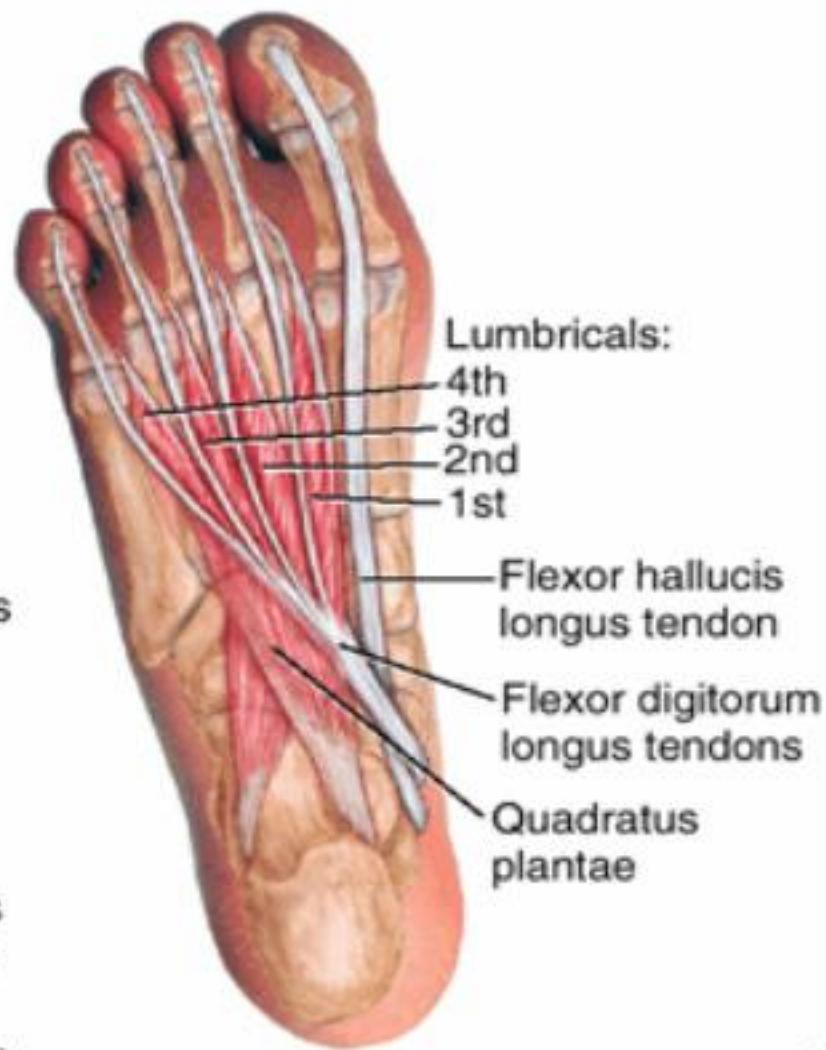
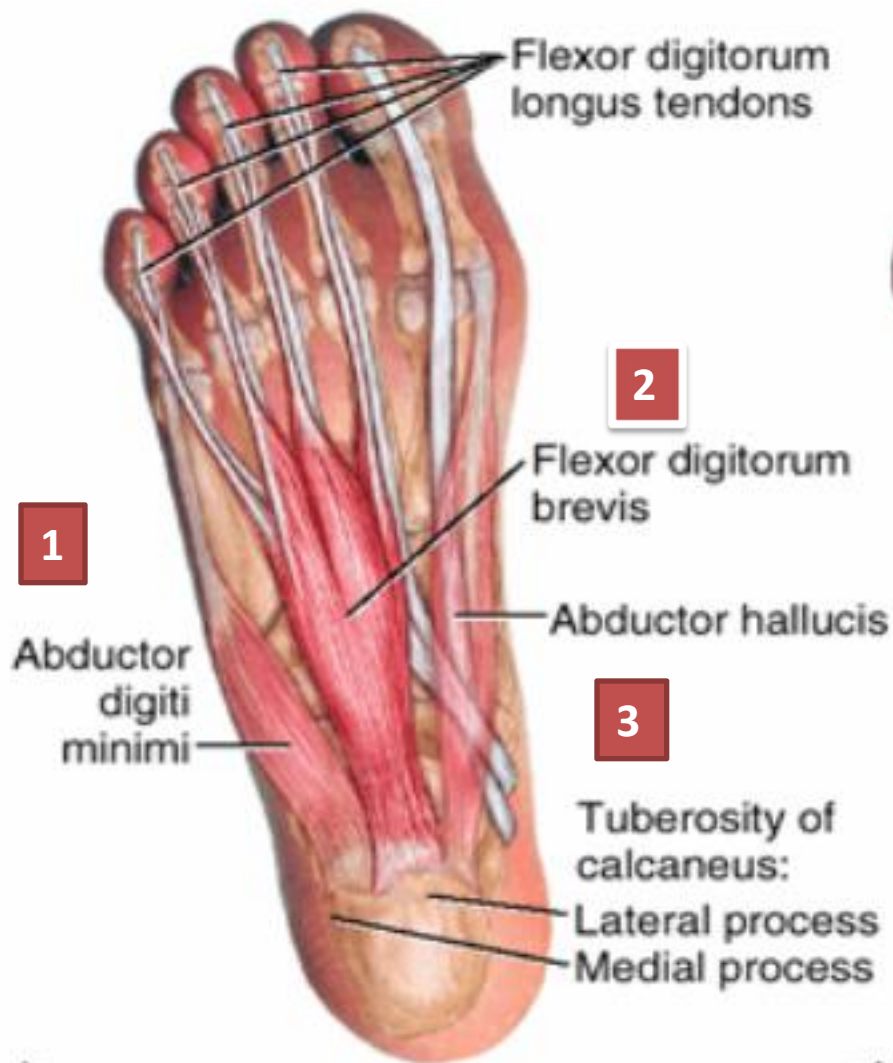
- There are 4 layers of plantar muscles, organized as:

□ **“3232”**

- First layer: 3 muscles
- Second layer: 2 muscles and 2 tendons
- Third layer: 3 muscles
- Fourth layer: 2 muscles and 2 tendons

First layer

- **Formed by 3 muscles which arise from the calcaneus**
- Consists of a flexor between 2 abductors
- **Abductor hallucis**
 - innervated by the **medial plantar nerve**
- **Abductor digiti minimi**
 - is innervated by the **lateral plantar nerve**
- **Flexor digitorum brevis**
 - is innervated by the **medial plantar nerve**

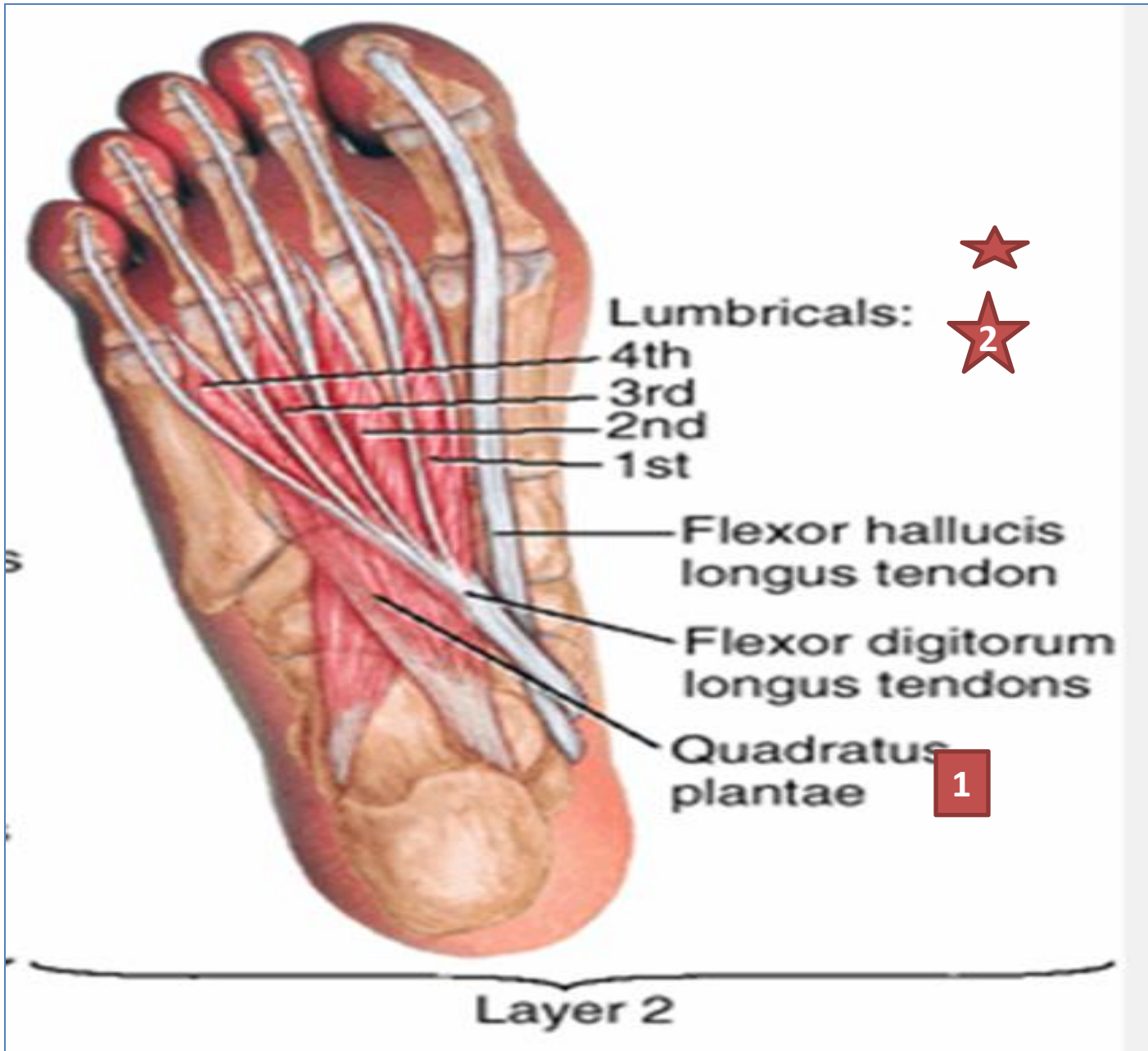


Layer 1

Layer 2

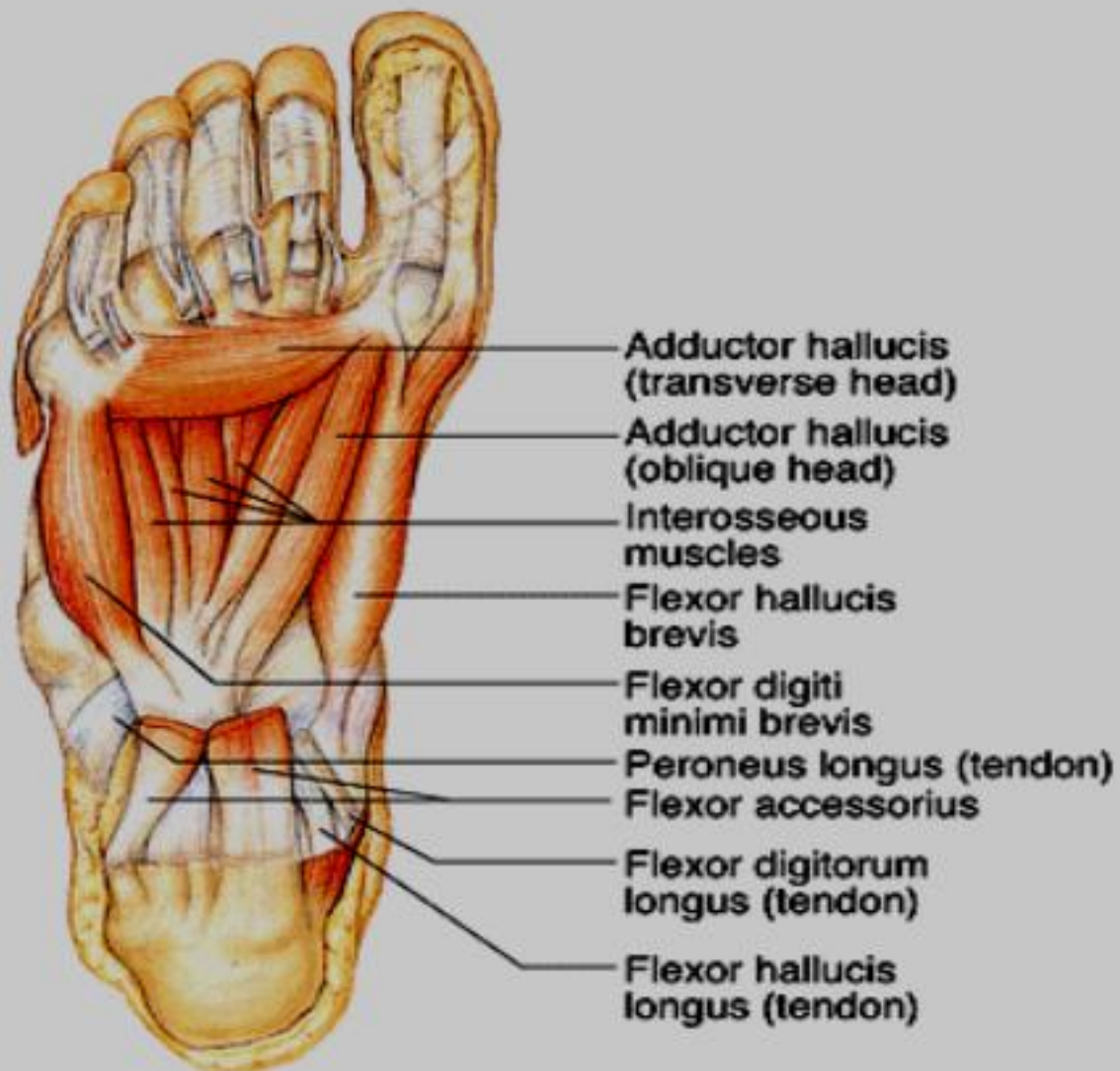
Second layer

- Muscles of the second layer flex, adduct, and abduct the toes
- Formed by 2 muscles and 2 tendons
 - **Flexor digitorum longus tendon**
 - **Flexor hallucis longus tendon**
 - **Quadratus plantae** (flexor digitorum accessorius)
 - It is innervated by the **lateral plantar nerve**
 - **Lumbricals**
 - lumbrical 1 is innervated by the **medial plantar nerve**; lumbricals 2, 3 and 4 are innervated by the **lateral plantar nerve**



Third layer

- Formed by 3 muscles:
- **Adductor hallucis**
 - Origin
 - oblique head: from the tarsal bones
 - transverse head: from the metatarsal heads
 - Insertion: lateral base of proximal phalanx of the hallux
- Innervation: *lateral plantar nerve*
- **Flexor hallucis brevis**
 - Innervation: medial plantar nerve
- **Flexor digiti minimi**
 - Innervation: lateral plantar nerve

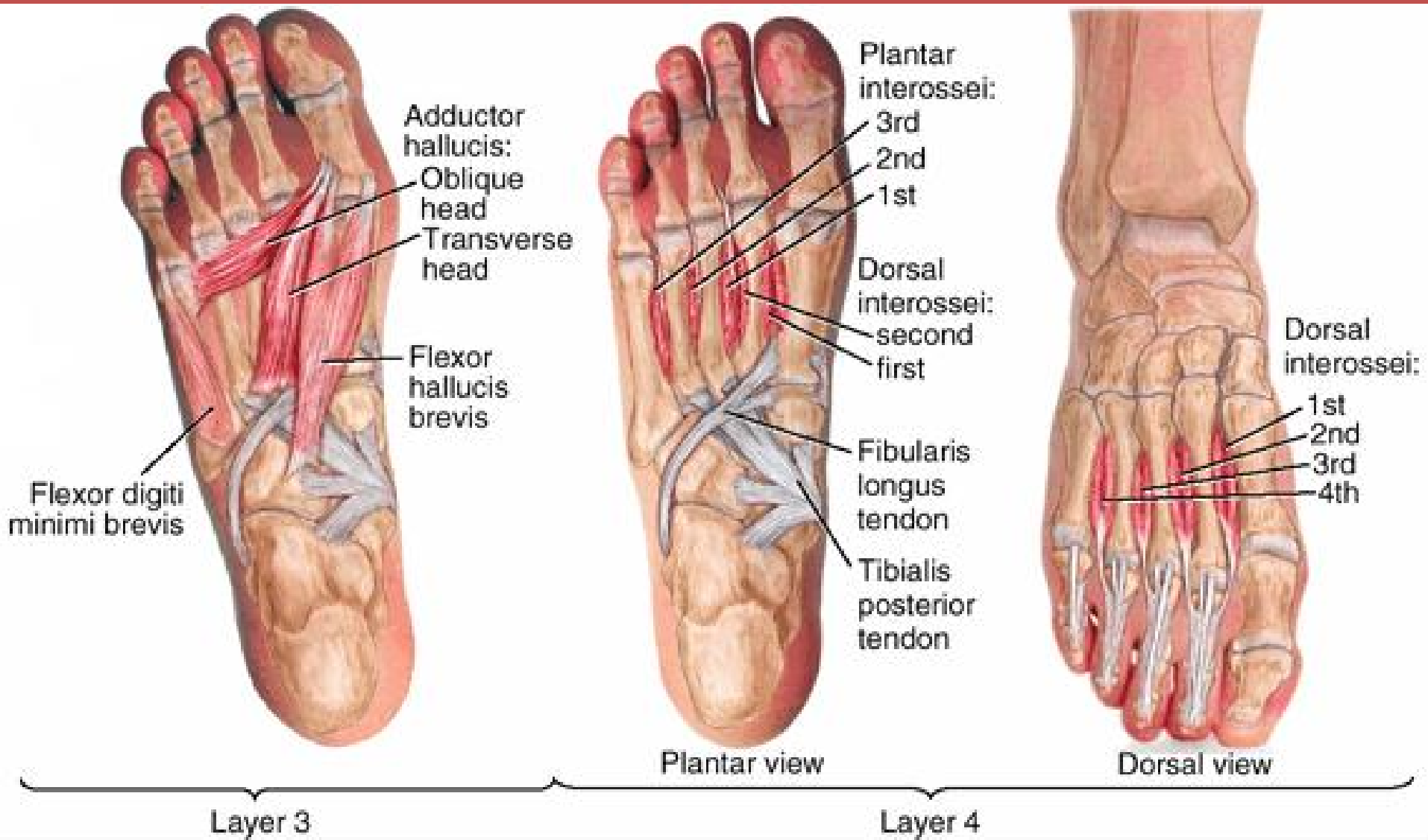


(c) Third layer

© BENJAMIN/CUMMINGS

Fourth layer

- Formed by 2 muscles and 2 tendons
- **7 interossei (4 dorsal and 3 plantar)**
 - keep the foot from spreading by holding the heads of the metatarsals together
 - also flex the metatarsophalangeal joints and extend the interphalangeal joints
 - are innervated by the *lateral plantar nerve*
 - Abduction and adduction are in reference to toe #2.
- **Tibialis posterior tendon**
- **Peroneus (fibularis) longus tendon**



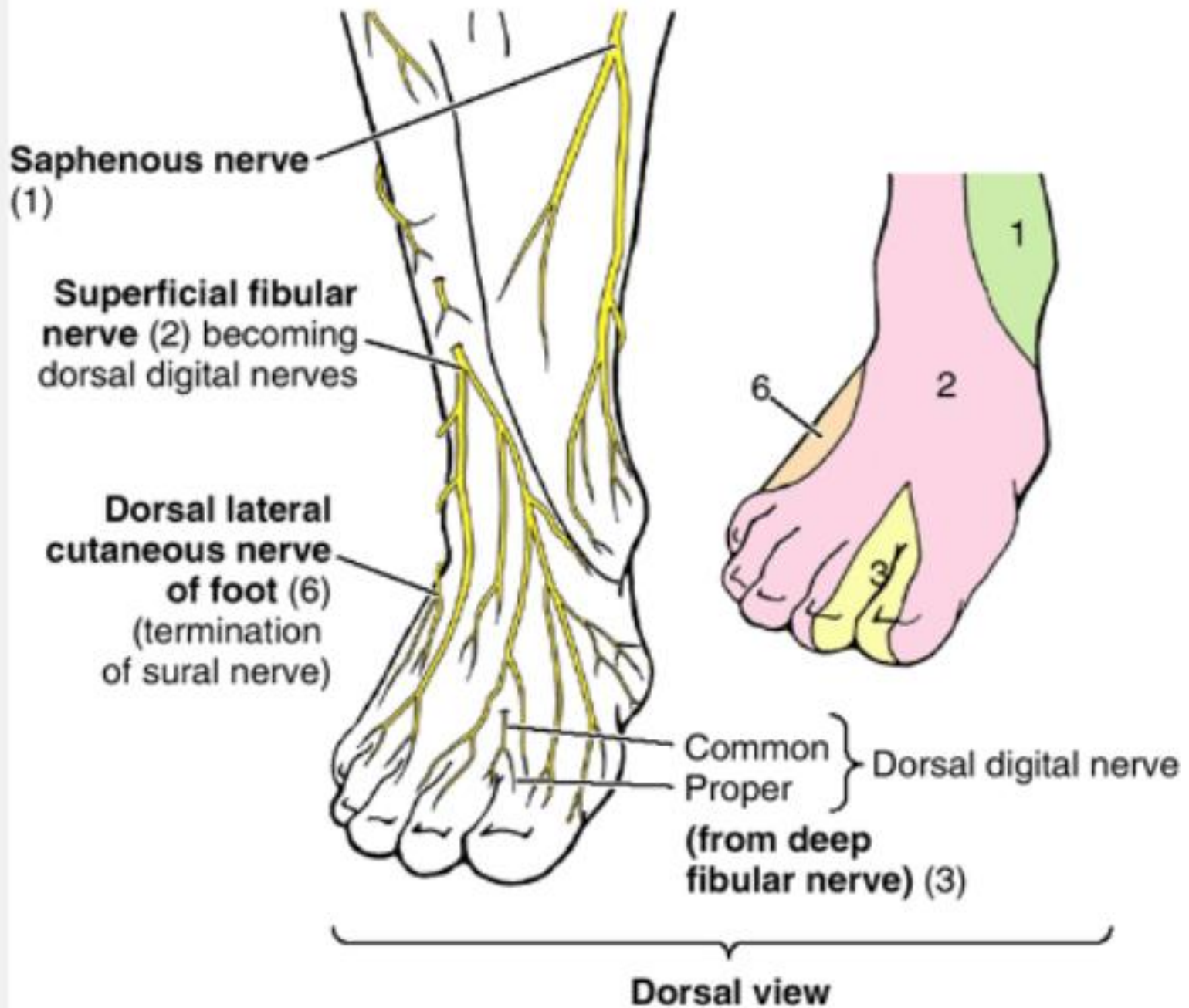
Nerves of foot

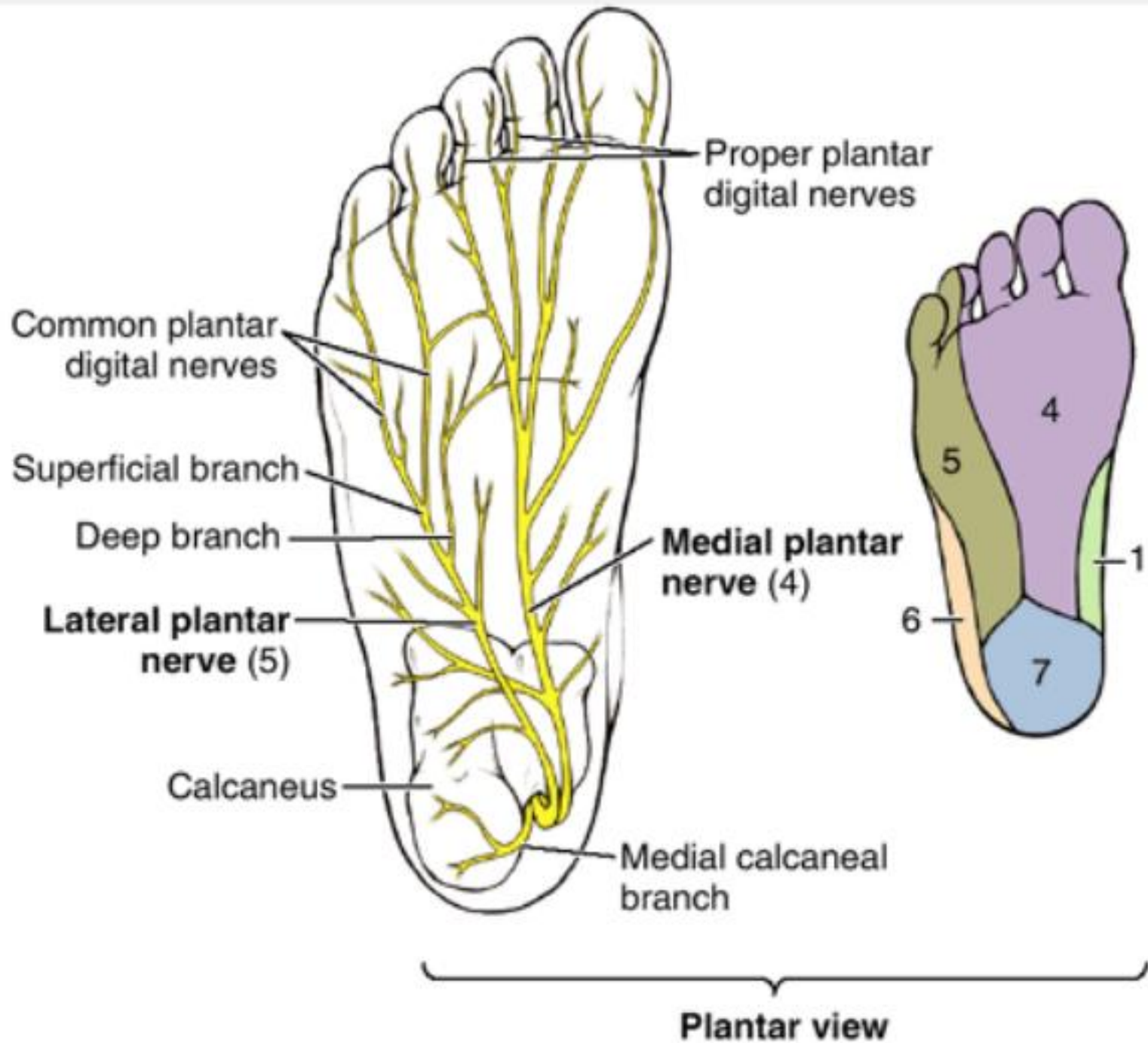
Lateral plantar nerve

- cutaneous branches to the lateral 1 and half digit
- motor branches to
 - quadratus plantae
 - abductor digiti minimi
 - flexor digiti minimi
 - 3 lateral lumbricals
 - 7 interossei
 - adductor hallucis

Medial plantar nerve

- cutaneous branches to the medial 3 and half digits
- motor branches to
 - abductor hallucis
 - flexor digitorum brevis
 - flexor hallucis brevis
 - lumbrical 1





From
tibial nerve

Medial calcaneal
branches
(S1, 2)

Medial
plantar nerve
(L4, 5)

Lateral
plantar nerve
(S1, 2)

Saphenous nerve
(L3, 4)

Sural nerve
(S1, 2) via
lateral calcaneal
and lateral dorsal
cutaneous
branches



Cutaneous innervation of sole

Blood vessels of foot

Dorsalis pedis artery

- It is Continuation of anterior tibial artery
- Branches
 - Medial tarsal arteries
 - Lateral tarsal arteries
 - Arcuate artery
 - between tendons of long and short extensor muscles
 - give dorsal metatarsal arteries which in turn gives dorsal proper digital arteries
 - joined by perforating branches of the deep plantar arch
 - anastomosis with lateral tarsal
- Termination – in the sole of the foot, join plantar arch

Perforating branch of fibular
(peroneal) artery

Lateral malleolar artery

Lateral tarsal artery

Arcuate artery

Perforating arteries

Anterior tibial artery

Medial malleolar artery

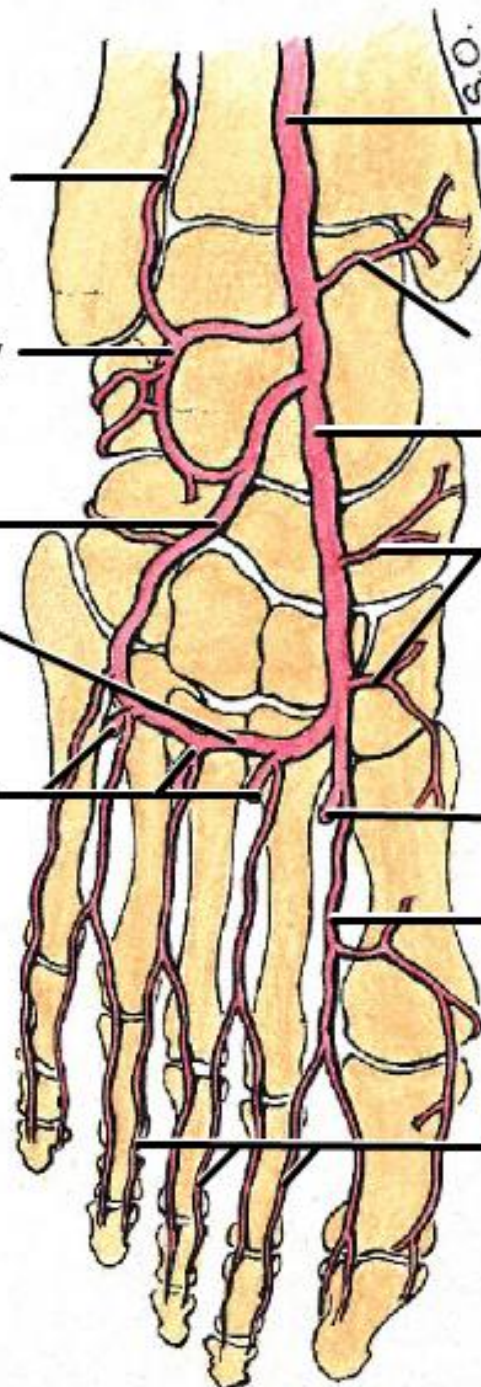
Dorsalis pedis artery

Medial tarsal arteries

Deep plantar artery

1st dorsal metatarsal artery

Dorsal digital arteries



Palpation of the dorsalis pedis pulse

- The pulse of the dorsal artery of the foot (dorsalis pedis pulse) is evaluated during a **physical examination of the peripheral vascular system**
- Dorsalis pedis pulses may be palpated with the feet slightly dorsiflexed
- The pulses usually are easy to palpate because the dorsal arteries of the foot are subcutaneous and pass along a line from the extensor retinaculum to a point just lateral to the extensor hallucis longus (EHL) tendons
- A diminished or absent dorsalis pedis pulse usually suggests vascular insufficiency resulting from **arterial disease**

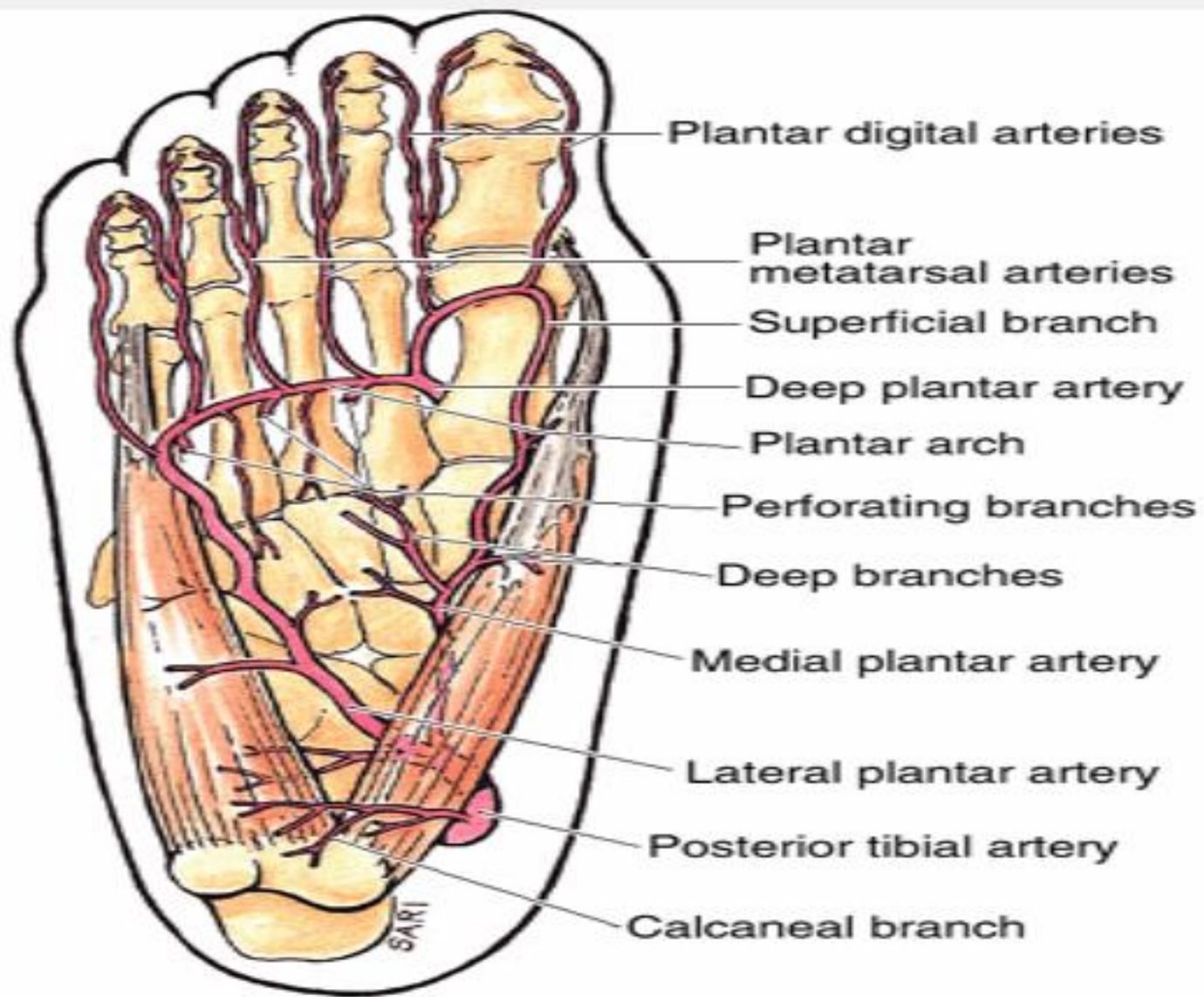
Blood supply of sole of foot

- **Medial plantar artery**
 - ❖ **The smaller terminal branch of the posterior tibial artery**
 - ❖ arises deep to the flexor retinaculum
 - ❖ runs on the medial side of the foot between the abductor hallucis and flexor digitorum brevis
 - ❖ supplies the medial side of the hallux
 - ❖ Branches - muscular, cutaneous, articular & join 1st, 2nd & 3rd plantar metatarsal
 - ❖ Completes plantar arch

- **Lateral plantar artery**
 - **the larger terminal branch of the posterior tibial artery**
 - runs between layers 1 and 2 and then between layers 3 and 4
 - gives off calcaneal, cutaneous, muscular and articular branches
 - forms the deep plantar arch by joining with the deep plantar branch of the dorsalis pedis artery

- **Plantar arterial arch**

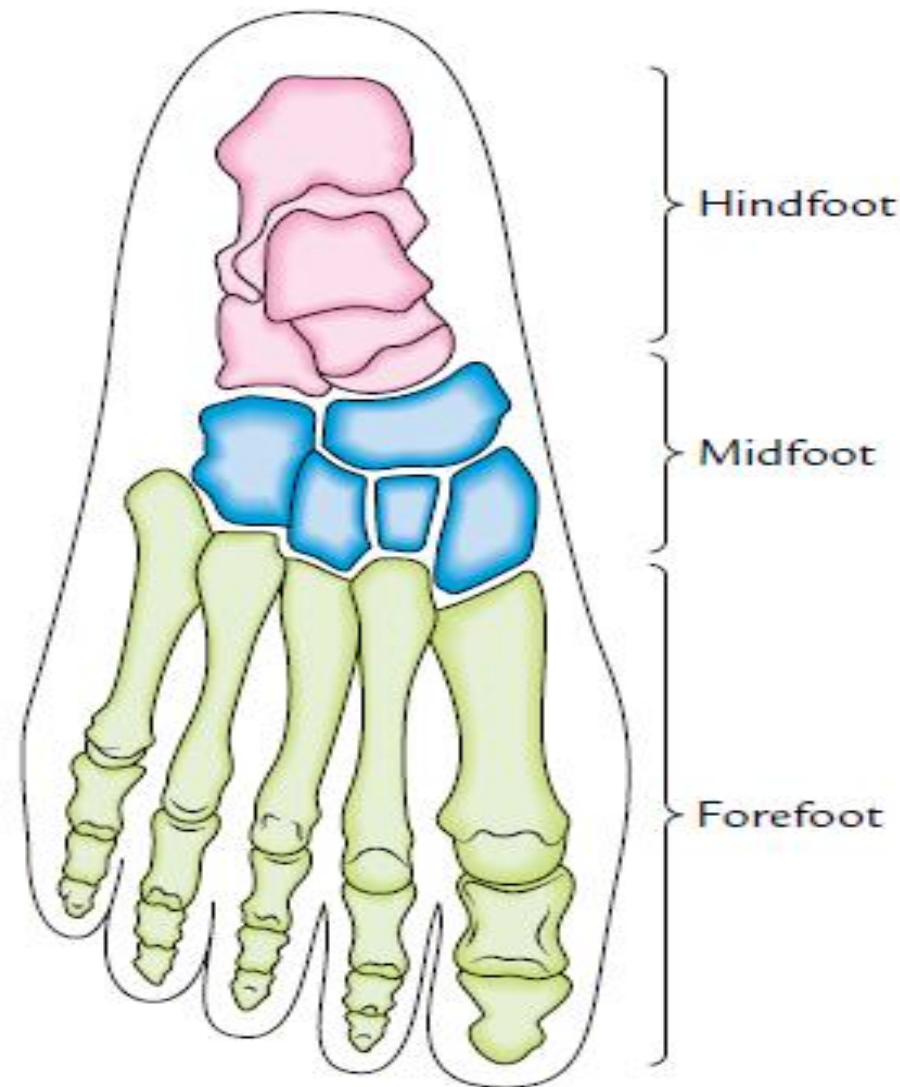
- 3 perforating arteries from the plantar arch join with the dorsal metatarsal arteries
- 4 plantar metatarsal arteries bifurcate into plantar digital arteries



Plantar aspect of foot

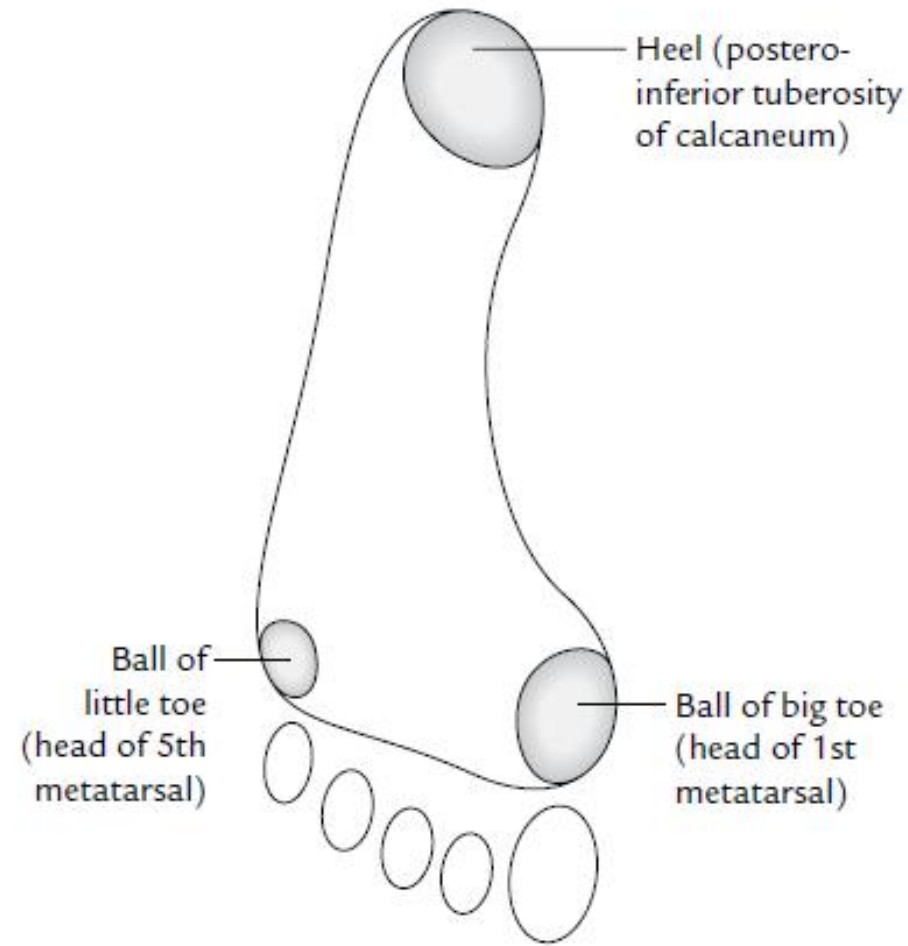
Arches of the Foot

- It is distinguishing feature of man from other primates
- Provides platform for supporting the weight of the body during standing position
- Provides lever to propel the body forward during running, jumping
- The foot and its bones are considered in terms of three anatomical and functional parts:
 - **The hindfoot:** talus and calcaneus
 - **The midfoot:** navicular, cuboid, and cuneiforms
 - **The forefoot:** metatarsals and phalanges



Arches of the Foot...

- Distribution of body weight among three points
 - At posterior inferior tuberosity of calcaneum
 - At head of 1st metatarsal
 - At head of 5th metatarsal



Types of arches

- There are two types of foot arches

1. longitudinal foot arches

2. Transverse foot arches

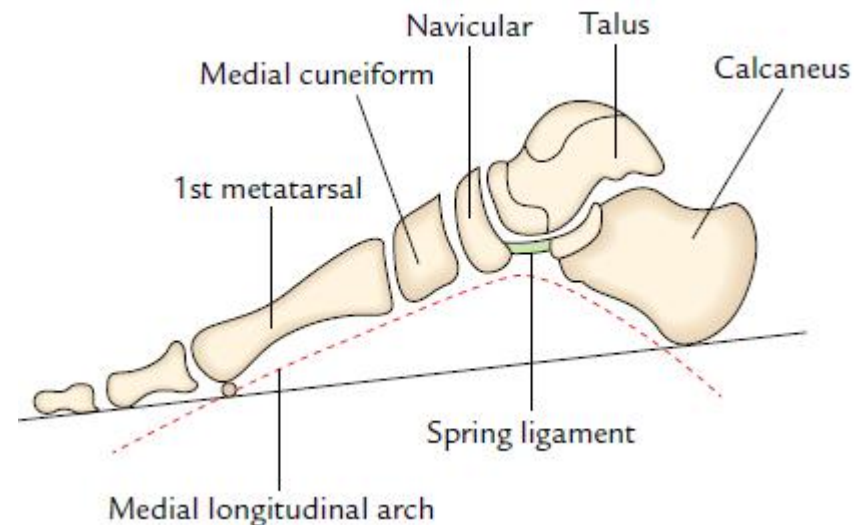
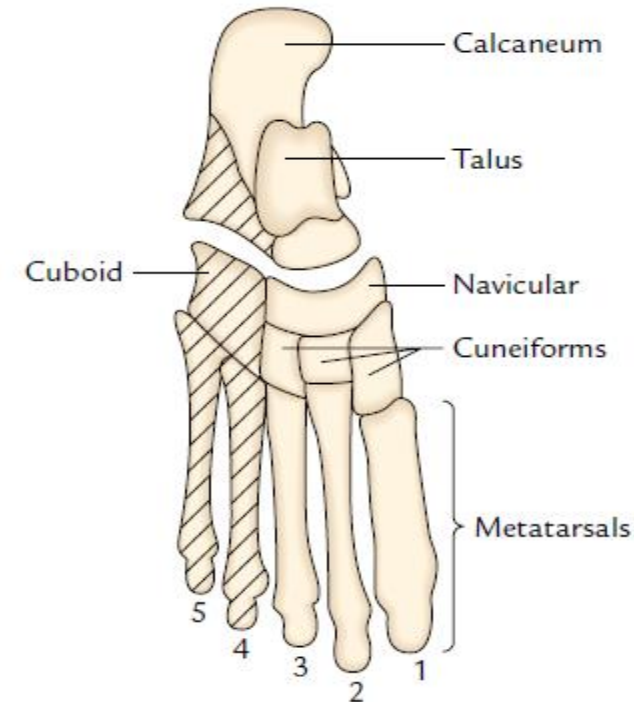
- There are two **longitudinal** arches in each foot

- Medial

- Lateral

There are a series **of transverse** arches in each foot.

At the **heads of metatarsals**, the transverse arch is complete



Types of arches...

1. longitudinal foot arches

Each longitudinal foot arches has pillar, summit and joints

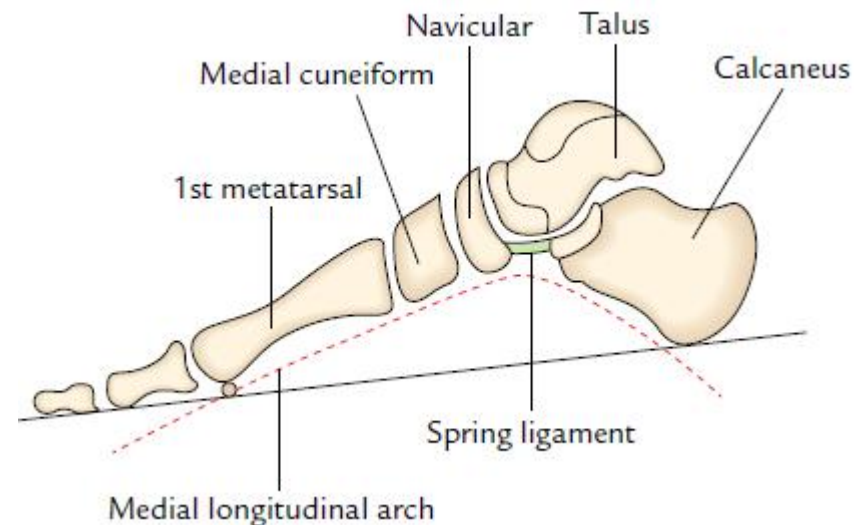
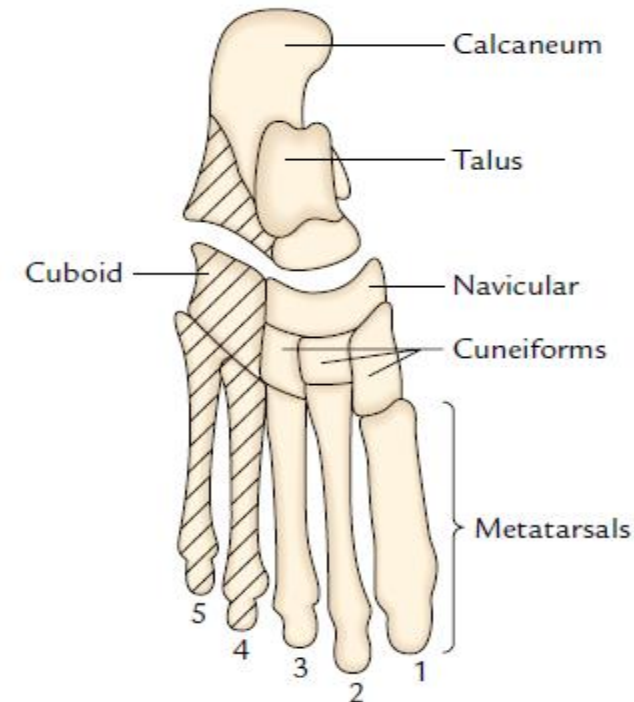
❑ Medial Longitudinal Arch

- It is formed by the **calcaneum, talus, navicular, three cuneiforms, and medial three metatarsals.**

❑ Pillars

The medial half of the calcaneum forms the **posterior pillar** of the medial longitudinal arch.

2. The heads of the medial three metatarsals form the **anterior pillar** of the medial longitudinal arch.



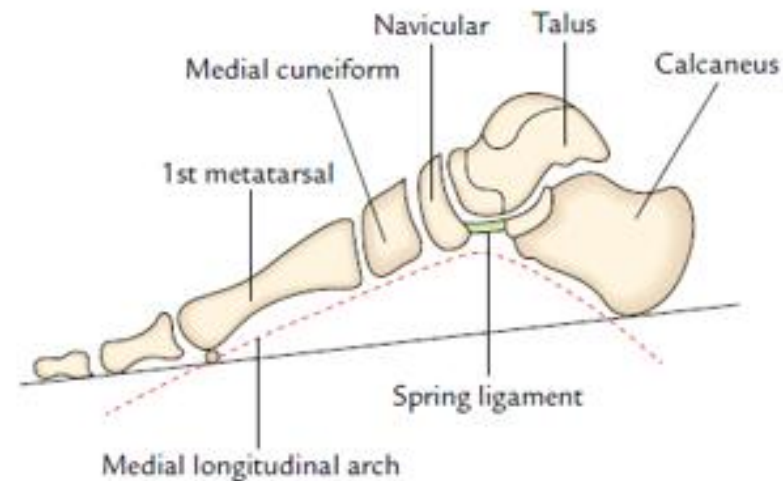
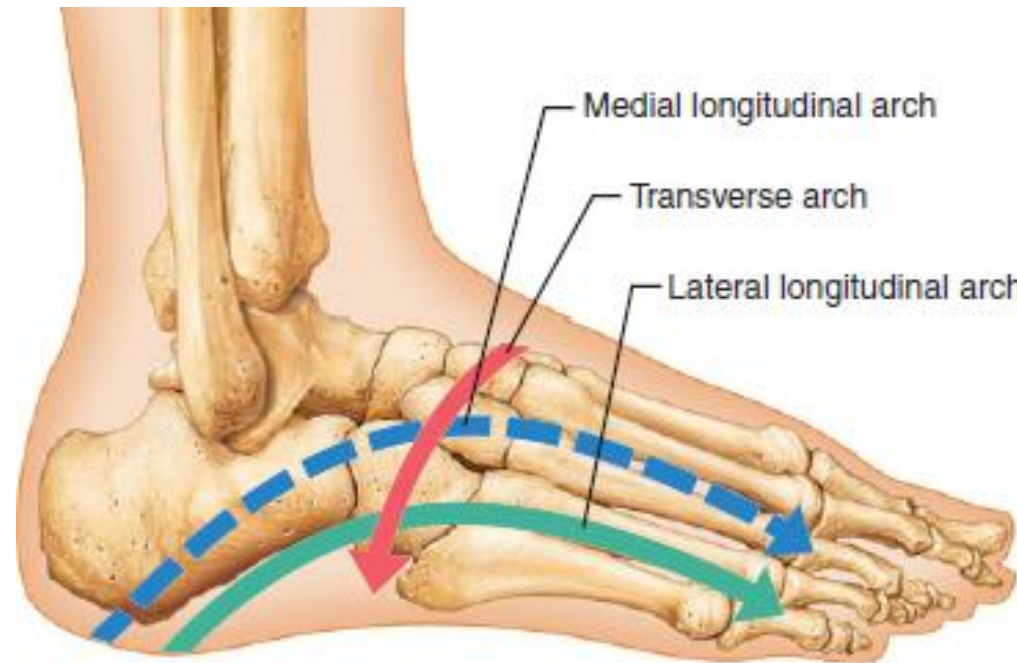
Types of arches...

□ Medial longitudinal arch's

The **talus** is the keystone of this arch.

□ Joints

- The main joints of the **medial longitudinal arch** are
- **Talocalcaneonavicular and subtalar joints.**



Types of arches...

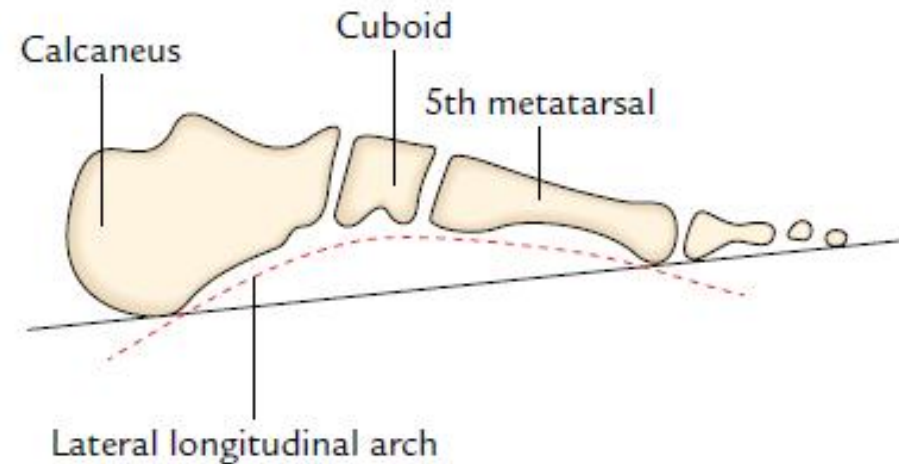
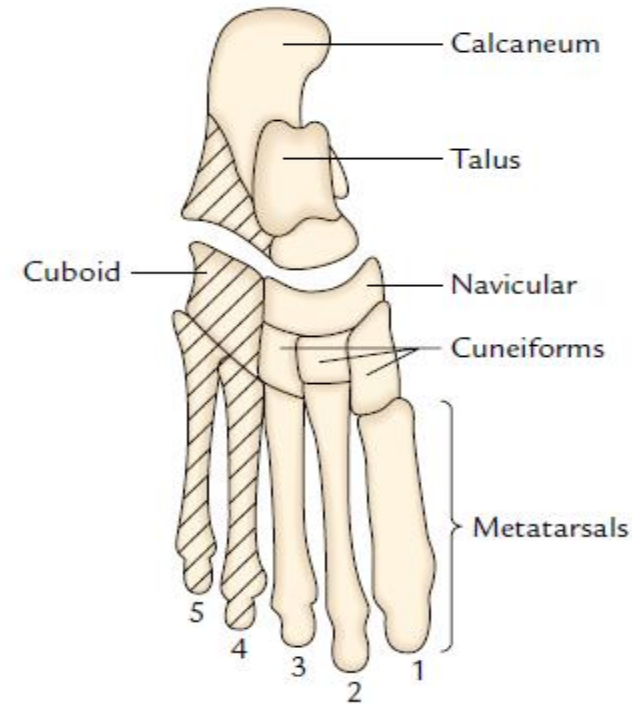
❑ Lateral Longitudinal Arch

- It is formed by the **calcaneum, cuboid, lateral two metatarsals**.

❑ Pillars

The lateral tubercle of the calcaneum forms the **posterior pillar** of the lateral longitudinal arch.

2. The heads of the lateral two metatarsals form the **anterior pillar** of the lateral longitudinal arch.



Types of arches...

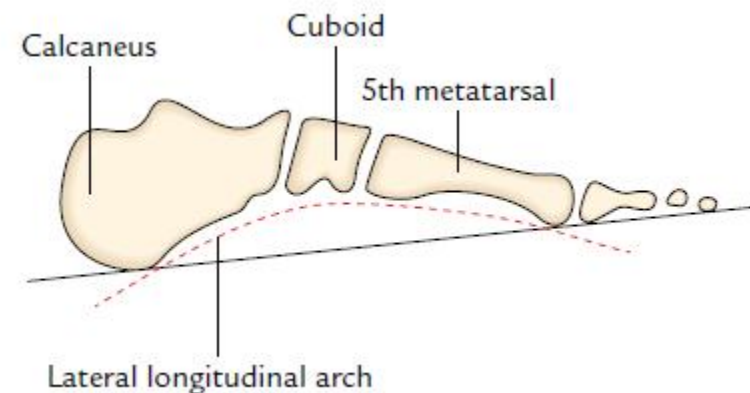
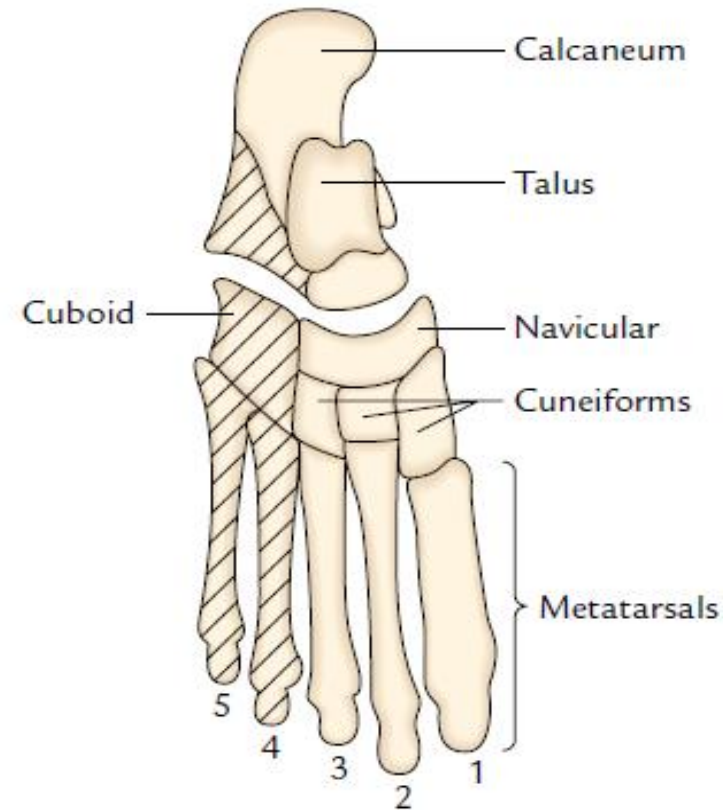
❑ Lateral longitudinal arch's...

❑ Summit

- The lateral longitudinal arch lies at the level of articular facets on the superior surface of the calcaneum (i.e., at the level of **subtalar joint**)..

❑ Joints

- The main joints of the lateral longitudinal arch are **Calcaneo-cuboid joints**.
- **NB:** The **lateral longitudinal** arch being **lower and less mobile** than the medial longitudinal arch is adapted for transmission of weight and thrusts

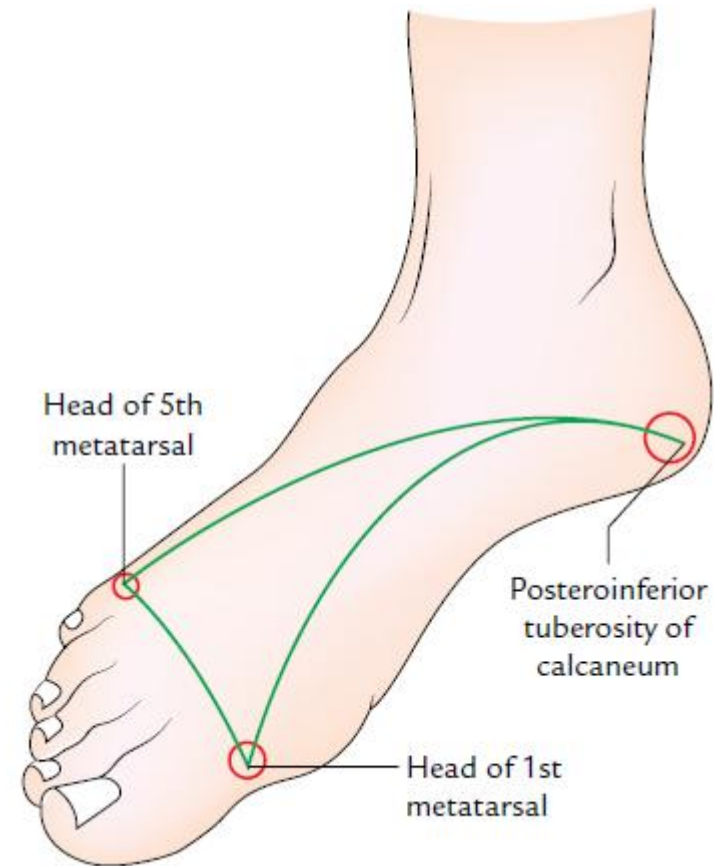


Types of arches...

❑ Transverse arch

❑ Anterior Transverse Arch

- The heads of the metatarsals form the **anterior transverse arch**.
- **It is a complete arch** because during standing position the heads of first and fifth metatarsals come into **contact to the ground** and form the two ends of the arch

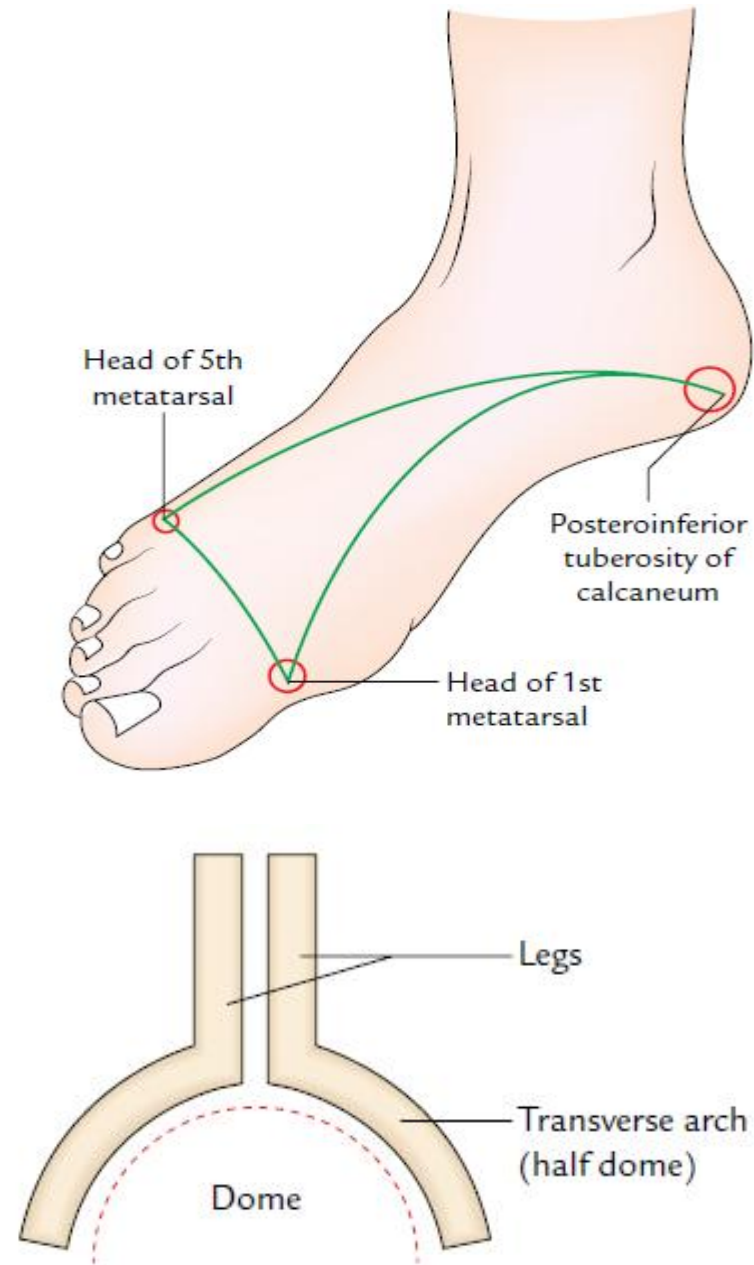


Types of arches...

❑ Transverse arch

❑ posterior Transverse Arch

- is formed by greater parts of the tarsus and metatarsus.
- It **is an incomplete arch** because only its lateral end comes into contact with the ground during standing position.
- It forms only half of the dome in one foot.
- The complete dome is formed when the two feet are brought together

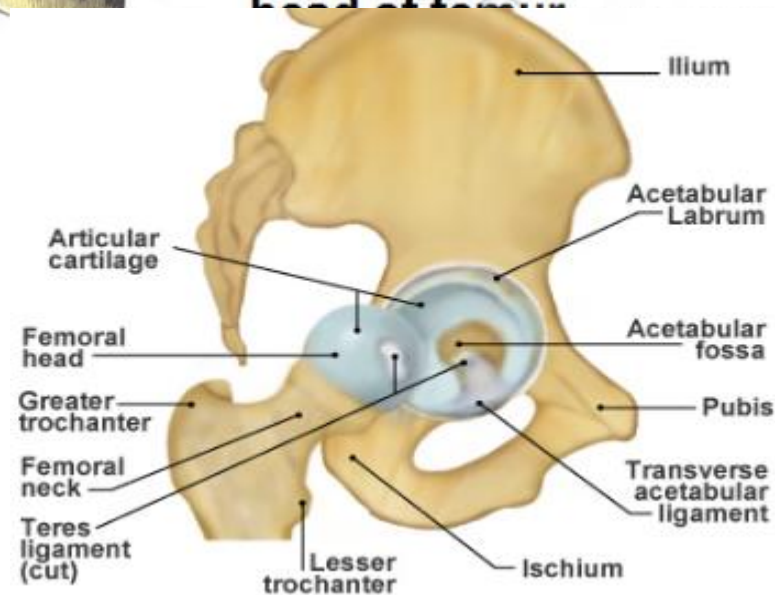
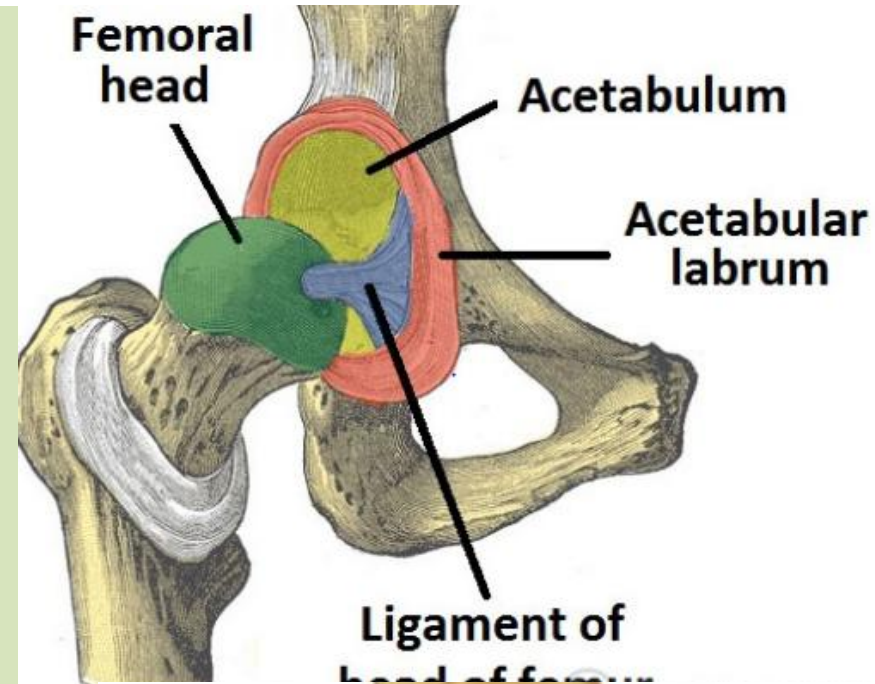


Hip joint& ligaments

- Synovial joint ball-socket variety
- The head of the femur articulates with the **horse-shoe shaped** acetabulum of the hip bone to form the hip joint
- The *head of femur* forms more than half of a sphere.
- The ***fovea capitis*** for **ligamentum teres**.

2. **The acetabulum has**

- **Lunate surface**
- **Acetabular notch**
- **Acetabular fossa**

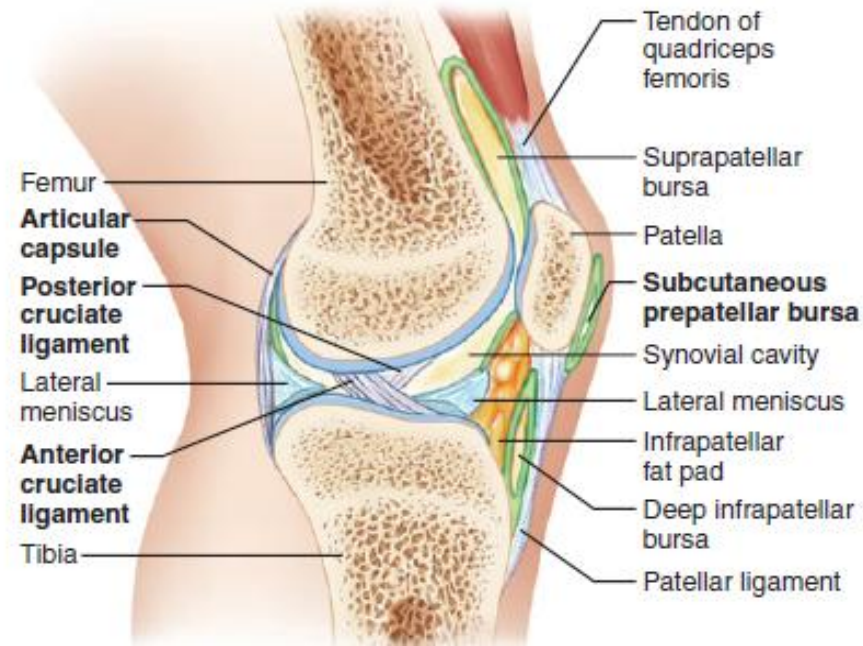
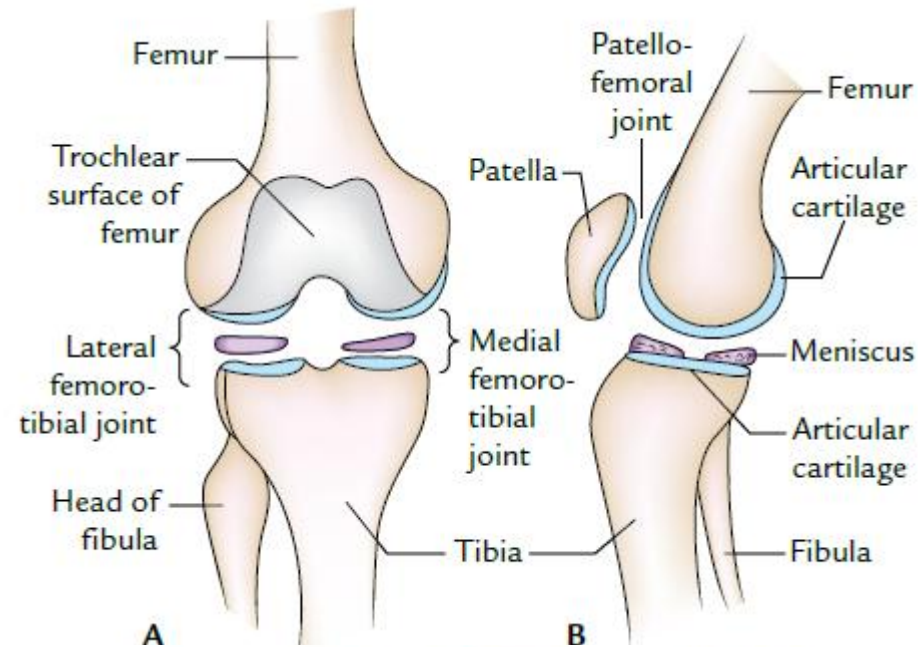


Ligaments of hip

Ligament Name	Main Function*
Iliofemoral inverted “Y” shape	Prevents hyperextension during standing
Pubofemoral	Limits abduction and extension
Ischiofemoral	Prevents hyperextension
Lig. of Head	Weak, contains artery, may limit adduction
Transverse Acetab.	Helps hold head in acetabular fossa

Knee joint & ligaments

- The knee joint is the **largest and most complicated joint in the body.**
- It is the major weight-bearing joint in the body.
- It is prone to undergo degenerative changes with advanced age
- There are two ligaments in the knee joint
- **Extra capsular ligament**
- **Intracapsular ligament**



Knee joint& ligaments

❑ Extra capsular Ligaments

■ Ligamentum Patellae

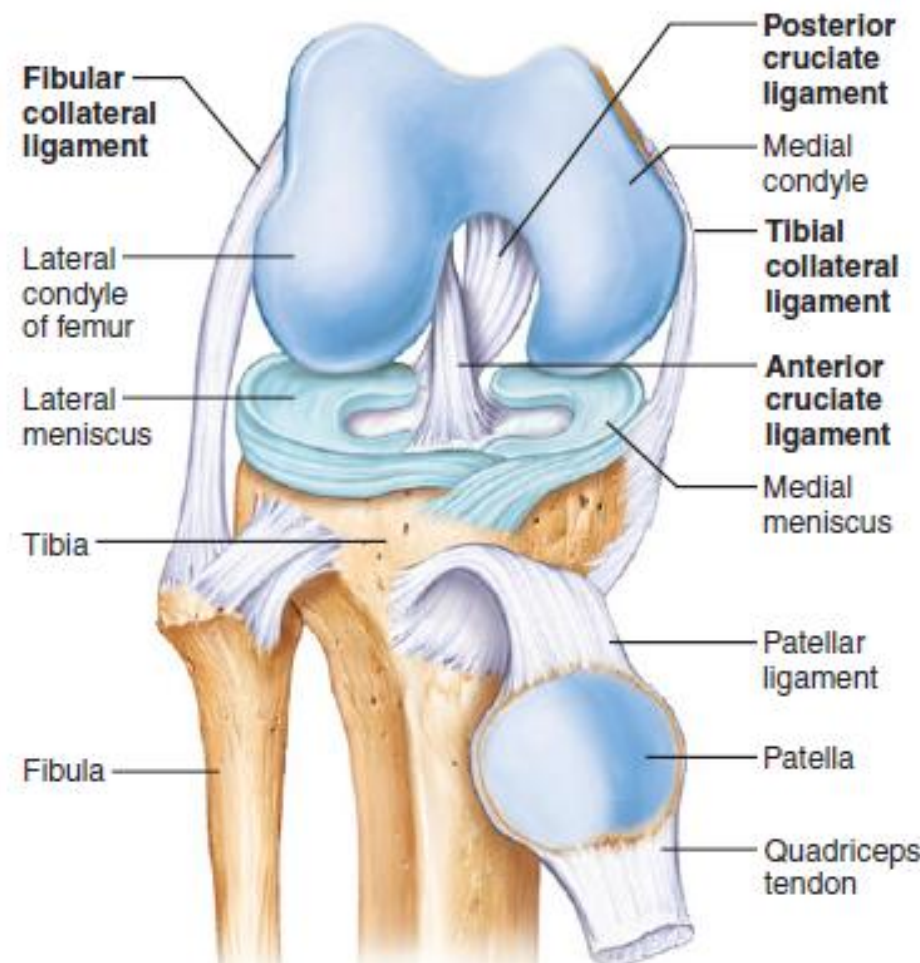
(a continuation of the Quadriceps Femoris muscle)

■ Lateral Collateral Lig.= FCL

■ Medial Collateral Lig.= TCL

■ Oblique Popliteal Lig.

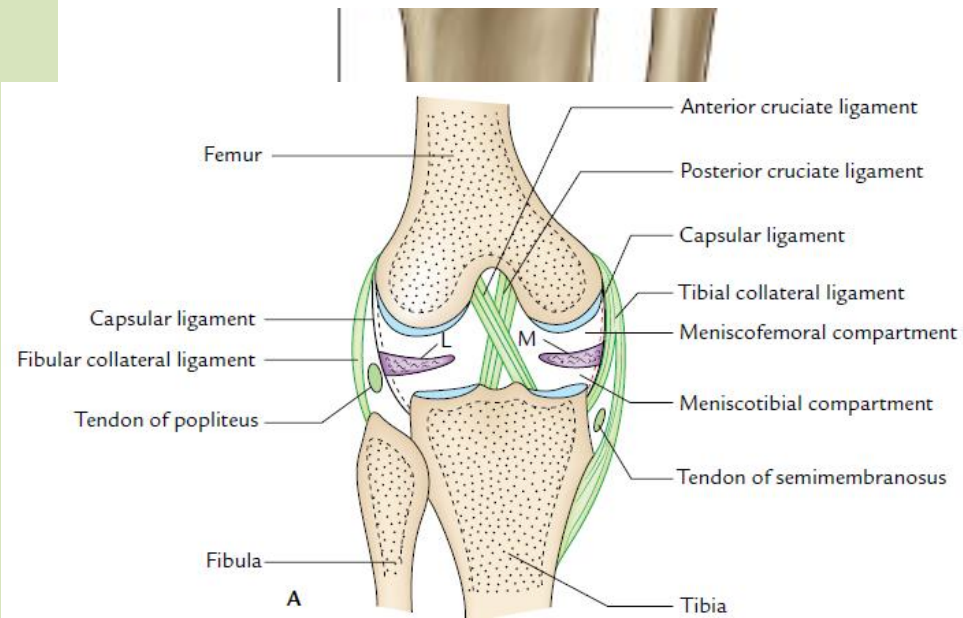
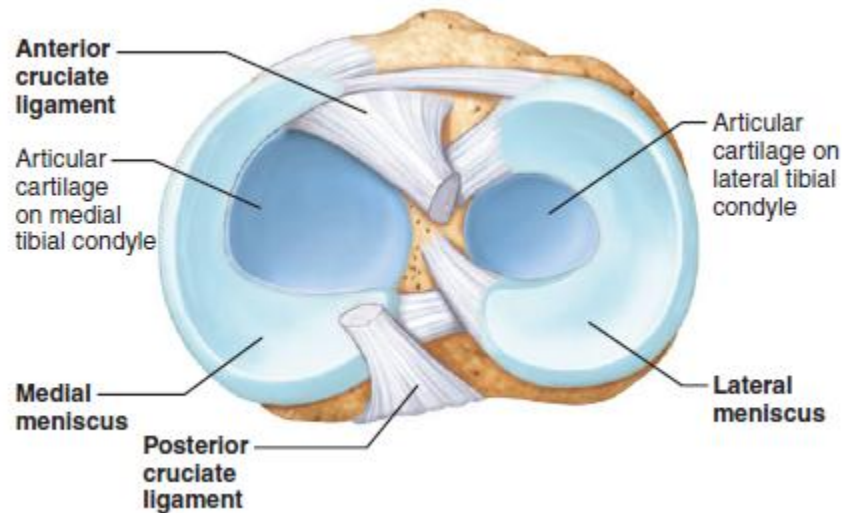
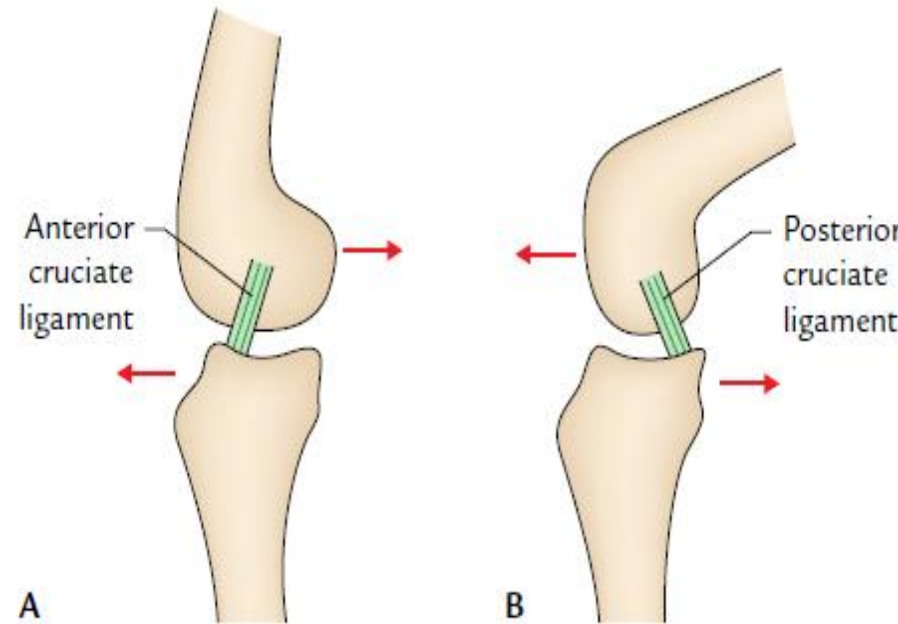
(derived from the Semimembranosus muscle).



Knee joint& ligaments

❑ Intracapsular Ligaments

- **Cruciate Ligaments** : 2 strong ligaments that cross each other within the joint cavity .
 - ACL
 - PCL
- **The two menisci (medial and lateral)**

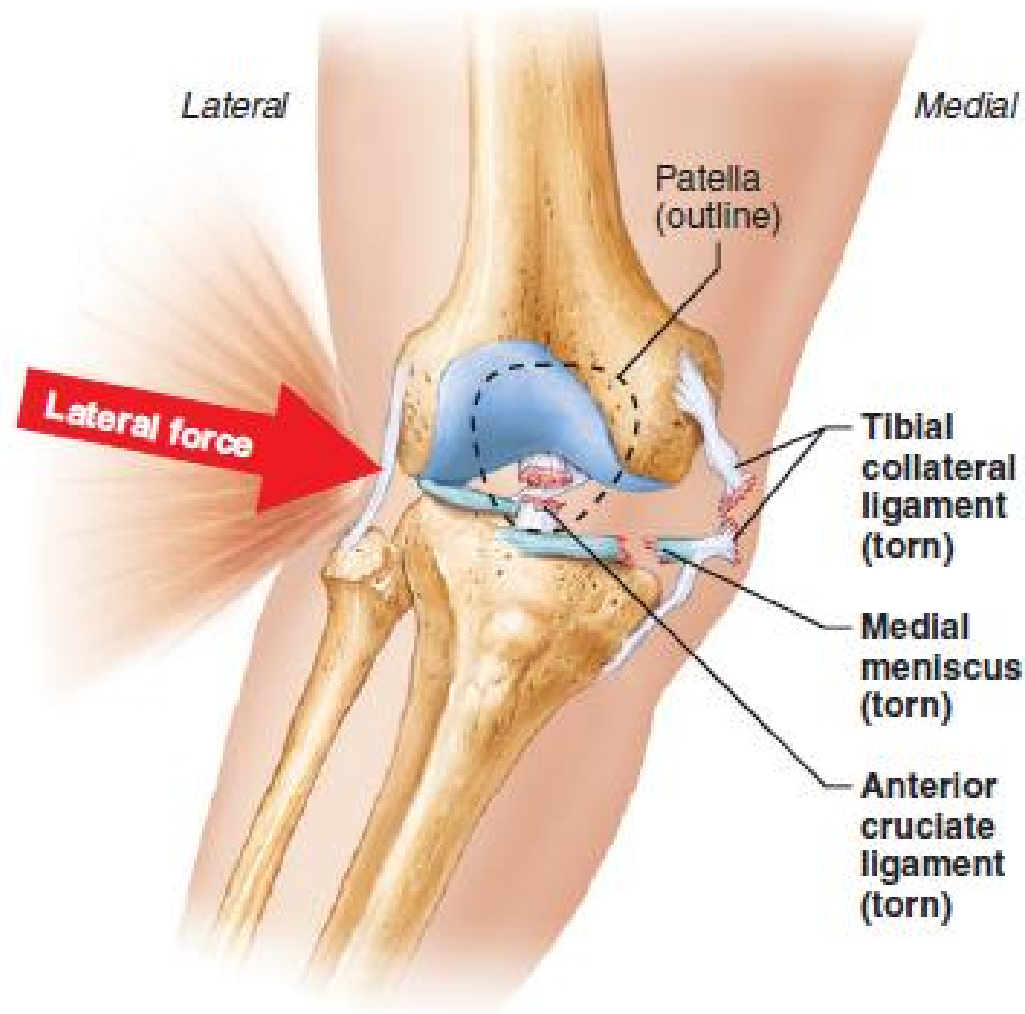


Knee joint injury

- Injury of three structures at the same time

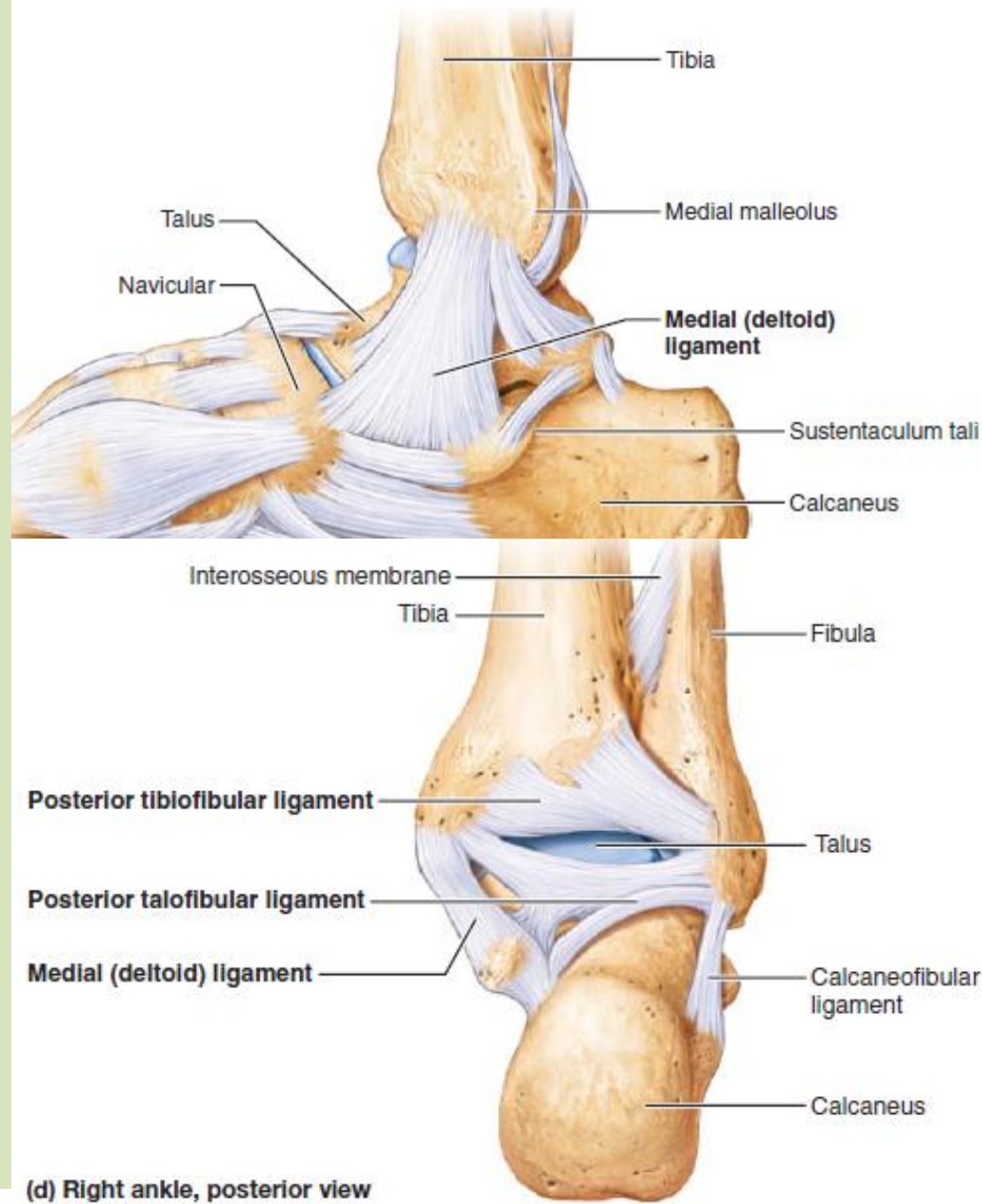
☐ UN Happy tried injury

- ACL
- PCL
- M Meniscus



Ankle joint

- The joint b/n the **inferior end of tibia & fibula with talus**
- It is **hinge type synovial joint**
- It **allow dorsiflexion & plantar flexion**
- The **ankle joint has capsule anteriorly & posteriorly is thin**
- **Medially & laterally thickened form ligaments**
- **Medially(Deltoid)- strong**
- **Laterally –weak ligaments**



Clinical correlation on lower limb

fracture associated with nerves

1. Fracture of the Fibular Neck:

- May cause injury to *common peroneal nerve* which laterally winds around the neck of the fibula.
- This results in paralysis of all muscles in the anterior & lateral compartments of the leg and presents with Foot Drop.

2. Bumper Fracture:

- Is a fracture of the lateral tibial condyle, caused by an automobile bumper, and is associated with a *common peroneal nerve injury*

Piriformis Syndrome & Sciatica

Piriformis Syndrome & Sciatica:

- Is a common condition in which the piriformis muscle irritates and places pressure on the sciatic nerve: Sciatica
→ pain across sensory distributor of sciatic nerve
- Sciatica can be caused IVD prolapse, pressure on one or more roots of sacral spinal nerves or it can be intra-pelvic tumor or inflammation of sciatic nerve or its terminal branches.

Gluteal Gate / Gluteus Medius Limp

- Is a tilting (dipping) or waddling gait characterized by the pelvis falling toward the unaffected side when the opposite leg is raised. It results from paralysis of the gluteus medius muscle, and damage to the superior gluteal nerve.

Positive Trendelenburg's Sign:

- Is seen in paralysis of gluteus medius, a fracture of the femoral neck or in dislocated hip joint.
- → *If the right gluteus medius is paralyzed then:*
 - 1). The unsupported left pelvis will drop below horizontal level when the patient is asked to stand on his right leg & raise his left leg clear of the ground;
 - 2). when the patient is asked to walk, he/she tilts to the right (paralyzed side) to raise his left leg clear of the ground → Tilting/Dipping Gait.

Nerve Injuries of Lower Limb

Tarsal Tunnel Syndrome:

- Results from compression of the tibial nerve in the tarsal tunnel.
- May be caused by flat feet and excess weight.

Damage to Obturator Nerve:

- Causes a weakness of adduction and lateral swinging of the limb, because adductor and lateral rotator muscles of the thigh will be paralyzed.

Nerve Injuries of Lower Limb

Femoral nerve injury

- **Motor** – Quadriceps muscle paralyzed Causes impaired flexion of the hip an impaired extension of leg
- **Sensory**– Skin sensation lost in anterior & medial side of thigh, over medial side of leg & medial border of foot.

Injury to Superior Gluteal nerve:

- Weakened gluteus medius, gluteus minimus, tensor fascia lata resulted in gluteal gait.
- It is also caused by fracture of greater trochanter & hip joint dislocation

Gluteus maximus & injection

- **To avoid injury of the underlying sciatic nerve the injection should be given on upper outer quadrant of the buttock on gluteus maximus**
- **Additionally acute or chronic trauma of gluteus maximus muscle can cause bursitis**

Sciatic nerve injury

- **Motor** – Hamstring muscles paralyzed, all muscles below the knee are paralyzed
- Causes impaired extension at hip & impaired flexion at the knee, loss of dorsi-flexion and plantar flexion at the ankle, **foot drop**
- but weak flexion of knee possible because sartorius (femoral nerve) & gracilis (obturator nerve).
- **Sensory** – Sensation below the knee except medial, lower part of leg & medial border of foot (saphenous from femoral nerve)

Damage to Common Peroneal Nerve

- The most injured nerve in the lower limb.
- May occur from fracture of fibula neck, knee joint injury or dislocation
- **Motor**- muscles of anterior & lateral compartments of the leg paralyzed, namely: (**TA, EDL, EHL, EHB, EDB, & PT** supplied by **deep peroneal nerve**); and **peroneus longus & brevis** (supplied by **superficial peroneal nerve**):
 - ➔ loss of dorsiflexion & eversion of foot results.
 - ➔ as a result, opposing muscles: (plantar flexors of flexion of ankle joint & invertors at subtalar & transverse tarsal joints) cause the foot to be plantar flexed (foot drop) & inverted: this condition is known as **Equinovarus**.
- **Sensory** - Loss of sensation down the anterior & lateral aspect of leg & dorsum of the foot & digits.

Damage to Superficial Peroneal Nerve

- loss of **eversion** of foot because the lateral compartment muscles (**peroneus longus & brevis**) paralysed.
- But no foot drop
- **Sensory** - Loss of sensation down the lateral aspect of leg & dorsum of the foot & digits.

Damage to Deep Peroneal Nerve

Motor:

- Results in **Foot Drop** and a **high stepping gait**, because muscles of anterior compartment of the leg (*tibialis anterior, extensor digitorum longus & extensor hallucis longus*) are paralyzed.
- **Sensory:**
- Loss of sensation of cleft b/n big toe & 2nd toe.

Damage to Tibial Nerve

- **Motor**: All muscles of posterior compartment of leg & sole of the foot are paralysed: ➔ this causes ***loss of plantar flexion of foot & impaired inversion***
- ➔ Foot is dorsiflexed at ankle joint & everted at subtalar & transverse tarsal joints by opposing muscles: ➔ this condition is known as **calcaneo-valgus**.
- **Sensory**: loss of sensation of skin in sole of foot

Arterial Palpitation

- It is necessary to know the precise position of main arteries within the lower limb.
 - It may be necessary to arrest severe hemorrhage, or palpate different parts of the arterial tree in patients with arterial occlusion
1. **Femoral artery:** enters the thigh behind the inguinal ligament at a point midway the anterior superior iliac spine & symphysis pubis

A. Arterial Palpitation

2. **Popliteal artery**: felt by gentle palpation in depth of popliteal space by flexion of knee
3. **Dorsalis pedis artery**: b/n the tendons of extensor hallucis longus & extensor digitorum longus, and midway b/n the medial & lateral malleoli, on the front of the ankle.
4. **Posterior tibialis artery**: behind medial- malleolus & beneath flexor retinaculum and lies b/n flexor digitorum longus & extensor digitorum longus
 - pulsation b/n medial malleoli & heel

B. Arterial Injury

1. Injury to large femoral artery:

- **can cause rapid exsanguination of the patient.**
- **if the collateral circulation also is inadequate & the vein around is damaged: ➔ damage /necrosis/ to lower limb occurs.**

B. Arterial Injury

2 . Arterial occlusive disease of the leg:

- is common in men; ischemia of the muscle produce a cramp like pain
- with exercise, femoral artery obstructed, supply of blood to the calf muscle is inadequate, patient forced to limp or stop walking after limited distance,
- with rest, the ischemia (O₂ depletion) is corrected & pain disappears: ➔ this is called intermittent claudication

B. Arterial Injury

3. Anterior Tibial Compartment Syndrome:

- **Ischemic necrosis of the anterior muscles.**
- **It results from a compression of the anterior tibial artery by swollen muscles.**

Unhappy Triad of the Knee injury

Unhappy Triad of the Knee:

- May occur when foot is firmly planted and knee is struck from lateral side:-
 - a . Rupture of the **tibial/medial collateral ligament**: excessive abduction of tibia.
 - b. Tearing of **anterior cruciate ligament**: forward displacement of tibia.
 - c. Injury to **medial meniscus**. Knock-Knee/Genu Valgum:
 - Is a deformity in which the tibia is bent or twisted laterally.
 - It may occur as a result of collapse of lateral compartment of knee & rupture of tibial collateral ligament.

The end